

Finite congruence lattices: characterization and undecidability

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Abstract

We give an explicit colored-graph characterization of the finite nonempty lattices that occur as full congruence lattices of finite algebras, and prove that deciding this representation property is undecidable. In particular, the finite lattice representation problem has a negative answer. We also prove that recognition of full subgroup intervals in finite groups is undecidable.

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1 Introduction

Grätzer and Schmidt proved that every algebraic lattice is the full congruence lattice of an algebra [16, Theorem I]. The carrier in that theorem need not be finite. The finite lattice representation problem asks whether every finite lattice is the congruence lattice of a finite algebra. The operations of the algebra are unrestricted: their number, arities, and tables may depend on the lattice. Thus a negative answer must exclude all finite algebras, rather than only groups, rings, or algebras in a fixed variety.

There is also a decision question: given the finite order table of a lattice, can one determine whether such a finite representation exists? McNulty records this precise question separately from the universal representation problem [32, p. 30]. We prove that no such decision procedure exists, and hence that not every finite lattice has a finite representation. We also construct a nonrepresentable family directly. Both conclusions must be distinguished from the effective search for affirmative witnesses, which is possible.

The relation with finite groups is particularly close. Pálffy and Pudlák proved that every finite lattice is a congruence lattice of a finite algebra if and only if every finite lattice is an interval in a subgroup lattice of a finite group [37, Theorem 2]. This is an equivalence between two universally quantified statements. It does not assert that these two kinds of representation are equivalent for each individual lattice. Our reductions below are proved for the particular lattice families to which they are applied.

Finite partition embeddings do not settle the problem: every finite lattice embeds in the equivalence lattice of a finite set [41, Theorem 7.10], but the image need not be the full congruence lattice of an algebra on that set. Positive representation results include all finite lattices of width at most two [44, Theorem 4.2]. Small examples likewise give little indication of the obstruction.

DeMeo’s analysis leaves only one possible exception among lattices with at most seven elements [8, Theorem 6.1.1 and Figure 6.2]. We represent that exception explicitly in [Section 3](#). The obstruction in this paper instead uses long systems of affine subgeometries and the restrictions that their subgroup labels impose simultaneously in several ambient simple groups.

1.1 Statements

A finite algebra means a nonempty finite set A , together with a finite list of total operations of finite nonnegative arity. The list may be empty. An operation of arity zero specifies a constant and imposes no additional condition on an equivalence relation. Write $\text{Con}(A)$ for the lattice of all congruences of the algebra. For a nonempty finite lattice L , let

$$\mathcal{R}(L) \iff L \cong \text{Con}(A) \text{ for some finite algebra } A.$$

Every representation in this paper is onto the entire congruence lattice. An embedding into a proper sublattice is not a representation in this sense.

Theorem 1.1. *The following statements hold.*

- (i) *A finite nonempty lattice is representable if and only if it satisfies the finite colored-graph criterion in [Theorem 2.2](#). For each candidate graph size the criterion is decidable by a finite test.*
- (ii) *There is no deterministic Turing machine that, given the order table of a finite nonempty lattice, always halts and decides $\mathcal{R}(L)$.*
- (iii) *There is a nonrepresentable finite lattice L_0 of minimum cardinality, specified by the finite formulas in [Section 17](#). Every finite lattice of smaller cardinality is representable, and $|L_0| > 7$.*
- (iv) *Given the order table of a finite nonempty lattice L , it is undecidable whether $L \cong [H, G]$ for some finite group G and subgroup $H \leq G$, where $[H, G]$ is the full subgroup interval.*

Part (ii) resolves the finite lattice representation problem negatively, since otherwise the constant affirmative algorithm would be a decider. We also construct a nonrepresentable family directly, before the undecidability argument. Part (iii) specifies an actual finite order table by an unambiguous finite description. Its constants include one ordinary Busy Beaver integer. The description does not evaluate that integer, the minimum cardinality, or the entries of the order table.

The distinction between the three congruence-lattice assertions is useful. The criterion in Part (i) gives a search that terminates on representable lattices; it does not provide a computable stopping rule for negative instances. The proof of Part (ii) establishes precisely that there is no uniform such rule. In Part (iii), a single finite search range is specified mathematically, using the Busy Beaver constant; no uniform algorithm for evaluating these constants is asserted.

1.2 Main ideas and proof organization

We first make the affirmative search explicit. Edge colors of a finite complete graph encode principal congruences, and label-decreasing selfmaps replace the choice of an algebraic signature. A path condition ensures that the recovered relations exhaust the congruence lattice. For fixed graph size this is a finite test. Enumerating graphs therefore finds every affirmative witness, and a decider exists if and only if a computable bound on witness sizes exists ([Section 2](#)). This is the effective bridge used in the undecidability proof.

The graph method itself is established background. Finite unary closure already gives positive semidecidability [8, Section 3.2, Theorem 3.2.1]. Pudlák’s valuation-colored construction uses graph paths and stable maps, which preserve an edge color or collapse the edge; its general representation can have an infinite carrier [40, pp. 270–272]. Kenney’s graphical-algebra framework already uses label-decreasing maps, principal-congruence complete graphs, edge-image paths, and reconstruction by all unary graph endomorphisms [22]; see also Kenney [21, pp. 51–58]. We present the finite criterion directly in terms of the lattice and one finite graph, with the proof and bounded arithmetic needed for the final minimum specification.

The first structural construction produces a negative family, without using a hypothetical decider. We assemble a simple self-dual lattice from private truncated affine charts. On a least carrier, every unary operation is a permutation or a constant, but the permutation action need not be transitive. A full-diagonal construction in a semidirect power represents the dual lattice as an ordinary subgroup interval; self-duality gives the required orientation. Normal-subgroup tests then leave either an almost simple interval or extension-kernel data. A four-dimensional affine grid excludes the latter, yielding an ordinary almost simple interval for the whole lattice or an initial point link (Section 4).

The ordinary-interval step develops methods used by Pálffy and Pudlák, including restriction along idempotents on a least carrier and, in the later subdirect descent, the correspondence between transitive actions and stabilizer intervals [37, Lemmas 1 and 3]. The subsequent extension analysis is related to the twisted-wreath and signalizer approaches of Baddeley and Lucchini, Baddeley, Börner, and Aschbacher [1–3, 5]; Pálffy compares their hypotheses and interval reductions [36, Section 4]. These results concern different classes of lattices and do not supply a pointwise reduction for the present family. Here the normal tests, compatibility of actual simple factors, and extension-kernel grid provide that reduction directly.

Recognizing abstract simple-group types at consecutive labels would not suffice for the obstruction. The affine patterns force actual nested simple subgroups, while exclusions are imposed relative to every earlier ancestor at once (Sections 5 to 7). Bounded Lie rank is excluded by algebraic subgroup tests and the sufficiently-general subgroup method of Larsen and Pink [25, Theorem 0.5]; Section 16 supplies an effective version for the later size ceiling. In high rank, one must further exclude field roots and tensor decompositions in every ancestor. These protections make the nested actions compatible with rational composition. Highest-weight theory then produces universal cuts in the full weight multiset, and multiplicities rule out a sufficiently long protected path (Sections 8 and 9). Choosing more affine generations than this bound gives the nonrepresentable family.

Undecidability requires a different uniformity statement. A weighted template is a subgroup interval built from finitely many permutation sets, called units, whose cardinalities encode integer values. A positive solution of an arithmetic circuit equation supplies such a template, already finitely representable by the coset construction. The number of units depends only on the circuit’s syntax, not its solution. We prove that any finite representation yields a new reading of all the encoded values in one common partition into cells. Equal totals encode addition, and square cardinalities encode multiplication; the new values need not equal the defining ones.

Two independent bounds make this useful. First, cell sizes are bounded computably in the size of a representation. Second, the number of elements of a template is bounded computably in its number of units, independently of their weights. A hypothetical decider would thus bound the sizes of affirmative witnesses for every template arising from a fixed circuit. Reading such a bounded witness would replace any positive solution by a computably bounded one. Searching the resulting finite box contradicts Diophantine undecidability. No algorithm is required to construct the unknown-weight template from the circuit (Sections 14 and 15). The same replacement argument, using bounds on top-group orders instead of algebra carriers, proves undecidability of full subgroup-interval recognition.

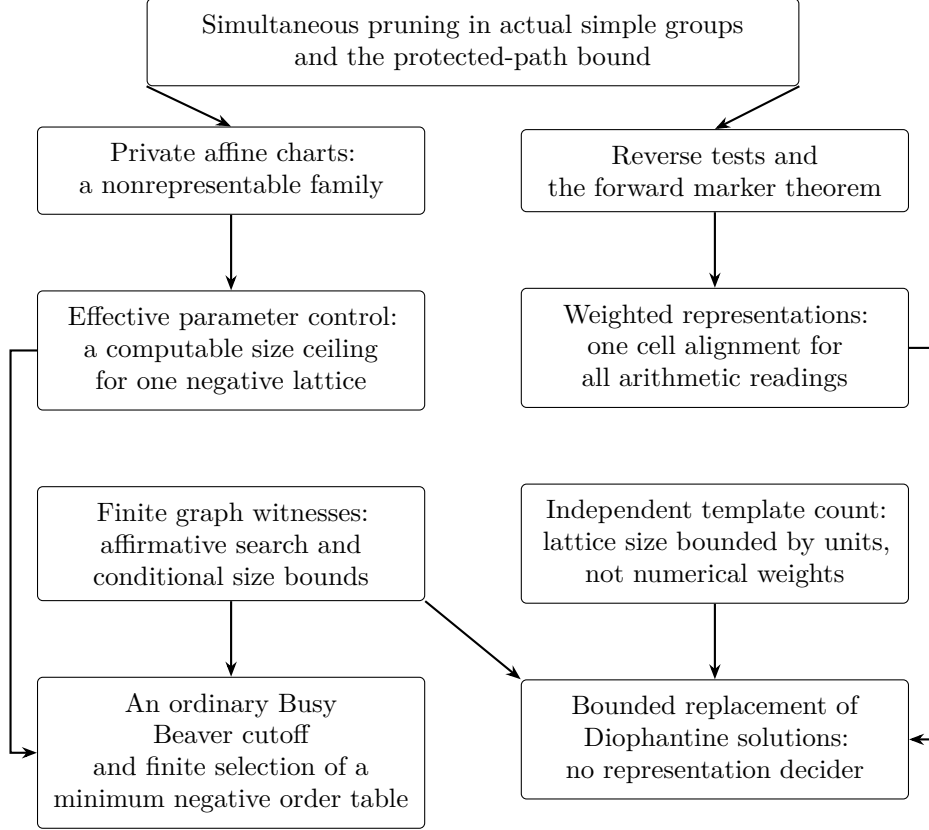


Figure 1: The two uses of the structural argument. The affine branch gives a negative lattice and a computable size ceiling independently of any hypothetical decider. The weighted branch combines simultaneous arithmetic readings with a separate, weight-independent count; a decider would then bound positive Diophantine solutions. Finite graph witnesses also supply the bounded searches used, with the size ceiling, in the minimum-table specification.

The common reading is the structural burden of this second branch. Reverse alternating tests reuse the pruning argument through elementary-linear charts. A forward marker theorem instead uses marker omissions and their three-coatom constraints to reach the protected-path bound (Sections 10 to 12). Together with the weighted normal tests, they force the remaining units into an actual alternating power. Normal and parity tests then recover one cell alignment in which every total and square comparison holds, even after the nontrivial lattice images needed by the least-carrier reduction (Section 13). The separate counting argument concerns the literal templates, not those readings. It uses nonabelian coordinate compression, effective integer profiles, bounded-tuple encodings of diagonal matchings, and Maróti’s primitive-group order theorem [28, Theorem 1.1].

Finally, Section 16 gives an ordinary computable size ceiling for one member of the nonrepresentable affine family. This step is independent of both the hypothetical decider and the later Busy Beaver constant. The latter constant bounds the witness sizes returned by the prescribed affirmative graph searches up to the ceiling. Explicit bounded arithmetic then selects the first negative order table, ordered first by cardinality (Section 17).

1.3 Conventions and dependencies

All finite lattices and algebra carriers are nonempty. The one-element lattice is included. A subgroup interval $[H, G]$ always means the whole interval. Invariant-subgroup intervals in a normal direct power will be identified explicitly and are not silently substituted for ordinary intervals. The notation $\text{core}_G(H)$ denotes the largest normal subgroup of G contained in H . Quotienting by a normal subgroup already contained in the base preserves the full interval above the base.

Labels such as K_X always denote actual subgroups in the realization or quotient currently under discussion. In a reverse affine chart, a smaller flat has a larger subgroup label. A normal subgroup is *retained* when it lies in the current base; it is *missing* when the relevant normal test forces the base to supplement it. An element y of a lattice is left-modular if

$$x \vee (y \wedge z) = (x \vee y) \wedge z \quad \text{whenever } x \leq z.$$

The field of an affine chart and the field of a group's natural module need not be the same. Group bases, field letters and local ranks are reintroduced when their ambient objects change.

We use the classification and standard structure of finite simple groups, algebraic groups and their representations. The exact conventions and applicable ranges are recorded in [Section 4](#) and at their later uses. Established results are cited; the structural tests, compatibility arguments, weighted readings and effectivity deductions specific to the constructions are proved below.

2 A finite colored-graph characterization

We first give a criterion whose individual witnesses can be checked by finite combinatorial calculations. Throughout this section, L is a finite nonempty lattice, with least element 0.

The graph language follows the valuation and path approach of Pudlák [40, Definitions 1–3 and pp. 271–272]; the unary-closure criterion is recorded in DeMeo [8, Theorem 3.2.1]. We give the full finite argument, including its witness-size consequence. Kenney's graphical-algebra formulation contains the label-decreasing-map and edge-image-path reconstruction used here [22][21, pp. 51–58]. Our statement isolates a finite test without an auxiliary graphical composition operation.

Definition 2.1 (Admissible placements). An L -colored complete graph consists of a finite nonempty vertex set A and a symmetric function $d : A \times A \rightarrow L$, with $d(p, p) = 0$. Loops are included among its edges. A placement into another such graph (A', d') is a map $f : A \rightarrow A'$; it is *admissible* if

$$d'(f(p), f(q)) \leq d(p, q) \quad (p, q \in A).$$

For a nonempty set S of edges of (A, d) , let Γ_S be the undirected graph on A whose marked adjacencies are the pairs $\{f(p), f(q)\}$, where $\{p, q\} \in S$ and $f : A \rightarrow A$ is admissible. Collapsed pairs are allowed and do not affect connectivity. Write $a_S = \bigvee_{\{p, q\} \in S} d(p, q)$.

Theorem 2.2 (Graph criterion). *The lattice L is the full congruence lattice of a finite algebra if and only if there is an L -colored complete graph (A, d) satisfying all three conditions below.*

- (i) *If $p \neq q$, then $d(p, q) \neq 0$, and*

$$d(p, q) \leq d(p, u) \vee d(u, q) \quad (p, q, u \in A).$$

- (ii) *Whenever $a \not\leq b$ in L , there are vertices p, q with $d(p, q) \leq a$ and $d(p, q) \not\leq b$.*

- (iii) *For every nonempty edge set S , vertices p, q satisfying $d(p, q) \leq a_S$ lie in the same connected component of Γ_S .*

For each fixed $|A|$, these conditions involve only lattice inequalities, finite vertex labelings, and paths.

Proof. Suppose first that $\phi : L \rightarrow \text{Con}(\mathcal{A})$ is an isomorphism, where $\mathcal{A}$ has finite carrier A . Define

$$d(p, q) = \phi^{-1}(\text{Cg}_{\mathcal{A}}(p, q)),$$

where $\text{Cg}_{\mathcal{A}}(p, q)$ is the least congruence containing (p, q) . Equality is the bottom congruence, so the diagonal colors are zero and the other colors are nonzero. The congruence $\text{Cg}_{\mathcal{A}}(p, u) \vee \text{Cg}_{\mathcal{A}}(u, q)$ contains (p, q) by transitivity, proving (i). If $a \not\leq b$, order reflection of ϕ gives a pair in $\phi(a) \setminus \phi(b)$, and that pair proves (ii).

A *one-variable translation* of an operation of $\mathcal{A}$ is obtained by leaving one argument free and fixing all its other arguments. Every such translation preserves every congruence, and is therefore an admissible self-placement. Admissible self-placements are closed under composition. Consequently a translation sends a marked adjacency of Γ_S to another marked adjacency, possibly a loop. It follows, by applying the translation to paths, that the equivalence ρ_S of belonging to the same connected component is stable under all translations. An equivalence with this property is a congruence: change the entries of an operation's input tuple one at a time. Nullary operations impose no further condition.

The identity placement is admissible, so ρ_S contains each pair in S . Its corresponding lattice element is thus at least a_S . If $d(p, q) \leq a_S$, then $(p, q) \in \rho_S$, which proves (iii).

Conversely, suppose (A, d) satisfies the three conditions. Give A one unary operation for each admissible self-placement. There are finitely many such maps, so this defines an allowed finite algebra $\mathcal{B}$. For $a \in L$ set

$$\theta_a = \{(p, q) \in A^2 : d(p, q) \leq a\}.$$

The diagonal convention, symmetry, and (i) show that θ_a is an equivalence relation. Admissibility makes it a congruence of $\mathcal{B}$. The map $a \mapsto \theta_a$ is order preserving, and (ii) makes it order reflecting.

Let θ be any congruence of $\mathcal{B}$, and take for S all its edges, including all loops. This is a nonempty set, even if A has one vertex. Put $a = a_S$. By construction $\theta \subseteq \theta_a$. Each marked adjacency of Γ_S belongs to θ , since every admissible placement is an operation of $\mathcal{B}$; hence so does every connected pair. Condition (iii) gives $\theta_a \subseteq \theta$. Thus $\theta = \theta_a$, proving surjectivity and the desired isomorphism. This argument includes the one-element lattice, represented by the one-vertex graph and the one-element algebra. $\square$

Corollary 2.3 (Enumeration and bounds). *Representability is semidecidable. Moreover, a total algorithm deciding representability of finite lattice order tables exists if and only if there is a total computable function $b : \mathbb{N}_{>0} \rightarrow \mathbb{N}_{>0}$ such that every representable lattice L has a graph witness in Theorem 2.2 with at most $b(|L|)$ vertices.*

Proof. For each positive integer k , enumerate all colorings on a k -element vertex set and check the three finite conditions. Running through $k = 1, 2, \dots$ halts affirmatively exactly on representable lattices. A computable bound b makes this search finite and decides representability. Conversely, given a decider, enumerate the finitely many order tables on each fixed n -element set, discard tables that are not lattices, and use the decider to retain the representable ones. Run the witness search to completion for each retained table, and take the maximum of the resulting witness sizes and 1. This computes the required $b(n)$. $\square$

The later undecidability argument will therefore also exclude every such computable witness bound.

3 Representations through seven elements

DeMeo's small-lattice reduction [8, Theorem 6.1.1 and Chapter 6] provides finite-algebra representations for every lattice of at most seven elements, with one possible exception. We now represent that remaining lattice directly. The closure constructions used for the other small lattices include Snow's parallel-sum construction: take a disjoint union and add a new bottom and top. A one-element summand is allowed [44, Lemma 3.9]; in particular, the parallel sum of a four-element chain and a singleton is represented [44, Lemma 4.1].

Definition 3.1. Let L_7 have elements $0, 1, a, b, c, d, x$, with 0 least, 1 greatest, and with the comparabilities among the five interior elements exactly

$$a < c, \quad b < c, \quad b < d.$$

In DeMeo's group-interval labeling these elements are J_1, J_2, M_2, M_1, K , respectively.

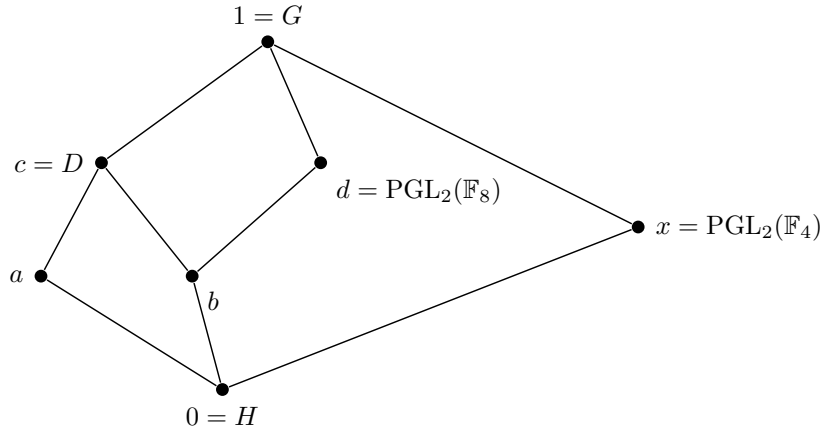


Figure 2: The lattice L_7 realized as the subgroup interval $[H, G]$, where $G = \text{PGL}_2(\mathbb{F}_{64})$ and $H = \text{PGL}_2(\mathbb{F}_2)$. The group D is the dihedral eigenpair stabilizer of order 126; its subgroups a and b containing H have orders 42 and 18, respectively. Among the five interior elements, the only strict comparabilities are $a < c$, $b < c$, and $b < d$.

Proposition 3.2. *With their usual subfield inclusion, the groups*

$$H = \text{PGL}_2(\mathbb{F}_2) \leq G = \text{PGL}_2(\mathbb{F}_{64})$$

have subgroup interval $[H, G] \cong L_7$.

Proof. We first classify the overgroups $H \leq K \leq G$ by their Sylow two-subgroups and odd-order eigenpair stabilizers, and then determine their intersections. We use the projective-line actions, extending scalars to the algebraic closure when speaking of eigenlines. For any finite field Q , counting invertible matrices and dividing by scalar matrices gives

$$|\text{PGL}_2(Q)| = |Q|(|Q| - 1)(|Q| + 1).$$

In particular $|H| = 6$ and H induces S_3 on its subfield line. It has no common fixed point: $z \mapsto z + 1$ fixes only infinity, and its conjugate by inversion fixes only zero. Also

$$C_G(H) = 1. \tag{3.1}$$

Indeed, a transformation commuting with these two involutions fixes zero and infinity individually, so is a scaling; commuting with $z \mapsto z + 1$ then makes that scaling the identity.

Fix an overgroup $H \leq K \leq G$. In characteristic two, a nonidentity element of G either has one Jordan block up to scalar, and has order two, or has two distinct eigenlines, and has odd order. The latter assertion follows by diagonalizing over a finite extension and taking the ratio of the two eigenvalues. For each unordered eigenpair, its individual stabilizer in K is cyclic of odd order, since the diagonal ratio embeds it in the multiplicative group of a finite field. The nonidentity parts of the nontrivial such stabilizers partition the nonidentity odd-order elements: each such element determines its unique eigenpair. Choose representatives I_i for their K -conjugacy classes and put $t_i = |I_i| \geq 3$. Their normalizers stabilize the eigenpairs, so

$$|N_K(I_i)| = n_i t_i, \quad n_i \in \{1, 2\}.$$

The full translation group is a Sylow two-subgroup of G , by the order formula; all Sylow two-subgroups are its conjugates. A Sylow two-subgroup of K has order $s \geq 2$ and hence lies in a translation group, fixing a unique point of the $\mathbb{F}_{64}$ -line. Distinct Sylow two-subgroups have distinct fixed points: at any point all involutions fixing it lie in its one translation group, whose intersection with K is itself a two-group. Let Ω be the set of these fixed points. It is a K -orbit of size $h > 1$, since otherwise H would fix its unique point.

At a point of Ω , take coordinates putting that point at infinity. The stabilizer K_∞ is affine. Its translation kernel has order s ; its multiplier image has odd order j and preserves the additive subgroup of translation parameters. Nonidentity multipliers act freely on nonzero parameters, so $j \mid s - 1$. Therefore $|K| = hsj$, and the number of involutions is $h(s - 1)$. Counting the identity, the involutions, and the odd-order elements yields

$$\sum_i \frac{1}{n_i} \left(1 - \frac{1}{t_i}\right) = 1 - \frac{s-1}{sj} - \frac{1}{hsj} = 1 - \frac{1}{j} + \frac{h-1}{hsj}. \quad (3.2)$$

The sum is nonempty because H contains an element of order three.

The case $j = 1$. The right side of Equation (3.2) is less than $1/s$; each summand is at least $1/3$. Thus $s = 2$, and there is exactly one summand. It must have $n_i = 2$, because a summand with $n_i = 1$ is at least $2/3$. Substitution gives $t_i = h$, so $|N_K(I_i)| = 2h = |K|$. Hence K lies in the setwise stabilizer of one eigenpair. The order-three subgroup of H must fix the pair individually, since it cannot act nontrivially on two points. This eigenpair is therefore the one of that order-three subgroup.

The case $j > 1$. Now $s \geq 4$. The number j is one of the t_i : a generator of the multiplier image lifts to an affine map with nonidentity multiplier, two fixed points, and order j . The individual stabilizer of its two fixed points embeds in the multiplier image and contains that cyclic subgroup, so has order j . If the sum in Equation (3.2) had only this term, it would be at most $1 - 1/j$, contrary to the equation. The whole sum is less than 1. Since each summand is at least $1/3$, there are exactly two summands, and both have $n_i = 2$; otherwise their sum would be at least $2/3 + 1/3$. Writing t for the other order gives

$$1 - \frac{j}{t} = \frac{2(h-1)}{hs}. \quad (3.3)$$

In particular $t > j$, and $s(t - j) < 2t$ implies

$$0 < t - j < \frac{2j}{s-2} \leq \frac{2(s-1)}{s-2} \leq 3.$$

The two orders are odd, so $t = j + 2$. Equation (3.3) now gives $h(j + 2 - s) = j + 2$. Since $h > 0$, we have $j > s - 2$, and $j \leq s - 1$ therefore forces $j = s - 1$ and $h = s + 1$.

The multiplier realization gives $s - 1 \mid 63$. As s is a power of two with $4 \leq s \leq 64$, it follows that $s \in \{4, 8, 64\}$. The multiplier group is then exactly $\mathbb{F}_s^\times$ inside $\mathbb{F}_{64}^\times$. The translation-parameter subgroup, of size s , is stable under these scalars, so is a one-dimensional $\mathbb{F}_s$ -space, say $v\mathbb{F}_s$. Translations act freely on the other s points of Ω , and hence those points form one coset of $v\mathbb{F}_s$. An affine coordinate change thus makes $\Omega = \mathbb{P}^1(\mathbb{F}_s)$.

Its full setwise stabilizer in G is $\text{PGL}_2(\mathbb{F}_s)$: the images of infinity, zero, and one determine a projective transformation, and every ordered triple of distinct subfield points is the image of that triple under a subfield transformation. The equality $|K| = (s + 1)s(s - 1)$ now gives $K = g \text{PGL}_2(\mathbb{F}_s) g^{-1}$ for some $g \in G$. This conjugate is in fact the standard subfield group. Let σ be the entrywise s -power automorphism. It fixes both H and $g^{-1}Hg$ pointwise. Consequently $g\sigma(g^{-1})$ centralizes H , so Equation (3.1) gives $\sigma(g) = g$ in the projective group. Normalize one nonzero matrix entry of a representative of g to 1; all its entries are then fixed by σ . Thus $g \in \text{PGL}_2(\mathbb{F}_s)$, as required.

It remains to record the incidences between the groups in this list. The eigenpair of the order-three subgroup of H splits over $\mathbb{F}_{64}$. Its full setwise stabilizer D is dihedral of order $2 \cdot 63$ and contains H , generated there by a rotation of order three and a flip. Subgroups between H and D correspond to rotation subgroups containing the order-three one, after adjoining that same flip. Their orders are 6, 18, 42, 126, so there are precisely two strict intermediate groups, of orders 18 and 42.

The other proper overgroups are the standard subfield groups over $\mathbb{F}_8$ and $\mathbb{F}_4$, whose intersection is H : normalize one nonzero matrix entry to 1 and use $\mathbb{F}_8 \cap \mathbb{F}_4 = \mathbb{F}_2$. Over $\mathbb{F}_4$ the eigenpair is split, so its setwise stabilizer has order $2(4 - 1) = 6$; consequently $D \cap \text{PGL}_2(\mathbb{F}_4) = H$. Over $\mathbb{F}_8$ the pair is nonsplit, because $3 \nmid 7$. Its individual stabilizer has order $8 + 1 = 9$. To see the latter count directly, matrices fixing the two eigenlines centralize a matrix with irreducible quadratic minimal polynomial. Its invertible centralizer is the multiplicative group of the quadratic field extension, and quotienting by subfield scalars gives order $(8^2 - 1)/(8 - 1) = 9$. The flip from H doubles this order. Hence $D \cap \text{PGL}_2(\mathbb{F}_8)$ is exactly the order-18 group. We obtain the asserted interval by taking

$$c = D, \quad d = \text{PGL}_2(\mathbb{F}_8), \quad x = \text{PGL}_2(\mathbb{F}_4),$$

and taking a, b to be the groups of orders 42, 18, respectively. □

Lemma 3.3 (Coset-action correspondence). *For finite groups $H \leq G$, the full congruence lattice of the unary algebra given by the left G -action on G/H is isomorphic to $[H, G]$.*

Proof. For a G -invariant equivalence θ on G/H , put $K_\theta = \{g \in G : gH \theta H\}$. This contains H . If $g, g' \in K_\theta$, invariance applied to $gH \theta g'H$ gives $H \theta g^{-1}g'H$, so K_θ is a subgroup. Conversely, $H \leq K \leq G$ gives the invariant equivalence obtained by pulling equality back along $G/H \rightarrow G/K$. These constructions are inverse and preserve order. □

Theorem 3.4. *Every finite nonempty lattice of at most seven elements is representable.*

Proof. Proposition 3.2 and Lemma 3.3 represent L_7 by a finite unary algebra. DeMeo's small-lattice reduction supplies the other lattices [8, Theorem 6.1.1]. □

4 Affine test lattices and almost simple intervals

The negative examples will come from two opposite orientations of a finite tree of truncated affine geometries. The common lattice is simple and self-dual. Its reverse links have several horizontal branches, and their bottom diamonds exclude particular normal-subgroup configurations. We first

construct these lattices and reduce a hypothetical representation to an ordinary almost simple subgroup interval. Subsequent sections prune paths through alternating and classical groups; bounded Lie rank gives a separate contradiction, and additional path protections bound the number of remaining stages.

Theorem 4.1. *There are choices of the finite parameters in [Definition 4.4](#) for which the resulting finite lattice is not the congruence lattice of any finite algebra.*

The construction and reductions in this section are uniform under the inequalities stated below. The successive choices of larger parameters and the completion of the proof of [Theorem 4.1](#) are given in [Sections 5 to 9](#). This theorem gives negative examples and, after bounding the parameters, a cardinality ceiling. It does not assert that the chosen family member has minimum cardinality; the minimum problem is treated separately.

4.1 Structural inputs and conventions

We use the classification of finite simple groups and its standard structural consequences, including Schreier’s Theorem, the automorphisms and central covers of finite groups of Lie type, and the classification of finite two-transitive groups. The algebraic-group inputs include Lang–Steinberg, highest-weight theory, Steinberg’s theorems, and the algebraic Peter–Weyl filtration. The bounded-rank argument will use the sufficiently-general subgroup Theorem of Larsen–Pink [\[25\]](#). To specify the scope of these inputs, we fix the following conventions.

Convention 4.2 (Lie-theoretic realizations). We use split root-system realizations over algebraic closures, with both simply connected and adjoint root data, the Chevalley root and parabolic structure, and the isogeny and Steinberg-map classification. Field maps, pinned diagram maps, and, in the relevant special characteristics, the special isogenies of bounded types are combined with inner automorphisms. A Steinberg map of positive height supplies the finite fixed group. For a simple adjoint group at sufficiently large parameter, or in the high-rank cases used here, the associated simple finite group is the derived fixed group. Outside the finitely many exceptional ranks and parameters for simplicity and Schur multipliers, the simply connected fixed group is its perfect universal central cover. Inner-diagonal, field, graph, and applicable special actions induce the usual automorphisms of the simple projective group. No uniform covering assertion is made at the excluded small cases. For high-rank natural signed types, type B is used only in odd characteristic; in characteristic two we use the symplectic realization of type C . These are the usual structure and automorphism conventions [\[13, 46\]](#).

Convention 4.3 (Representation-theoretic inputs). Over the algebraic closure in defining characteristic, we use rational highest-weight theory: Weyl groups, Weyl and induced modules, weight convex hulls, and the triangular and distribution-algebra properties of highest weights. We also use the restricted-simple theorem for simply connected groups and their Frobenius kernels, Steinberg’s Tensor Product Theorem and finite-group restriction and parametrization theorems on the simply connected finite cover, and the two-sided algebraic Peter–Weyl good filtration of the regular function algebra [\[18, 19, 45\]](#). The precise finite P -restricted convention at high rank is given in [Section 9](#). These inputs concern rational modules or the specified full finite covers; they do not assert that arbitrary almost simple subgroup inclusions arise from algebraic-group inclusions.

On the permutation side we use Jordan’s prime-cycle Theorem: a primitive group of degree d containing a cycle of prime length at most $d - 3$ contains A_d . We use Bochert’s primitive-index bound, in particular an exponential lower bound on the index in S_d of a primitive group not

containing A_d in sufficiently large degree, and Burnside's prime-degree Theorem. The finite two- and multiply-transitive classification gives the following consequences: a sufficiently large alternating simple group has only its natural degree as a nontrivial two-transitive action; in high degree a six-transitive group contains the alternating group; and a non-affine two-transitive group of degree 37 contains A_{37} [10]. For the last assertion, the projective degrees $1 + s + \cdots + s^j$, with s a prime power, never equal 37: for $j = 1$, one would need $s = 36$; for $j = 2, 3, 4$ consecutive possible integer values bracket 37 without equality; and for $j \geq 5$ even $s = 2$ gives at least 63. The unitary, Suzuki, Ree, and symplectic-over-two degrees $s^3 + 1$, $s^2 + 1$, and $2^{e-1}(2^e \pm 1)$ also give no permitted parameter of degree 37, and the additional small and sporadic actions on the classification list do not have that degree. The simple-group identifications in the classification exclude spurious high-degree alternating overlaps. We use ordinary finite subset Ramsey theory, finite multidimensional Hales–Jewett for a fixed alphabet, number of colors, and number of free slots, and later the finite vector-space Ramsey Theorem for linear subspaces [14, 15, 17, 43]. More specialized uses of these inputs are stated where they occur.

4.2 Charts, branches, and prepared links

Definition 4.4 (Affine test lattice). Let $q \geq 5$ be a prime power such that $q - 1$ is not a prime power, and let r, D, g, m, n, N be positive integers satisfying

$$D \geq 2, \quad g \geq 2, \quad 3 \leq m < n - 2, \quad Dr < m, \quad N \geq n + 2. \quad (4.1)$$

There are arbitrarily large allowable q , for example powers of 7. A *chart* is a copy of $\text{AG}(N, q)$. The rank of a nonempty affine flat is its affine dimension plus one; the empty flat has rank zero. To attach a fresh chart at a specified flat W , identify a flat of the new chart affinely with W , and take all its other points fresh. Its intersection with the union of the old point sets is exactly W . Attachment at the empty flat makes no identifications.

Build the trunk charts recursively, beginning with the path ending at $W = \emptyset$.

- (i) At a path ending at W , attach g fresh *private charts* at W .
- (ii) In each such chart, reserve a child path extending this path by U for every flat $U \supset W$ with rank increment between 1 and r . Perform stages for D generations: the empty path has a stage, as does every child path reached in fewer than D steps.
- (iii) Finally, at every nonempty flat of rank at most m in a trunk chart, attach at least one additional fresh *leaf chart*, with no stages of its own.

Repeated reservations may use distinct fresh charts. There are only finitely many charts. The primal interior labels are the point sets that are nonempty affine flats of dimension at most n in some chart, ordered by inclusion. Adjoin the empty bottom and one formal top above all these labels, obtaining a bounded poset L^+ . Let L^- be an order-reversed copy of L^+ . The *test lattice* L is their horizontal sum: identify the two bottoms and the two tops, with no other cross-comparabilities. The interior labels on L^- are called *reverse flats*, named by their affine point sets; their lattice order reverses inclusion. On this side the top may accordingly be called the empty flat.

A comparability component below means a connected component of the undirected comparability graph on the strict interior of a bounded lattice. Distinct such components are horizontal: two elements from different components have meet zero and join one.

Proposition 4.5 (Geometry of the test lattice). *The construction in Definition 4.4 gives a finite self-dual lattice L . It is graded of length $n + 2$, atomistic, coatomistic, and simple. Any two charts*

intersect in a flat of each, of rank at most m , with the same affine structure on the intersection. Within one chart the primal join of two flats is their affine span if that span has dimension at most n . Attachment preserves all old meets and joins. In particular, flats in two distinct private charts attached at the same W , neither contained in W , have no common proper primal upper bound. There are at least three horizontal comparability components in the interior of L , each containing a chain of distinct comparable interior elements.

Proof. Induct on the attachments. When a chart is attached at W in an old chart, its intersection with any other old chart is exactly that chart's intersection with W . By the inductive hypothesis this is a flat in the old chart containing W and in the other old chart, with agreeing affine structures. It is therefore a flat of the new chart as well. This proves the pairwise-intersection assertion. Trunk attachment flats have rank less than $Dr < m$, and leaf attachment flats have rank at most m , so every intersection of distinct charts has rank at most m . Affine subflats of a shared flat are consequently identified consistently.

It follows that intersections of labels are again flats, allowing the empty set. Moreover, whenever one nonempty label is contained in another, it is an affine subflat in every chart containing the larger label. Thus L^+ has finite meets and a top, which suffice to give all joins. A new proper upper bound of two old labels, not itself already an old flat, can contain those old labels only if both lie in the attachment flat W ; it then contains their old affine span in W . Hence no old join changes. Meets are set intersections and do not change either. This also proves the claim about private charts, first at their attachment and then after all subsequent attachments. Within a chart the stated affine span is the least proper upper bound. The horizontal sum with the opposite lattice is a self-dual lattice.

Every proper chain in one orientation can be refined one affine dimension at a time, up to dimension n , before reaching the formal top. Thus L is graded of length $n + 2$. A flat is generated by its points, so L^+ is atomistic; intersecting its n -dimensional extensions inside one chart recovers the flat, so it is coatomistic. The same properties hold in the opposite orientation and hence in the horizontal sum.

The $g \geq 2$ initial charts are disjoint. An attachment meets all old points in a flat contained in one old chart; it cannot connect two old initial branches by comparabilities. Each orientation therefore has at least two horizontal interior components, and each component contains long affine chains. In particular L has at least three such components.

Finally, let a nonidentity lattice congruence identify two different comparable elements $u < v$; such a pair exists whenever any two distinct elements are identified, by meeting and joining them. By atomisticity there is an atom $a \leq v$ with $a \not\leq u$. Meeting the identification with a gives $a \equiv 0$. Choose an interior element b in another component. Joining with b gives $1 \equiv b$. Meeting this with an interior element c in a third component gives $c \equiv 0$. Since $a \vee c = 1$, we conclude $1 \equiv 0$. The congruence is total, proving simplicity. $\square$

Proposition 4.6 (Prepared reverse links). *If X is a nonempty flat of rank at most m in a trunk chart, the reverse ideal with top X , denoted $[0, X]$, is graded, atomistic, and coatomistic. Its strict interior has at least two horizontal comparability components, each containing comparable distinct elements. The following additional properties hold for L and for every such prepared link.*

- (i) *Every bottom interval of length two is a diamond M_b with $b \geq 3$ atoms, where $b - 1 \geq 2$ is not a prime power.*
- (ii) *In every interior component there is an element y two ranks below the interval's top whose upper interval is a diamond with at least two middle elements.*

In addition, L has the following separation property:

- (iii) For any atom u and any proper upper bound $\omega \geq u$, there is a rank-two element $z \geq u$ with $z \not\leq \omega$.

Proof. A leaf attached at X supplies a branch of strict superflats of X incomparable with the previous strict superflats. Later attachments cannot connect two old branches in this reverse link: any connection through a new chart passes through superflats of X lying in its attachment flat; if any exist, they are already connected within that flat. Every superflat of X lies in a chart containing X , and in that chart it has extensions and intermediate flats. Each branch therefore contains comparable distinct elements and chains out to n -flats. Gradedness follows from these dimension chains. The flats extending X by one dimension, and the intersections of n -extensions, establish atomisticity and coatomisticity in the reversed order.

On the primal side of L , a bottom interval of length two is a line, with $b = q$ points; the choice of q gives the claimed condition on $b - 1$. On the reverse side, including a prepared link, its top flat has dimension $n - 1$. Such a flat belongs to exactly one chart, since distinct charts meet in rank at most $m < n - 2$. Its n -dimensional extensions are counted by one-dimensional subspaces of a vector space of dimension $N - n + 1$, so

$$b = \frac{q^{N-n+1} - 1}{q - 1}, \quad b - 1 = q(1 + q + \cdots + q^{N-n-1}).$$

The second factor is greater than one because $N \geq n + 2$, and is congruent to 1 modulo the characteristic prime. Thus $b - 1$ has at least two prime divisors. This proves (i).

For (ii), choose a primal $(n - 1)$ -flat, or, on the reverse side, a flat whose rank exceeds that of the empty flat or the prepared X by two. The appropriate upper interval is a diamond, with at least two middle flats. Such choices can be made in every branch.

For (iii), a primal atom is a point. In one chart through it there is a line escaping the given proper flat ω . A reverse atom is an n -flat, belonging to a unique chart; its proper upper bound corresponds to a nonempty subflat. Choose a hyperplane of the n -flat that does not contain that subflat. It is the required rank-two reverse label. $\square$

4.3 From algebras to ordinary group intervals

We write $[H, G]$ for the ordinary interval of all subgroups between finite groups $H \leq G$. The adjective *ordinary* distinguishes such intervals from invariant-subgroup intervals and from the extension posets introduced below.

Lemma 4.7 (Subdirect products of simple groups). *A subdirect subgroup of a finite power of a nonabelian simple group is a product of full, possibly automorphism-twisted, diagonal subgroups on disjoint parts partitioning the coordinates. Containment between two such subgroups means refinement of the tied parts, with agreement of the diagonal identifications on each refined part. If a subgroup contains the untwisted full diagonal, all its diagonal identifications are untwisted.*

Proof. Induct on the number of coordinates. Project away the last coordinate. The kernel in the last simple factor is normal in that factor, by surjectivity, so is either the whole factor or trivial. In the first case the last factor splits off. In the second, the subgroup is the graph of a surjective homomorphism from the shorter subdirect product to the last simple group. By induction the shorter group is a product of nonabelian simple diagonal factors. The images of those factors are commuting normal subgroups of a simple group; exactly one has nontrivial image, and on that factor the map is an isomorphism. This ties the last coordinate to just that part. The description

of containment follows by projecting to pairs of coordinates: a required tie must hold in the smaller subgroup with the same identifying automorphism. Containing the untwisted full diagonal forces each such automorphism to be the identity. $\square$

Proposition 4.8. *If the test lattice L is representable by a finite algebra, it is isomorphic to an ordinary subgroup interval of a finite group.*

Proof. We begin with the least-carrier restriction argument of Pálffy and Pudlák [37, Lemma 1]. The final diagonal construction below will turn the resulting permutation action into an ordinary interval. Choose a representing algebra with smallest possible carrier A . Let F be the monoid of all maps $A \rightarrow A$ preserving its represented congruences. The unary algebra $(A; F)$ has exactly those congruences: each represented congruence is preserved by definition, while F contains all one-variable translations of the original operations, whose preservation forces an equivalence to be an original congruence.

Let $e \in F$ be idempotent. On eA use the unary operations $ef|_{eA}$, $f \in F$. Restriction induces a surjective lattice homomorphism

$$\text{Con}(A; F) \longrightarrow \text{Con}(eA; \{ef|_{eA} : f \in F\}). \quad (4.2)$$

Meets are preserved by restriction. To check joins, retract an equivalence path by e : its vertices then lie in eA and each step remains in the same congruence. For surjectivity, if δ is a congruence on the displayed algebra on eA , define

$$x \hat{\delta} x' \iff (efx, efx') \in \delta \text{ for every } f \in F.$$

This is an F -invariant equivalence. Its restriction to eA is exactly δ : one direction uses its compatibility with all $ef|_{eA}$, and the other uses $f = 1$ and $e|_{eA} = 1$.

If $|eA| > 1$, the target of (4.2) is nontrivial. Simplicity of L makes the homomorphism injective, and minimality of $|A|$ forces equality of carrier sizes. A surjective idempotent is the identity. Thus every nonunit idempotent of F is constant.

Let I be the ideal of nonunits of F . Every sufficiently long product of elements of I is constant. Indeed, otherwise take more than $|F|$ successive prefixes in order of application of a product whose final image has size greater than one. Every prefix image then has size greater than one, and two prefix maps coincide. The intervening nonempty product fixes their common image pointwise. An idempotent power of that product is a nonunit idempotent fixing at least two points, a contradiction.

For $\theta \in \text{Con}(A; F)$ let $\Phi(\theta)$ be the equivalence generated by

$$\{(f(x), f(y)) : f \in I, (x, y) \in \theta\}.$$

This is a congruence, by the ideal property of I , and $\Phi(\theta) \leq \theta$. It preserves joins: both equivalence generation and images of paths show that applying it to a join gives the join of the images. Iterating Φ generates precisely the pairs obtained from products of the corresponding length in I . The inserted equivalence closures contribute nothing extra, since an image of a path is a path; diagonal pairs accommodate collapsed steps. The product observation makes some iterate of Φ identically zero. For an atom α , the deflation $\Phi(\alpha) \leq \alpha$ must therefore be zero, since the only other possibility would fix α forever. Atomisticity now gives $\Phi(1) = 0$. Hence every map in I sends every pair to a diagonal pair: all nonunits are constant.

The units consequently form a permutation group J with exactly L as its lattice of invariant equivalences; constants impose no conditions. Choose any finite nonabelian simple group C , and form the coordinate permutation semidirect product $C^A \rtimes J$. Then

$$[\Delta(C) \rtimes J, C^A \rtimes J] \quad (4.3)$$

is dual to that invariant-equivalence lattice. Indeed, intersection with C^A identifies this interval with the J -invariant subgroups above the full diagonal. By Lemma 4.7 these are the diagonal products determined by partitions of A , and subgroup containment reverses refinement of the corresponding equivalences. Since L is self-dual, (4.3) represents L itself. $\square$

4.4 The monolith and extension description

For a subgroup $H \leq G$, its core is $\text{core}_G(H) = \bigcap_{g \in G} gHg^{-1}$. A group with a unique minimal nontrivial normal subgroup is called *monolithic*; that minimal normal subgroup is its *monolith*. The normal-supplement and unique-monolith argument follows Pálffy [36, Lemma 3.6]. The horizontal components of our prepared intervals give the required lattice obstruction directly.

Proposition 4.9 (Normal supplements). *Suppose $[H, G]$ is isomorphic either to L or to a prepared reverse link. After quotienting by $\text{core}_G(H)$, every nontrivial normal subgroup is supplemented by H . The quotient has a unique minimal normal subgroup $M = S^s$, with S nonabelian simple, and H acts transitively on its simple factors. Moreover $C_G(M) = 1$ in the core quotient. In that quotient the subgroup interval identifies with*

$$[P, S^s]^H, \quad P = H \cap S^s, \quad (4.4)$$

the H -invariant subgroups in this range, by intersection with S^s ; the inverse sends Y to HY .

Proof. If $R \trianglelefteq G$, then HR permutes as a set with every overgroup of H . Dedekind's law consequently makes HR left-modular in $[H, G]$. There is no proper interior left-modular label: for a putative such label u , choose comparable distinct interior labels $v < w$ in another horizontal component. Then

$$(v \vee u) \wedge w = w \neq v = v \vee (u \wedge w).$$

Thus either $R \leq H$ or $HR = G$. In the core quotient the first alternative forces $R = 1$.

An abelian minimal normal supplement would identify the interval with an invariant-subgroup interval in that abelian group. Such an interval is modular, contradicting the displayed failure of the modular identity. If two distinct minimal normal subgroups existed, they would commute. Using one to supplement H , the conjugation action of H on the other would induce all its inner actions. The interval would again identify with a modular interval, now of invariant normal subgroups of that other minimal normal subgroup. Both possibilities are impossible. The unique minimal normal subgroup is therefore a power S^s of a nonabelian simple group. Its factor orbits under H are normal in $HM = G$, so minimality makes the action transitive. Its centralizer is normal; if nontrivial it would contain M , which is impossible because M is centerless. Finally $HM = G$, and the elementary intersection-product correspondence gives (4.4). $\square$

For a subgroup $Y \leq S^s$, write $c(Y)$ for the product of its coordinate projections. It is H -invariant whenever Y is. We will need a precise parametrization of the invariant subdirect groups. We use the invariant-subdirect/extension correspondence in Pálffy [36, Proposition 2.2 and Lemma 2.3], expressed here for the transitive action on the actual simple coordinates.

Lemma 4.10 (Extension description). *Suppose H acts by automorphisms on S^s , transitively on the simple coordinates. Fix a coordinate o , identify its group with S , and let $A_0 \leq H$ be its stabilizer, with induced action $\alpha : A_0 \rightarrow \text{Aut}(S)$. The H -invariant subdirect subgroups of S^s are in bijection with extensions*

$$A_0 \leq T \leq H, \quad \beta : T \longrightarrow \text{Aut}(S), \quad \beta|_{A_0} = \alpha. \quad (4.5)$$

This bijection reverses extension order: increasing the extension means increasing its domain and agreeing on the smaller domain. If one restricts to subdirect groups containing a specified P , then above any retained group in actual subgroup order all restrictions of its extension to intermediate domains still occur.

Proof. By Lemma 4.7, a subdirect group has tied diagonal parts that form an H -invariant partition of the coordinates. The part containing o is a block for the transitive action; it equals To , where T is its setwise stabilizer and $A_0 \leq T$. The action of T on that diagonal, identified with S by projection to o , gives β extending α .

Conversely, given (4.5), define the diagonal on To using left-action notation. At coordinate ho , for $h \in T$, send $u \in S$ to the transport by h of $\beta(h)^{-1}(u)$ from coordinate o . Replacing h by ha , with $a \in A_0$, gives the same element because $\beta(a) = \alpha(a)$. This defines a T -invariant full diagonal. Transport it to the other blocks and multiply the resulting disjoint diagonal factors. The result is H -invariant and subdirect, and recovers the original data. Refinement of tied parts, with agreement of identifications, is exactly restriction of the extension. The last assertion follows because such restrictions give larger actual subgroups, still containing P . $\square$

Proposition 4.11 (Cases for a prepared interval). *Use the notation of Proposition 4.9, in the core quotient. Let $\Lambda \leq \text{Aut}(S)$ be the image of a coordinate stabilizer in H , and put $I = \Lambda \cap S$, identifying S with its inner automorphisms. The following are the only possibilities.*

- (i) $c(P) = S^s$. All labels are subdirect. If the interval represents the whole L , its opposite has an ordinary subgroup representation with top-group order at most $|H| < |G|$.
- (ii) $P = c(P) \neq 1$. The interval is isomorphic to the ordinary almost simple interval $[\Lambda, S\Lambda]$.
- (iii) $P = 1$ and $I = S$. Every nonbottom label is subdirect, and the dual interval is the extension poset of Lemma 4.10 with one artificial top adjoined.

In particular the cases $P < c(P) < S^s$, $P = 1$ with $I = 1$, and $P = 1$ with $1 < I < S$ are impossible.

Proof. We check all the cases, retaining the prepared-interval hypotheses.

Subdirect bottom. If $c(P) = S^s$, every label is subdirect and refines the fixed diagonal ties of P . Restrict to one tied part of P . Its setwise stabilizer in H acts transitively on its coordinates. A refinement of its coordinates that is invariant under this stabilizer extends uniquely to the other parts by H , keeping the twists fixed by P . Thus the interval is dual to the invariant-equivalence lattice of that transitive action. By Lemma 3.3, the latter is an ordinary interval above a point stabilizer in the part stabilizer. Its top has order at most $|H| < |G|$, proving (i).

A proper nonproduct bottom. Suppose $P < c(P) < S^s$. Choose a horizontal interior component different from that of $c(P)$. Every label Y in it has $Y \vee c(P) = S^s$ and $Y \cap c(P) = P$. Its coordinate projections already contain those of P , so adjoining $c(P)$ does not enlarge them. The join condition therefore makes Y subdirect. In its diagonal description, intersection with $c(P)$ restricts each coordinate to the corresponding nontrivial projection of P , with the same ties. These restrictions project onto every such subgroup, because the intersection contains P . There are comparable distinct labels in the chosen component. Their strict containment changes at least one diagonal tie, and releasing a tie strictly enlarges the intersection: its tied coordinate subgroup is nontrivial and may now be chosen independently on the separated parts. This contradicts the fact that both intersections equal P .

A nontrivial product bottom. Suppose $P = c(P) \neq 1$, necessarily a proper product. A subdirect group containing independent nontrivial subgroups in each coordinate cannot have any nontrivial ties. Thus the only subdirect label above P is S^s . For a coatom Y , the product $c(Y)$ is a label

containing Y ; maximality and the preceding observation imply $Y = c(Y)$. Since the interval is coatomistic, every label is a product. At one coordinate let Q be the projection of P . Then $1 < Q \leq I$: the coordinate elements of $P \leq H$ induce their inner automorphisms. Products identify the interval with $[Q, S]^\Lambda$. For any coatom T of this invariant interval, I normalizes T . If $TI = S$, then T is normal in TI , hence in S , impossible because $1 < Q \leq T < S$. Thus $TI < S$, and maximality gives $I \leq T$. Intersecting all coatoms yields $I \leq Q$, so $I = Q$. Intersection with S and multiplication by Λ now identify $[Q, S]^\Lambda$ with $[\Lambda, S\Lambda]$ inside $\text{Aut}(S)$, proving (ii).

Trivial bottom and no inner operators. Suppose $P = 1$ and $I = 1$. Schreier's Theorem makes Λ soluble, since it embeds in $\text{Out}(S)$. We first show that a soluble operator group on a finite nonabelian simple group has a nontrivial proper invariant subgroup. We may replace it by its effective image. If that image is trivial, choose any subgroup of prime order. Otherwise choose a minimal normal elementary abelian p -subgroup A of the operator group. Its fixed group is invariant, and if nontrivial it is proper by effectiveness. If its fixed group is trivial, orbit counting on the simple group gives $p \nmid |S|$. For each prime divisor ℓ of $|S|$, the number of Sylow ℓ -subgroups divides $|S|$ and is prime to p , so the p -group A fixes at least one such Sylow subgroup. There is at most one: the set of conjugators between two A -invariant Sylow subgroups is A -invariant and has size $|N_S(P_\ell)|$, prime to p , so has an A -fixed conjugator. The trivial fixed group forces that conjugator to be 1. Uniqueness makes the Sylow subgroup invariant under the whole operator group. This proves the assertion.

Choose a minimal nontrivial Λ -invariant subgroup Q of S . Its characteristic subgroups are invariant, so Q is characteristically simple. If it were a power of a nonabelian simple group, apply the preceding assertion to the soluble operator image of a factor stabilizer, and transport the resulting proper invariant subgroup through the transitive factor action. This would give a nontrivial proper invariant subgroup of Q , a contradiction. Therefore Q is elementary abelian. Its normalizer is proper because S is simple. It strictly contains Q : if Q is not Sylow, use the normalizer property in a Sylow subgroup containing it; if it is Sylow and were self-normalizing, Burnside transfer would give a normal prime complement, again contradicting simplicity.

Transport Q and $N_S(Q)$ to all coordinates and take their products. They are labels, the latter strictly above the former. If the product of the Q 's has lattice rank at least two, choose a rank-two label below it and any atom in that interval. Otherwise the product itself is an atom; choose a rank-two label above it and below the normalizer product. In both cases the chosen atom A is an elementary abelian normal subgroup of the rank-two group R . Every other atom is an H -invariant complement to A in R , and conversely every such complement is an atom: intersection with A is trivial, and the join with A is R in the diamond. Fix one invariant complement C . In the semidirect product $R = A \rtimes C$, other complements are graphs of derivations $C \rightarrow A$. Since H preserves both factors, the invariant complements correspond precisely to the H -equivariant derivations. These form a vector space over the prime field of A . Their number is a prime power, allowing the zeroth power. This is the operator-invariant version of the complement-counting argument in Pálffy and Pudlák [37, Theorem 3]. The number equals $b - 1$ in the bottom diamond M_b , contradicting Proposition 4.6(i).

Trivial bottom and proper nontrivial inner operators. Suppose $P = 1$ and $1 < I < S$. Transporting I gives a proper interior product label. A label Y in another horizontal component is subdirect: its nontrivial coordinate projection R satisfies $RI = S$, because its join with the product of the I 's is full. The subgroup I normalizes R , so $R \leq RI = S$ and hence $R = S$. Write (T, β) for the extension describing Y in Lemma 4.10. Its image contains all inner automorphisms. Otherwise $\beta(T) \cap S$ is a nontrivial proper invariant subgroup of the diagonal, containing I . Transporting it over the tied parts and multiplying gives a nonbottom label below Y with proper coordinate projections. It would lie in the same component, contradicting the preceding conclusion about all its labels.

Choose y in this component two below the top with a diamond above it. The filter above y identifies dually, by restriction of β_y , with $[A_0, T_y]$. Every domain strictly larger than A_0 has image containing S . Apply this observation to $A_0 \ker \beta_y$: its image is still Λ , which does not contain S . Hence $\ker \beta_y \leq A_0$. After quotienting by this kernel, every proper increase over Λ contains $S\Lambda$, so there is a unique minimal proper increase. This contradicts the diamond's having at least two middle elements.

Trivial bottom and all inner operators. Finally, if $P = 1$ and $I = S$, every nonbottom label has nontrivial coordinate projections invariant under all inner, so those projections equal S . The nonbottom labels are exactly the invariant subdirect groups, and [Lemma 4.10](#) identifies their dual with the extension poset. The bottom contributes the artificial top. This proves (iii) and completes the case analysis. $\square$

Choose now an ordinary representation of the whole L with smallest possible top-group order, which exists under the hypothesis of [Proposition 4.8](#). Self-duality and [Proposition 4.11\(i\)](#) exclude the subdirect-bottom case. Consequently either L already has an ordinary almost simple representation, or it is isomorphic to an extension poset with an artificial top. We eliminate the latter alternative unless a prepared reverse point link has the desired almost simple representation.

4.5 Pure extensions and their kernels

The full-inner extension case and its kernel description appear in Pálffy [[36](#), Proposition 2.5 and Remark 2.6]; see also the attribution to Aschbacher's signalizer formulation there in Section 4. We now use the affine lattice to constrain these extensions further.

Definition 4.12. An extension datum consists of a subgroup H of a finite ambient group K , a nonabelian simple group B , and a homomorphism $\psi : H \rightarrow \text{Aut}(B)$ whose image contains B , identified with its inner automorphisms. Its extension poset consists of all homomorphisms $\beta : T \rightarrow \text{Aut}(B)$, $H \leq T \leq K$, restricting to ψ , ordered by extension, together with an artificial top. Put

$$D_0 = \ker \psi, \quad P_0 = \psi^{-1}(B).$$

An extension is *pure* if $\beta(T) = \psi(H)$, equivalently $T = H \ker \beta$.

Lemma 4.13 (Purity and kernel joins). *If the full extension poset with artificial top of [Definition 4.12](#) is isomorphic to L , every proper label is pure. These labels are then parametrized by all H -normalized subgroups $E \leq K$ satisfying*

$$D_0 \leq E, \quad E \cap H = D_0. \tag{4.6}$$

The domain is $T = HE$, and its entire lower ideal is the ordinary interval $[H, T]$. Proper joins of kernel labels are generated subgroups. If two kernel labels join to the artificial top, their generated subgroup contains P_0 .

Proof. Subextensions of pure extensions are pure. If finitely many pure labels have a common extension, their join is its restriction to the subgroup generated by their domains, and is pure as well: its image is generated by copies of $\psi(H)$.

Suppose an atom (T, β) is not pure. The intermediate restriction to $H \ker \beta$ is pure, so atomhood forces $H \ker \beta = H$ and $\ker \beta = D_0$. Since H already maps onto the inner subgroup B , it also follows that $\beta^{-1}(B) = P_0$. Thus T normalizes both P_0 and D_0 . The simultaneous normalizer

$$N_K(P_0) \cap N_K(D_0)$$

acts on $P_0/D_0 \cong B$. Its conjugation homomorphism restricts to ψ on H , since action on the identified inner automorphisms determines an automorphism of the centerless group B . On each nonpure atom it similarly agrees with β . This gives one extension label ω , strictly below the artificial top, bounding every nonpure atom.

Choose a nonpure atom u . By [Proposition 4.6\(iii\)](#), there is a rank-two label $z \geq u$ not below ω . At most one atom of its diamond lies below ω , since two distinct atoms join to z . The diamond has at least three atoms, so at least two are pure. Their proper join is z , which is consequently pure; its subextension u is pure, a contradiction. All atoms, and hence all proper labels by atomisticity and purity of proper joins, are pure.

A pure extension has a kernel satisfying (4.6) and domain HE . Conversely, such a subgroup E is normal in HE , and $HE/E \cong H/D_0$ defines the unique extension killing E and agreeing with ψ . This proves the parametrization. If $H \leq U \leq HE$, then $U = H(U \cap E)$, so restricting the extension identifies its lower ideal with all of $[H, HE]$. For a proper join, restriction of a common extension to the generated domain shows that the kernel is the subgroup generated by the kernels: their product with H is the generated domain, and their intersection with H is D_0 . If two labels instead join to the artificial top and their generated kernel E still met H in D_0 , the same construction would give a common proper extension, impossible. Therefore $E \cap H$ properly contains D_0 . It is normal in H , so its image is a nontrivial normal subgroup of the almost simple group $\psi(H)$; it contains B . As $D_0 \leq E$, this implies $P_0 \leq E$. $\square$

4.6 The lower affine-grid contradiction

We compare pure extensions indexed by the points, lines, and planes of an affine four-space. Four dimensions allow two planes to meet in exactly one point; after using this to choose a suitable plane, the contradiction is obtained inside that plane. The next lemma requires purity only for the labels of dimension at most two.

Lemma 4.14 (Lower-grid obstruction). *Fix an extension datum as in [Definition 4.12](#), and an affine four-space over a finite field with at least three elements. Suppose each nonempty flat X of dimension at most two supplies a pure extension with domain $T_X > H$ and kernel E_X , ordered in reverse containment, with the following properties.*

- (i) *If $X \cap Y = \{i\}$, their labels have proper join the point label i ; thus $T_i = \langle T_X, T_Y \rangle$ and $E_i = \langle E_X, E_Y \rangle$. Distinct point labels have incompatible join, and their generated kernels contain P_0 .*
- (ii) *For each X , modulo $L_X = \text{core}_{T_X}(H)$ the interval has the normal-supplement and unique nonabelian monolith properties of [Proposition 4.9](#), including transitivity of the base on the simple factors and the trivial monolith centralizer. Denote its monolith by W_X , with preimage V_X in T_X .*
- (iii) *For every point $i \in X$, the subgroup $(T_X \cap V_i)/L_i$ is subdirect in W_i and is strictly above the intersection of the base with W_i .*

These data cannot exist.

Proof. We organize the argument into the compatible monolith embeddings, the selection of a plane with small cores, and the final factor action.

Monolith embeddings. For $i \in X$ put

$$Z_{X,i} = (T_X \cap V_i)/L_i \leq W_i. \quad (4.7)$$

The base H is transitive on the simple factors of W_i . It is therefore transitive on the tied diagonal factors of its invariant subdirect group $Z_{X,i}$. This makes $Z_{X,i}$ a nontrivial minimal normal subgroup of

$$T_X/L_i = (H/L_i)Z_{X,i}.$$

It is not contained in H/L_i , by the strictness in the hypothesis. Since $T_X \leq T_i$, we have $L_i \leq L_X$. Minimal normality and $L_X \leq H$ now give $L_X \cap V_i = L_i$. Thus the image of $Z_{X,i}$ in T_X/L_X is still injective and minimal normal, so equals W_X . We obtain a canonical H -equivariant subdirect injection $f_{X,i} : W_X \rightarrow W_i$, and

$$V_X = (T_X \cap V_i)L_X. \quad (4.8)$$

For $i \in X \subseteq Y$ we have $H \leq T_Y \leq T_X \leq T_i$ and $L_i \leq L_X \leq L_Y$. Intersecting (4.8) with T_Y gives

$$V_X \cap T_Y = (V_i \cap T_Y)L_X, \quad V_Y = (V_X \cap T_Y)L_Y, \quad L_Y \cap V_X = L_X.$$

The last equality follows also by using $L_Y \cap V_i = L_i$. These formulas give injections $f_{Y,X} : W_Y \rightarrow W_X$ composing with the $f_{X,i}$ as expected: lift representatives in $V_i \cap T_Y$. They are subdirect. Indeed, every simple factor of W_X is visible by projection to some simple coordinate of W_i through $f_{X,i}$, and the composite image of W_Y projects onto every such coordinate. The kernel of the action of H on W_X is exactly L_X , by the trivial centralizer of the monolith in T_X/L_X .

If $X \cap Y = \{i\}$, the proper-join condition gives $T_i = \langle T_X, T_Y \rangle$. Writing each domain as H times its intersection with V_i shows that

$$W_i = \langle Z_{X,i}, Z_{Y,i} \rangle, \quad L_i = L_X \cap L_Y. \quad (4.9)$$

For completeness, the subgroup generated by the two Z 's contains $(H \cap V_i)/L_i$, is normalized by H/L_i , and multiplying it by H/L_i gives T_i/L_i . Intersecting that product with W_i therefore recovers all of W_i . An element of H acts trivially on this generated group exactly when it acts trivially on both embedded monoliths; this proves the asserted equality of cores.

A plane whose cores are killed by ψ . For any affine plane U , write

$$L_* = \bigcap_{i \in U} L_i.$$

Every line domain T_Y , for a line $Y \subset U$, normalizes L_* . To see this, fix a target point $j \notin Y$ and choose two distinct lines through j meeting Y , at points k_1, k_2 . An element of $T_Y \leq T_{k_t}$ normalizes L_{k_t} , so sends $L_* \leq L_{k_t}$ into the core of the line jk_t , because $L_{k_t} \leq L_{jk_t}$. Equation (4.9) for those two lines puts the resulting image inside L_j . For $j \in Y$, use $T_Y \leq T_j$ directly. The same argument for inverses proves normalization. Two distinct lines through a point generate its domain, so L_* is normal in the group generated by all point domains of U .

Since $L_* \leq H$, we have $L_* \cap E_i = L_* \cap D_0$ for every point i . Both E_i and L_* are normal in T_i , so $[E_i, L_*] \leq L_* \cap D_0$. Thus every E_i centralizes $L_*/(L_* \cap D_0)$. The group generated by kernels of distinct points contains P_0 , by hypothesis. Therefore $\psi(L_*)$ centralizes B , whose centralizer in $\text{Aut}(B)$ is trivial. Hence

$$L_* \leq D_0. \quad (4.10)$$

There is a point i with $L_i \leq D_0$. Otherwise every $\psi(L_i)$ is a nontrivial normal subgroup of $\psi(H)$ and contains B . Take an iterated commutator of the original normal subgroups L_i inside H . It lies in $\bigcap_{i \in U} L_i$ and its ψ -image still contains B , because B is perfect. That contradicts (4.10). Choose two affine planes in the four-space meeting exactly at this point i . Their cores intersect in L_i by (4.9), so their two core images cannot both contain B , by the same commutator argument. Hence

one of these planes, henceforth denoted U , has $L_U \leq D_0$. All point cores of U then lie in D_0 , since $L_i \leq L_U$ for $i \in U$.

Quotient the group generated by the point domains of U by $\bigcap_{i \in U} L_i$. This subgroup is normal there as shown above, lies in the base, and is contained in all the point cores and all extension kernels in the plane. The core in each quotient is the quotient of the old core. Thus the inclusions, join statements, and monolith identifications survive. We retain the notation and may assume

$$\bigcap_{i \in U} L_i = 1, \quad L_U \leq D_0. \quad (4.11)$$

Compatible perfect subgroups. For a nonempty affine subflat $X \subseteq U$ set

$$M_X = E_X \cap \bigcap_{i \in X} V_i. \quad (4.12)$$

Every factor $V_i \cap T_X$ is normal in T_X , as is E_X , so M_X is normal in T_X . It lies in V_X by (4.8). It maps onto W_X : each $V_i \cap T_X$ maps onto W_X , while the normal image of E_X modulo L_X is nontrivial, since $HE_X = T_X > H$ and $L_X \leq H$, and therefore contains the monolith. Taking nested commutators of these finitely many normal subgroups gives a subgroup in their intersection whose image still contains W_X , by perfection of W_X . As $M_X \leq V_X$, its image is exactly W_X . The supplement condition then yields

$$HM_X = T_X, \quad D_0 M_X = E_X. \quad (4.13)$$

The second equality follows by intersecting the first with E_X .

For $X = U$, the kernel of $M_U \rightarrow W_U$ is contained in every L_i : it lies in $L_U \cap V_i = L_i$. Equation (4.11) makes it trivial. Consequently $M_U \cong W_U$ is an embedded product of nonabelian simple groups; write its factors as C_a . The normal subgroups L_U and M_U of T_U have trivial intersection, so L_U centralizes every C_a .

The subdirect embedding $f_{U,X}$ assigns to each factor C_a a disjoint set of factors of W_X on which it has its tied diagonal, and these sets partition all factors of W_X . Let $M_{X,a}$ be the preimage in M_X of the entire product of W_X -factors assigned to a . These groups generate M_X , are permuted by H , and each is normalized by L_U : the latter centralizes M_U , so fixes its factors and their assigned sets. For $i \in X$, the images of $M_{X,a}$ in W_i have support only over the same index a . For $a \neq b$ it follows that

$$[M_{X,a}, C_b] \leq L_i \leq L_U. \quad (4.14)$$

In fact $M_{X,a}$ centralizes C_b . Since L_U centralizes C_b , (4.14) and the three-subgroups lemma imply $[M_{X,a}, [C_b, C_b]] = 1$; perfection gives the claim. Likewise, for meeting lines X, Y (allowing $X = Y$) and $a \neq b$, a common point i gives

$$[M_{X,a}, M_{Y,b}] \leq L_i \leq L_U. \quad (4.15)$$

The final action on the simple factors. Choose four pairwise nonparallel lines in U , two through a point i and two through a distinct point j . There are at least four directions, since the field has at least three elements; the lines can be chosen with the required distinct directions. Set

$$R' = \langle M_Y : Y \text{ is one of these four lines} \rangle.$$

It is normalized by H . The two proper point joins and then the incompatible join of i, j , together with (4.13), imply $P_0 \leq D_0 R'$. Hence

$$\psi(P_0 \cap R') = B. \quad (4.16)$$

It remains to show that this image is soluble. We first show that $P_0 \cap R'$ fixes every factor C_a . Schreier's Theorem then controls its outer action, and we show that ψ kills the subgroup inducing inner automorphisms.

Fix an index b . A word in the generating groups $M_{Y,a}$ can be reordered as

$$\ell uv,$$

where $\ell \in L_U$, all letters of u have indices $a \neq b$, and all letters of v have index b . For adjacent letters in the wrong order, put $e = xyx^{-1}y^{-1} \in L_U$, by (4.15). If p is the preceding word, then

$$pxy = peyx = e(e^{-1}pe)yx.$$

Conjugating the preceding letters by e preserves their indices because L_U normalizes every indexed generating subgroup. Thus the swap reduces the inversion count. Induction gives the required reordering.

In right-conjugation notation, $C_b^{\ell uv} = C_b^v$, since ℓ and u centralize C_b . The element v centralizes every other C_a , and C_b already commutes with them, so C_b^v commutes with every C_a for $a \neq b$. If the word lies in $P_0 \cap R' \leq H$, it permutes the factors of M_U ; its image of C_b cannot be another C_a , because that would make a nonabelian simple group commute with itself. It must therefore be C_b , proving the assertion.

The action of $P_0 \cap R'$ on $W_U \cong M_U$ consequently has soluble quotient modulo its inner action, by Schreier's Theorem on each fixed simple factor. An element inducing an inner action on W_U lies in V_U : after multiplying by the corresponding monolith element its image in T_U/L_U centralizes W_U , and that centralizer is trivial. On the other hand, purity and $L_U \leq D_0$ give

$$H \cap V_U \leq H \cap (E_U L_U) \leq D_0 L_U = D_0,$$

because the nontrivial normal image of E_U contains W_U and hence $V_U \leq E_U L_U$. Thus the inner-action kernel within $P_0 \cap R'$ is killed by ψ . Consequently $\psi(P_0 \cap R')$ is a quotient of a soluble group, contradicting (4.16) and the nonabelian simplicity of B . $\square$

Theorem 4.15 (Ordinary almost simple starting interval). *If the test lattice L is representable, then either the whole L or a prepared reverse link with top an initial trunk point has an ordinary almost simple subgroup-interval representation. On the reverse side, there are at least $D - 1$ further generations of prepared stages starting at that top. This conclusion uses only the field condition and (4.1), independently of later choices of larger parameters.*

Proof. Choose the least-top-order ordinary representation furnished by Proposition 4.8. The reduction after Proposition 4.11 either already gives the whole L as an ordinary almost simple interval, or gives an extension poset with artificial top isomorphic to L . In the latter case all proper labels are pure by Lemma 4.13. For each nonempty reverse flat X , write E_X, T_X for its kernel and domain. Its lower ideal is the ordinary interval $[H, T_X]$.

Take an affine four-subspace of one initial trunk chart and consider all its points, lines, and planes. Their ranks are at most three, hence at most m , so all their reverse links are prepared. The proper and incompatible joins have exactly the domain and kernel properties of Lemma 4.14, by Lemma 4.13. Each ordinary domain interval has the normal-supplement and monolith properties of Proposition 4.9. If at some point i the nontrivial-product case (ii) of Proposition 4.11 occurs, it gives the required almost simple representation of that point's link. Otherwise, the remaining possibilities at every point are the subdirect-bottom and the trivial-bottom full-inner cases. In both, every domain label strictly above the base intersects the point monolith subdirectly modulo

the core, as required in Lemma 4.14. The strictness in that lemma follows from $HV_i = T_i$: every incident domain satisfies $T_X = H(T_X \cap V_i)$, so $T_X > H$ gives strict containment above the point-base intersection after quotienting by L_i . That lemma gives a contradiction.

For the whole lattice take the empty reverse flat as the starting top. For a point in an initial trunk chart, its rank increment from the empty flat is one, which lies between 1 and r ; its child path was therefore reserved at the first stage. In either case at least $D - 1$ generations remain by the construction. $\square$

5 Simultaneous path pruning and affine witness patterns

We now organize the local group arguments into a finite pruning construction on the test lattices of Section 4. The construction is simultaneous over all ancestors of a path. Its output will be used with the classical and alternating exclusions in Sections 6 and 7, Proposition 8.6, and Theorem 9.21.

5.1 Path labels and compatible simple groups

We work in the ordinary interval supplied by Theorem 4.15, with fixed base group H . Write K_X for the subgroup labeled by a reverse flat X . Thus $X \subseteq Y$ means $K_Y \leq K_X$. A downward path chooses, at its current flat W , a private trunk chart and then a flat $U \supsetneq W$ of rank increment between 1 and r in that chart. The local group arguments will be applied both in the current group's almost simple quotient and in the quotients belonging to all earlier path nodes.

At a path node W we maintain an actual nonabelian simple group $B_W \trianglelefteq K_W$ such that

$$HB_W = K_W, \quad C_W := C_{K_W}(B_W) \leq H,$$

where C_W is soluble and K_W/C_W is almost simple with socle the faithful image of B_W . The starting interval has this property. The next lemma explains why the local conclusions in several such quotients give one subgroup in the original interval.

Lemma 5.1 (Compatibility of ancestor socles). *Suppose $H \leq K \leq K_W$, and a local conclusion in K_W/C_W supplies a nonabelian simple normal subgroup $\bar{B} \trianglelefteq K/C_W$ contained in the image of B_W . Suppose also that $(H/C_W)\bar{B} = K/C_W$, and that its centralizer in K/C_W is soluble and contained in H/C_W . Then $\bar{B}$ has a unique lift $B \leq B_W$, with*

$$B \trianglelefteq K, \quad HB = K, \quad C_K(B) \leq H,$$

and $C_K(B)$ is soluble. The quotient by this centralizer is almost simple. If the earlier actual socles contain B_W , the image of B in every earlier ancestor quotient agrees with any nonabelian simple normal subgroup supplied there with soluble centralizer. All the simultaneous structural conclusions therefore concern this same B .

Whenever two comparable candidate labels have these simple and centralizer properties, their actual simple groups are nested, strictly if the labels are distinct.

Proof. The quotient map is faithful on B_W , since $B_W \cap C_W = 1$. Its inverse on the image of B_W defines B , which lies in K and is normal there. The supplement identity lifts because $C_W \leq H$. A lift of an element centralizing $\bar{B}$ centralizes B : its commutators with B lie both in B and in C_W , so are trivial. Hence the new centralizer is the inverse image of the quotient centralizer and is soluble and contained in H . Quotienting by it gives a faithful automorphism group of B containing its inner automorphisms.

For an earlier node W' , the inclusion $B \leq B_W \leq B_{W'}$ makes its image faithful in $K/C_{W'}$. The same commutator argument identifies its centralizer there with $C_K(B)/C_{W'}$, which is soluble. Distinct nonabelian simple normal subgroups would commute, contradicting that solubility. Thus the simple supplied relative to W' must be this image. All these quotient maps preserve the exact interval, since their kernels lie in H .

For comparable candidates with $K_U \leq K_L$, the simple B_U acts on B_L . Schreier's Theorem makes its outer action trivial, so $B_U \leq B_L C_{K_L}(B_L)$. This is a direct product because B_L is centerless. Projection of B_U onto the soluble centralizer is trivial, giving $B_U \leq B_L$. If the labels differ, their simples differ because $HB_U = K_U$ and $HB_L = K_L$. $\square$

5.2 One pruning round and its failure witnesses

A structural exclusion forbids containment in specified proper full stabilizers in an ancestor's almost simple quotient. Such a full stabilizer is pulled back to an actual subgroup of the original interval. Avoiding it is preserved when a tested subgroup is enlarged.

At the current path flat W , fix one category of these stabilizers, allowing finitely many ancestor and case choices. A round tests flats $U \supsetneq W$ of a prescribed rank increment s in each available private chart. A successful flat has a label K_U contained in none of the stabilizers under test. After the first round, tests take place inside a previously selected successful flat V , with $s < \text{rk } V - \text{rk } W$. The subgroup $E = K_V$ then satisfies $H < E < K_U$ and retains all preceding exclusions. We call it the *anchor* for that chart.

Lemma 5.2 (Prepared full-stabilizer witnesses). *Suppose K_W already avoids the final exclusion list relative to every strict ancestor. A failed test at U supplies an actual label K_X with*

$$W' \subseteq X \subseteq U, \quad X \not\subseteq W,$$

where W' is the ancestor at which the failure occurs. If all trial ranks lie below m , this witness has a prepared reverse lower interval. In a round with anchor E , it also satisfies $H < E < K_X$.

Proof. The proper full stabilizer contains K_U , so its label lies between K_U and $K_{W'}$. It is therefore a subflat X of U containing W' . If $X \subseteq W$, its subgroup contains K_W , contrary to the earlier exclusions; for $W' = W$, properness gives the same conclusion. Such an X is nonempty and lies in the trunk chart containing U , so [Proposition 4.6](#) applies. Finally $E < K_U \leq K_X$. $\square$

5.3 Patterns forced by a failed round

Lemma 5.3 (Affine patterns in failed charts). *Suppose every increment- s test in a sufficiently large available flat fails, all trial ranks lie below m , and a witness $X(U) \subseteq U$, with $X(U) \not\subseteq W$, is chosen for each test. Color it by its ancestor, category and case. Given any finite number of parameter directions, sufficient available dimension produces a family X_t , indexed by an affine space over $\mathbb{F}_q$, with these properties:*

- (i) *All witnesses have the same color and rank. For distinct t, t' , their span has rank one larger, contains every X_u on the parameter line through t, t' , and gives the subgroup intersection $K_{X_t} \cap K_{X_{t'}}$. Each witness label covers this intersection.*
- (ii) *An affinely independent parameter list increases the span rank by one at each new point, giving strictly descending subgroup intersections for as long as the spans remain within the truncation.*
- (iii) *In an anchored round every label contains the same $E > H$. Families in different private charts have pairwise subgroup intersections exactly H .*

The first round can supply more labels than the full interval height plus any fixed constant, without requiring their total span to be proper.

Proof. Choose a fixed model for the horizontal increment and many unused vertical coordinates, with coordinate space V_{vert} . For nonempty W , take an origin in W and graph linear maps from its s new horizontal directions to the vertical directions; these maps vanish on W . For empty W , use $s - 1$ horizontal affine dimensions and graph affine maps instead. Every graph is an increment- s test flat.

Color each map by its witness's failure color and the pullback of $X(U)$ into the fixed horizontal model. There are finitely many colors for the fixed horizontal data. View the maps as words whose letters are individual-coordinate linear or affine functionals. Multidimensional Hales–Jewett [17] gives a monochromatic family with fixed coordinates and any prescribed number k of disjoint nonempty variable batches. Thus the pulled-back witness is one fixed flat $X_{\text{hor}} \not\subseteq W$.

Choose a horizontal functional λ nonzero at some point of X_{hor} ; in the nonempty case it vanishes on W . Restrict the functional in each variable batch to a scalar multiple of λ . Let $\iota : \mathbb{F}_q^k \rightarrow V_{\text{vert}}$ send each parameter coordinate to the sum of the vertical basis vectors in its batch. The disjoint nonempty batches make ι injective. The witness family is

$$X_t = \{(x, f_0(x) + \lambda(x)\iota(t)) : x \in X_{\text{hor}}\}, \quad t \in \mathbb{F}_q^k, \quad (5.1)$$

where f_0 vanishes on the variable batches and records the fixed graph coordinates elsewhere. At a point where $\lambda \neq 0$, distinct parameters have different vertical coordinates, so the witness flats are distinct. The span of two graph flats adds exactly the vertical direction $\iota(t' - t)$, and therefore contains the witnesses on the parameter line. More generally,

$$\text{rk}\langle X_{t_0}, \dots, X_{t_h} \rangle = \text{rk } X_{\text{hor}} + \dim \text{aff}\{t_0, \dots, t_h\}. \quad (5.2)$$

The chart join rule converts these spans, when proper, into the asserted subgroup intersections and covers. Pair spans are proper because the trial ranks are below $m < n - 2$. The anchor inclusion follows from the round setup. Across private charts the witness flats lie outside W and have no common proper primal upper bound, so their reverse meet is the base H .

Taking sufficiently many parameter directions supplies the first-round family of more labels than the interval height. Only pair spans and the bounded independent lists used later must fit the truncation; the span of the whole family need not be proper. In subsequent rounds the number of parameter directions is bounded independently of n . $\square$

Assign each failed chart the ancestor/category/case of one homogeneous pattern extracted by Lemma 5.3. Suppose the exclusion argument for each such choice prohibits either one such pattern or a pair in distinct charts. In the first case no chart can fail in that case; in the second at most one can. Thus the number of charts lost in a round is at most the number of cases requiring two charts. These are the group-theoretic failure colors; the pullback-flat colors used inside Hales–Jewett do not enlarge the chart-loss count. Retaining one successful flat in every surviving chart provides the anchors for the next round.

If F is successful after the last round, every $W \subsetneq X \subseteq F$ has $K_X \geq K_F$ and therefore avoids every completed-category stabilizer relative to every ancestor tested. The local conclusions consequently apply to all comparable candidates in F , before a next path node is chosen. Lemma 5.1 makes their simple groups actual and nested. This is the property used for simultaneous root and tensor protection in Proposition 9.12.

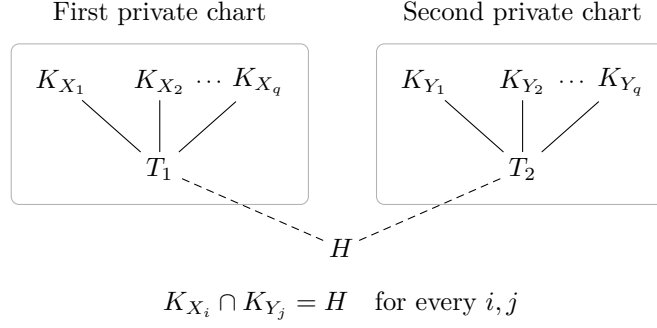


Figure 3: The subgroup relations supplied by [Lemma 5.3](#). In the first chart, the witnesses on one parameter line satisfy $\Sigma_1 = \langle X_i, X_j \rangle$ for every $i \neq j$, $\text{rk } \Sigma_1 = \text{rk } X_i + 1$, and $T_1 = K_{\Sigma_1} = K_{X_i} \cap K_{X_j}$; each K_{X_i} covers T_1 . The second chart has the analogous relations with Y_j, Σ_2, T_2 . Solid segments denote these covers, and dashed segments denote inclusions, with intervening labels omitted. The diagram is schematic in rank and shows only the stated relations.

Proposition 5.4 (Finite schedule of simultaneous exclusions). *Suppose that the local exclusions below prohibit their prescribed failed patterns with absolute case and chart-loss bounds. For a bounded path length, finitely many rounds then produce a flat of final increment r such that the subgroup label of every proper extension in it is contained in none of the prescribed proper full stabilizers relative to any high-type ancestor. The next local simple has the corresponding transitivity or full-spanning properties relative to all those ancestors.*

Proof. Use a simultaneous initial round for all initial classical subspace exclusions relative to any classical ancestors (the ones allowing a more-than-height requirement); their pattern proofs need no earlier anchor. The remaining categories of the structure arguments can then be ordered as there, requiring only bounded line/orbit or independent-list bounds: e.g. direct line systems and then other direct systems, prime fields then ordered factors for linear type, partial ordered factors then prime fields then paired idempotents and remaining ordered factors for isometry types, elementary subgroups then tensor systems; in alternating ancestors subsets and block partitions then affine, Cartesian and diagonal structures. Subcases rendered impossible by an already forbidden structure need no round of their own. There are only boundedly many categories/ancestor instances altogether for bounded path length. We thus obtain a pruned flat of final increment r from W ; all labels at proper extensions within it avoid the exclusions, giving the needed next almost simple socles by the structural arguments, and giving transitivity or natural full spanning as applicable relative also to earlier ancestors. Selection of which extension to take (additional closure-field tensor/root protections within a same-characteristic run) uses [Proposition 9.12](#). If the current parent instead is of bounded Lie rank, use [Proposition 8.6](#) (same graph pattern with a bounded independent list and no prior anchor, relative just to this parent). $\square$

Proposition 5.5 (Order of the finite parameter choices). *The test-lattice parameters can be chosen in an order that meets all the pattern requirements and all the later path requirements. In particular the initial height-dependent pattern requirement does not create a circular dependence between m, n, N . Once the local group results are applied, these choices force either the bounded-type contradiction or a path longer than the protected-path bound.*

Proof. Here is one parameter order, always finite. Fix the high thresholds, the resulting bounded-type constants, and the absolute bound on protected high-type path lengths from the group arguments.

Choose D exceeding this bound by several stages (allowing the displaced point start and a stage for the bounded-type contradiction). Choose q , a power of 7 for instance, above all absolute line-count requirements. Choose r as in the final flag selection, large for bounded-type requirements and the exponential-gap union bound over at most D steps, so the path ranks are at most Dr . Choose larger intermediate trial increments backwards from r , the increment after the first round thus a constant independent of m, n : each preceding increment merely must allow the Hales–Jewett bound for the next bounded-slot round, at any path rank up to Dr . Choose m above the ranks of these trials, then n sufficiently larger, including for residual interval length below all of them and bounded required spans. Now N can allow the first-round Hales–Jewett bound including the larger initial slot requirement depending on n , and any no-anchor bounded-type trial. Finally take g sufficiently large for the failed-chart bounds at every stage. Thus the separate arguments imply contradiction on this finite lattice: either a bounded-type parent gives the direct contradiction, or too long a protected path is forced. $\square$

6 Pruning inside a high-rank classical ancestor

All assertions in this section concern the prepared configuration of [Section 5](#). A structure is excluded when its full stabilizers cannot supply the failed-chart patterns required by [Lemma 5.3](#); this is precisely the input needed by [Proposition 5.4](#). The statement concerns these prepared configurations, and includes the preceding exclusions specified at each stage.

6.1 Standing interval and matrix conventions

Convention 6.1 (Prepared classical configuration). Here we give the high-rank classical structure exclusions for the reverse-flat path argument. We work in one of the current almost simple quotients, with interval base again denoted by H , top group M , and simple socle $S \leq M \leq \text{Aut}(S)$. We have $HS = M$: S is not in the base (the outer quotient is soluble and the base-core quotient must have a nonabelian monolith), and use the normal supplement property. (For a strict ancestor the top here still means that ancestor group in its almost simple quotient.) A bad structure has for label its **full stabilizer in M** . We consider labels containing the tested group and proper in M ; by irreducibility/fullness of the ambient simple groups below (of sufficiently high rank), the stabilizers we need are proper. Equivalently one can discard full-group stabilizers, checking properness when using the exclusions. Here is the same preparedness observation as for the witness flats in the path construction: any such proper full stabilizer containing a chart test label, or the chosen previously good anchor in the chart, for a structure on the full list of intended exclusions, also labels a trunk subflat within the relevant trial-rank range. Indeed for a strict ancestor the current path node has already avoided *all* these final exclusions relative to it (even if the pruning order now starts over); for the current ancestor use properness. This applies also when testing or using a case condition on the structure before its exclusion is complete (such as which tensor side is retained). Each witness label K_i thus has full interval $[H, K_i]$ a prepared reverse link. Thus a normal subgroup of K_i is either retained (contained in H) or missing (supplemented by H), soluble normals are retained, and two commuting normals cannot both be missing (also valid modulo normal subgroups contained in H). These consequences of [Proposition 4.9](#) will be denoted by $(*)$.

Recall the patterns supplied when pruning fails: many distinct labels in a private chart, including a line of q distinct labels with $K_i \cap K_j \leq K_l$ for all l on the line and $i \neq j$ there. Moreover K_i covers $K_i \cap K_j$, by the hyperplane-in-span property. We may require a bounded number of independent pattern points with strictly descending successive intersections, and in the initial round more distinct labels than the total height plus any fixed constant. We may use another such pattern in a separate

chart for the same category, with $K_i \cap K_j = H$ across the two. Any needed bounded subdivision into colors will be indicated; all patterns used together for a category can be taken homogeneous. Except in the initial round we use a common previously good anchor $H < E < K_i$ for labels within a chart, with $[H, E]$ itself a prepared reverse link. Earlier exclusions imply E lies in none of the previously forbidden stabilizers. The prepared links have height exceeding any required absolute constants, interior horizontal branching, coatomisticity, and the bottom diamond parameters used in the interval reductions. Constants and bounded choices here are independent of the field and dimension of the ambient finite group; q denotes the **lattice** field size and will exceed all required absolute orbit/count bounds.

Convention 6.2 (Natural matrix groups and their small parameters). Let V have dimension v over the finite ground field F . The natural types are linear (PSL), unitary, symplectic, and orthogonal (nonsingular, using odd dimension only in odd characteristic); in isometry types write b for the defining nondegenerate bilinear or hermitian form, and include the defining quadratic form Q for orthogonal groups. Adjoints for b are denoted by $*$. We use the standard descriptions of finite simple classical groups, their covering matrix groups (SL, SU, Sp, Ω), orders and automorphisms. Here v is above a sufficiently large absolute threshold. In the isometry cases M acts by semilinear similarities modulo scalars. In the linear case it acts on $\mathcal{A} = \text{End}_F(V)$ by semilinear automorphisms and possibly anti-automorphisms (transpose with conjugation/field action); on the projective group the latter action uses also inverse. There is also the corresponding action on subspaces, order-reversing for an anti-action. Write $+$ for the collineation/automorphism part (the entire group in the isometry cases). Isometry actions on $\mathcal{A}$ respect the adjoint involution. All of this uses sufficiently high rank (so no triality or small exceptional automorphism issue).

We give a few more specifics on the standard classical-group facts being used (reference points are Taylor [47] and Kleidman and Liebeck [23] for classical groups and their natural geometry, together with Conventions 4.2 and 4.3). In recalling the group lists use parameter $s = |F|$ except $s = \sqrt{|F|}$ in unitary type. Forms and subspaces considered over the ground field use the finite-field Witt theory; “nondegenerate” for a subspace used in an orthogonal decomposition requires nondegeneracy of the restricted polar pairing also in characteristic two.

- In orthogonal type use Ω as the kernel in the isometry group of determinant and spinor norm in odd characteristic, and of the Dickson invariant (rank of $u - 1$ modulo two) in even characteristic with nondegenerate quadratic space of even dimension. In particular ordinary isometry commutators are available inside it. These kernels are preserved by semilinear similarities (for the spinor norm on determinant one, scaling the form does not change the norm modulo squares, by the reflection formula with an even number of reflections). In the other types use determinant one in linear and unitary, and the full symplectic group. The quotients of linear similarity groups over the respective special groups are soluble; adjoining field automorphisms still gives soluble remaining quotients. For unitary and high orthogonal types the automorphisms needed in M , including graph actions if present, are covered by the ground semilinear similarity description (no D_4 here).
- Recall for natural degree at least two the usual simplicity/perfection list: for linear and unitary the special group is perfect, simple projectively, outside degree two at $s = 2, 3$ and the unitary degree three at $s = 2$; the analogous exceptions for symplectic are degree two at $2, 3$ and $\text{Sp}_4(2)$. Ω in the above convention/range of polar spaces has this high structure (perfect, simple projectively) in degree at least five. In degree three (odd characteristic) use the projective rank-one structure via the symmetric square of $\text{SL}_2(s)$; in four the split determinant space has the two independent rank-one tensor actions (projectively), and nonsplit uses the

natural module and its field- s twist tensoring for $\mathrm{SL}_2(s^2)$. Thus the split Ω_4^+ is perfect with two projective simple factors for $s \geq 4$, and Ω_4^- is perfect, simple projectively. The plane and soluble cases needed will also be noted below.

- The natural actions of the special groups in these assertions, apart from orthogonal planes, are absolutely irreducible, so their matrix lifts generate the full natural matrix algebra. Equivalently one can use the standard irreducible defining-characteristic vector modules and Steinberg restriction (restricted highest weight at the natural parameter; using B only for odd characteristic); in degree four orthogonal type use the rank-one tensor descriptions, and $\mathrm{SU}_2(s)$ gives the split $\mathrm{SL}_2(s)$ module over the extended field. For this last description one can write the defining hermitian form with matrix a suitable scalar times the standard nondegenerate alternating matrix in degree two (choose the scalar to make it hermitian and use equivalence of nondegenerate hermitian forms); the special group then includes $\mathrm{SL}_2(s)$, of the full special order. When using a derived replacement at $\mathrm{Sp}_4(2)$ we check it separately. The size comparisons below use only ordinary classical-group order formulas.

The exclusion order is as follows, within the prepared configuration and under the full hypotheses of the cited statements. Initial subspace exclusions precede direct-system pruning (Propositions 6.4, 6.5 and 6.12). In linear type, fields precede ordered factors (Propositions 6.15 and 6.16). In isometry types, partial ordered-factor pruning leaves the two stated retained-plane cases; field transfer and prime-degree fields precede paired idempotents and completion of the ordered-factor exclusions (Propositions 6.17, 6.22 and 6.23 and Lemma 6.19). Full spanning and elementary normalizers then precede permuted tensor systems and the next actual simple socle (Lemma 6.24, Propositions 6.25 and 6.29, and Theorem 6.30).

Lemma 6.3 (Properness of the stabilizers used below). *For sufficiently large ambient natural degree, each nontrivial structure used in the exclusions below has proper full stabilizer. The same properness check applies when an earlier exclusion is invoked for a structure found from the anchor.*

Proof. For properness throughout, the high-rank covering matrix group of S is perfect and absolutely irreducible. It cannot preserve the initial subspace, signed-flag or individual-projector structures below (its own action is linear), nor even an adjoint-exchanged complementary pair of projectors up to swap in isometry type (perfection). It cannot normalize a proper direct-sum system below: irreducibility would force a uniform split by transitivity, then apply the direct-split order comparison below on V itself. Similarly use the tensor order comparison for ordered tensor factors or tensor systems; normalizing a field would force centralizing it by perfection, and a nontrivial elementary abelian subgroup inside S cannot be normal there. Thus when an earlier exclusion is invoked for a structure found using an anchor, its full stabilizer is indeed proper. $\square$

6.2 Initial subspace exclusions

Proposition 6.4 (Linear subspaces, signed flags and idempotents). *In linear type, the initial round excludes stabilizers of proper nonzero subspaces when there are no anti-actions. In the presence of anti-actions it excludes signed flags $0 < I \leq J < V$ and individual proper nontrivial idempotents. These exclusions require no prior anchor and make the collineation part of every subsequent good group irreducible.*

Proof. First suppose linear type, without anti-actions. Exclude subspace stabilizers of nonzero proper U . In each such label the shears from V/U to U are in the normal unipotent radical, hence in H . Any two subspaces in the pattern must therefore be comparable. Stabilizers of a descending

list along that chain have strictly descending intersections (special linear maps inside the smallest previous space can move a new proper subspace). The total interval height bounds their number, contradiction.

If M in linear type contains anti-actions, so does H ; fix one, $a \in H$. Exclude the signed flag stabilizers for $0 < I \leq J < V$ (automorphisms fix the two terms, anti-actions exchange them). For labels containing H , $J = aI$. Again the normal unipotent radical of the corresponding parabolic in S (preserved also by signed anti-actions) contains the shears from V/I to I , and is in H . Thus the I 's form a chain, the J 's the reversed chain above it. Intersections are strict taking descending I 's, just as before.

Also exclude stabilizers of individual nontrivial proper idempotents e (in the linear case with anti-actions). The two supported special linear groups on the complementary spaces are individually normal, commute projectively, and the quotient over their product is soluble. Thus exactly one is retained by $(*)$; designate the retained space U , with complement determined by a (anti-action on an idempotent sends its image to the subspace-dual transform of its kernel). In a pattern with $\dim U \geq 2$, the supported group acts irreducibly nontrivially there and fixes the complement pointwise. Any other retained space, being H^+ -invariant, projects to zero along the complement or contains U (use differences under the supported group). By mutual irreducibility and distinctness, all retained spaces are pairwise in one another's complements. Their projectors are thus orthogonal idempotents in the ring sense. Intersecting the stabilizers strictly decreases until possibly the last one, by taking special linear maps on the unsplit remainder. Again this is height-bounded.

The retained line case. For the color with retained dimension 1, take a cross-chart pair. There is a subspace B_0 of codimension at most 4 in both designated complementary hyperplanes (that is, contained in both and of codimension at most 4 in V), disjoint from the span of both image lines; take a complement C_0 containing both lines. The supported $\mathrm{SL}(B_0)$, fixing C_0 , belongs to $K_i \cap K_j = H$. Thus every designated line in these patterns lies in C_0 and every corresponding hyperplane contains B_0 , by large-dimension irreducibility. Both the total line span and the total span of dual functional lines have bounded dimension. Now a boundedly long list of independent pattern points already contradicts strict descent of intersections. Indeed descent must be strict also on intersection with S , since $HS = M$ and all groups contain H . Track the growing two spans along the list; they each increase only boundedly often. Between increases, invariance is already imposed on the current spans, and further constraints are linear conditions on their endomorphism spaces (restrict matrices to the first and their transposes to the second). Each strict descent shrinks the solution subspace, by testing a witness matrix above the descent in S . This bounds the length absolutely.

These exclusions imply irreducibility of the collineation part: if it preserves U , then with anti-actions also aU ; the full group in question preserves these two with signed exchange. Nonzero intersection gives a signed flag (intersection and sum), and zero intersection a stabilized idempotent. $\square$

Proposition 6.5 (Singular and nondegenerate subspaces). *In isometry type, the initial round excludes stabilizers of nonzero totally singular subspaces, proper nonzero nondegenerate subspaces, and nonsingular radical lines in orthogonal characteristic two. It requires no prior anchor. The resulting good groups act irreducibly by semilinear transformations.*

Proof. For isometry types first exclude stabilizers of nonzero totally singular subspaces U (totally isotropic for unitary or symplectic). The standard unipotent radical is again in H . Here are details of the resulting comparability. Use Witt coordinates $U \oplus Z \oplus U^-$ with last coordinate λ paired

dually with U . The radical acts by

$$(u, z, \lambda) \mapsto (u + f(z) + h\lambda, z + g\lambda, \lambda)$$

with arbitrary f and accordingly adjoint g , and a form-preserving correction h . Such corrections exist by representing the quadratic correction as the diagonal of a bilinear pairing (orthogonal), filling an alternating correction (symplectic), or by surjectivity of quadratic-field trace on the diagonal in the hermitian case. With $f = g = 0$ the available h include the symmetric shears in symplectic type, skew-hermitian in unitary type, alternating in orthogonal type. These unipotent groups are inside the defining special groups (the additional determinant/spinor restrictions have order prime to the characteristic in the relevant unitary or odd-characteristic orthogonal cases). In orthogonal characteristic two the Dickson rank parity is even: the rank of the displacement is $2 \operatorname{rank} g$ plus the rank induced by h from $\ker g$ to its dual $U/\operatorname{im} f$, and the latter is alternating by the quadratic isometry condition.

Another invariant totally singular space within $U^\perp$, if not within U , must contain U by the arbitrary f . A vector in such an invariant space with $\lambda \neq 0$ outside $U^\perp$ is impossible in unitary/symplectic type by the pure h -shears (one can have nonzero pairing with $h\lambda$). In orthogonal type it forces the space to contain the annihilator hyperplane of λ in U ; also $Z = 0$, else choose a shear with nonsingular $g\lambda$ and take the difference vector. Thus the only incomparable possibility consists of two split maximal singular spaces adjacent in a hyperplane. At most two distinct maxima can so occur (through their common hyperplane there are just two; a third adjacent to both and not through it contains two vectors with nonzero pairing modulo that hyperplane). The rest are in a chain. Intersections down a descending chain of singular spaces are strict by elementary special linear transvections inside the smallest previous singular space preserving the larger flag, extended dually to isometries (also in S). This again contradicts the pattern size.

Next exclude stabilizers of proper nonzero nondegenerate subspaces in isometry types. On the two orthogonal parts there are commuting normal supported special groups with soluble quotient over their product, so designate the unique retained side. For sufficiently large retained dimension its supported derived classical group acts irreducibly there, hence distinct retained spaces are pairwise orthogonal by the same mutual argument as above. Intersections strictly decrease until possibly the last (use the large irreducible supported group on the remaining orthogonal space). For bounded retained dimension use a cross-chart pair. Their two small spaces embed in a nondegenerate subspace of bounded dimension (complete the radical of their span by dual partners); the supported large irreducible classical group on its complement lies in H . All the small spaces in the patterns thus have bounded total span, and strict descent for independent pattern points is bounded by the growing-span linear-constraint argument above. Use this same small-space argument for the stabilizers of nonsingular radical lines in orthogonal characteristic two. These exclusions ensure ordinary semilinear irreducibility: an invariant degenerate space has an invariant radical, totally singular except possibly in that quadratic case, where the kernel of the quadratic form on the radical is totally singular and otherwise the radical is a nonsingular line.

All initial categories needing more than height-boundedly many labels required no previously good anchor. $\square$

6.3 Direct systems of imprimitivity

Convention 6.6 (Direct systems after the subspace exclusions). Next exclude stabilizers of sets of complementary nontrivial projectors giving a direct decomposition (at least two spaces), orthogonal nondegenerate decompositions in the isometry cases. Use the irreducible previously good E^+ . It is transitive on the blocks, all of common dimension k , and a block stabilizer in E^+ is semilinearly

irreducible on its block (transport any supposed subspace). In isometry type the blocks consist of at most two isometry classes, with independent alternating groups on each class induced by the inner stabilizer: permute isometric copies, then take commutators. Similarly there is at least the alternating permutation group in linear type. If two isometry classes occur, they have equal sizes and the class partition is preserved by the full stabilizer (this uses only finite-field form types and semilinear similarities).

Lemma 6.7 (Two rich permutation actions). *Suppose a finite group has two permutation actions, each preserving a partition into at most two classes and inducing at least the independent alternating groups on its classes. The number of orbits on ordered pairs of pairs, with one index from each action in each pair, is bounded absolutely. Replacing indices by equivariant fibers of absolutely bounded size preserves this conclusion.*

Proof. Here and below we use an orbit bound. Suppose a group has two permutation actions, each preserving a partition into at most two classes and inducing on the points at least the independent alternating groups on its classes. Then the number of orbits on ordered pairs of pairs (one index from each action per pair) is bounded absolutely. Indeed choose a size gap among the at most four class sizes, with all small sizes bounded and every large size at least 8 with alternating order exceeding the product of the quotient-index bounds from the small factorials, class permutations and parities (choose five successive absolute thresholds rapidly increasing, or more to leave a gap). Intersect the joint action with the product of all large alternating factors. This is normal of bounded index, projecting onto each large alternating factor by the index comparison and simplicity (one can restrict for this check to the preimage of the single alternating factor on its action side, whose image there is onto by hypothesis). By the subdirect description these actions split as independent diagonals, with identifications natural up to relabeling (automorphisms of alternating groups in large degree). The orbits on the specified positions are then bounded by equality patterns and small-class choices. Bounded equivariant fibers keep the bound absolute. $\square$

The orbit bound will control the number of invariant partitions, rather than the number of points. Suppose a group has at most C orbits on $\Pi \times \Pi$, and $\Omega \rightarrow \Pi$ is an equivariant map with fibers of size at most d . Over a representative of each orbit on $\Pi \times \Pi$ there are at most d^2 ordered pairs, so there are at most Cd^2 orbits on $\Omega \times \Omega$. An invariant equivalence relation is a union of these orbits. Consequently

$$\#\{\text{invariant partitions of } \Omega\} \leq 2^{Cd^2}. \quad (6.1)$$

In the applications, Ω consists of common refinement cells or tensor factors. We must recover each original system by grouping those actual objects before this bound can count its full stabilizer labels.

Lemma 6.8 (Direct line systems). *Under [Convention 6.6](#), the full stabilizers of direct line systems are excluded. The only possible distinct linear systems reduce to matchings over $\mathbb{F}_3$; the corresponding odd-characteristic orthogonal case also reduces to matchings. In both cases an absolute orbit bound contradicts the pattern-line size.*

Proof. First exclude line systems, $k = 1$. In linear type the determinant-one diagonals in S are normal soluble in a label, thus in H . A line of another system with support s on this one has diagonal orbit size $(|F| - 1)^{s-1}$ (divide by $\gcd(v, |F| - 1)$ for full support), at most s by independence. For $|F| = 2$ the monomial collineations would preserve the all-ones vector line contrary to irreducibility. Thus distinct systems force $|F| = 3$ and support two throughout (v large and use E^+ -transitivity). They then pair the lines of the first system by a perfect matching, putting both the sum and

difference lines on each pair, uniquely. Indeed each support pair has orbit spanning its full plane, so distinct pairs are disjoint. In unitary type use determinant-one norm-one diagonals similarly; the base $\sqrt{|F|} + 1 \geq 3$ allows only support one. In odd-characteristic orthogonal type use sign flips even in each norm-square class, lying in Ω , again a normal soluble group. Orbit size at least 2^{s-3} forces $s \leq 5$; each occurring type is large, allowing arbitrary support signs, so actually $s \leq 2$. Distinct systems again give matchings, uniquely per matching since the two lines on a pair must be orthogonal: they are generated by $x + y, x - y$ from the two first-system lines with equal nonzero norms, forcing matching within one square class and the relative identification up to sign. There are no nondegenerate lines for alternating polar form. In the matching cases fix two pattern points i, j on a pattern line. Their intersection group contains alternating permutations of matched configurations independently in each of the (at most two) isometry types, so has a subgroup preserving both with absolutely bounded number of orbits on ordered pairs of first-system indices. Every other system on the pattern line gives a matching invariant by that subgroup. This bounds the number absolutely, contradiction for large lattice q . $\square$

Lemma 6.9 (Supported block groups and exceptional planes). *For direct blocks of dimension $k \geq 2$, the supported special groups described below give a normal projective product Y , with retained block centers. Modulo those centers its nonsoluble parts are simple powers, with at most two simple sites per block. The orthogonal-plane cases in which the supported rotation group has order at most two are either already impossible or are excluded by the graph-isometry matching argument below.*

Proof. We explain the supported inner group choices for $k \geq 2$. Use products across blocks of their natural special groups, actually supported (for orthogonal spaces we can take Ω in the sense of the determinant-one spinor kernel in odd characteristic, and the kernel of the Dickson parity in even characteristic; thus in split four-space over two we use the group with the two independent natural SL_2 tensor actions); replace $\mathrm{Sp}_4(2)$ by its simple derived group. Denote the resulting projective product by $Y \trianglelefteq K$, and include its normal block-center subgroup among the soluble normals retained by $(*)$. Except for soluble cases, modulo these centers Y is a direct product of isomorphic nonabelian simple groups, one per block, or two per block in orthogonal split dimension four. We recall details to control small cases. The standard natural special modules are absolutely irreducible in these dimensions except the orthogonal planes. This follows also from the usual natural modules in defining characteristic (orthogonal 4^- uses the two Frobenius-conjugate natural SL_2 factors tensoring after scalar extension, 4^+ the external tensor; orthogonal odd dimension here is in odd characteristic). For $\mathrm{Sp}_4(2)' \cong A_6$, restriction stays irreducible by normality in $\mathrm{Sp}_4(2)$ and the orders too large for $\mathrm{GL}_3(2)$; the end field cannot enlarge by the order in $\mathrm{GL}_2(4)$ either. Soluble low exceptions include natural SL_2 (also SU_2 or Sp_2) at parameters 2,3, $\mathrm{SU}_3(2)$ (its cycle and determinant-one cube-root diagonals act absolutely irreducibly), $\Omega_3(3)$, and Ω_4^+ at 2,3; the indicated irreducibility still holds. For an orthogonal plane Ω is the rotation group of order $(|F| \mp 1)/\gcd(2, |F| - 1)$, acting semisimply without fixed space or invariant nondegenerate line when that order exceeds two.

Small orthogonal planes. For order at most two in those planes: split over two gives a canonical nonsingular vector per block, impossible by irreducibility; split over three gives a canonical orthogonal nonsingular line refinement (the two nonsingular lines in the split plane), now excluded. The remaining cases (split over five and nonsplit over three) have individual supported signs $z_a = -1$ on block a in H . Treat patterns of these cases separately. If two systems are compatible (commuting projectors), distinctness and transitivity give a common orthogonal line refinement, impossible. Otherwise some sign of one moves blocks of the other (if they all fix them the projections split the blocks). By rank two it transposes precisely two and is identity on the rest. Those two planes must be graphs $\pm f$ of an isometry from the sign block U_a to a plane in its orthogonal complement: their

sum contains U_a , they project isomorphically in both directions there, and they are orthogonal. Every other sign preserves the graph pair by commutation with z_a , so the partner plane is precisely some U_b (each sign outside a acts on it as a scalar). Transitivity gives a perfect matching and a graph-isometry choice on each edge. The intersection group for two fixed systems contains alternating permutations of identically aligned matched configurations (all are isometric). Every further system on the pattern line must give a matching invariant by this group, boundedly many possibilities, and boundedly many isometry choices because the field here is bounded and only one choice per edge orbit needs specifying. The edge orbit bound again follows from bounded rank on the block indices. This contradicts large lattice q . $\square$

Lemma 6.10 (Supported groups in a prepared anchor). *For every remaining direct system, each block a has an actually supported matrix subgroup R_a , projectively contained in E^+ , of order greater than two. It is normalized semilinearly by the block stabilizer of E^+ . Its block module is semisimple and has no trivial constituent. This includes the case in which the supported simple power is missing from the base.*

Proof. In the remaining nonexceptional direct systems, there exist for every block a actually supported matrix subgroups R_a projectively in E^+ , normalized semilinearly by the block stabilizer in E^+ , of order greater than two acting nontrivially there. For soluble Y or retained Y , this follows from the full supported choices above. If $HY = K$ is missing, H induces the full top permutation image (on blocks and simple sites), since Y acts trivially there. If two simple-site orbits occur (two per block), at most one orbit product is missing modulo centers, and the other supplies supported groups as required. Otherwise there is a transitive orbit, and we have the exact interval over P in the H -invariant subgroups of the simple power $Y/(\text{block centers})$, where P is the base intersection image. In fact after the centers have been quotiented this transitive normal simple power is minimal normal, hence meets the base core trivially (being missing) and maps to the monolith in the core quotient. By Proposition 4.11, either P is a nontrivial product and all overgroups here are products, or all nonbottom labels are subdirect. The latter contradicts the height: strict comparable subdirect groups give strict refinement of invariant ties, whose chain length is absolutely bounded by the top orbit bound (simple sites have fibers of size at most two over blocks). Thus $E > H$ gives an intersection product with all projections strictly larger than the corresponding nontrivial projections of P . Lifting with the retained centers supplies the required singly supported R_a (one can take the full available preimage on that block in the specified supported group, including both sites if they are permuted there). Indeed all centers lost in passing from the supported choices in Y to the simple-site images have projective images in H ; so independent tuples in this intersection product lift with their supported representatives projectively in E . Conjugating by a semilinear collineation of E^+ preserving the block preserves the supported matrix choice as well, acting on representatives which are the identity on the other blocks. In all cases the block module for R_a is semisimple with no trivial constituent, since its module socle and its fixed space are invariant under the semilinearly irreducible block stabilizer. $\square$

Lemma 6.11 (Compatibility of direct systems). *Any two remaining direct systems in one homogeneous pattern have commuting projectors and hence a common direct refinement.*

Proof. Any two systems in such a pattern must be compatible. If all R_a 's of one fix the other's blocks individually, each projection onto a of such a block lies within the block (use differences and absence of a trivial quotient), giving compatibility. Thus incompatibility would force movement each way by supported matrices. A supported matrix moving blocks has displacement rank at most its support-block dimension and at least the moved block dimension. Hence both dimensions are

equal and equality holds. On the moved system it then transposes just one pair of blocks with square the identity, and acts identically elsewhere (from the fixed dimension). Its displacement has full rank on the support block, so an involution there is impossible in characteristic two and can only be -1 in odd characteristic. But a group of order greater than two cannot have only one element inducing a nonidentity block permutation, contradiction. $\square$

Proposition 6.12 (Completion of direct-system pruning). *The remaining direct systems are excluded, both when their supported normal product is retained and when it is missing. The bounds are absolute, independently of the ground field and natural degree.*

Proof. Separate retained and missing colors. For retained systems, any two compatible systems must refine one another in some direction: incidence degrees are constant on each block side by transitivity, and if both exceed one, each touching block extends outside, hence is fixed setwise by the other's supported group there (fixed outside points). This gives an invariant proper nonzero block intersection, nondegenerate when relevant, impossible by the retained actions above. A coarse block of large dimension cannot admit the resulting uniform finer split under its full supported special group by orders. Uniformly, with base size $s_0 = |F|$ ($\sqrt{|F|}$ in unitary type), and $c = 1$ in linear/unitary, $c = 1/2$ in orthogonal/symplectic, the large m -dimensional special group order is at least $s_0^{cm^2 - O(m)}$, whereas the isometry (or linear) stabilizer of a corresponding uniform t -block direct split is at most $t!s_0^{cm^2/t + O(m)}$ by standard orders, smaller uniformly for large m and $2 \leq t \leq m$. Bounded coarse dimensions permit only boundedly many comparable refinements. This excludes the retained patterns.

For missing systems take all points of a pattern line. Compatibility gives their common refinement into nonzero fine cells. Fix two systems i, j and group the fine cells by their pair of block indices. Each aggregate cell has a supported special group inside $K_i \cap K_j$. If its dimension is large, no further system can split it: a splitting block extending outside the cell would give a proper invariant intersection under the supported group, whereas a block contained in the cell would give a uniform finer split contrary to the same order bound. A small aggregate has at most its dimension many nonzero fine cells. Thus the number of fine cells over each occupied pair is bounded by an absolute constant d .

The map from a fine cell to its block-index pair is H -equivariant. Indeed H stabilizes both systems and permutes their common refinement; anti-actions also permute products of commuting projectors. The two rich top actions and Lemma 6.7 give an absolute bound C on the orbits of ordered pairs of occupied pairs. Each original system is recovered by grouping the fine cells and taking their direct sums. Distinct full stabilizer labels therefore give distinct H -invariant partitions of the fine cells. By (6.1) there are at most 2^{Cd^2} such partitions, contradicting the chosen pattern-line size. This completes direct-system pruning. $\square$

6.4 Tensor geometry and order comparisons

Lemma 6.13 (Ordered factors and their forms). *Let A be a unital proper nonscalar central simple F -subfactor of $\mathcal{A} = \text{End}_F(V)$, normalized by the current previously good anchor. Then $V = X \otimes_F Y$, with $A = \text{End}(X) \otimes 1$ and commutant $A' = 1 \otimes \text{End}(Y)$. Both factors have dimension at least two. If $A = A^*$, the ambient form is the tensor of factor forms. In bilinear characteristic two the factor forms are alternating; an orthogonal quadratic form is the canonical form with that polar pairing and zero on pure tensors.*

Proof. We next treat subfields and ordered tensor structures. All algebras below inside $\mathcal{A}$ are unital F -algebras (“prime” extension or field algebra in these exclusions refers to prime relative degree,

not to the characteristic field). Recall that a central simple matrix subfactor A gives

$$V = X \otimes_F Y, \quad A = \text{End}(X) \otimes 1, \quad A' = 1 \otimes \text{End}(Y)$$

with A' its commutant (central simple algebras over finite fields split). For A proper and nonscalar both factor dimensions are at least two. Linear normalization acts by tensor matrices up to scalars, by the matrix algebra automorphism theorem; similarly with field automorphisms. If $A = A^*$, both factors have defining forms, with b their tensor form up to a scalar: use the restrictions of the matrix involution to the two factors. These are both hermitian for unitary type, and each symmetric/alternating bilinear otherwise. Semilinear similarities preserving the ordered tensor act by semilinear similarities on the factors. In bilinear characteristic two, an ordered factor normalized by such an irreducible E cannot have nonalternating form (the diagonal contraction has an invariant hyperplane, then tensor with the other space). With both alternating, any orthogonal quadratic form in question must be the canonical one for the polar tensor, vanishing on pure tensors. Such a quadratic form exists (impose zero on basis tensors and expand, using alternation on both factors: for a pure tensor the potentially nonzero pair terms have both indices different and cancel by interchanging which second indices are matched to the two first indices); it transforms under E like the given one, and a nonzero difference, the square of a linear functional, contradicts irreducibility. We will use these facts whenever there is a common previously good anchor. $\square$

Lemma 6.14 (Uniform tensor order obstruction). *For sufficiently large dimension m , a full supported perfect natural classical group cannot normalize a proper nonscalar tensor factor or a system of at least two nontrivial tensor factors, even allowing permutations of the factors.*

Proof. Useful also is the uniform order obstruction: a full supported perfect natural classical group on a space of sufficiently large dimension m cannot normalize a proper nonscalar central simple tensor factor, or a tensor system splitting the space into at least two nontrivial slots even up to permutations. Its order (also modulo scalars) is at least $|F|^{m^2/2 - O(m)}$; the linear tensor-system order is bounded by

$$t! |F|^{\sum_j x_j^2}, \quad \prod_{j=1}^t x_j = m, \quad t \geq 2, \quad x_j \geq 2,$$

which is uniformly smaller ($t \leq \log_2 m$, $\sum x_j^2 \leq m^2/4 + 4$, e.g. by merging factors). Use the linear action of the covering matrix group for this comparison. This proves also properness of such stabilizers in the large ambient group. Properness for field algebras follows directly from perfection (fix them pointwise if normalized by the socle) and absolute irreducibility on the natural module. $\square$

6.5 Field and ordered-factor exclusions in linear type

Proposition 6.15 (Prime-degree field algebras in linear type). *After the direct-system exclusions, the full normalizers of field algebras D/F of prime relative degree are excluded in linear type, including when the ambient group contains anti-actions.*

Proof. Exclude stabilizers of field algebras D_i/F of prime degrees r_i . Their soluble normal projective torus $\mathbf{P}(D_i^*) \cap S$ is in H , of order

$$\frac{|F|^{r_i} - 1}{|F| - 1} / \frac{\gcd(v, |F| - 1)}{\gcd(v/r_i, |F| - 1)}$$

by the norm determinant. This exceeds r_i except possibly $|F| = 3, r_i = 2$, when it is at least two. If a pair of fields do not commute, each torus injects by normalization into the other's relative field

Galois group (a nontrivial torus element has lifts generating its prime field extension). This forces the exceptional case on both sides. If two fields commute, their generated algebra is commutative semisimple, necessarily a field since the primitive idempotent system would otherwise contradict the direct-system exclusion at E . Distinct commuting fields therefore have distinct prime degrees. In a nonexceptional line take their compositum; the intersection group contains the natural determinant-one group over it, perfect if module dimension is at least two, which forces any third prime field on the line into the compositum by centralization (use elementary matrices). If dimension one, pairwise commutation alone does the same. This contradicts distinctness. In a line of the exceptional type, distinct fields must be noncommuting. They have generators $J_i^2 = -1$ with projective class in the indicated torus (norm one), and pairwise normalization gives anticommutation. Two generate a split quaternion algebra $M_2(F)$; their intersection group includes the large determinant-one group on its tensor complement. Every line field lies in that bounded pair algebra by centralization, impossibly many. We separate the types by color here. $\square$

Proposition 6.16 (Ordered factors in linear type). *After Proposition 6.15, the full stabilizers of proper nonscalar ordered central simple factors are excluded in linear type.*

Proof. Now exclude ordered proper nonscalar central simple factors A . The projective supported special linear groups on both tensor spaces are commuting normals in the label, even allowing anti-actions, with soluble quotient. Exactly one is retained; designate its factor algebra. Its natural projective group is simple except dimension two over two or three. Over two, the characteristic order-three subgroup would give an invariant quadratic field, impossible. Over three, the A_4 has a unique Klein four subgroup, normal and spanning the factor algebra by its quaternion lifts. Treat this last color separately from the simple case. For a pattern of designated algebras the groups are normal in H ; intersection nontrivial gives equality (or equality of the Kleins in the exceptional case), hence equality of algebras, since their lifts span. Otherwise they commute projectively and also linearly on special-group lifts (perfection, or no scalar characters for $SL_2(3)$). Thus distinct designated algebras commute elementwise. For a pair on a line the new algebras lie in the centralizer matrix factor, on whose space the intersection contains the full supported special linear group. In large dimension this prohibits any proper new tensor subfactor there by orders (the whole centralizer factor then permits at most one distinct new designated algebra); in bounded dimension only boundedly many distinct commuting nonscalar central simple factors can fit there anyway (their tensor-product matrix algebra embeds there by simplicity, so the product of degrees is bounded). Contradiction. $\square$

6.6 Partial ordered-factor pruning in isometry type

Proposition 6.17 (Ordered factors apart from two retained plane types). *In isometry type, after the direct-system exclusions, the full stabilizers of ordered proper nonscalar selfadjoint central simple factors are excluded except possibly when the retained support is an odd-characteristic bilinear orthogonal plane or a symplectic plane over $\mathbb{F}_2$. These exceptions concern the retained side of the ordered factorization.*

Proof. First try to exclude ordered proper nonscalar selfadjoint central simple factors A , leaving two possible remaining types temporarily. In the stabilizer the natural projective special groups supported on X, Y are commuting normals with soluble quotient. Again use $Sp_4(2)'$ in its small case. The supports indicated lie in S . Besides perfection in the usual cases, note unitary determinant one, and in orthogonal type: in characteristic two both factors are alternating as above and the tensor action from a factor has even displacement rank; in odd characteristic Ω on a symmetric factor lies

in Ω on the tensor (diagonalize the other symmetric factor and use orthogonal sums; scaling does not change the spinor norm on determinant one). If both factors alternating in odd characteristic, the symplectic supports also lie projectively there by perfection or absence of a 2-quotient for $\mathrm{SL}_2(3)$. Quotient solubility follows by the two tensor projections and the standard similarity groups. Exactly one support is retained, and we designate its factor algebra.

We exclude the labels for which this retained type is neither a bilinear orthogonal plane in odd characteristic nor a symplectic plane over two. Here are the details of the supported projective groups within the designated algebra:

- Usually it is a single simple group with perfect special lift and absolutely irreducible spanning natural action. In split orthogonal dimension four (odd characteristic here) there are two projective PSL_2 components on symplectic slots, with their matrix algebras tensoring to the factor. They are permuted by the label, even at parameter three: in the determinant-form model the singular rank-one quadric and its two rulings give these two tensor slots up to exchange (the maximal linear rank-one spaces are precisely of the two fixed-line kinds). Indeed the endomorphisms preserving each member of a ruling individually form the matrix algebra on the other coordinate (use two independent first-coordinate lines and their sum-vector line). Nonsplit four has the standard simple projective group with absolutely irreducible tensor-twist action instead.
- The degree-two symplectic or special unitary groups at parameter three (also the symplectic slots just described at three) give projective A_4 with Klein lifts spanning the slot and special lift $\mathrm{SL}_2(3)$ having no ground-field scalar characters. Indeed the Klein preimage in the rank-one special group at three is quaternion of order eight with commutator -1 acting as negative identity in degree two, thus absolutely irreducible there (semisimplicity for this 2-group, no linear constituent possible), and the abelianization of $\mathrm{SL}_2(3)$ has order three whereas the ground scalar group here has order two or eight. A degree-two unitary factor at parameter two is impossible: its characteristic order-three subgroup in the projective special has two orthogonal nondegenerate eigenlines over four, giving a forbidden nondegenerate direct decomposition on tensoring. An orthogonal 3-space over three is likewise impossible: for the form up to similarity a sum of three squares, the three norm-one lines are the axes preserved as a set even by similarities, again a direct decomposition.
- For the special unitary 3-space over four (parameter two) the projective special C of order 72 has a unique normal Sylow 3-group D_9 elementary of order nine, contained in every nontrivial normal subgroup, and its lifts span the slot. Indeed in an orthonormal basis use the cycle and $\mathrm{diag}(1, \omega, \omega^2)$ for a cube root $\omega \neq 1$. Their commutator generates the scalar special center; projectively they give D_9 . In the projective full unitary group of order 216, the Fourier matrix and $\mathrm{diag}(1, 1, \omega)$ normalize it inducing generators of $\mathrm{SL}_2(3)$ (rotation and shear). Thus the whole 216 normalizes it with centralizer the nine. The special image contains the nine and induces order eight acting irreducibly there (a line stabilizer in $\mathrm{SL}_2(3)$ has order six). Thus the nine is minimal normal and a normal subgroup disjoint from it must centralize it, proving the assertion. Its linear special lift has no character to the scalars of order three by this action and the center commutator.

These lists follow from the same natural classical groups as for direct blocks, now with odd characteristic on symmetric factors. The individual component matrix algebras here are selfadjoint.

Color according to simple components, A_4 components, or the single C . The projective components lie in H , which permutes at most two of them in each system. In a simple or A_4 pattern they

cannot swap the other's pair (no quotient of order two), so components across systems mutually normalize. A nontrivial intersection, normal in each, gives equality or common Klein, hence the same component algebra; otherwise they commute projectively and linearly on special lifts as above. Likewise for the C pattern, using a common D_9 if intersection is nontrivial (it contains the normal Sylow on each side, which must then be the same). Thus all the distinct component algebras are commuting tensor slots. For two labels i, j on a line their components (at most four) have a tensor centralizer factor, itself normalized by E ; any new component lies there. In the pair intersection is the large supported perfect classical group on that centralizer space if of large dimension, by the tensor form observations. It must stabilize individually the at most two components for each further point of the line. The order bound prohibits a proper nontrivial central simple subfactor there being new (the whole factor itself would add only one); if of bounded dimension only boundedly many commuting new ones fit. Thus the list of distinct component algebras on the whole line is absolutely bounded (at most four from the original two labels plus those new ones). Each designated algebra consists of one or two such components tensoring and determines the stabilizer (it and its tensor commutant determine the ordered tensor). Only boundedly many labels then fit on the line, contradiction. Consequently any remaining previously good-normalized proper ordered selfadjoint factorization must have the **retained** side one of the two remaining plane types. $\square$

6.7 Transfer to normalized field algebras

The remaining field obstruction is commutation. In linear type the retained scalar tori nearly settled this directly. In isometry type we instead use classical groups over the extension fields. The next setup identifies the algebra whose center determines which transfer argument is needed.

Convention 6.18 (A pair of field witnesses). Let D_i, D_j be selfadjoint field algebras of prime relative degree, with distinct full normalizer labels K_i, K_j on a pattern line, and put $T = K_i \cap K_j$. The algebra $R = F[D_i, D_j]$ is selfadjoint semisimple: its radical is normalized by the common irreducible E and must annihilate the space. Its primitive central idempotents are either just one or exactly two exchanged by $*$, since sums on $*$ -orbits are orthogonal projectors contrary to the earlier direct-system exclusion if there is more than one orbit.

When the center is a field larger than F , or has two exchanged components, we will find a perfect group inside T whose centralizer forces D_i and D_j to commute. In the scalar-center case we will rule out $R = \mathcal{A}$ and use the resulting ordered tensor factor to return to the two retained plane types. We first establish the form and centralizer calculations needed for these alternatives. The two-dimensional unitary exception in the centralizer calculation will be handled by a norm-one scalar count; exchanged components require the paired special linear group below.

Lemma 6.19 (Transferred forms). *Let $D/F \subseteq \mathcal{A}$ be a normalized selfadjoint field algebra, of arbitrary finite degree, and put $\delta = *|_D$. In the prepared irreducible-anchor configuration there is a nondegenerate transferred form B satisfying*

$$b(dx, y) = \text{Tr}_{D/F}(dB(x, y)).$$

The associated bilinear, hermitian and quadratic structures, and their transformations under the normalizer, are as described below. The perfect natural special group of the transferred form, when available, lies in S .

Proof. For a normalized field (degree possibly composite), let $\delta = *|_D$. There is a nondegenerate transferred form B given by

$$b(dx, y) = \text{Tr}_{D/F}(dB(x, y)).$$

It is linear over D on the first slot and δ -semilinear on the second. It is symmetric/skew or hermitian/skew-hermitian as appropriate (skew-hermitian may be rescaled). In particular if b is alternating and $\delta = 1$ then so is B , also in characteristic two by testing all square scalars. For orthogonal characteristic two and $\delta = 1$ transfer $Q(dx) = \text{Tr}_{D/F}(d^2 Q_D(x))$, valid since the left side is additive in d (the cross term vanishes by alternation of B) and scales by squares over F , so gives a linear function of d^2 over F ; testing the trace against these same squares gives D -quadratic homogeneity of Q_D and polar B . For orthogonal characteristic two and $\delta \neq 1$ use instead

$$Q(x) = \text{Tr}_{D_0/F}(B(x, x)), \quad D_0 = D^\delta.$$

Indeed the right side has polar b and transforms the same way under any previously good anchor normalizing D . The difference from Q has zero polar form, hence is the square of a linear functional over the finite, therefore perfect, ground field. If nonzero, its kernel would be a proper nonzero invariant subspace for that anchor. Irreducibility therefore makes the difference zero. All form transformation assertions follow from trace uniqueness under semilinear maps normalizing D . For instance for a similarity u with multiplier μ on b and induced field power τ on D , testing at d^τ gives $B(ux, uy) = \mu B(x, y)^\tau$. Thus the stabilizer acts projectively by corresponding D -semilinear similarities (extra scalar kernel soluble), and the standard D -linear special group, when perfect, lies inside S . $\square$

Lemma 6.20 (Centralizers of perfect transfer groups). *Suppose the field normalizer in Lemma 6.19 is nonsoluble and the total ground-field dimension is sufficiently large. There is a perfect natural classical group P_D , absolutely irreducible over D . Its centralizer in ground-field endomorphisms is exactly D , except possibly for a unitary D -module of dimension two with $\delta \neq 1$ fixing F . This exception does not occur for prime extension degree in sufficiently large total dimension.*

Proof. If this normalizer is nonsoluble and total ground-field dimension large, we can use a perfect such classical group P_D on the natural D -module, absolutely irreducible over D . Indeed either module dimension or extension degree/field size is large, avoiding the small parameter exceptions; in the permanently soluble form dimensions nonsolubility of the normalizer would be impossible. Its centralizer in ground-field endomorphisms is exactly D , except possibly the unitary module of dimension two over D with $\delta \neq 1$ fixing F . To see this, extending from F to algebraic closure splits the module into copies twisted by the different embeddings of D over F . They are irreducible and pairwise inequivalent since matrix traces generate D over F : symplectic type uses an SL_2 block; orthogonal type can use split 4-space with tensor trace there an arbitrary product, or norm traces from the 4^- tensor twist, and in dimension three the symmetric-square traces (odd characteristic). Unitary type has traces generating at least D_0 via SU_2 , and in dimension at least three a norm-one diagonal with entries $a, b, (ab)^{-1}$, where $a, b, ab \neq 1$, has trace differing from inverse trace, giving all of D . In dimension two, $D_0 F = D$ if δ is nonidentity already on F . Here on a split nondegenerate orthogonal 4-space use the two $\text{SL}_2(D)$ tensor actions, supported there (available also within higher dimensions of Witt index at least two); in the nonsplit dimension-four case the trace on the tensor twist ranges over norms of arbitrary SL_2 -traces, and in dimension three the values $u^2 - 1$ from the symmetric square generate the field in odd characteristic. For SU_2 the natural action over D realizes the split rank-one traces over D_0 . The indicated three-entry unitary diagonal exists even when the norm-one group has order three (take $a = b$); the difference of its trace from inverse trace, multiplied by ab , is $-(a - 1)(b - 1)(ab - 1)$. Thus the embedding distinction (and absolute irreducibility in each copy, as for natural classical groups above) gives scalar centralizer on each over closure whenever claimed, hence the centralizer assertion over F . In particular the exception does not arise for prime degree in sufficiently large ground-field dimension (it would require degree two and module dimension two). $\square$

Lemma 6.21 (Paired transfer spaces). *In the high-dimensional prime-degree setting of [Convention 6.18](#), suppose an unordered complementary pair $e, e^* = 1 - e$, normalized by E , commutes with D_i . Its two spaces are dually totally isotropic and, in orthogonal type, totally singular. Their simultaneous field-and-pair stabilizer contains the paired group $\mathrm{SL}_m(D_i)$. In the nonsoluble case this group is perfect in S , and every ground-field endomorphism centralizing it commutes with D_i .*

Proof. We will also use the following paired situation (still with the prime-degree D_i/F and high total dimension). Suppose two central or commuting projections to be used below are exchanged by $*$ (a complementary pair $e, e^* = 1 - e$ normalized as a pair by E), and commute with D_i . Their spaces are dually totally isotropic, and totally singular also for Q when relevant (in characteristic two each quadratic restriction with zero polar is a linear square, and the sum of the two kernels would be a proper nonzero invariant space if a restriction were nonzero; the dimensions are large). The full simultaneous stabilizer of D_i and the pair thus includes paired $\mathrm{SL}_m(D_i)$ on these spaces, acting by inverse adjoint on the second, where m is their D_i -dimension; if $m = 1$ the stabilizer would be soluble. For $m \geq 2$ this group is perfect inside S . Every endomorphism centralizing its matrices commutes with D_i . Indeed over closure the constituents are natural at embeddings τ over F and dual at $\tau'\delta_i$ with τ' over F (by the transferred pairing); the field scalars act there by τ, τ' , respectively. Natural twists are distinguished by traces, as are dual twists; a natural and dual identification could only have $\tau = \tau'\delta_i$ by traces on an SL_2 block. It cannot occur for $m \geq 3$ (diagonals $a, b, (ab)^{-1}$ again; $|D_i| \geq 4$). If $m = 2$, either the two identified constituents have the same field-scalar action, or $\delta_i \neq 1$ fixes F , forcing prime degree two and bounded total dimension. This proves the claim. $\square$

6.8 Prime-degree field pruning in isometry type

Proposition 6.22 (Prime-degree selfadjoint fields). *After direct-system pruning and the partial ordered-factor exclusion of [Proposition 6.17](#), the full stabilizers of selfadjoint field algebras of prime relative degree are excluded. The pattern is colored by whether the adjoint restriction on the field is the identity.*

Proof. *A larger center field.* If the center of R is a field larger than F , take a central prime extension D' invariant under $*$ and T . If $K_i \leq N_M(D')$, then P_{D_i} forces $D' \subseteq D_i$ by centralization, so $D_i = D'$ commutes with D_j . Otherwise the covering property gives

$$T = K_i \cap N_M(D')$$

exactly. Consider the commutative semisimple $L_f = F[D_i, D']$. If a field, the degrees must be distinct primes with compositum degree their product, and the perfect transfer group over it is available in T (the subfields are unique, and T is nonsoluble since it contains E). This perfect group centralizes D_j , forcing commutation with D_i by the centralizer assertion. In the exceptional unitary dimension two case here, the compositum degree is $2s$ with s a large odd prime (total dimension $4s$). Write $a = |F|$ and identify the compositum with $C = \mathbb{F}_{a^{2s}}$. Work in the actual scalar group

$$U = \{z \in C^\times : z^{a^s+1} = 1\}, \quad |U| = a^s + 1.$$

The subgroup U_S whose projective classes lie in S has index at most two in U : these are actual determinant-one isometries, with at most the spinor or Dickson restriction still to impose. Their classes belong to T . Let $U_0 \leq U_S$ be the scalar preimage of the kernel of their action on D_j . Since the matrices are F -linear, this action has order at most $[D_j : F]$. That prime degree divides $4s$, so it is either 2 or s , and

$$|U_0| \geq \frac{a^s + 1}{2s}.$$

The two maximal proper relative subfields of C/F are $\mathbb{F}_{a^s}$ and $\mathbb{F}_{a^2}$. Their intersections with U have sizes at most 2 and $a + 1$, respectively: the first satisfies $z^2 = 1$, while on the second the oddness of s turns the norm-one condition into $z^{a+1} = 1$. For sufficiently large s , the displayed lower bound exceeds $2 + (a + 1)$. Hence U_0 contains a generator of C/F . Ground-field scalar changes do not affect conjugation on D_j , so this is a count in the scalar preimage throughout. Such a generator centralizes D_j , forcing all of C , and in particular D_i , to commute with D_j .

A split compositum. If L_f here splits, there are exactly two exchanged components, and D_i, D' must have the same prime degree r (tensor over F). The possible primitive components are indexed by relative identifications of the two degree- r fields; E acts on these by translations of a cyclic r -set (the two field automorphisms extend the same base automorphism), transitively on the occupied subset. Thus $r = 2$. Now apply the paired situation to the two components: the indicated paired special linear group also centralizes D' (on each component that field acts through D_i). Hence it lies in T , centralizes D_j , and gives commutation with D_i .

Two exchanged central components. If the center of R instead has two exchanged components, use their projection-pair stabilizer in place of $N_M(D')$. It cannot contain K_i by perfection and the centralizer of P_{D_i} , so its full intersection with K_i is T , by covering. The paired situation applies directly, yielding the same commutation.

Scalar center. It remains to consider center F . Here $R \neq \mathcal{A}$, else T 's actions on the two fields have trivial projective kernel and soluble image. Thus R gives a proper ordered selfadjoint factorization, necessarily with one of the retained remaining plane factors by the partial exclusion. If R itself is that plane matrix algebra, then for bounded field size the intersection contains the high-dimensional perfect classical group on its tensor complement, forcing every line field into bounded R , contradiction. For an odd orthogonal plane factor at large field size the projective supported rotations (from Ω_2) have order greater than four, and lie in H as a normal soluble group for the stabilizer label of R (which contains E). Normalizing both D_i, D_j (quadratic within R), some nonscalar rotation centralizes both. In the plane matrix algebra this gives their equality (the centralizer of a nonscalar matrix there has dimension two), impossible. If instead R' (the matrix commutant of R) is the remaining plane algebra, K_i cannot preserve R , since then P_{D_i} would act trivially on R' by its soluble similarity projection, contrary to its field centralizer. Thus again T is the full intersection $K_i \cap N_M(R)$. On the large R -factor space apply transfer to D_i there, selfadjoint for its factor form. Its field normalizer there must be nonsoluble (otherwise both factor projections of T soluble). We get its supported perfect transfer group in T , with centralizer in R exactly D_i ; in characteristic two here use the alternating factor form, the total tensor preserving its given orthogonal quadratic if necessary as explained earlier. This group centralizes $D_j \subset R$, impossible.

Completion of the line argument. Consequently the fields on a pattern line all commute unless we already obtained a contradiction. Two distinct ones of the same prime degree would give a split commutative algebra with exactly two exchanged components as above, degree two, and the two adjoint restrictions must differ (to exchange the relative-identification idempotents). Use homogeneous color of the adjoint type (identity or not on the prime-degree field algebras) to prevent this. Thus two have distinct primes and generate a field. Its transfer group in the nonsoluble intersection forces any third into the compositum, impossible by prime degrees. There is no unitary dimension-two exception here: if the involution fixes F but not the compositum, the two prime fields have different adjoint types. This completes prime-field pruning. $\square$

6.9 Paired idempotents and the remaining ordered factors

Proposition 6.23 (Completion of the algebra exclusions). *After Proposition 6.22, the full stabilizers of unordered proper idempotent pairs $e, e^* = 1 - e$ are excluded. The two remaining retained-plane*

types in [Proposition 6.17](#) are then impossible, completing the ordered-factor exclusions.

Proof. Exclude stabilizers of unordered pairs $e, e^* = 1 - e$ of proper idempotents. For two such labels, choosing idempotents e, f , the generated selfadjoint algebra is again semisimple. A center field larger than F is now forbidden. If there are two central components exchanged, their projector pair commutes with each of the given pairs. Distinct commuting such pairs are impossible: they would give all four projector cells (or already be the same pair, using adjoint exchange), with forbidden orthogonal direct sums on adjoint orbits. For center F , note $(e - f)^2$ is central, and the algebra is spanned over this scalar algebra by $1, e, f, ef$, since the relation rewrites fe in terms of e, f, ef and scalars and the generators are idempotent. It is therefore $M_2(F)$, which must have remaining plane type. An orthogonal plane has at most one such pair. For the bounded symplectic plane case, the high perfect tensor-complement group in the intersection fixes individually each further pair on the line, forcing them into the bounded plane algebra. Thus all paired stabilizers are excluded.

Finally the remaining ordered factors themselves cannot occur: an odd orthogonal plane factor has the unique pair of isotropic lines over the closure, giving a forbidden projector pair if split, or a forbidden quadratic field (the diagonal algebra descended to F) if nonsplit, selfadjoint and normalized by all the factor similarities. A symplectic plane over two gives a normalized selfadjoint quadratic field via its characteristic order-three subgroup. This completes the stated field and ordered-factor exclusions. $\square$

6.10 Full spanning and elementary normalizers

Lemma 6.24 (Full spanning). *After the subspace, direct-system, field, ordered-factor and paired-idempotent exclusions, every nontrivial projective subgroup $P \leq S$ normalized by a previously good anchor satisfies $F[\text{lifts of } P] = \mathcal{A}$.*

Proof. After the preceding exclusions we have the following full spanning property: any nontrivial projective subgroup $P \leq S$ normalized by a previously good anchor has

$$F[\text{lifts of } P] = \mathcal{A}.$$

Indeed the radical vanishes by anchor irreducibility, and in isometry type the algebra is selfadjoint. Multiple center components are now impossible by the direct-system exclusions (and the exchanged pair exclusion for adjoints). Center-field enlargement and a proper nonscalar central simple factor are also excluded. Semilinear anti-actions in linear type respect the algebra as well (their group action on lifts uses inverse, harmless here). $\square$

Proposition 6.25 (Elementary abelian normalizers). *Under the full-spanning conclusion of [Lemma 6.24](#), full normalizers of nontrivial elementary abelian subgroups of S are excluded.*

Proof. Exclude full normalizers of nontrivial elementary abelian subgroups $P \leq S$. In a pattern each such subgroup is contained in $H \leq E$ by (*). Its scalar commutator pairing from linear lifts is nondegenerate by full spanning. Thus it identifies P with its ground-field character dual (nondegeneracy already forces the exponent prime to divide $|F^\times|$). Lifts are linearly independent by their distinct simultaneous conjugation characters, and their linear span is the generated algebra. Hence $|P| = v^2$ and its projective linear centralizer is itself (match the scalar conjugation character using the pairing, then use full spanning). There cannot be two distinct ones here normal in E : a nontrivial intersection would itself satisfy the same properties, and a trivial intersection would give projective commutation. This excludes the pattern. $\square$

6.11 Tensor systems and missing normal powers

Lemma 6.26 (Reduction to one simple group per tensor slot). *After the preceding exclusions, a tensor system has transitive equal-dimensional slots and the appropriate tensor forms. Its stabilizer induces at least the alternating group on the slots. To exclude all tensor systems, it suffices to exclude those having a normal supported projective product $Y = B^t \leq K \cap S$, with one nonabelian simple group per slot whose lifts span the slot algebra.*

Proof. Now exclude stabilizers of tensor systems of at least two nontrivial matrix factors tensoring to $\mathcal{A}$, allowing permutations of slots, with factors individually selfadjoint in isometry type. For systems under consideration the anchor acts transitively, else an orbit product contradicts ordered-factor exclusion. Thus we have t slots of common dimension k . The tensor forms in isometry type can be aligned to identical forms on the factors up to scalars by transitivity/similarity (finite-field form classes). In bilinear characteristic two all are alternating (by transitivity and alternation on V); again any orthogonal quadratic here is the canonical one zero on simple tensors by irreducibility. Consequently the stabilizer label K induces at least A_t on the slots by taking commutators of actual tensor-slot permutations inside the full linear or isometry group.

We can reduce to the case of a normal supported projective product $Y = B^t$ inside $K \cap S$, one nonabelian simple group per slot (embedded projectively), whose linear lifts span each slot matrix algebra. Use the natural special groups as above, including $\mathrm{Sp}_4(2)'$. Here is the treatment of other possibilities. Small degree-two parameter cases at two and three in the linear, symplectic and unitary families give canonical nontrivial elementary abelian projective groups per slot (order three or the Klein, characteristic in the projective special). Their product is elementary abelian projectively inside S , now forbidden; the same applies for $\mathrm{PSU}_3(2)$ using D_9 . Containment uses the same special supports as for ordered factors. $\Omega_3(3)$ again gives direct orthogonal line cells from the axes. Odd-characteristic orthogonal planes of split type give pairs of singular lines per slot, whose tensor cells paired with their duals give a forbidden nondegenerate direct decomposition. In nonsplit type use the canonical quadratic fields per slot; their tensor algebra has 2^{t-1} components, each central projector selfadjoint (conjugation on all the fields simultaneously), again forbidden. Split orthogonal four-space factors canonically refine into pairs of symplectic plane tensor slots as above, giving a normalized tensor system of the latter instead. Thus it suffices to exclude the single-simple-per-slot systems (first treating just these labels; the refinement observation then handles the four-space case). $\square$

Lemma 6.27 (Retained and missing tensor products). *Retained products from Lemma 6.26 give at most one system in a pattern. For a homogeneous missing pattern, every $N_i = E \cap Y_i$ is a nontrivial slotwise product $\prod_a R_{i,a}$, strictly above the base intersection.*

Proof. For a pattern with Y retained, these products are minimal normal subgroups of E by slot transitivity. Two must be equal (commutation impossible by perfection and spanning), which recovers the same system by the lifts of the simple factors. For missing patterns $HY = K$, H induces the full top action, transitive and containing the natural alternating group. Exactly as for missing direct simple-site products, the interval over $H \cap Y$ in Y of invariant subgroups must be the nontrivial product case by the reductions and subdirect tie-chain bound. In particular on a pattern line, for each system i ,

$$N_i = E \cap Y_i = \prod_a R_{i,a} \trianglelefteq E,$$

a slotwise projective product strictly above the base intersection. $\square$

Lemma 6.28 (Common refinement of missing tensor systems). *A finite pattern line of the missing systems in Lemma 6.27 has a common tensor refinement formed from the lift algebras of tuple intersections $\bigcap_i R_{i,a_i}$. After scalar factors are removed, those algebras tensor to $\mathcal{A}$ and group to recover every original slot. They are permuted transitively by E . In bilinear characteristic two their forms are alternating; if an orthogonal quadratic form is present, it is canonical also for every grouping into two nonempty batches.*

Proof. Write bars for the quotient by $\text{core}_E H$, and let W be the nonabelian simple-power monolith of $\bar{E}$. Each $\bar{N}_i$ contains W . The simple factors of W partition among the $\bar{R}_{i,a}$: their commutators with W generate W by perfection, lie in the respective intersections with W , and are normal there and mutually commuting.

We need these assignments to give nontrivial intersections in the actual group, not only compatible images in the core quotient. Put $N = \bigcap_i N_i$. Since the N_i are normal in E , an iterated commutator of them lies in N ; its image contains W by perfection. Thus $\bar{N} \supseteq W$. Fix a simple factor C of W , and let a_i be its assigned slot in system i . Because $R_{i,a_i} \trianglelefteq N_i$ and $N \leq N_i$, we have

$$[N, R_{i,a_i}] \leq N \cap R_{i,a_i}.$$

The image of this commutator contains C . Each $N \cap R_{i,a_i}$ is normal in N , so a second iterated commutator shows that

$$M_\gamma := \bigcap_i R_{i,a_i}, \quad \gamma = (a_i),$$

has image containing C . In particular it is nontrivial.

Now take the intersections M_γ for all slot tuples, allowing trivial intersections. They are permuted by E and generate a nontrivial E -normalized subgroup of S , whose lifts span $\mathcal{A}$ by Lemma 6.24. For distinct tuples the lifts commute linearly, because they lie in distinct tensor slots of at least one original system. Their lift algebras therefore commute and jointly generate $\mathcal{A}$. The center of each algebra centralizes all of $\mathcal{A}$, hence is scalar. Its radical, multiplied by the commuting remaining algebras, would give a nilpotent ideal of $\mathcal{A}$, hence vanishes. The non-scalar lift algebras are thus central simple; their tensor product maps isomorphically onto $\mathcal{A}$.

This also recovers the original embedded factors. Group the tuple algebras by their assigned slot in any one system. Each grouped algebra is contained in that slot, and the product of all grouped algebras is still $\mathcal{A}$. Comparing dimensions in the original tensor decomposition forces equality in every slot. We have obtained a common tensor refinement, not merely abstract isomorphic factors.

The fine factors are selfadjoint when appropriate and are permuted transitively by E , by the ordered-factor exclusion. Thus the same form and quadratic observations apply. In particular in bilinear characteristic two the fine-slot forms are all alternating, by transitivity and alternation of the total form. Grouping them into two nonempty batches still gives alternating forms on both. The orthogonal quadratic, if present, is zero also on pure tensors for every such two-batch grouping: its canonical quadratic with the given polar form exists by alternation and is zero on fine pure tensors too. The two quadratics therefore agree on a basis and, having the same polar form, agree everywhere. $\square$

Proposition 6.29 (Completion of tensor-system pruning). *The missing tensor systems of Lemma 6.27 are excluded by an absolute bound on the possible invariant partitions of their common fine slots. Hence all tensor systems are excluded.*

Proof. Fix two systems i, j and send each nontrivial fine factor to its pair of original slot indices (a_i, a_j) . Grouping factors with the same pair gives an aggregate matrix factor. On a large aggregate

space, $K_i \cap K_j$ contains the full supported perfect classical group. No further system on the line can split this aggregate: its fine factors would give a tensor system normalized by that group, contrary to [Lemma 6.14](#). Thus a large aggregate contains only one nontrivial fine factor. On a small aggregate of degree h , there are at most $\log_2 h$ such factors, because their degrees multiply to h and each is at least two. The fibers of the displayed map are therefore bounded absolutely.

The map is H -equivariant, since $H \leq E$ stabilizes both original systems and permutes all tuple intersections; discarding scalar factors preserves this action. By [Lemma 6.7](#), the two rich top actions give an absolute bound on ordered-pair orbits of occupied pairs, hence on ordered-pair orbits of fine factors. Each original system is recovered by grouping those embedded factors, as proved in [Lemma 6.28](#). Distinct stabilizer labels therefore give distinct invariant partitions. The bound (6.1) contradicts the pattern-line size and completes tensor-system exclusion. $\square$

Theorem 6.30 (The next classical-ancestor socle). *Let K be a tested descendant with a prepared interval, good for all the exclusions in this section, or containing the final good anchor. There is an actual nonabelian simple subgroup $B \trianglelefteq K$ inside the ancestor socle S , with full natural spanning, such that*

$$HB = K, \quad C_K(B) \leq \text{core}_K H,$$

and $C_K(B)$ is soluble. The quotient by this centralizer is almost simple with socle B , and the exact subgroup interval is preserved.

Proof. Finally consider a tested descendant K with prepared interval, good for these exclusions (or containing the final good anchor). It has $K \cap S \neq 1$ since K is nonsoluble. Take a minimal K -normal subgroup there. It cannot be elementary abelian, and if a power of nonabelian simples, their lift algebras commute linearly by perfection, and jointly span fully. As in the refinement argument they are central simple tensor factors, individually selfadjoint when relevant; multiple ones would be a forbidden tensor system. Thus there is an actual simple $B \trianglelefteq K$ in S , absolutely irreducible projectively. Its projective linear centralizer is trivial by perfection and spanning, so $C_K(B)$ injects in the soluble outer quotient and lies in $\text{core}_K H$ by (*). Also B is not retained (else base-core quotient soluble), hence $HB = K$. Quotienting its centralizer gives the required next almost simple group with socle B , keeping the exact interval. $\square$

7 Pruning inside an alternating ancestor

We use the same failed-chart construction to force the next almost simple group inside a high-degree alternating ancestor. The successive anchors become transitive, primitive, non-affine and non-Cartesian as the exclusions proceed.

Convention 7.1 (Prepared alternating configuration). Use the pattern, prior-anchor and full-stabilizer-label conventions of the structure exclusions. Thus the present almost simple ancestor quotient is $A_v \leq M \leq S_v$ of high degree on Λ ; labels under test and witnesses are in the interval above H . For a witness $K_i > H$, $[H, K_i]$ has the prepared link properties, including (*): a normal subgroup is retained in H or supplemented by H (missing), soluble normals are retained, and two commuting normals cannot both be missing (in the core quotient they would both contain the nonabelian monolith). Where earlier exclusions are used we have the previously good $H < E < K_i$ common within a pattern, with link properties at E as well. Patterns can have a whole line of q distinct labels with $K_i \cap K_j \leq K_l$ for distinct i, j and all l on the line; cross-chart pairs when invoked have $K_i \cap K_j = H$.

7.1 Permutation orbit bounds and primitivity

Lemma 7.2 (Two alternating permutation images). *If a group acts on two finite sets, inducing at least the alternating group on each, then its number of orbits on ordered pairs of pairs, one coordinate from each set in each pair, is bounded absolutely.*

Proof. We use the following orbit bound for two finite set actions on sizes s, t , each inducing at least the alternating group on that set. The number of orbits on ordered pairs of pairs (one coordinate from each set per pair) is absolutely bounded. Indeed the joint action is subdirect on the two induced groups. In large degrees either the product of the alternating groups is contained, or the two alternating simples are linked diagonally, hence the same degree and natural identification up to relabeling. This follows by the normal kernels of subdirect projections and simplicity (normal subgroups not containing the alternating simple must centralize it); use also the usual automorphism theorem for alternating groups in high degrees. If only one degree is large the large alternating factor acts independently by the same argument. Equality patterns and the bounded small sizes give the assertion. $\square$

Proposition 7.3 (Subsets and equal-block partitions). *In the prepared alternating configuration, full stabilizers of proper nonempty subsets and nontrivial equal-block-size partitions are excluded. Subsequent good anchors are primitive.*

Proof. First exclude stabilizers of proper nonempty subsets. For a cross-chart pair, the intersection contains supported alternating groups on at most four cells. Thus H has boundedly many point orbits, and every subset whose stabilizer is a witness is a union of these orbits. Only boundedly many subsets, and hence witness labels, are possible.

Next exclude nontrivial equal-block-size partitions, which suffice for primitivity once transitivity is known. Write s for the number of blocks and k for their common size. In a witness label, the intersection Y of M with the base block group is normal. If Y is retained and $k \geq 3$, every individually supported A_k lies in H . Its primitivity makes the restriction of any other tested H -invariant equivalence to that block either universal or discrete. In the discrete case a point has an equivalent point outside the block, since the other partition also has block size at least two. The supported transitive group fixes that outside point and would make it equivalent to every point of the original block, a contradiction. Consequently each original block lies in a block of the other partition. Applying this in both directions to two retained systems with block sizes at least three forces their equality. For retained $k = 2$, the even simultaneous flips on any two blocks lie in H . A distinct matching fails invariance by flipping a block while leaving one of its points' outside partner fixed, using the high degree.

If $HY = K_i$, then H induces the full symmetric top action on the blocks. For a cross-chart pair of missing partitions, Lemma 7.2 bounds the orbits on ordered pairs of block cells. Moreover H contains the supported alternating groups on the points of each cell, so each such orbit has boundedly many point-pair orbits above it, including when both points lie in the same cell. Thus H has boundedly many orbits on Λ^2 . Every invariant equivalence is a union of those orbits, giving the absolute partition bound used in (6.1). This is too small for all the distinct pattern labels. Here and below the chart field size exceeds these absolute bounds, after coloring by the indicated finitely many cases. $\square$

Proposition 7.4 (Elementary abelian regular groups). *For a primitive previously good anchor, normalizers of elementary abelian regular groups are excluded.*

Proof. Exclude next normalizers of elementary abelian regular groups. Such groups lie in A_v (parity of translations, high degree) and would thus be retained soluble normals. They would be minimal normal in the primitive previously good E , by transitivity of every nontrivial normal subgroup and regularity. Distinct ones must commute, impossible for regular self-centralizing groups. $\square$

7.2 Cartesian decompositions

Lemma 7.5 (Retained Cartesian products). *For a primitive anchor avoiding the affine exclusion, a normalized nontrivial Cartesian decomposition has transitive slots of common size $k \geq 5$. Its full stabilizer has a normal product $Y = A_k^s \leq A_v$ and induces at least A_s on slots. Retained products determine at most one system in a pattern.*

Proof. Next exclude stabilizers of nontrivial Cartesian decompositions of the point set (product coordinates $\Lambda \simeq \prod_{a=1}^s \Lambda_a$, $s \geq 2$, $|\Lambda_a| \geq 2$; stabilize their system, i.e. equality-of-coordinate partitions, allowing slot permutations). Primitivity of E gives slot transitivity (an orbit of slots otherwise gives an invariant nontrivial equivalence), and thus common slot set size k , say on $s \geq 2$ slots. Sizes 2, 3, 4 would give canonical elementary abelian regular groups (S_2, A_3, V_4 , respectively, multiplied over slots). For $k \geq 5$ let $Y_i = A_k^s$ acting slotwise, inside A_v and normal in the label K_i . The induced slot action of K_i contains A_s by commutators of true slot permutations. These stabilizers are proper (Cartesian coordinate agreement prevents full 2-transitivity). If $Y_i \leq H$, it is minimal normal in E , and has trivial centralizer on Λ (point stabilizer in Y_i self-normalizing). Thus all retained ones in the pattern must coincide, determining the slots themselves by the orbits of factor subproducts. $\square$

Lemma 7.6 (Missing normal powers with primitive factor action). *Let $[H, K_i]$ be a prepared interval and $Y_i \trianglelefteq K_i$ a direct power of nonabelian simple groups with $HY_i = K_i$. If the action of H on its simple factors is primitive, then $H \cap Y_i$ is a nontrivial coordinate product, and every group between H and K_i intersects Y_i in a coordinate product.*

Proof. Here and in the next exclusion we use a simple consequence of [Proposition 4.11](#). If $HY_i = K_i$ for a direct normal power of nonabelian simples and H 's action on the factors is primitive, then $P = H \cap Y_i$ is a nontrivial coordinate product and all groups between H, K_i intersect Y_i in coordinate products. Indeed Y_i is minimal normal, missed by the base so embeds as the minimal normal modulo core, and the exact interval is that of invariant subgroups above P . In the earlier reductions, the subdirect case and the all-nonbottom-subdirect case are now both impossible: ties must be factor-index invariant, so primitivity allows no chain of comparable distinct proper nontrivial subdirect groups. This leaves precisely the asserted product case. $\square$

Lemma 7.7 (Common Cartesian refinement). *A finite pattern line of missing Cartesian labels has a common refinement into finite coordinate sets. It is constructed from the tuple intersections of the slotwise products $E \cap Y_i$. The original Cartesian systems are recovered by grouping the fine coordinates, and E permutes the fine coordinate directions.*

Proof. For missing Cartesian labels $HY_i = K_i$, H induces the full top image, transitive and containing A_s , hence primitive on slot indices. On a line of these labels write

$$N_i = E \cap Y_i = \prod_a R_{i,a} \trianglelefteq E$$

a slotwise product strictly above $H \cap Y_i$. In $\bar{E} = E / \text{core}_E H$, each $\bar{N}_i$ contains the nonabelian simple-power monolith W . A simple factor C of W lies in one of the $\bar{R}_{i,a_i}$ for each i : the commutators of those slot groups with W generate W , are normal in W , and commute for distinct slots.

Apply the two commutator steps from the proof of [Lemma 6.28](#). First, since the N_i are normal in E and their images contain the perfect group W , their intersection $N = \bigcap_i N_i$ still has image containing W . Next, $[N, R_{i,a_i}] \leq N \cap R_{i,a_i}$ has image containing C . The subgroups $N \cap R_{i,a_i}$ are normal in N , so iterating commutators among them shows that the actual tuple intersection

$$T_\gamma = \bigcap_i R_{i,a_i}, \quad \gamma = (a_i),$$

has image containing C .

Take these intersections for all slot tuples. They are permuted by E and generate a nontrivial normal subgroup, which is transitive because E is primitive. Distinct tuple groups commute, since they act in distinct slots of at least one original system. Consider their action through the abstract product $\prod_\gamma T_\gamma$ and fix a point. If a product of tuple elements fixes that point, retain only the components assigned to one prescribed slot a of an original system i . This partial product still fixes the point: it acts only on coordinate a , where its action agrees with that of the original fixing product. Repeating this restriction through every original system isolates one tuple component at a time. Thus the stabilizer in the abstract product is exactly the product of the individual point stabilizers.

Transitivity now identifies the point set with the product of the individual coset sets. Each original coordinate depends only on the tuple components assigned to its slot. The maps from these separate batches to the original coordinate sets have a bijective product map, so each is itself a bijection. Grouping the fine coordinates therefore recovers the original equality-of-coordinate partitions. Discard the trivial coset sets; the remaining coordinate directions are still permuted by E . $\square$

Proposition 7.8 (Completion of Cartesian pruning). *Missing Cartesian labels are excluded. Together with [Lemma 7.5](#), this excludes all nontrivial Cartesian decompositions.*

Proof. Fix two distinct systems i, j and group the fine coordinates by the occupied pairs of their original slot indices. On an aggregate set of size u , the supported A_u lies in $K_i \cap K_j$, hence in every label on the pattern line. If another system split that aggregate, then on a slice with all outside coordinates fixed its nonconstant coordinate partitions would give a nontrivial Cartesian decomposition invariant under the natural A_u . Such a split has $u \geq 4$ and distinguishes pairs agreeing in some coordinate from pairs agreeing in none, contrary to 2-transitivity. Thus there is only one nontrivial fine coordinate over each occupied pair.

The map from fine coordinates to occupied pairs is H -equivariant: $H \leq E$ permutes the tuple groups and their coordinate directions, and stabilizes both original systems. By [Lemma 7.2](#), the two slot actions bound the number of ordered-pair orbits on this set. Each original system is recovered by an H -invariant grouping of the fine coordinates, as proved in [Lemma 7.7](#). Distinct full stabilizer labels therefore give distinct invariant partitions. The bound (6.1), with fibers of size one, contradicts the number of labels on the pattern line. $\square$

7.3 Diagonal actions and the almost simple conclusion

Lemma 7.9 (Primitive groups after Cartesian pruning). *Let E_0 be a primitive group satisfying the preceding affine and Cartesian exclusions. If it is not almost simple, it normalizes a transitive group $Y = B^s$, with B nonabelian simple and $s \geq 2$, whose point stabilizer is a full possibly twisted diagonal. Its degree is $|B|^{s-1}$. This includes the pair of opposed regular simple groups.*

Proof. Recall a short primitive-group observation after these exclusions. A minimal normal $P = C^u$ of a primitive group E_0 here is transitive with C nonabelian simple (an abelian one would be regular), $E_0 = P(E_0)_x$ at a point x . Its point stabilizer $Q = P_x$ has no proper intermediate increase towards P normalized by $(E_0)_x$. Taking its coordinate product closure, Q is either a product or subdirect. In the product case $u > 1$ would give a Cartesian coset product; in the subdirect case the same holds for multiple tie parts (uniform sizes by factor transitivity of $(E_0)_x$). Indeed the point stabilizer of E_0 is transitive on factors (minimal normality, with P itself not permuting them). In the product situation, including when taking batches given by multiple tie parts, the cosets P/Q accordingly split as a product of coset sets. Their sizes are uniform and nontrivial, P respects this product coordinatewise, and $(E_0)_x$ respects it by its conjugation/transport of the factors or batches with their respective stabilizers in Q . Thus for $u > 1$ we have just a full diagonal, possibly with twists, and $v = |C|^{u-1}$. For $u = 1$ with nontrivial $C_{E_0}(P)$, that centralizer must be transitive so both are regular opposite simple groups, again giving a transitive power C^2 with full diagonal point stabilizer. Consequently any non-almost-simple case here normalizes a transitive $Y = B^s$, $s \geq 2$, B nonabelian simple, with full diagonal point stabilizer. $\square$

Proposition 7.10 (Diagonal normalizers). *Under the successively primitive, non-affine and non-Cartesian anchor hypotheses, the full normalizers of the groups Y in Lemma 7.9 are excluded. Both retained and missing normal powers are covered.*

Proof. Finally exclude the normalizers K_i of all such Y . They are proper since $Y \leq A_v$ by perfection, and induce at least A_s on factors by diagonal-coordinate factor permutations and commutators. For retained $Y \leq H$, transitivity of E on factors, when present, makes Y minimal E -normal with trivial centralizer (the diagonal is self-normalizing), leaving no distinct such system. If instead E has multiple factor orbits, their products are each transitive so use at least $s - 1$ factors each, forcing just two opposed regular simple groups, the only minimal normals in E , again determining Y . If K_i 's top action is intransitive, likewise $s = 2$, and at least one individual factor is retained by (*); such factors are regular simple minimal normals of E , determining Y by taking also their opposite regular group (distinct such minimal normals commute so determine the same pair).

The missing transitive case. It remains that $HY = K_i$ and H induces a transitive top action containing A_s . By the interval product consequence, $N = E \cap Y = \prod R_a$ is a nontrivial slot product with each projection proper (since $E < K_i$). Choose a minimal E -normal $P' = C^u$ inside it. Each factor of P' is contained actually in one R_a , since $[P', R_a] \leq P' \cap R_a$, normal in P' , and those commutators generate P' by perfection. This assignment is E -equivariant with some common positive number h per slot, so $u = sh$. The primitive observation applied to E gives $Q = P'_x$ a full diagonal. If $h > 1$, take inside P' the product of the projections of Q along the factor blocks of the assignment. It is a proper intermediate increase, normalized by E_x , impossible. If $h = 1$, point degrees give $|C|^{s-1} = |B|^{s-1}$, contrary to containment in the proper R_a . $\square$

Theorem 7.11 (The next alternating-ancestor socle). *After the exclusions, every tested descendant with its prepared interval is primitive almost simple. Its actual simple socle is transitive, lies in A_v , and is supplemented by H .*

Proof. These arguments use only successively primitive, non-affine, and non-Cartesian prior anchors as needed and finite case subdivisions. Thus after the exclusions the tested descendant with its prepared interval is primitive almost simple, with an actual simple transitive socle in A_v , supplemented by H via (*) (it cannot be retained, since then factoring by the base core leaves a soluble quotient by Schreier, inconsistent with the prepared interval), as desired. $\square$

8 Excluding bounded algebraic types

Other than the high alternating and classical cases, simple socles of bounded order can be discarded immediately by the available interval length: $M = HS$ at an almost simple current parent with socle S , so $|S| \geq |M : H|$ tends uniformly above any prescribed constant. By CFSG the remaining cases use a finite list of root systems, with parameter as large as needed, $S = (G^{F_A})'$ for a simple adjoint algebraic group over $k = \overline{\mathbf{F}}_p$.

Throughout this section we work at a current almost simple parent in the prepared reverse-flat interval of [Section 5](#). Thus $H \leq M \leq \text{Aut}(S)$, S is the simple socle, and $M = HS$. The root systems under consideration belong to a fixed finite list. Every complexity bound below depends only on this list and on the stated degree bound; it does not depend on the characteristic or the finite-field parameter.

8.1 Uniform algebraic input

Definition 8.1 (Bounded equation complexity). Fix closed affine split models for the root systems in the finite list, with a bounded number of coordinates and bounded-degree group operations. A closed subset has equation complexity at most d if it is the common zero set, inside the chosen group, of polynomials of degree at most d . There need not be a previously specified bound on the number of equations: their linear span lies in the finite-dimensional space of polynomials of degree at most d .

Theorem 8.2 (Sufficiently general finite subgroups). *For the fixed finite list of simple adjoint types, there is a uniform degree bound d_0 with the following property. If a finite subgroup $\Gamma \leq G(k)$, where $k = \overline{\mathbf{F}}_p$, is contained in no proper closed subset of equation complexity at most d_0 , then there is a Steinberg map Ψ such that*

$$(G^\Psi)' \leq \Gamma \leq G^\Psi.$$

Here the degree bound follows from the uniform constructible-family formulation of the sufficiently-general subgroup theorem of Larsen–Pink [25, Theorem 0.5 and Metadefinition 2.2]: the exceptional family has uniformly bounded defining equations in the fixed affine models.

This formulation applies to the fixed split models in all characteristics. Its hypothesis is sufficient generality, expressed by avoidance of bounded-complexity proper closed subsets. The alternative argument in [Section 16](#) will also establish the implication needed over the indicated algebraic closures.

Convention 8.3 (Point maps and height). We also use the standard classification of the Steinberg maps and of automorphisms and orders of the finite Lie type groups. On points take the group of maps generated by inner and diagram automorphisms, split Frobenius powers and inverses, and special isogeny powers where applicable (square split p -Frobenius in special small characteristics). It has a height homomorphism to $\mathbf{Z}$ with unit of parameter growth p , or $\sqrt{p}$ in the special case. Height zero maps are ordinary algebraic automorphisms, and positive height maps are Steinberg maps with fixed groups, of parameter $P = (\text{unit})^{\text{height}}$. The normalizer of S here induces the automorphisms required for our large-parameter S . Modulo inners one uses the standard pinned maps here: split Frobenius commutes with diagram maps, and in the cases with the special self-isogeny one uses its powers (square as above, no other nontrivial ordinary diagram needed in those types). Thus arbitrary words simplify to an inner twist of these combinations; the positive-height ones have standard fixed-group types (inner twists removed by Lang–Steinberg). The exceptional generators only concern fixed small characteristics. All coordinates, variety complexities etc. may use fixed closed

affine split models of bounded size/degree, with the group operations and pinned ordinary diagram maps of bounded complexity uniformly in the characteristic. Images of bounded-degree-equation varieties by the whole point-map group have bounded equation degree (field powers just transform coefficients, inner and diagram have bounded complexity, special map of fixed degree has inverse using itself and inverse field power). Positive maps are given algebraically of degree $O(P)$. The classification and fixed-group facts used here are those of Gorenstein et al. [13], Steinberg [46]. We write $\mathcal{A}(G)$ for this group of maps on $G(k)$, ht for its height, and δ for the parameter unit, so $\delta = p$ or $\sqrt{p}$.

Lemma 8.4 (Order, degree, and pointwise rigidity). *For the finite type list, let $b = \dim G$ and $B(\Psi) = (G^\Psi)'$. For sufficiently large parameter $P = \delta^{\text{ht}(\Psi)}$, the following hold.*

1. $B(\Psi)$ is simple, and its order exceeds $|G^\Theta|$ for every $\Theta \in \mathcal{A}(G)$ with $0 < \text{ht}(\Theta) < \text{ht}(\Psi)$.
2. For each fixed equation complexity d , $B(\Psi)$ lies in no proper closed subset of complexity at most d , once P is sufficiently large for d .
3. If a positive-height point map fixes $B(\Psi)$ pointwise, it is a positive integral power of Ψ .

Proof. Two useful uniform comparisons follow. Write $b = \dim G$. For P large, $B(\Psi) = (G^\Psi)'$ is simple of order at least $P^b(1 - o(1))/z$ inside G^Ψ of order $P^b(1 + o(1))$, with index at most a center bound z from the root system. In particular $|B(\Psi)|$ exceeds the entire fixed-group order at any smaller positive height, since $z < (\text{unit})^b$ by the usual center sizes. This applies uniformly to comparison with smaller heights: below parameter $\sqrt{P}$ even the rough order bounds suffice, and otherwise the two product formulas both have small relative error. Also $B(\Psi)$ lies in no proper closed subvariety of bounded degree, for P sufficiently large for that bound. On any irreducible component of dimension $e < b$ and bounded degree, the fixed-point equations of degree $O(P)$ cut a finite set of size $O(P^e)$, by affine Bézout (combine the equations successively generically to cut to dimension zero). Hence a point map of positive height fixing $B(\Psi)$ pointwise must be a power of Ψ : divide its height by the given height and compose off that power on points. Positive remainder is impossible by orders, zero-height nonidentity by the bounded equations of its fixed locus.

For the last assertion, if α has height $q \text{ht}(\Psi) + r$ with $0 \leq r < \text{ht}(\Psi)$, the point map $\Psi^{-q}\alpha$ still fixes $B(\Psi)$ pointwise. If $r > 0$, its fixed group is too small. If $r = 0$, a nonidentity algebraic automorphism has a proper fixed locus with uniformly bounded equations, contradicting the second assertion. Hence $r = 0$ and $\alpha = \Psi^q$. $\square$

8.2 Witness stabilizers and the divisor contradiction

Lemma 8.5 (A bounded-complexity failure witness). *Let $H \leq K < M$ be a tested overgroup with $\Gamma = K \cap S \neq 1$, and suppose Γ fails the sufficient-generality hypothesis of Theorem 8.2. For sufficiently large parameter of S , there is a proper full stabilizer label K' with $K \leq K' < M$ such that*

$$K' \cap S = N_G(Z) \cap S$$

for a proper closed subset $Z \subset G$ whose equation complexity, and that of its conjugation normalizer $N_G(Z)$, are uniformly bounded.

Proof. Consider a proper K in the interval, above a tested large-index label, and $\Gamma = K \cap S \neq 1$. If insufficiently general for the sandwich, it lies in a proper bounded-degree-equation closed $X \subset G$ by Larsen–Pink as used above. Take the subgroup of our point-map group preserving S and inducing

M -elements on S ; it maps onto M by the automorphism description. Let $\tilde{K}$ be the full preimage of K there (so each such map normalizes Γ). Then

$$Z = \bigcap_{\alpha \in \tilde{K}, \gamma \in \Gamma} \alpha(\gamma X)$$

contains Γ , is proper closed with equations of uniformly bounded degree, left- Γ -invariant and $\tilde{K}$ -invariant. Even for the whole intersection one can just span all the equations occurring in the fixed bounded-degree space. Its point-map stabilizer (within that preimage of M) thus defines a label $K' \geq K$, since the stabilizer before projection contains the kernel there (already in $\tilde{K}$). Identifying S with its inner actions on itself, $K' \cap S = N_G(Z) \cap S$ by using the actual inner lifts to G . The conjugation normalizer here has bounded-degree equations (test the equations of Z on all conjugates of its points by g, g^{-1}). It is proper since otherwise all conjugates of Γ left-stabilize Z , while their normal closure is Zariski-dense by simplicity and trivial center. The degree comparison for S thus makes K' proper.

Here $N_G(Z)$ denotes the conjugation stabilizer, not the left stabilizer. To justify its properness explicitly, if every $g \in G(k)$ normalizes Z , left invariance under Γ implies left invariance under every $g\Gamma g^{-1}$. The Zariski closure of the group generated by these conjugates is a nontrivial closed normal subgroup of the simple adjoint group G , hence is G . Since Z is closed and nonempty, left invariance by this dense subgroup forces $Z = G$, a contradiction. The nonidentity of Γ suffices because the adjoint group has trivial center. $\square$

Proposition 8.6 (No bounded-type parent with a stage remaining). *Fix the finite list of bounded root systems. There is a fixed trial increment t such that a prepared current parent with a remaining private-chart stage cannot have socle of one of these types, provided the residual interval length is sufficiently large. All choices are uniform in the characteristic and parameter.*

Proof. At the current parent flat W , test a sufficiently large fixed increment t in a private chart. We first find a flat U of that increment for which every $K \in [K_U, M]$ admits the sandwich for $K \cap S$. If every trial failed, [Lemma 8.5](#) would supply full stabilizer witnesses above the trial labels. The affine-pattern construction gives an independent parameter list whose successive label intersections decrease strictly for more steps than the dimension of the relevant polynomial space. Descent remains strict after intersection with S , since $M = HS$ and every overgroup of H is determined by its intersection with S . Each strict descent must add a defining polynomial outside the span of the preceding equations. But all conjugation-normalizer equations lie in one bounded-dimensional space, a contradiction. This step needs only a bounded independent list, no prior anchor, and tests relative to the current parent.

Put $E = K_U$ and $\Gamma_K = K \cap S$ for $E \leq K \leq M$. We now have

$$B(\Psi_K) \leq \Gamma_K \leq G^{\Psi_K}.$$

The reserved interval length below E makes every parameter here large enough for [Lemma 8.4](#). Indeed, if that residual height is h , then $|\Gamma_K| \geq |\Gamma_E| \geq [E : H] \geq 2^h$, whereas all fixed groups below a prescribed parameter have bounded order for the fixed type list. Write $B_E = B(\Psi_E)$. Perfection and the abelian quotient of a fixed group by its derived subgroup give $\Gamma'_E = B_E$.

Since $B_E \leq \Gamma_K$, the map Ψ_K fixes B_E pointwise. Rigidity therefore gives $\Psi_K = \Psi_E^{a_K}$ for a positive integer a_K . The ambient map F_A likewise fixes B_E , so $F_A = \Psi_E^d$. It also fixes $B(\Psi_K)$ pointwise, and rigidity applied to that simple group makes F_A a positive power of Ψ_K . Consequently

$$F_A = \Psi_E^d, \quad \Psi_K = \Psi_E^{a_K}, \quad a_K \mid d.$$

These are equalities of point maps, so each exponent determines an actual embedded fixed group.

Every lift of E normalizes Γ_E and hence $B_E = \Gamma'_E$. Conjugating Ψ_E by such a lift gives a map of the same height fixing B_E pointwise; rigidity makes it equal to Ψ_E . Thus the lifts commute with Ψ_E . For each divisor a of d , put $B_a = B(\Psi_E^a)$. This group lies in S : its elements are fixed by F_A , and perfection puts it in $(G^{F_A})'$. It is normalized by E , so EB_a is an overgroup label. Moreover

$$(EB_a) \cap S = \Gamma_E B_a \subseteq G^{\Psi_E^a}, \quad EB_1 = E, \quad EB_d = M.$$

If $a \mid b$ with $a < b$, then $B_a \leq B_b$, and Lemma 8.4 gives $|B_b| > |G^{\Psi_E^a}|$. Hence $EB_a < EB_b$. A saturated divisor chain therefore gives $\Omega(d)$ strict steps in the exact height- t interval $[E, M]$, where $\Omega(d)$ counts prime factors with multiplicity. Thus

$$\tau(d) = \prod_{\ell^a \parallel d} (a+1) \leq 2^{\Omega(d)} \leq 2^t.$$

For a fixed exponent a_K , the sandwich leaves only boundedly many choices for Γ_K : its quotient by B_{a_K} is a subgroup of the bounded-order group $G^{\Psi_E^{a_K}}/B_{a_K}$. If z bounds that order, $C = 2^z$ bounds the number of choices. Since $K = H\Gamma_K$, no additional choices of outer automorphisms occur. We conclude that $||[E, M]|| \leq C2^t$.

The reverse-flat interval, however, contains at least $q^{\lfloor (t-1)^2/4 \rfloor}$ intermediate subflats, by the ordinary linear-subspace count; this bound also covers an empty starting flat. For sufficiently large fixed t it exceeds $C2^t$, already using $q \geq 5$. This is the required contradiction, uniform in characteristic and parameter, whenever a bounded-type parent has a stage remaining. $\square$

9 Protections and the length of high-type paths

The structural exclusions give nested simple groups whose actions remain transitive or absolutely irreducible in every earlier ancestor. We now need two further conclusions. First, within each consecutive same-characteristic classical run, the chart construction can select descendants whose actions on each ancestor's natural module admit no nontrivial factorization into individually invariant tensor factors over the algebraic closure, and that are fixed by no prime compositional root of its defining field map. Second, any path with these protections has absolutely bounded length. The selection uses the affine geometry; the length bound uses only the group properties stated below, so it can also be applied to the later reverse and forward constructions.

All high-degree and high-rank thresholds in this section are sufficiently large absolute constants. Bounded types have already been treated in Section 8.

Definition 9.1 (Prepared paths and ancestor properties). A *prepared path* is a strict descending chain of finite groups K_W containing a common subgroup H , together with actual strictly nested nonabelian simple groups $B_W \trianglelefteq K_W$ such that

$$HB_W = K_W, \quad C_{K_W}(B_W) \leq H,$$

where each centralizer is soluble and the quotient by it is almost simple with socle B_W . Fix the usual natural realizations of the alternating and classical socles. For every ancestor-descendant pair require the following properties.

1. In an alternating ancestor, the descendant socle is transitive on the natural set.

2. In a high classical ancestor, the descendant socle is projectively absolutely irreducible on the natural module. Its label normalizes no proper ground-field tensor system of the endomorphism algebra of the kind excluded in Section 6; in isometry types the factors are individually selfadjoint.

The pruning construction of Section 5 supplies these properties by Theorems 6.30 and 7.11. Lemma 5.1 makes the simple groups actual and compatible in all ancestor quotients, whose centralizer kernels lie in H . Consequently a full stabilizer containing a tested label pulls back to a label of the exact original interval. In the selection argument the same properties hold for all comparable candidate subflats of a successful pruned flat before the next node is chosen: if $W \subset L \subset U$, then $B_U \leq B_L \leq B_W$.

9.1 Classical conventions and changes of type

Convention 9.2 (High classical groups). We use the standard structure theory of finite simple classical groups, simply connected coverings, Schur multipliers and automorphisms. In sufficiently high rank there are no small covering exceptions: the universal central extension is the usual fixed group $\hat{S} = \hat{G}^\Phi$ of simply connected type. Write $k = \overline{\mathbf{F}}_p$ in characteristic p defining the group. We also use the adjoint algebraic group G over k , with $S = (G^\Phi)'$, and regard it on points as a projective group on a natural v -space. Here Φ is standard of parameter $P = p^f$ (split Frobenius composed possibly with diagram/inner automorphisms); ground field F on the finite natural module is of size P , or P^2 for unitary type. We use types A (linear and unitary), C, odd orthogonal B only in odd characteristic, and even orthogonal D; by the bilateral (bilinear) form types we mean B,C,D, not the unitary case. The usual automorphisms of S lift to point maps in the group on adjoint $G(k)$ generated by inner and diagram automorphisms and integral powers of a split Frobenius Fr_p (including inverse on points). Modulo inner this group is abelian, with at most two diagram choices and field exponent called height. A positive-height element is an ordinary Steinberg map (inner conjugate to standard form by Lang–Steinberg). We shall use that in this adjoint group the order of the derived fixed simple group at positive height f_2 exceeds the order of the whole fixed group at $0 < f_1 < f_2$, uniformly in high rank. Indeed the classical order formulae give order $(p^f)^{\dim G}$ times factors bounded above and away from zero (products of $1 \pm (p^f)^{-j}$); the derived index costs at most a polynomial in the rank. When finite types B_e, C_e coincide in characteristic two we use the symplectic realization (type C); the odd-dimensional B_e convention in this section is only for odd characteristic. In particular all Steinberg maps needed *in this high classical section* are ordinary graph/field ones, not the special square-root-field maps. We use the standard high-rank multiplier and automorphism results in Gorenstein et al. [13], Kleidman and Liebeck [23].

We use the standard highest weight theory of simply connected semisimple algebraic groups, including Steinberg’s Tensor Product Theorem and his parametrization of the irreducibles of $\hat{S}$ in defining characteristic as restrictions of $L_{\hat{G}}(\lambda)$ for P -restricted dominant weights (including standard diagram twists). Projective representations of S lift to ordinary ones on its universal cover, uniquely once the projective matrices are fixed (perfection).

More explicitly, we use the universal-central-extension and automorphism description in the high-rank range of the standard multiplier/automorphism theorems for finite groups of Lie type. Thus lifting a projective representation means pulling back the scalar central extension and then using universality on $\hat{S}$. The Steinberg theorems we use here are the ordinary p -adic tensor product for rational simples and the defining-characteristic Restriction Theorem on the full simply connected fixed group: highest weights with fundamental coefficients $0, \dots, P-1$ give irreducibles, pairwise inequivalent, and give all of them, also for these standard twisted forms. With pinned map a

diagram twist times split Fr_p^f , a further field twist rotates digit positions on restriction modulo f , with wrapping applying a diagram (since the defining map is the identity on the finite points). One obtains this by twisting the tensor product digit factors separately. Equivalence of linear lifts up to a projective intertwiner on the perfect cover already implies ordinary equivalence (the scalar discrepancy would be a character). These are the ordinary tensor-product and finite-restriction theorems, including ordinary diagram twists, of Steinberg [45, Theorems 1.1, 7.4 and 9.3]; see also Jantzen [19].

Lemma 9.3 (A lower bound for projective degree). *There is an absolute constant $c_0 > 0$ such that every nontrivial projective representation in defining characteristic of a sufficiently high-rank simple classical group with natural dimension v has dimension at least $c_0 v$.*

Proof. Any nontrivial irreducible projective representation of S in characteristic p has dimension at least a positive absolute constant times v in high rank: its highest-weight Weyl orbit already has at least that size (permutations of a nonconstant weight word in type A; signed permutations, possibly with even sign changes, in B,C,D). This bound suffices here and also bounds any nontrivial projective representation, using perfection if the lift has only trivial composition factors. In the last case a lift with all composition factors trivial has unipotent, hence soluble, image. A perfect group has no nontrivial soluble image, so a nontrivial representation must have a nontrivial composition factor. $\square$

Lemma 9.4 (Alternating subsequences). *A prepared path contains at most two alternating socles of sufficiently large degree.*

Proof. Any alternating subsequence of the path has bounded length. Suppose there are three socles $A_n > A_m > A_e$ as subgroups ($n > m > e$), with the transitivity for each ancestor pair. For large e , $m \geq \binom{e}{2}$. In fact an intransitive point stabilizer in natural A_e -coordinates giving index $> e$ has index at least $\binom{e}{2}$ (if within a point group A_{e-1} properly, use simplicity for minimum proper index at least $e-1$). Proper transitive subgroups of A_e have index at least exponential in e for large e , by Bochert's Theorem in the primitive case, and the block partition count in the imprimitive case. For instance with b blocks of size a , both at least two, pair the blocks disjointly and interchange corresponding positions arbitrarily in each pair, giving at least $2^{(a-1)\lfloor b/2 \rfloor}$ partitions. Thus $|A_e| = \exp(o(m))$. If D is the stabilizer in A_m on the n -set, $A_m = A_e D$ by transitivity, so D cannot be transitive on the natural m -set by the same index bound. Nor can it have just orbits of sizes 1 and $m-1$: the bound on proper transitive subgroups on $m-1$ would force $D = A_{m-1}$, contrary to $n > m$. It thus stabilizes an h -set, $2 \leq h \leq m-2$, making A_e h -homogeneous on m . This implies 2-homogeneity: the count inside an h -set of edges from any one pair orbit is constant; comparing sets consisting of i or j and $h-1$ other points shows the adjacency rows at i, j agree away from both (all such sums of differences vanish). Thus there is just one edge orbit. Even order then gives 2-transitivity by an involution flipping an ordered pair. By the classification of finite 2-transitive groups, alternating simple groups of sufficiently large degree have no such nonnatural action. This proves the bound. The permutation-group inputs used here are the primitive subgroup bound and the classification of finite doubly transitive groups; see Dixon and Mortimer [10]. $\square$

Lemma 9.5 (Changes of defining characteristic). *For a classical ancestor i and classical descendant j in a prepared path with distinct defining characteristics, there is an absolute $c > 0$ such that*

$$v_i \geq P_j^{c v_j}.$$

There cannot be three high classical nodes $i < j < l$ with $p_i \neq p_j$ and $p_j \neq p_l$. Consequently, after splitting at alternating nodes and changes of defining characteristic, the path is the union of an absolutely bounded number of consecutive classical runs of one characteristic.

Proof. For high classical pairs in distinct characteristics, absolute constants give $v_i \geq P_j^{cv_j}$ for ancestor i , descendant j , with $c > 0$. Here is a direct bound. The high classical S_j contains images of paired special linear groups $\mathrm{SL}_h(P_j)$, $h \geq c'v_j$, using a subspace in linear type or paired maximally isotropic coordinates in the form types (one may restrict scalars of matrix entries in the unitary coordinates to $\mathbf{F}_{P_j}$). These act nontrivially including on column shear elements. The restricted projective representation lifts linearly on that $\mathrm{SL}_h(P_j)$, centrally closed for such high h . Its abelian column shears of size P_j^{h-1} are diagonalizable here and have some nontrivial character occurring; the SL_{h-1} action is transitive on all their nontrivial characters. This proves the assertion. Thus we cannot have classical nodes $i < j < l$ (indices increasing downwards) with $p_i \neq p_j$, $p_j \neq p_l$, since then

$$\log |S_l| = O(v_l^2 \log P_l) = O(\log^2 v_j)$$

whereas v_i is at least exponential in v_j . But $|S_l| \geq v_i^2$ by its absolutely irreducible projective action in i : lift that action to its perfect central cover. Absolute irreducibility implies that its matrices span the full $v_i \times v_i$ matrix algebra. Central lifts of one projective element are scalar multiples, so at most $|S_l|$ matrices are needed in this span. This contradicts the two exponential lower bounds for sufficiently high rank.

Hence there are boundedly many interruptions of consecutive high-classical runs in the *same* characteristic, on paths avoiding bounded types. It suffices to bound those runs. We now work in such a run, notation $S_i = B_{W_i}$. $\square$

9.2 Canonical tensor factors over the algebraic closure

Lemma 9.6 (Canonical tensor atoms). *Let S_i be a high classical ancestor, and let B be a candidate simple subgroup of high classical type in the same defining characteristic. Assume that B acts projectively absolutely irreducibly on the ancestor natural module V over $k = \overline{\mathbf{F}}_p$. Among decompositions of $\mathrm{End}_k(V)$ into nontrivial full matrix factors fixed individually by B , there is a unique finest one as a collection of embedded matrix subalgebras. Call its factors the atoms. There is at least one atom for each nonzero ordinary p -digit in the restricted highest weight of the lifted representation of $\widehat{B}$. The only possible further separation within a digit is across the multiple bond in types B and C . Every semilinear algebra action, or algebra anti-action with inversion on the group, normalizing the projective B -action permutes these atoms.*

Proof. To verify this, in any tensor factorization into nontrivial slots with B fixing the factor algebras individually, lift each projective action to $\widehat{B}$. They tensor to the original lift by perfection, and are all irreducible. Regard them as restrictions of rational simples with restricted highest weights for the finite-group parameter of B . Their tensor is then irreducible rationally as well. By Steinberg decomposition, even the subproducts of the contributing digit modules at any one exponent (untwisted) must be irreducible.

Recall the top Levi face of a rational simple $L(\lambda)$ for nodes I , using weights differing downwards from highest only by I -roots, is a simple highest module for the Levi derived group with the corresponding coefficients. Indeed it is generated by the highest vector there by distribution-algebra triangular decomposition (PBW); a nonzero Levi submodule on that face has a highest unipotent-fixed vector, fixed also by outside raising groups since they go off the weights. In the simple module this forces the highest line. The face of a tensor product with the sum as highest weight tensors the corresponding faces. Now at any one digit, no two modules from different slots can have active coefficients in the same simple-bond connected component of the Dynkin diagram. Otherwise take a shortest simple-bond path between active nodes of two different slots, possibly just one node (minimize over such pairs in a single simple-bond component). For the two modules the faces on

that path have endpoint highest weights $a\omega_1, b\omega_\ell$ of type A_ℓ , $1 \leq a, b < p$, with no other supports there by minimality. These are symmetric powers of natural and dual modules (the same natural type for A_1), simple for these exponents. For example simplicity follows using torus monomials of distinct weights and elementary root transfers of powers with nonzero coefficients at degree less than p . Their tensor is reducible by contraction (one derivative on each side paired), a nonzero equivariant map to a strictly smaller space. This contradicts the face property.

Thus coefficients are disjoint with no carry, and the digitwise sum highest weight is restricted and must be exactly the given tuple for V . Take one atom per nonzero digit except for the following possible refinement in B,C: if the regular chain and the node beyond the multiple bond both contribute and can be separated in a factorization, split them. Indeed in such a separation their two rational pieces tensor irreducibly to the digit module, so all such splits at all digits can be made together. This gives a factorization, and any other one as above has its slot representations given by grouping the atoms (by the coefficient argument and uniqueness of the simples). This is true of the actual matrix subalgebras: equivalent factorizations with fixed slot representation classes have their tensor identifications differing, after intertwiners, by a scalar since V is absolutely irreducible for $\hat{B}$. In more detail, separated active components at a digit give actual rational factor modules by taking the digit pieces of the slot modules of that separation; their subproduct, untwisted, is irreducible as above, with the individual component weights and their sum all forced. Thus it realizes the indicated split even as rational modules of that digit, independently of the other digits. This proves the claim. Any semilinear or anti-action on the algebra normalizing the projective B -action (using inverse with an anti-action) permutes the atoms, by the same intrinsic description. $\square$

Lemma 9.7 (Descent and the single Frobenius cycle). *Retain the hypotheses of Lemma 9.6, and assume the candidate label has the ancestor tensor-system exclusion in Definition 9.1. Let σ be arithmetic Frobenius for the ancestor ground field, except in unitary type where it is the extended semilinear adjoint anti-action with coefficient exponent P_i . Then σ permutes the atoms in one cycle. If its length is $t = t(B)$, all atoms have a common degree $u \geq 2$ and*

$$u^t = v_i.$$

The lifts of the candidate normalizer used to act on the endomorphism algebra commute with σ . In bilateral form types the factors are individually selfadjoint, with alternating factor forms in characteristic two.

Proof. The map σ permutes the atoms because it fixes the projective B -action, using inversion with the anti-action in unitary type. We first specify lifts of the ancestor automorphisms. Use finite-field projective semilinear representatives, similarities in the form types, and also graph anti-actions in linear type. Lift their field exponents by integral powers of split Frobenius on k . These lifts commute with σ : in unitary type this is the hermitian-similarity identity extended to k , and otherwise it follows from the finite-field matrices.

The finite projective representatives are faithful on the ancestor simple group. Indeed, if a collineation with coefficient exponent $0 \leq a < \log_p |F|$ centralizes it, perfection identifies its natural finite lift with the coefficient twist of highest weight $p^a \omega_1$. Restricted-weight uniqueness forces $a \equiv 0 \pmod{f_i}$. In unitary type it also rules out $a = f_i$, since wrapping produces the distinct diagram-dual natural weight. With the field part removed, absolute irreducibility leaves only a scalar. The same weight comparison rules out a linear-type duality in the kernel. Thus closure lifts differ only by arithmetic ground-field Frobenius powers. A system stable under those powers as a collection has a well-defined full stabilizer in the finite ancestor. In unitary type the representatives here are collineations; σ itself is the semilinear adjoint anti-action and σ^2 is arithmetic Frobenius for F .

We next check the forms needed for descent. In bilateral types the perfect lift of B preserves the ambient form. Restricted-weight uniqueness makes its digits self-dual, and any pieces separated across the B or C multiple bond are self-dual as well. Their invariant forms tensor to the ambient form by uniqueness, so the atom algebras are individually selfadjoint. In characteristic two each factor form is alternating: a nonalternating symmetric invariant form would give the nonzero invariant functional $x \mapsto \sqrt{b(x, x)}$ on a nontrivial irreducible module.

If a quadratic form is present in characteristic two, every grouping into at least two aggregate factors has the canonical quadratic with the given polar form and value zero on simple tensors. To construct it, prescribe zero on tensor-basis vectors. On a simple tensor, polar terms with a repeated index vanish and the remaining terms cancel in pairs by exchanging one index. This quadratic is invariant under the lifting group. Its difference from the given invariant quadratic is the square of an invariant linear functional, hence zero.

Suppose σ has more than one orbit on atoms, and tensor together the atoms in each orbit. The aggregate algebras descend under arithmetic Frobenius to central simple F -factors, which split over the finite field. In unitary type first descend under σ^2 ; stability under σ then gives selfadjointness for the hermitian adjoint. Bilateral selfadjointness was just proved. Candidate lifts commute with σ and therefore permute these aggregate factors. This is exactly a ground-field tensor system forbidden by the preceding exclusions.

Its full stabilizer is proper. Otherwise S_i acts trivially on at most $\log_2 v_i$ slots, by simplicity and comparison of its order with the corresponding factorial. Absolute irreducibility makes each slot action nontrivial. A smallest slot has degree at most $\sqrt{v_i}$, contrary to [Lemma 9.3](#) at the fixed high threshold. There is therefore one σ -cycle. Its atoms have equal degree u , and $u^t = v_i$. $\square$

9.3 Counting proper containing stabilizers

The next two counts supply the finite lists needed for selection. Each full stabilizer contains the candidate label, so it is an actual label in the ancestor interval. To count distinct primes, we construct a subgroup lying in all previously imposed stabilizers but not in the next one. Their intersections then descend strictly, and interval height bounds the number of such steps. Tensor coarsenings leave a bounded small-slot remainder; fixing roots do not.

Proposition 9.8 (Counting prime tensor coarsenings). *Under the hypotheses of [Lemma 9.7](#), each prime $s \mid t(B)$ defines a canonical congruence coarsening $\mathcal{P}_s$ into s matrix factors, normalized by the candidate label. Its full stabilizer in the ancestor is proper. There is an absolute constant c_1 such that*

$$\#\{s : s \text{ is prime and } s \mid t(B)\} \leq \text{ht}[K_{\text{cand}}, K_{W_i}] + c_1.$$

The height is taken in the actual subgroup interval, or equivalently in its prepared ancestor quotient.

Proof. For each prime $s \mid t$, group the atoms by their residue classes modulo s around the σ -cycle. Candidate lifts commute with the cycle, so they normalize this canonical coarsening $\mathcal{P}_s$. Its full stabilizer is well defined by [Lemma 9.7](#). If the entire socle S_i preserved it, its permutation action on the s slots would commute with an s -cycle. That image is cyclic, hence trivial by perfection. A slot of degree $u^{t/s} \leq \sqrt{v_i}$ would then carry a nontrivial projective action, contradicting [Lemma 9.3](#). Thus each coarsening gives a proper containing label.

Let d be the product of distinct primes already imposed. Their common refinement is $\mathcal{P}_d$, with d slots of degree $h = u^{t/d}$. In $G_i(k)$ take the product X_d of the supported connected projective groups on these slots: PGL_h in ambient type A, and the connected projective classical groups of the factor forms in bilateral types. Put $X_1 = G_i(k)$ itself. In particular, at $d = 1$ we keep the orthogonal

ambient group in characteristic two. For $d > 1$ the tensor construction embeds these products in the connected ambient group; the canonical-quadratic observation in [Lemma 9.7](#) justifies the characteristic-two orthogonal case.

The map Φ_i cycles the d factors through σ , using inversion on units in the anti-case. On one factor Φ_i^d is an ordinary positive Steinberg map of parameter P_i^d : after this coefficient twist, its matrix-algebra action is an isomorphism or anti-isomorphism preserving the factor group. The fixed group $X_d^{\Phi_i}$ therefore projects isomorphically to that one-factor fixed group. While h is sufficiently large, its derived group gives a full high-rank natural simple Y on the chosen slot. It lies in S_i by taking commutators and preserves every previously imposed coarsening.

For a new prime s , suppose Y also preserves $\mathcal{P}_s$. Perfection and commutation with the induced s -cycle make it fix those slots individually. Inside the chosen d -slot, the two coarsenings intersect in s tensor factors of degree $h^{1/s}$, by the Chinese remainder description of their atom indices. At least one factor action is projectively nontrivial, since their tensor is the natural action of Y on that slot. Its degree contradicts [Lemma 9.3](#). Thus the next stabilizer makes the intersection strictly smaller.

Interval height bounds the primes added while h is high. At stopping, $h \geq 2^{t/d}$ and t/d is divisible by every unused distinct prime. Bounded h leaves only absolutely many unused primes. This gives the claimed additive constant c_1 in the actual interval, whose height is unchanged by the centralizer quotient. $\square$

Proposition 9.9 (Counting fixing prime roots). *Let B be as in [Lemma 9.7](#), in the high classical ancestor with adjoint group G_i and ordinary Steinberg map Φ_i of height f_i . Consider pairs (s, Θ) where s is prime,*

$$\Theta^s = \Phi_i, \quad \Theta|_B = 1,$$

and Θ belongs to the inner-diagram-Frobenius point-map group. The full normalizer labels of $S(\Theta) = (G_i^\Theta)'$ are proper, contain the candidate label, and number at most

$$2 \operatorname{ht}[K_{\text{cand}}, K_{W_i}].$$

There are at most two maps at each possible height, one for each diagram choice, and all such roots commute.

Proof. The inner centralizer of B is trivial by projective absolute irreducibility and perfection. Modulo inner automorphisms the point-map group is abelian, with at most two diagram choices at any height. Consequently two maps of the same height and diagram choice fixing B agree, and all fixing roots commute. Conjugating a root by a candidate lift changes no outer class and still fixes B pointwise. The same argument shows that the lift commutes with the root and normalizes $S(\Theta)$.

An inner element normalizing $S(\Theta)$ likewise commutes with Θ , by applying this centralizer argument to the natural absolutely irreducible fixed simple. Since G_i is adjoint, that element itself lies in G_i^Θ . The root has height $f_i/s < f_i$, so the order comparison makes its normalizer proper in the ancestor. We have thus obtained actual proper labels containing the candidate.

Choose one root Θ_s for each occurring prime. For a product d of primes already chosen, select integers a_s with $\sum_{s|d} a_s d/s = 1$ and put

$$\Theta_d = \prod_{s|d} \Theta_s^{a_s}.$$

The roots commute, and $\Theta_s = \Phi_i$ gives

$$\Theta_d^{d/s} = \Theta_s, \quad \operatorname{ht}(\Theta_d) = f_i/d.$$

For example, raising the product to d/s_0 gives $\Theta_{s_0}^{a_{s_0}d/s_0}$ times $\Phi_i^{\sum_{s \neq s_0} a_s d/(ss_0)}$, which is exactly Θ_{s_0} . This verifies the identity also with diagram components. For the empty product set $\Theta_1 = \Phi_i$.

The group $S(\Theta_d)$ is fixed by each previous root and lies in all their normalizer labels. If it normalized the fixed simple for a new prime root Θ_l , the preceding inner-normalizer argument would make it fixed by Θ_l as well. It would then lie in the fixed group of the combined root Θ_{dl} , contradicting

$$|S(\Theta_d)| > |G_i^{\Theta_{dl}}|.$$

Each new prime therefore makes the label intersection strictly smaller. There are at most as many primes as the interval height, and at most two roots at each associated height. The number of normalizer labels is consequently at most twice that height. $\square$

Corollary 9.10 (Equal natural modules force a fixing root). *Suppose $B < S_i$ is a same-series high classical candidate of the same natural dimension as S_i , and its projective action on V_i is its own natural module, allowing a Frobenius twist and, in type A, a diagram natural/dual twist. Then some prime root of Φ_i fixes B pointwise. Thus the absence of such fixing roots excludes this case.*

Proof. We record why root-free protection precludes equal-natural embeddings within one root series. If the action of B on V_i is projectively its own natural module (allowing Frobenius twists, also the diagram natural/dual twist in A), both groups in the same classical series and dimension, then $B = S(\Psi)$ inside G_i for a standard Steinberg map up to conjugacy. Indeed identify the natural spaces over k by the representation equivalence (split Frobenius twists preserve the standard finite images); forms match up to scalar by uniqueness, also the quadratic when needed by square-difference uniqueness on the finite group. This identifies the corresponding algebraic adjoint groups. Divide the height of Φ_i by that of Ψ ; after composing off that power of Ψ we still fix B pointwise, thus cannot have positive remainder by the order comparison. At height zero, an inner centralization is trivial; so is a possible D-diagram centralization (acting in $\text{PGL}(V_i)$); and a non-inner A-graph centralization would give projective self-duality of the finite natural representation, impossible by the distinct restricted weights and perfection. So $\Phi_i = \Psi^z$, $z > 1$ by strict inclusion, and some prime root fixes B . $\square$

9.4 Simultaneous protection in a pruned chart

Definition 9.11 (A protected classical run). A consecutive high classical run of one defining characteristic is *protected* if for every ancestor–descendant pair within the run the descendant has one tensor atom on the ancestor natural module and no prime root of the ancestor Steinberg map fixes it pointwise. The path also retains all properties of [Definition 9.1](#).

Proposition 9.12 (Selecting all ancestor protections). *Assume that the current same-characteristic classical run is already protected through its current node W . Fix an upper bound D on the number of path steps and a fixed number c of continuation colors. There is a finite extension rank r , depending only on D, c and earlier fixed requirements, such that from every suitably pruned private chart at a current high classical node one can select a next proper extension. If that extension continues the current same-characteristic run, it has both protections relative to every ancestor in that run, including the current node. Earlier structural exclusions are preserved.*

Proof. Work inside a successful pruned flat of extension rank r at the current node W . All its comparable candidate labels have the prepared properties and actual nested simples, by [Section 5](#). Color its proper extensions with at most c colors, recording in particular whether their socle continues

the current same-characteristic high-classical run. Consider a uniform random flag at extension ranks

$$\ell, 2\ell, \dots, (c+1)\ell, \quad r = (c+1)\ell.$$

Every flag has an equal-colored pair. Hence some two ranks $a < b$ have matching probability at least $\binom{c+1}{2}^{-1}$. Conditioning on the b -extension gives a particular U for which at least that fraction of the a -extensions $L \subset U$ have its color. If this color does not continue the run, any such L needs no additional protection.

Otherwise apply the two counts to B_U in each ancestor i of the current run, including W itself. These lists detect every failure of protection at a matching L . Indeed, if $t(B_L) > 1$, a prime congruence coarsening of its atoms has a stabilizer containing K_L . Since $B_U \leq B_L$ fixes its factors individually, each factor groups atoms of B_U . The partition is invariant under the single Frobenius cycle; a cycle has a unique invariant partition into s congruence classes. It is therefore one of the coarsenings counted for B_U . A prime root fixing B_L likewise fixes B_U and occurs on its root list.

None of the counted labels contains K_W . For an earlier ancestor, such containment would make B_W fixed by the root, or individually fix the coarsening slots: its slot permutation commutes with the cycle and is trivial by perfection. Either conclusion contradicts the protection already imposed on W . For the current ancestor, properness of the counted labels gives the same noncontainment. Thus each bad label is K_Z for a subflat $Z \subseteq U$ with $Z \not\subseteq W$.

For a uniform a -extension L in U , the probability of $Z \subseteq L$ is at most $2q^{a-b}$: it suffices to require one fixed point of $Z \setminus W$ in the ordinary affine-flat count. Here q is the chart field size. The two counts give $O(1 + \text{ht}[K_U, K_{W_i}])$ bad labels per ancestor. Reverse rank differences bound this height by Dr over at most D path steps. The total number is therefore at most $CD(1 + Dr)$ for an absolute C after the colors and thresholds are fixed. Since $b - a \geq \ell$, their union occupies a fraction at most

$$2CD(1 + D(c+1)\ell)q^{-\ell}$$

of those extensions. This tends to zero with ℓ , whereas the matching-color fraction is at least $\binom{c+1}{2}^{-1}$. Choose ℓ large enough also for the earlier fixed requirements. A matching L outside the union has one atom and no fixing prime root relative to every ancestor in the run. It retains the structural exclusions because it lies in the successful pruned flat.

There is no circular choice of parameters: the following path bound concerns paths already satisfying these group protections. Its absolute constant is obtained first, then D is fixed, and only then are ℓ and r chosen. $\square$

9.5 From finite inclusions to rational compositions

The tensor protection makes each edge a single ordinary p -digit. To compose two edges rationally, the lower algebraic group must preserve the middle group's defining form. Bilinear forms follow from restricted-weight uniqueness. Quadratic forms in characteristic two require the next two lemmas. Root protection will have a separate role: it excludes natural coefficient-one edges by [Corollary 9.10](#).

Lemma 9.13 (Restriction is injective in degree-one cohomology). *Let J be a simply connected simple classical algebraic group in the high-rank range of [Convention 9.2](#), over $k = \overline{\mathbf{F}}_p$. Let $\Phi = \tau \text{Fr}_p^f$ be an ordinary pinned diagram-field Steinberg map with $f \geq 1$, parameter $P = p^f$, and diagram action τ on weights. If $D' = L_J(\xi)$ is a nontrivial P -restricted simple rational J -module, then*

$$H^1(J, D') \longrightarrow H^1(J^\Phi, D')$$

is injective. The finite group here is the full simply connected fixed group.

Proof. Put $\mathcal{G} = k[J]^{J^\Phi}$, with right finite invariance and left translation. It suffices to show

$$(D' \otimes (\mathcal{G}/k))^J = 0.$$

Indeed, a finite-fixed vector splitting an extension of k by D' defines the equivariant function sending g to its translate of that vector. It is right finite invariant, and its image in the quotient module is constant. Modulo constant functions it therefore lies in the displayed invariant space. Vanishing makes the function itself constant, giving a rationally fixed lift.

The Lang map $g \mapsto g\Phi(g)^{-1}$ identifies the right quotient J/J^Φ with J [24, 46] [34, Definition 17.95 and Theorem 17.96]. It is surjective and étale since $d\Phi = 0$, hence faithfully flat. Its fibres are exactly the right cosets of the finite reduced fixed-point scheme: $h = gx$ with $x = \Phi(x)$. Descent of regular functions identifies $\mathcal{G}$ with the regular function algebra carrying the twisted left-right action.

The Donkin–Koppinen filtration begins with constants and has tensor pairs of induced highest-weight modules as its sections [29, Theorem 4.3 and its preceding construction] [4, Section 2.4]. Pulling this two-sided filtration back to the twisted diagonal gives an exhaustive filtration of $\mathcal{G}/k$ with sections

$$\nabla(\nu) \otimes \nabla(\nu^*)^{[\Phi]}, \quad \nu \neq 0 \text{ dominant.}$$

Here $*$ denotes highest-weight dual, $[\Phi]$ pullback, and Δ below is a Weyl module. A nonzero invariant after tensoring a section by D' gives a nonzero map

$$\Delta(\nu)^{[\Phi]} \longrightarrow L(\xi) \otimes \nabla(\nu).$$

The source is still generated by its highest vector, since Φ is surjective on algebraic-group points. The map is nonzero there, so

$$P\tau(\nu) \leq \xi + \nu$$

in dominance order. The source is also trivial on the f -th Frobenius kernel J_f . Its nonzero image therefore gives $\text{Hom}_{J_f}(L(\xi)^*, \nabla(\nu)) \neq 0$, with kernel and invariance understood as group schemes.

Since ξ is P -restricted, the restricted-simple theorem and Steinberg decomposition force a rational composition factor of $\nabla(\nu)$ whose highest weight has the form

$$\xi^* + P\gamma \leq \nu, \quad \gamma \text{ dominant.}$$

Indeed, every rational simple restricts to copies of the J_f -simple for its P -restricted part, and those restricted simples are pairwise distinct. Pairing the two inequalities with $\rho^\vee$ gives the diagram- and dual-invariant height comparisons

$$(P - 1) \text{ht}(\nu) \leq \text{ht}(\xi) \leq \text{ht}(\nu).$$

Only $P = 2$ and equality remain. Then $p = 2$, $f = 1$, and ξ has an active coefficient one. Equality gives $P\tau(\nu) = \xi + \nu$, so the image of the source highest vector lies on the unique tensor highest line. Let $v_\xi \otimes v_\nu$ span it and choose α with $\langle \xi, \alpha^\vee \rangle = 1$. The lowering operator has value

$$(f_\alpha v_\xi) \otimes v_\nu + v_\xi \otimes (f_\alpha v_\nu).$$

The first summand is nonzero by the simple-root bracket, and the first-factor weights $\xi - \alpha$ and ξ differ. The summands cannot cancel, including in characteristic two. The line is not J_1 -fixed, contradicting the kernel-trivial source action. Thus no section has an invariant. Every invariant would occur in the exhaustive filtration and have a nonzero successive image, so the required vanishing follows.

The restricted-kernel and regular-bimodule results are the simply connected theorems in Jantzen [19, Part II, Chapter 3 and II.4.20]; see also Lin and Nakano [26, Section 1.4] and Mathieu [29, Theorem 4.3]. The pullback to the ordinary diagram-twisted diagonal uses no complete reducibility. $\square$

Lemma 9.14 (The quadratic form in characteristic two). *Let J be simply connected of sufficiently high type D_e in characteristic two, with ordinary pinned map $\Phi = \tau \text{Fr}_2^f$, $f \geq 1$. Let $\eta \neq 0$ be 2-restricted, and consider $L_J(2^y\eta)$. Suppose a quadratic form Q on this module is invariant under the full finite cover J^Φ , and its polar form is rationally J -invariant. Then Q itself is rationally J -invariant.*

Proof. Untwist 2^y to $L_J(\eta)$; split Frobenius permutes J^Φ because Φ is a standard pinned graph/field map. Exact invariance under the chosen finite linear lift holds by perfection: its similitude scalar is a character. Rational invariance of the polar form makes

$$c(g) = gQ - Q$$

a regular cocycle with values in the subspace of squares of linear forms. We will take its coefficientwise square root to obtain a cocycle in $L_J(\eta)^*$.

The target Lie action on squares is zero, so the differential of c at the identity kills commutators. The Lie algebra of simply connected type D is perfect also in characteristic two: simple coroots span the torus part and are opposite-root brackets, and every D-root pairs oddly with some coroot. Thus this differential, and by translation every differential of c , vanishes.

Each coordinate of c is consequently a square in the function field, using the kernel of differentials over a perfect field. Its square root is regular by normality. Taking roots of the coefficients in a fixed dual basis therefore gives a regular linear-form cocycle, zero on J^Φ . Additivity, injectivity and equivariance of squaring on k -points give its cocycle identity.

The square-root cocycle takes values in the nontrivial restricted dual module after untwisting. By Lemma 9.13, its class is zero, so it has the form $g\ell - \ell$ for a linear functional ℓ . Its vanishing on J^Φ makes ℓ finite-fixed. The restricted finite module is nontrivial and irreducible, so $\ell = 0$. Thus the square-root cocycle, and therefore $gQ - Q$, is identically zero. The perfect-Lie-algebra argument above is used here only for type D in characteristic two. $\square$

Proposition 9.15 (Composing the rational highest-weight actions). *Let $i < j < l$ lie in one protected run, and suppose these three nodes have the same series A, B, C , or D . Use consistent natural realizations and simply connected covers of the actual nested simple subgroups. Write $G_s = \widehat{G}_j$ and $J = \widehat{G}_l$. The lifted edge representations ij and jl have highest weights*

$$p^x\lambda, \quad p^y\eta,$$

where $\lambda, \eta \neq 0$ are p -restricted. The latter representation induces an algebraic homomorphism $J \rightarrow (G_s)_{\text{ad}}$ which lifts to $J \rightarrow G_s$. The resulting rational composition of $L_{G_s}(p^x\lambda)$ is irreducible. Its highest weight μ is one nonzero p -digit times a power of p , and that digit has the shape, up to diagram, of the direct edge il .

Proof. Consider path nodes $i < j < l$ in one protected run, of the same root series A, B, C or D . Use consistent finite-group identifications at each node (actual simple B_{W_j} , its projective natural realization, and simply connected cover, for both its incoming and outgoing edges). Thus the projective restriction along $i \rightarrow j \rightarrow l$ is just via inclusion of the same nested simples as $i \rightarrow l$; the weight colors below also respect diagram changes. Write $G_s = \widehat{G}_j$ and $J = \widehat{G}_l$ temporarily for the

simply connected algebraic groups. Choose pinned coordinates with standard graph/field Frobenius on each (conjugating the finite realizations accordingly if necessary). The edges ij, jl , evaluating natural modules on descendants projectively, have highest weights $p^x\lambda, p^y\eta$ respectively for the lifts, $\lambda, \eta \neq 0$ p -restricted, since each has one atom. We claim $L_J(p^y\eta)$ on the natural space for j gives a morphism $J \rightarrow (G_s)_{\text{ad}}$ inducing the given projective action on finite points. This is immediate in A. In the remaining types the finite representation preserves the bilateral form (perfection); by restricted-weight uniqueness it is rationally self-dual, and the invariant form rationally therefore agrees up to scalars. This suffices except for D in characteristic two, where [Lemma 9.14](#) supplies rational invariance of the quadratic form as well.

The morphism to the adjoint group (in form types one can first map by the given linear representation to the connected classical isometry group, using connectedness of J , then take its adjoint quotient) lifts to G_s by simply-connectedness. For example pull back the central isogeny; the reduced identity group in the pullback maps surjectively to J (finite surjection, finitely many components), by a central isogeny with kernel of multiplicative type; its radical has trivial image and hence dimension zero, so it is semisimple and maps isomorphically by the simply connected root datum of J . Its natural action agrees linearly with the above one (no scalar characters). On finite points the lift differs at most by central elements from lifts to the usual finite covering group at node j . Indeed both give the prescribed projective matrices on the natural space (the image of the finite cover surjects projectively onto the given socle), and the kernel of the projective natural map on simply connected algebraic-group points is central. Thus composing $L_{G_s}(p^x\lambda)$ yields on restriction to J^Φ projectively the direct-edge action il . The rational composition is irreducible by finite irreducibility. Its highest weight μ for J is again a single p -digit times a power of p , by Steinberg decomposition and the protected individual tensor indecomposability on finite points. Indeed multiple rational digits here would give multiple nontrivial-dimensional factors fixed individually by the finite cover, hence also as matrix factor algebras by the finite projective S_l -action (and this projective action already agrees as just explained). Moreover the digit shape agrees up to diagram with the direct edge's: reduce its Frobenius twist exponent on the finite group using the defining graph/field map and apply restricted-weight uniqueness (linear lifts agree by perfection). $\square$

9.6 Universal cuts of the full weight multiset

Lemma 9.16 (Chamber maxima and universal cuts). *Use the triple, groups, and weights of [Proposition 9.15](#). Let Ω be the full weight multiset of $L_J(\eta)$, with every multiplicity retained. After mapping a maximal torus of J into one of G_s , the natural weights restrict to $p^y\Omega$. One Weyl translate of λ maximizes throughout the positive cocharacter chamber of J , and on each active fundamental summand of λ .*

In type A an active node a of the upper weight λ therefore gives a cut selecting a entries of Ω , each dominating every unselected entry, with no equal-vector tie across the cut. In B and C the analogous signed fundamental cut selects a entries, including the last node up to the B spin scale; its selected weights are nonzero and in the positive cone. This also holds at the nonspin D nodes when λ has no spin support. At an active D spin node the maximum instead chooses one orientation u_a of each natural signed pair, and

$$u_a + u_b \text{ is nonnegative on the chamber whenever } a \neq b.$$

Distinct ambient orbit weights cannot restrict to the same highest weight. More generally, every weight of a simple highest module differing from highest by a multiple of one simple root has multiplicity at most one.

Proof. Map a maximal torus of J into one of G_s . The restricted natural multiset is $p^y\Omega$. By irreducibility of the composition and the highest-weight convex hull property, the maximum pairing on the orbit of $p^x\lambda$ is the linear function given by μ throughout the positive cocharacter chamber of J . Choose a maximizing orbit weight at a strictly interior cocharacter. Its restriction must equal μ as a vector, so that same Weyl translate $w\lambda$ maximizes everywhere in the chamber.

It also maximizes each active fundamental summand. For any one cocharacter a common Weyl translate maximizes all fundamental summands; equality in their positive sum therefore forces equality for each. Two distinct ambient orbit weights cannot both restrict to the composed highest weight, whose multiplicity is one.

Suppress the positive scales p^x, p^y for the cut description. In type A an active upper node a selects a natural coordinates at $w\lambda$. Exchanging one selected and one unselected coordinate shows, throughout the lower chamber, that every selected restriction dominates every unselected restriction. An equal-vector tie would give another maximizing ambient orbit weight across a strict coefficient boundary, contradicting highest multiplicity one.

In B and C the analogous choice takes a entries of the full signed multiset, also at the last node up to the B spin scale. In D the same argument applies before the spin nodes when λ has no spin support. Sign changes show that selected restrictions lie in the positive cone; in D compensate by an unused coordinate. A zero-vector restriction would permit a sign change in the full maximizing orbit weight, using a zero-coefficient coordinate for D parity, and again contradict multiplicity one. Thus the selected values are nonzero. Signed coordinate exchanges compare them with every unused value, including negative choices and the extra zero in B. An equal nonzero tie across the cut also changes the ambient orbit weight; parity can be preserved by changing both directed signs when needed. These cuts are therefore strict as asserted.

For D spin support, take an active spin fundamental summand instead. Its maximizing signed choices u_i satisfy the stated pair-sum condition, because flipping any two choices cannot increase the pairing on the chamber.

These comparisons concern all entries of Ω , including nonextremal weights and repeated entries. A cut of its Weyl orbit alone is only a necessary consequence. The single-root multiplicity assertion follows because the corresponding weight space of the highest-weight module is generated by the one divided-power monomial in that root; it does not assert multiplicity one for arbitrary weights. $\square$

9.7 The word argument in type A

Lemma 9.17 (Type A orbit cuts). *Let J have sufficiently high type A_{e-1} , and let $0 \neq \eta$ be p -restricted. Suppose a nonempty proper universal cut of the full weight multiset of $L_J(\eta)$ is given as in Lemma 9.16. Replace the cut by the longest-element image of its complement if necessary so its selected part of the extremal orbit has size at most half that orbit. Unless η is supported at one endpoint, the selected orbit words consist of the highest word alone, possibly its single boundary swap for a binary word, or the positive half of the orbit of the exact highest root. In the final case the highest weight is exactly $\epsilon_1 - \epsilon_e$, not a larger restricted multiple.*

Proof. Take J of type A_{e-1} . Its orbit on η consists of permutations of a descending nonconstant word, ordered for dominance by prefix sums (coordinate sum zero). In a universal cut of Ω , the upper orbit words form a nonempty proper subset U , since the orbit contains the full highest and lowest weights. We may study the smaller orbit side, replacing the full cut by the longest-element reversal of its complement when needed (this is again an upper cut of the weight multiset, now of complementary size) so $|U|$ on the orbit is at most half. Then U is an initial segment both for lexicographic decreasing order and for lexicographic increasing order starting from the right. This

follows by the first differing coordinate comparisons from strict cone domination and equality of total sums.

Here is the description needed of such segments. Let M be the largest letter, multiplicity b , and m the smallest, multiplicity c . If U contains a word not starting with M , it contains all starting with M . If $b \geq 2$ this includes one also ending with M , forcing all ending with lower letters; the two fractions (and extra words) contradict the half bound. Thus $b = 1$. With second-largest letter T , a starting- M word ending in T forces in all ending below T . If there are at least two such lower letters counting multiplicity, these include one also starting below T , forcing all starting at or above T . Again the fractions contradict the bound. Namely if l letters are below T , the proportions ending below T and starting at or above T are l/e and $1 - l/e$; in the indicated situation U contains each such class entirely and strictly. We obtain only a natural two-value shape with singleton extreme against constant, or M, T^{e-2}, m . Symmetrically these are the exceptions if some selected word does not end with m .

Otherwise all words in U start with M and end with m . Until the first word not ending with m , left lex order fixes the first $e - c - 1$ coordinates of the descending word: the greatest word not so ending already has those coordinates. Right lex order confines changes to the first $b + 1$. Only positions in $[e - c, b + 1]$ can therefore change. This interval has at most one position unless $b + c = e$, the binary case; then its two positions permit only the single boundary swap. Thus, outside the exceptional shapes, U consists of η alone or of η and that swap.

For M, T^{e-2}, m , use positions a, d of the largest and smallest letters. Both orders begin with the half $a < d$, one prioritizing a increasing there and the other d decreasing. A proper nonempty common prefix of the half has only $(1, e)$ (taking $a > 1$ forces $(1, 2)$, then the full half; similarly for $d < e$). Dominance across the cut consisting of the half requires $M - T = T - m$, by comparing words with both defects early and both defects late, in both choices of early side. At a prefix separating the two defects of one word (both before the boundary) from those of the other (both after) this compares $(M - T) + (m - T)$ to zero, in both directions. Thus $\eta = z(\epsilon_1 - \epsilon_e)$ in that case (ϵ_i standard coordinates). In the full weight set each selected weight then has all prefix sums nonnegative (dominance over negative scaled simple roots on the other side), and each unselected one has them nonpositive. If $z > 1$, the weight $\eta - \alpha_1$ occurs by simple-root lowering ($z < p$, $\alpha_1 = \epsilon_1 - \epsilon_2$), and can be permuted to mixed prefix signs. So this half exception requires $z = 1$. $\square$

For edges whose descendant has type A, color the nonzero restricted digit by the following five possibilities:

1. one active endpoint with coefficient one;
2. one active endpoint with coefficient greater than one;
3. one active interior node;
4. multiple active nodes, excluding the exact highest root;
5. the exact highest root.

This palette is invariant under the A diagram reversal.

Proposition 9.18 (No homogeneous type A configuration). *Four nodes of a protected run of type A cannot have all pair edges of one color in the preceding palette.*

Proof. Take the first three of four homogeneous nodes, with the notation of [Proposition 9.15](#). For each color we first constrain the extremal orbit cut, then use actual full weights or their multiplicities.

Multiple support and the highest root. In the multiple nonexact color, the orbit lemma leaves only η on the smaller side of every cut. A full selected weight ν must dominate every active simple reflection $s_h\eta$. Thus the nonnegative difference $\eta - \nu$ is supported only at h for each active h . Multiple support forces $\nu = \eta$, whose multiplicity is one. Every active node of the upper digit λ must consequently be an endpoint; since it has multiple support, both endpoints are active. Apply the same argument to the last three nodes to obtain that conclusion for η as well. In the exact highest-root color both digits already have these two active endpoints.

Choose natural coordinates at a maximum of λ whose first, second, penultimate and last restrictions, before the lower twist, are

$$\eta, \quad \eta - \alpha_1, \quad w_0\eta + \alpha_1, \quad w_0\eta,$$

respectively. Endpoint activity supplies these distinct weights in the lower module. The middle coefficients of λ are constant, so their placement preserves the maximum. Write $\theta = w\lambda$ for this maximizing weight, and β_1, β_{v_j-1} for the two endpoint simple roots in that coordinate order. Restricted activity makes $\theta - \beta_1$ and $\theta - \beta_{v_j-1}$ distinct weights of the upper module. Their restrictions satisfy

$$\text{res}(p^x(\theta - \beta_1)) = \text{res}(p^x(\theta - \beta_{v_j-1})) = \mu - p^{x+y}\alpha_1.$$

The two ambient weight spaces stay independent on restriction. They give multiplicity at least two along one simple-root ray from the composed highest weight, contrary to [Lemma 9.16](#).

One interior node. Write $\lambda = c\omega_a$ and $\eta = z\omega_h$, with a and h interior in their respective diagrams. Replace the full cut by the longest-element image of its complement if required by [Lemma 9.17](#). Its selected orbit words are η , possibly together with $s_h\eta$.

Every full selected weight ν differs from η only in the α_h direction. If $s_h\eta$ is outside, domination of that word proves this. Otherwise use the two outside words $s_{h-1}s_h\eta$ and $s_{h+1}s_h\eta$: their downward supports intersect only at h . The actual module also contains

$$\zeta = \eta - \alpha_h - \alpha_{h-1}.$$

To obtain it, lower at h and then across the simple bond to its left, where the new coefficient is one. This weight is outside the cut, because its downward support is not confined to h . Domination of ζ leaves only η and $\eta - \alpha_h$ on the selected side, each of multiplicity at most one.

The cut has a entries, or $v_j - a$ in the complemented case. Both numbers are at least two, so these are exactly its two entries. Since the full multiset Ω sums to zero, complementation and the longest element replace a sum by its highest-weight dual. Hence

$$\mu = cp^{x+y}(2\eta - \alpha_h) \quad \text{or} \quad \mu = cp^{x+y}(2\eta - \alpha_h)^*.$$

Both neighboring fundamental coefficients of $2\eta - \alpha_h$ are positive. No common Frobenius rescaling removes either support node, contradicting the one-node digit shape of the direct edge.

One endpoint. In the coefficient-one color, the edge is a natural or dual-natural module of equal dimension, excluded by [Corollary 9.10](#). Otherwise the upper digit module, up to duality, is a symmetric power of degree c with $2 \leq c < p$. The lower natural or dual-natural restriction has highest weight η_0 , equal to η or η^* , and a highest simple-root string containing vectors a_0, a_1, a_2 of weights $\eta_0, \eta_0 - \alpha, \eta_0 - 2\alpha$, since its restricted endpoint coefficient is also greater than one. The distinct monomials

$$a_0^{c-1}a_2, \quad a_0^{c-2}a_1^2$$

both have weight $c\eta_0 - 2\alpha$ before the common twists. They contradict the same single-root multiplicity bound. Thus every color is impossible. $\square$

9.8 The signed series

Lemma 9.19 (Signed orbit cuts). *Let J have sufficiently high type B , C , or D , and suppose $L_J(\eta)$ is self-dual, with $\eta \neq 0$ restricted. In type D suppose for this lemma that no spin node of η is active. Let a signed fundamental cut as in Lemma 9.16 be given, including the last B and C nodes and excluding only the D -spin case, and let U be its selected extremal orbit words. Use nonnegative dominant magnitudes, write h for their nonzero count, and let O^+ be the unsigned permutation orbit.*

1. *If some unsigned word is unselected, then $U \subset O^+$. For $2 \leq h < e$, $|U| \leq 2$, with the binary possibility $\eta, s_h\eta$; for $h = e$, needed only in B and C , $U = \{\eta\}$.*
2. *If every unsigned word is selected and $h < e$, then either $h = 1$ or $h = 2$ with the two nonzero magnitudes equal.*
3. *If every unsigned word is selected and $h = e$ in B or C , all magnitudes are equal and $|U| \leq 3$.*

Proof. Now J is B, C or D of rank e , and dominance in particular requires the prefix comparisons in the usual ordered signed coordinates. The orbit of η is antipodal by self-duality; make its dominant magnitudes nonnegative using the D diagram if necessary. First suppose no D spin nodes active (case of spin activity handled separately). For a universal fundamental cut above in Ω , upper orbit words U include η , and each has all prefix sums nonnegative and is nonzero. Write h for the number of nonzero magnitudes in a word, and O^+ for all unsigned permutations.

If some unsigned word is outside U , comparison of total sums gives $U \subset O^+$. If $h < e$, we have all choices of signs (even in D , by zeros). Before the earliest negative word in decreasing lex order the first $h - 1$ positive coordinates are fixed, hence $|U| \leq e - h + 1$. Indeed in this case all negative words are outside U , and one can put the largest $h - 1$ magnitudes first positive, with the final nonzero one negative later; left prefix domination gives the lex bound. For $2 \leq h < e$ in high rank this is at most half the unsigned words (there are at least $\binom{e}{h}$). In this range apply the A -word prefix description among those equal-total words. We obtain at most two upper words (the adj-shaped half for a word M, T^{e-2}, m of three distinct magnitudes would be too large by $e - h + 1$, and the remaining binary singleton-extreme exception in this range has a singleton zero, already giving bound two by $e - h + 1$). For a binary shape in that range with equal nonzero coordinates they are η and possibly $s_h\eta$, also when $h = e - 1$ by left lex order. If instead $h = e$, here only needed for B, C , the first $e - 1$ positives are fixed before the first negative word, giving just η .

Suppose all unsigned words are selected. If $h < e$, every signed word with positive first coordinate is forced in: otherwise it fails the first-prefix comparison with a selected unsigned word starting with zero. For $h \geq 3$, put a smallest nonzero magnitude first positively and two others negatively; the third prefix is negative. For $h = 2$ with unequal magnitudes, the smaller positive followed by the larger negative gives the same contradiction. Thus only $h = 1$ or two equal nonzero magnitudes can occur.

For $h = e$ in B or C , suppose the smallest and largest magnitudes are $P < M$, and let S be the sum of all e magnitudes. The signed word with M first and all other entries negative must be selected, by comparison with a selected unsigned word starting with P . The word starting with $-P$ and having all other entries positive is unselected, since its first prefix is negative. Domination of their total sums would require

$$2M - S \geq S - 2P, \quad \text{hence} \quad S \leq M + P,$$

impossible with at least three positive coordinates. All magnitudes are therefore equal. A selected word cannot have two or more negatives, since total-sum comparison would then force in every

single-negative word, including one with negative first prefix. Left lex order before the first two-negative word now leaves only the all-positive word and the words with one negative in either of the last two positions. Thus $|U| \leq 3$. $\square$

For a fixed series B, C, or D, use the following edge colors:

1. a single node 1, with coefficient one or greater than one as separate colors;
2. a single interior nonspin node, separating index 2 from larger indices;
3. the single last node in B or C;
4. any spin activity in D;
5. all remaining multiple support.

Here “interior” in B and C excludes the last node. B and C are separate vertex types, and B’s last-node color includes its spin weight. The D colors are invariant under interchange of its spin nodes.

Proposition 9.20 (No homogeneous signed configuration). *Four nodes of a protected run having one fixed series B, C, or D cannot have all pair edges of one color in the preceding palette.*

Proof. Use the triple notation of [Proposition 9.15](#), and consider the possible common edge color.

D spin activity. For D spin activity on the two edges, η has at least $e - 1$ nonzero coordinates. In its orbit take words starting with a smallest nonzero entry positive and two others negative, then many independent signs on the remaining nonzero ones respecting parity. If all e coordinates are nonzero, parity leaves at least 2^{e-4} choices; if one is zero, it absorbs the parity adjustment and gives the same lower bound. Their first prefix is positive and third prefix negative. No two constructed words are antipodes, since the first coordinate is positive in each. Thus for $e \geq 7$ this gives more than e distinct antipodal pairs with mixed prefix signs. Self-duality supplies antipodality even in odd D rank, where it forces the last coordinate to be zero when opposition interchanges the spin nodes. Each pair appears (multiplicity one) as a natural ambient signed pair in Ω . In the spin maximum for λ as above, its chosen orientation therefore contributes a negative prefix somewhere. But at any one prefix at most one coordinate choice can have negative sum, by the pair-sum condition. Contradiction.

Remaining multiple support. In the remaining multiple-support color, the orbit description forces U to have just η . Indeed $h = 1$, two equal nonzero coordinates only, and all equal full magnitudes are all single-node shapes here. And a bound of two words in the description reduces to just η under multiple support: any additional one would have to dominate the active reflections outside, impossible (its difference down from highest must be supported only at each such node; if itself an active reflection use a different active node). Then every full selected weight also dominates all those reflections, giving just η with multiplicity one. This cannot accommodate the active indices of λ .

One interior nonspin node. Write $\lambda = c\omega_a$ and $\eta = z\omega_h$, where a and h are interior nonspin nodes in their respective diagrams. Here $2 \leq h < e$, with $h \leq e - 2$ in D. If all unsigned words were selected, the two-equal-magnitude exception would force $h = 2$. The palette then also gives $a = 2$, whereas the cut would contain at least $\binom{e}{2} > 2$ unsigned words. Thus only η , possibly together with $s_h\eta$, occurs in the selected orbit.

As in type A, every full selected weight differs downward from η only at h . If $s_h\eta$ is selected, use reflected words from two distinct Dynkin neighbors; at the D fork either right neighbor suffices. Intersecting their downward supports does not require the right bond to be simple. The outside weight $\eta - \alpha_h - \alpha_{h-1}$ is present by the left simple bond. After the first lowering its vector is highest for that left rank-one subgroup and has coefficient one there, so the second lowering is nonzero in every characteristic. Domination of this outside weight leaves at most the two entries $\eta, \eta - \alpha_h$, each of multiplicity one. Since the cut has $a \geq 2$ entries, it has exactly two. The composed highest weight is a positive scalar multiple of $2\eta - \alpha_h$, with both neighboring nodes active, contrary to the direct edge's one-node digit shape.

The last B or C node. For the last single node of B,C the fundamental maximum in the upper group orients every natural signed pair to the upper side. It therefore chooses half the orbit of η , which has all equal positive magnitudes before sign changes. This contradicts the bound above.

Node one. Coefficient one gives the prohibited natural embedding. For a single node 1 with both coefficients larger than one, write $\lambda = z\omega_1$. Assign $\epsilon_1, \epsilon_2, \epsilon_3$ at a maximum respectively to $\eta, \eta - \alpha_1, \eta - 2\alpha_1$, up to twist (distinct, nonzero and nonopposite, adjusting unused signs if needed). The top A_2 -face in the upper module before twist is the natural symmetric power of degree z . Thus the distinct weights $(z-1)\epsilon_1 + \epsilon_3$ and $(z-2)\epsilon_1 + 2\epsilon_2$ both occur in these coordinates and restrict to $z\eta - 2\alpha_1$ before the common twists. Their two weight spaces give multiplicity at least two along one simple-root ray from highest after restriction, contradicting [Lemma 9.16](#).

All constructions require only fixed lower bounds on rank: four coordinates suffice for the type A equal-gap test, three for the signed total-sum comparison, and seven for the displayed D parity count. They therefore apply after fixing one sufficiently large absolute high-rank threshold. $\square$

9.9 The absolute path bound

Theorem 9.21 (Bound on protected high-type paths). *There is an absolute constant D_0 such that a prepared path all of whose socles are of high alternating or high classical type, and whose consecutive same-characteristic classical runs are protected, has fewer than D_0 nodes. This constant is independent of all chart dimensions, finite fields, and path-construction parameters.*

Proof. Within a protected classical run, first color vertices by the four series A, B, C, and D. The edge palettes above have an absolute number of colors and are compatible with diagram changes. Finite Ramsey theory [43] therefore bounds the number of nodes in the run: a sufficiently long run contains four same-series nodes whose pair edges are homogeneous, contrary to [Propositions 9.18](#) and [9.20](#). All pairs of the selected nodes remain protected because the protection was imposed for every ancestor–descendant pair, not just consecutive nodes.

By [Lemma 9.4](#), at most two high alternating nodes occur. By [Lemma 9.5](#), there cannot be two successive changes of characteristic among the classical nodes after ignoring the alternating interruptions. Thus there are only absolutely boundedly many same-characteristic classical runs. Applying the run bound to each gives D_0 . $\square$

Completion of the proof of Theorem 4.1. Use the almost simple reduction of [Theorem 4.15](#) and the prepared path construction of [Section 5](#). Choose the high thresholds and D_0 first, then take the number D of reserved generations greater than D_0 , with room for the initial displacement and one remaining bounded-type stage. Choose the lattice field size q above the finitely many absolute line requirements. Next choose the terminal chart increment r using [Proposition 9.12](#) and [Proposition 8.6](#).

Choose the earlier trial increments backwards from r as in [Section 5](#), then m above their ranks and n large enough for all residual interval lengths and bounded spans. Finally choose N for the initial pattern demand and enough private charts g for all finite losses.

At each stage, [Sections 6 and 7](#) supplies a pruned chart with the ancestor properties of [Definition 9.1](#). A bounded-order socle is excluded by the residual interval length. A bounded-type Lie socle is impossible by [Proposition 8.6](#). In every remaining case the next high-type node can be selected, with [Proposition 9.12](#) adding the protections whenever the current classical run continues. This produces more than D_0 high-type nodes, contradicting [Theorem 9.21](#). It proves the required finite large-family nonrepresentation statement. Together with [Section 3](#), this gives a finite minimum strictly greater than seven. Its explicit size ceiling and the description of L_0 are supplied in [Sections 16 and 17](#) after the uniform-decision argument. $\square$

10 Reverse alternating tests

The undecidability argument will use exact subgroup intervals in both orientations, and intervals of invariant subgroups in powers of a nonabelian simple group. We first prove the reverse exclusion needed for that reduction. Its hypothesis is an arbitrary exact realization of a reversed alternating subgroup lattice. From such a realization we recover an ordinary site interval, build the linear charts of the path argument from actual elementary subgroups, and reproduce the structural exclusions. The forward marker theorem in the next section has a complementary conclusion: after a bounded prefix, it supplies an actual alternating simple subgroup with many point-stabilizer labels.

In [Section 13](#), markers will be placed beside units carrying grids and full permutation groups. Normal and projection tests on nontrivial lattice images of that template will allow these two theorems to force a forward product configuration. Intersections of cuts then identify one common partition into actual orbit cells, in which equal totals and squares supply arithmetic. Finally, [Section 14](#) bounds the number of lattice elements computably in the number of units, independently of the formal alphabet sizes. A computable bound for carriers of realizable lattices would consequently bound positive Diophantine solutions whenever they exist.

Throughout, reversal means reversal of the entire displayed parameter lattice. Selecting its coordinate-product labels produces a separate ordinary site interval; it need not preserve all labels of the original invariant interval. We continue to use property (*): every normal subgroup of the upper endpoint is contained in, or supplemented by, the base, and the quotient by the base core has a unique minimal nontrivial normal subgroup, which is nonabelian. Fixed absolute thresholds need not be effectively evaluated: they may be fixed constants in an implication between the existence of algorithms. All groups in this section are finite.

Theorem 10.1 (Reverse alternating exclusion). *For all sufficiently large integers v , the lattice $\text{Sub}(A_v)^{\text{op}}$ has no exact realization as*

$$[P, S^b]^\Delta,$$

where S is nonabelian simple, Δ acts by automorphisms transitively on the b simple factors, P is a Δ -invariant coordinate product, and either $P \neq 1$ or $b = 1$.

10.1 Normal tests and the ordinary site interval

We first establish the normal test needed at every prepared link. The test applies to the full invariant interval through a semidirect product; we will then transfer its conclusion to the ordinary interval formed by the coordinate-product labels.

Lemma 10.2 (Semiregular alternating normal test). *Let $X \leq A_n$ be semiregular and elementary abelian, allowing $X = 1$. For n sufficiently large in terms of $|X|$, every exact finite group interval isomorphic to $[X, A_n]$ or $[X, A_n]^{\text{op}}$ has property $(*)$.*

Proof. We first show that a normal subgroup can give only an endpoint label. The unique nonabelian monolith will then follow from nonmodularity. Suppose $[H_0, G_0]$ is such an interval, and let $D \in [X, A_n]$ label H_0N , where $N \trianglelefteq G_0$. This label satisfies the left-modular identity, in either orientation. Moreover, projecting H_0N to $[L, K] \subseteq [H_0, G_0]$, by meeting and joining in either order, gives

$$L(N \cap K), \quad N \cap K \leq K.$$

The identity

$$z \vee (d \wedge y) = (z \vee d) \wedge y \quad (z \leq y) \tag{10.1}$$

is self-dual. Consequently neither D nor any of these projections can admit a test

$$D \cap y \leq z < y \leq \langle D, z \rangle. \tag{10.2}$$

We show that $D = X$ or $D = A_n$.

Choose 37 points in distinct X -orbits. Their X -translated 37-tuples are pairwise disjoint. Let T be the product of the alternating groups supported on these tuples. Thus $X \cap T = 1$, and $[X, XT]$ identifies with the X -invariant subgroups of T . For $L \leq A_{37}$ let T_L denote its product in the matched coordinates. Put $D_0 = D \cap T$, and write $V \leq A_{37}$ for a coordinate projection of D_0 . The following pair of conditions is impossible whenever $B_0 \leq a < c \leq A_{37}$:

$$\langle V, B_0 \rangle \cap c \leq a, \quad c \leq \langle a^{V, a} \rangle. \tag{10.3}$$

Indeed project once more to $D' = \langle D_0, T_{B_0} \rangle$. The first condition gives $D' \cap T_c \leq T_a$. Conjugating independently supported copies of a by elements whose site projection belongs to V gives $T_c \leq \langle D', T_a \rangle$, contrary to (10.2).

Suppose first that V is proper and transitive. Burnside's prime-degree Theorem [35] and the classification of finite two-transitive groups imply that V lies in the affine normalizer of a regular cyclic group of order 37; in particular it contains a 37-cycle. Here the non-affine possibilities at degree 37 all contain A_{37} . For completeness, 37 is not a projective point number $1 + s + \dots + s^j$ for a prime power s : for $j = 1$ one would have $s = 36$, for $j = 2$ the adjacent possibilities $s = 5, 6$ give 31, 43, and for $j \geq 3$ the small remaining possibilities likewise miss 37. The other classical and exceptional two-transitive degrees give none at 37; we use the permutation-group classification in Meagher et al. [33, Table 1].

Choose a cyclic 37-group a so that $c = N_{A_{37}}(a)$ intersects V trivially. To justify the choice, both V and c have order at most $37 \cdot 18 = 666$. A nonidentity affine permutation fixes at most one point. From its cycles one can choose nine disjoint directed moved pairs. A commuting permutation is determined on those eighteen points by the ordered images of the nine sources, and has at most $19!$ choices on the rest. Its centralizer in S_{37} therefore has order at most $(37 \cdot 36 \dots 29)19!$, so its A_{37} -class has size at least $28!/(2 \cdot 19!) > 666^2$. Averaging the intersections of V with conjugates of c now gives a conjugate with no nonidentity intersection. The groups V and a contain distinct regular cyclic 37-groups; a proper transitive subgroup would be affine, where the Sylow 37-group is unique. Hence $\langle V, a \rangle = A_{37}$. Its simplicity gives (10.3) with $B_0 = 1$, a contradiction.

Suppose next that V is nontrivial and intransitive. Choose complementary invariant sets A, F as follows: if there are at least two singleton orbits, use two of them as F ; if there is exactly one, use that one; if there are none, use a smallest orbit. In all cases $|A| \geq 19$. Remove two points from

A when $|F| \geq 2$, and three when $|F| = 1$, leaving a set Z meeting every orbit in A . In the first case with two fixed points removed to form F , this is possible because a nontrivial even action has $\sum_O(|O| - 1) \geq 2$ on its other orbits. In the remaining cases the orbits of A are nonsingletons, so the required removals are possible. Set $B_0 = A_Z$. The V -conjugates of B_0 generate A_A : their supports cover A and have large overlaps.

Take a 4-cycle on the removed points and enough points of F to make four. When two points come from each set, arrange the two from A consecutively and the two from F consecutively. Multiply this cycle by a disjoint transposition in Z , obtaining an even permutation g . Set

$$a = \langle B_0, g^2 \rangle, \quad c = \langle B_0, g \rangle.$$

No element of $c \setminus a$ preserves A , whereas $\langle V, B_0 \rangle$ does. Thus the first condition in (10.3) holds. The normal closure of a in $\langle V, a \rangle$ contains A_A and its conjugate by g^2 . Their overlapping supports generate the alternating group on A with the used points of F added. This group contains c , proving the second condition and yielding another contradiction.

The two exclusions show that every site projection V is 1 or A_{37} . We next rule out diagonal ties in the latter case. Then D_0 is subdirect and in fact equals T . Otherwise the description of subdirect simple powers as products of twisted diagonals gives a tied pair of sites. Choose $a = C_2 < c = C_6 \leq A_{37}$. Independent nontrivial supports from T_a break all ties, so $\langle D_0, T_a \rangle = T$. Modularity would give

$$T_c = \langle D_0 \cap T_c, T_a \rangle.$$

On a tied pair, quotient each C_6 coordinate by its C_2 subgroup. The image of $D_0 \cap T_c$ has size at most 6, whereas that of the left side has size $3^2 = 9$. The displayed equality is impossible. Thus every transversal product T is either contained in D or has trivial intersection with D .

If one such T is contained in D , then all are. Change one point of a transversal at a time, using spare X -orbits when needed. Consecutive transversals share supported three-cycles, so the alternative of trivial intersection is impossible at the next one. The supported A_{37} 's on the transversals themselves have connected, overlapping supports covering the letter set. They generate A_n , and hence $D = A_n$.

Otherwise $D \cap T = 1$ for every transversal product, also for shorter transversals after extending them to length 37. We will show that every element of D then belongs to X . First suppose $d \in D$ sends a point outside its X -orbit. Consider the directed non-loop moves on the set of X -orbits. If three have six distinct endpoint orbits, use a three-cycle u on their sources; its d -conjugate u' is supported on their destinations. If not, endpoints of a maximal disjoint matching comprise at most four designated orbits, and every non-loop move meets one of them. Since each designated orbit has size $|X|$, only boundedly many orbits, in terms of $|X|$, are touched. On sufficiently many untouched orbits, two points in different orbits have the same X -offset under d . Multiplying d by an element of $X \leq D$, make it fix those two points. Use a three-cycle u on them and one point moved outside its orbit, and again put $u' = u^d$. In either case one transversal product T contains u, u' , and

$$z = XT_{\langle u \rangle} < y = \langle z, u' \rangle \leq XT, \quad D \cap y = X, \quad y \leq \langle D, z \rangle.$$

This contradicts (10.2).

It remains to exclude $d \in D \setminus X$ preserving each X -orbit. Some transversal five-tuple has its matching of X -translated tuples disrupted by d ; preservation of every such matching would force all offsets to be one common element of X . On this matching use

$$z = XT_{C_5} < y = XT_{A_5}.$$

Inside $\langle D, z \rangle$, take the supported five-cycles on the original matching, their d -conjugates, and all X -translates. The connected components of their support hypergraph on the five orbits have size at least ten: the original matching partitions these orbits into five-tuples, a disrupted tuple joins at least two, and the X -action is transitive on the original matching tuples. On each component the generated group is primitive. Indeed a supported prime cycle cannot permute nontrivial blocks: if it moved blocks of size at least two, it would move at least ten points. Its support must therefore be within a single block, and connectedness of the support hypergraph leaves no nontrivial block system. Jordan's prime-cycle Theorem [20, Theorem 1.1] now supplies the full alternating group on each component, independently, because the generating cycles are independently supported. It follows that $y \leq \langle D, z \rangle$, again contradicting (10.2). Thus $D = X$ or A_n , proving the normal endpoint assertion.

Pass to the quotient by $\text{core}_{G_0}(H_0)$. Every nontrivial minimal normal subgroup supplements the base. An abelian one would identify the interval, by intersection, with an invariant-subgroup lattice of an elementary abelian group, hence a modular lattice. But $[X, A_n]$ is nonmodular, as already witnessed inside the transversal product labels by the intransitive test above. If two distinct minimal normals occurred, they would commute. Using one as the supplement identifies the interval with its invariant subgroups; supplementation by the other means that the base induces all inner actions on the first. All these intersections would then be normal subproducts of its simple factors, again giving a modular lattice. The same contradictions hold for the reversed lattice. The core quotient therefore has a unique nonabelian minimal normal subgroup. $\square$

Lemma 10.3 (Product labels and the site normal test). *Suppose the realization in Theorem 10.1 exists. After restricting its parameter lattice to a sufficiently large supported $A_n \leq A_v$, coordinate-product labels form an ordinary interval*

$$[H, M], \quad H = \Lambda Q, \quad M = \Lambda S,$$

where Q is the base projection at a fixed site and $\Lambda \leq \text{Aut}(S)$ is the site-stabilizer action. They are closed under meets and joins in the original interval. Write K_X for the ordinary label indexed by a prime-generated $X \leq A_n$. For every bounded-size semiregular elementary q -group X , including 1, with $q > 5$ prime, the interval $[H, K_X]$ has property (*), has base equal to a meet of strictly intermediate coatoms, and has arbitrarily large residual height when $n/|X|$ is sufficiently large. These conclusions persist under the core and centralizer quotients whose kernels lie in the base.

Proof. A proper subdirect label cannot contain a nontrivial invariant product P : factor transitivity makes every base projection nontrivial, and its independent supports normally generate each full simple site. For $b = 1$ the assertion is automatic. A coatom therefore has proper product closure and, by maximality, equals that closure. A subgroup of A_v generated by prime-order elements is a join of atoms in parameter order; its reversed label is a meet of coatoms, hence a product. In particular a supported A_n is a product label. It can be used as the new base; we keep the name P .

Identify S with its inner automorphisms and set $I = \Lambda \cap S$. A proper coatom product projects to a nontrivial proper group T normalized by I , maximal in the local proper invariant interval. Since $TI = S$ would make $T \trianglelefteq S$, we have $TI < S$, whence $TI = T$. Intersecting these coatoms gives $I \leq Q$, since the corresponding prime-order groups generate A_n . Consequently an overgroup of H in M is determined by its intersection with S ; that intersection transports equivariantly to every site. Conversely each invariant site projection containing Q gives such an overgroup. Coordinate products are closed under intersection and generation, proving the interval assertion.

Fix X as in the statement and put $Z = K_X \cap S$. The exact invariant interval from P to its product label is an ordinary interval with endpoints

$$P \rtimes \Delta, \quad (\prod Z) \rtimes \Delta, \tag{10.4}$$

in matched-site notation; replace Δ by its finite acting image if necessary. It has reversed parameter lattice $[X, A_n]$. A K_X -normal subgroup $N \leq Z$ transports well-definedly, by site-stabilizer invariance, to a normal product at the upper endpoint of (10.4). By Lemma 10.2, unless $N \leq Q$, it supplements the lower endpoint. Thus $QN = Z$, and $HN = K_X$.

We also need actual product coatoms with meet H . On three points in distinct free X -orbits, the product of translated supported C_3 's is elementary abelian and disjoint from X . Coprimality $q \neq 3$ splits it into irreducible X -modules. Adjoining one nonzero irreducible summand to X covers X , remains strictly below A_n , and is prime-generated. The joins of all these groups contain every supported transversal three-cycle, hence generate A_n . On reversing, their product labels are strictly intermediate coatoms whose meet is H . Using disjoint triples of orbits gives arbitrarily long chains of such prime-generated product labels.

Now let $R \trianglelefteq K_X$ satisfy $R \cap Z \leq H$. Then $[R, Z] \leq H \cap Z$. Therefore R normalizes the intersection with Z of each indicated coatom. If R is not contained in one of them, it and that coatom generate K_X , so the coatom's inner intersection is normal in K_X . That intersection properly contains $H \cap Z$, but its product with H is precisely the proper coatom. This contradicts the preceding normal test for normals within Z . Hence $R \leq H$.

Every missing normal thus has a missing intersection with Z . Two missing normals commuting modulo the base core, including a single one abelian modulo the core, would transport their inner intersections to missing normal products in (10.4). They still commute modulo the transported product of the core's intersection with Z . That product is normal at the upper endpoint and lies in its base. This contradicts the unique nonabelian monolith from Lemma 10.2. All assertions of property (*) follow. Factoring a core or centralizer kernel already in the base preserves the exact intervals, the coatom meets, and these conclusions. $\square$

10.2 Elementary charts and supported normal products

The site interval need not retain every label of the original invariant interval. We therefore build charts for which all comparable labels are products, and identify the precise replacements for the cross-chart and normal-product arguments used in the earlier pruning construction.

Lemma 10.4 (Elementary charts with projective separation). *Fix a prime $q > 5$, a bounded path depth D , a maximum increment r , and finite chart dimensions and numbers. There are arbitrarily large integers n for which one can assign semiregular elementary q -subgroups of A_n to all nodes and private charts in the finite recursive construction, compatibly at their attachments. At a node W , each private chart is $W \oplus \mathbf{F}_q^N$ and extends the prescribed action of W . Two nonidentity root-coordinate translations from distinct private charts attached at that same W generate a common $\mathrm{PSL}_2(q^u)$ factor. Here u may be chosen a sufficiently large odd prime at each attachment.*

Proof. At a stage over W take a replicated regular action of

$$W \times \mathrm{PSL}_2(\mathbf{F}_{q^u}). \quad (10.5)$$

Assign different projective points e to the private charts, and embed their new coordinates as N -dimensional $\mathbf{F}_q$ -subspaces in the corresponding unipotent root groups. For distinct e, f , unipotent determinant-one lifts with coordinates x, y have product trace

$$2 + c_{ef}xy, \quad c_{ef} \neq 0.$$

Choose the embeddings so that, for every pair of nonzero chart vectors from different charts, this trace has square outside $\mathbf{F}_q$. Such a choice exists: independent random linear maps from

the fixed-dimensional chart spaces to $\mathbf{F}_{q^u}$ are injective with probability tending to one. Nonzero input vectors in different charts have independent uniform images. For each fixed nonzero c_{ef} the probability that $2 + c_{ef}xy \in \mathbf{F}_q$ tends to zero, uniformly in the chosen points. There are only finitely many tests. Since u is odd, an element of $\mathbf{F}_{q^u}$ whose square belongs to $\mathbf{F}_q$ already belongs to $\mathbf{F}_q$; thus the stated squared-trace avoidance follows.

The Dickson subgroup Theorem for PSL_2 [12, Theorem 2.2] now implies that the two translations generate the full projective group. Their distinct fixed points exclude the point-stabilizer alternative; their order q excludes the dihedral groups, whose orders are $q^u - 1$ and $q^u + 1$, and $q > 5$ excludes the small groups A_4, S_4, A_5 . Since u is odd, the square-field PGL_2 alternative cannot occur. Since both q and u are prime, the only remaining proper subfield group is $\mathrm{PSL}_2(q)$. Squared traces of determinant-one lifts in a conjugate of that group belong to $\mathbf{F}_q$, contrary to our choice. The square is unchanged by changing the lift projectively.

Choose n divisible by all orders needed for this finite recursive construction and sufficiently large for every residual-height test. At a new attachment, the given semiregular W -action and the restriction of the replicated regular action (10.5) are both disjoint unions of regular W -sets of the same total size. An equivariant bijection identifies them, extending the existing action. The actions lie in A_n : W has odd order, and the projective factor is perfect, so neither has a nontrivial sign character. No disjointness between later subtrees is needed. The recursive construction is finite, and the orders and sizes can all be accommodated by the final choice of n . $\square$

Lemma 10.5 (Linear failure patterns). *Consider a pruning round at an elementary parameter W in the charts of Lemma 10.4, with actual ancestor socles as in Lemma 5.1. Assume K_W avoids the final exclusion list relative to every strict ancestor. In a failed chart every trial extension U of the prescribed increment has a chosen proper full-stabilizer witness at some tested ancestor W' , with parameter*

$$W' \leq X(U) \leq U, \quad X(U) \not\leq W.$$

In an anchored round, the available elementary extension B has K_B previous-good and $\dim(B/W)$ greater than the trial increment. Take q above the absolute line bounds for the initial exclusions, the available dimensions sufficiently large for the prescribed finite patterns, and $n/|X|$ sufficiently large for Lemma 10.3 at every elementary parameter used.

Then the graph construction of Lemma 5.3 supplies lines and bounded independent patterns in every failed chart. For two matched patterns in distinct private charts attached at the same current W , there is one ordinary subgroup J such that every cross-pair intersection is J , with $H \leq J$, and J lies in every witness label of both patterns. Every comparable chart interval is an exact reversed linear-subspace interval, and every tested witness and previous-good anchor has property (). In the respective high-type classical and alternating configurations, this replacement for a cross-chart intersection equal to H makes the initial exclusions apply using only a bounded line or independent list.*

Proof. All intermediate subgroups between comparable elementary chart parameters are elementary and prime-generated. By Lemma 10.3, their labels are products, so these intervals are exact even after passage to the ordinary site interval. The same is true between an ancestor and a comparable trial extension, and after the indicated ancestor core or centralizer quotients. Thus full stabilizers witnessing failure have labels K_{X_i} with all prepared-link normal tests available.

Within an available extension of W , choose horizontal and vertical complements from the new root coordinates. Color graphs of linear maps by the failure's ancestor, category, bounded case data, and horizontal witness subspace. The multidimensional Hales–Jewett Theorem supplies arbitrarily many affine parameter slots with all this data fixed [17]. The fixed horizontal witness

is not contained in W . Choose a horizontal functional nonzero on its image outside W , and vary each slot by multiples of that functional. Distinct parameters give distinct witness subspaces X_i . The span of two includes all witnesses on their parameter line and adds exactly one dimension; an affinely independent list adds a dimension at every step. Upon reversal, pair intersections lie below every label on that line, are covered by the two labels, and bounded independent lists give strictly decreasing successive intersections. In particular all pairs of distinct labels on one parameter line have the same intersection E , and E lies in every label on that line.

For two private charts attached at W , write $L_0, L_1 \leq W$ for the projections of their witnesses to W . These projections are constant within their respective patterns, because only the vertical root coordinates vary. Put $G = \mathrm{PSL}_2(q^u)$ for the attachment factor. Any cross pair generates a subgroup of $(L_0 + L_1) \times G$ projecting onto G , by [Lemma 10.4](#). Its commutator subgroup lies in G and projects onto $[G, G] = G$, since W is abelian. Hence the generated subgroup is exactly $(L_0 + L_1) \times G$. It is independent of the chosen pair, contains every witness parameter in both patterns, and is prime-generated. Its ordinary reversed label J is therefore the common cross-pair intersection, and

$$H \leq J \leq K_{X_i}$$

for every witness in either pattern.

A witness need only contain the subspace of its tested ancestor and extend outside the current W ; it need not contain W . The fixed W -projections suffice for the preceding argument. We now verify the uses of J in the initial exclusions. For subset stabilizers, J contains the supported alternating groups on the at most four intersection cells, so it has boundedly many point orbits; every subset in either pattern is J -invariant. For missing equal-block partitions, supplementation still gives H its rich actions on both block lists. Thus H has an absolute bound on orbits of ordered pairs of intersection cells. The supported alternating groups within those cells lie in J ; a cell-pair stabilizer therefore has an absolute bound on point-pair orbits, including coincident cells and cells of size at most two. Hence J has boundedly many point-pair orbits and boundedly many invariant equivalences. This gives the contradiction in [Proposition 7.3](#); the retained-block cases use the original normal tests. In the classical small-space cases, the supported large group on the common complement lies in J and preserves every structure in both patterns. The bounded-span and linear-constraint arguments of [Propositions 6.4](#) and [6.5](#) therefore apply to a bounded independent list.

The remaining initial classical cases need only a parameter line. For a descending list of distinct subspaces or signed flags, the intersection of the first two stabilizers fails to preserve the third, by the elementary maps in [Proposition 6.4](#). This contradicts $E \leq K_{X_3}$. For singular spaces, first discard the at most two exceptional incomparable maximal spaces, then choose three members of the remaining chain. For mutually perpendicular retained large spaces or orthogonal idempotents, choose four distinct members. After the first two structures are imposed, the third cannot exhaust the unsplit remainder because the fourth nonzero space remains. The supported classical or special linear group on that remainder gives the same strict third decrease. Thus a fixed number of labels on a line replaces every height-sized initial comparison. $\square$

We next recover the independently supported groups needed when a normal simple power is missing. We cannot assume that every intermediate intersection with that power is a coordinate product: the ordinary site interval may omit labels of the full invariant interval. The structural exclusions need only a previous-good common intersection containing enough independently supported normal subgroups. We find it by bounding the codimension of product closures and then selecting compatible affine parameters. The selection uses the following finite-vector-space observation.

Lemma 10.6 (Simultaneous affine kernels). *Fix a finite field $\mathbf{F}_q$ and integers $h, m \geq 0$. For sufficiently large d , any family of linear maps $\phi_i : \mathbf{F}_q^d \rightarrow \mathbf{F}_q^h$, indexed by $i \in \mathbf{F}_q^d$, has an affine m -space A such that*

$$\phi_i(j - i) = 0 \quad (i, j \in A).$$

Proof. Color every two-dimensional subspace, with its canonical reduced row-echelon basis (b_1, b_2) , by the finite table

$$(\phi_{ab_1+bb_2}(cb_1+db_2))_{a,b,c,d \in \mathbf{F}_q}.$$

The number of colors depends only on q, h . The finite-vector-space Ramsey Theorem gives a homogeneous subspace of dimension $m + 2$ [14, 15]. Let $e_1, e_2, \dots, e_{m+2}$ be its canonical basis, and put

$$A = e_1 + \langle e_3, \dots, e_{m+2} \rangle.$$

Reduced row-echelon coordinates are compatible with this passage to a subspace: multiplying a reduced row-echelon matrix in these coordinates by the reduced row-echelon matrix of the containing space again gives reduced row-echelon rows in ambient coordinates. For $i \in A$ and $w \in \langle e_3, \dots, e_{m+2} \rangle$, both (i, e_2) and $(i, e_2 + w)$ are such canonical plane bases. Their colors agree, so $\phi_i(e_2) = \phi_i(e_2 + w)$. Linearity gives $\phi_i(w) = 0$, which proves the claim. $\square$

Lemma 10.7 (Supported products in a selected pattern). *Let $X_i \leq B$ be the elementary parameters of a failure pattern in Lemma 10.5, with K_B a previous-good common anchor. Suppose the witness K_{X_i} has a missing normal simple power Y_i , possibly modulo a retained normal subgroup, and H is transitive on its simple sites with an absolute bound on its ordered-pair orbit count. After selecting a line, or an affine subpattern of any prescribed positive dimension, its common intersection label E has property $(*)$, contains K_B and hence remains previous-good, and, for each i , contains an E -normal product*

$$N_i = \prod_a R_{i,a} \leq E \cap Y_i$$

of independently supported proper subgroups, not contained in H . Its image in $E/\text{core}_E(H)$ contains the monolith. The supported factors are nonsoluble, and the same statement with preimages applies when retained centers have been factored out.

Proof. Write $Y = Y_i$ temporarily, so $HY = K_{X_i}$. Every intermediate overgroup is H times its intersection with Y . Take the coordinate-product closure of this intersection and multiply by H . For $X_i \leq Z \leq B$ this is an overgroup $K_{g_i(Z)}$, where

$$X_i \leq g_i(Z) \leq Z,$$

and g_i is monotone. The resulting group's intersection with Y is the product closure itself. Therefore $g_i(Z) = X_i$ forces the intersection at K_Z to be subdirect in Y . All intermediate intersections from K_Z to K_{X_i} are then subdirect. Their tie relations are unions of H -orbits on ordered pairs of sites; a strict inclusion forces strict refinement of ties, because comparable subdirects with the same ties coincide. The pair-orbit bound gives an absolute h with

$$g_i(Z) = X_i \implies \dim(Z/X_i) \leq h.$$

Choose a complement C of $g_i(B)/X_i$ in B/X_i . Monotonicity gives $g_i(C) \leq g_i(B) \cap C = X_i$ (using the preimage of C in B in this notation), so

$$\text{codim}_B g_i(B) \leq h. \tag{10.6}$$

Exactness between comparable elementary labels ensures that every closure label used here really is indexed by the indicated subspace. The same calculation can be made modulo a retained normal subgroup; taking its preimages does not alter the intervals.

In the graph pattern, X_j differs from X_i in the single vertical direction determined linearly by the parameter difference $j - i$. The preimage of $g_i(B)$ in these vertical directions has codimension at most h , by (10.6). Choose a map ϕ_i to $\mathbf{F}_q^h$ with that preimage as kernel. Apply Lemma 10.6, asking beforehand for enough Hales–Jewett slots. In the resulting affine subpattern, $X_j \leq g_i(B)$ for every i, j . Let U be the span of its witnesses and put $E = K_U$. Then $U \leq g_i(B)$ for all i , and consequently

$$H < K_{g_i(B)} \leq E < K_{X_i}.$$

Here $K_{g_i(B)} > H$ because $g_i(B) \leq B < A_n$. The label E is above K_B , so it inherits all previous-good exclusions, and Lemma 10.3 gives property $(*)$ for $[H, E]$.

The intersection $K_{g_i(B)} \cap Y_i$ is a coordinate product not contained in the base; otherwise supplementation would give $K_{g_i(B)} = H$. Each of its single-site components already lies independently in $E \cap Y_i$. Let $R_{i,a}$ be the full subgroup of $E \cap Y_i$ supported on site a and put $N_i = \prod_a R_{i,a}$. Conjugation by E permutes these support subgroups, so $N_i \trianglelefteq E$, and $N_i \not\leq H$. Site transitivity and $E < K_{X_i}$ imply that each factor is proper: one full simple factor would force all of them and then $E = K_{X_i}$. The factors are mutually conjugate under H , and if they were soluble, N_i would be a soluble missing normal, contrary to property $(*)$. Thus they are nonsoluble. The image of N_i contains the common nonabelian monolith of $E/\text{core}_E(H)$, again by property $(*)$.

We spell out the scope needed in the structure exclusions. For direct systems calculated modulo block centers, each block carries at most two simple sites, and the at-most-two block classes have independent alternating top actions induced already by H through supplementation; in linear type there is one list. These give the required absolute pair-orbit bound. In the case of several site orbits one instead uses the supports of a retained orbit as in Lemma 6.10. Lifting independent factors on a block to its natural supported matrix group gives projective image in E , since the lost centers were retained. The lift is preserved by the semilinear block stabilizer in E^+ and supplies the groups of order greater than two required in the block-module argument. If needed take products of the supported factors over one block.

For tensor and Cartesian systems the Y_i already have independent simple supports. The common-monolith and iterated-commutator arguments of Lemmas 6.28 and 7.7 apply to these E -normal products N_i , without requiring $N_i = E \cap Y_i$. Their tuple intersections still generate a nontrivial E -normalized subgroup. It spans in the tensor case, and is transitive in the Cartesian case, by the previous-good exclusions inherited from K_B . In the diagonal primitive case choose a minimal E -normal subgroup inside the nontrivial product of proper supported factors. This is exactly the containment needed by that argument. Retained centers, when present, are normal in E and lie in H , so these conclusions survive their removal and reinstatement. $\square$

Proof of Theorem 10.1. Prepare the charts in Lemma 10.4, and use Lemma 10.3 for every elementary parameter encountered. At every node the current almost simple quotient is the one from Lemma 5.1; initially it is M . The normal, residual-height, and exact comparable-link hypotheses have all been verified. Lemma 10.5 supplies the line, independent-list, and cross-chart configurations for the pruning rounds relative to every ancestor. Lemma 10.7 replaces the interval-product consequence exactly at its uses for missing direct, tensor, Cartesian, and diagonal systems. Thus each finite round either loses only the bounded number of charts allowed by the matched failure arguments or produces the next actual almost simple socle, with all ancestor-relative exclusions. These are the structural conclusions of Theorems 6.30 and 7.11.

If a parent has bounded type, its bounded-order case is excluded by residual height. In bounded Lie rank, the bounded-equation pattern argument of [Section 8](#) applies unchanged with a bounded independent list: strict subgroup descents are strict after intersection with the socle, whereas the spaces of equations have bounded dimension. Thus one obtains a sandwich throughout the interval over a selected increment t . Its height is t and the same divisor count gives at most $C2^t$ labels. The exact t -dimensional vector-subspace interval has at least $q^{\lfloor t^2/4 \rfloor}$ labels, a contradiction for sufficiently large fixed t . The sufficiently-general input is used here in precisely the scope established in [Section 8](#) [25].

For a high-type path, use linear flags in the final protection selection of [Proposition 9.12](#). All bad containing labels belong to the exact subspace intervals just proved. The same exponential-gap estimate for flag proportions applies; hence the closure-field tensor and root protections hold simultaneously for the required ancestor comparisons. The absolute group-theoretic bound [Theorem 9.21](#) on the length of such a protected path then contradicts a choice of depth D larger than that bound.

There is no dependence of the earlier finite choices on n . First fix the high thresholds and the protected path bound, and choose D beyond it. Next choose a prime q above every absolute line bound, then r for the bounded-type count and protection estimates. Choose trial dimensions backwards through the finite pruning rounds, allowing at each round all Hales–Jewett and [Lemma 10.6](#) slot requirements for the trial and bounded path ranks. The latter requirements depend only on q , desired fixed dimensions, and absolute orbit bounds. The initial comparisons require only a line or a bounded independent list, as proved in [Lemma 10.5](#); none requires comparison with $|A_n|$. Choose N and the number of private charts large enough for the finite color and chart-loss bounds. Only then choose the fields of (10.5) and finally n . All elementary groups X used are now bounded in size, so $n/|X|$ can exceed all normal-test and residual-height requirements while n accommodates every replicated regular action. This gives one fixed forbidden degree n . For every $v \geq n$, the supported A_n restriction of [Lemma 10.3](#) gives the same contradiction. $\square$

11 The forward marker theorem

This section concerns pure markers. Its target is a product-interval statement forcing a bounded prefix and many actual point stabilizers in an ordinary site group. The individual group-type arguments that follow are exclusions under the assumption that this point configuration is absent; their final contradiction again uses the protected path bound. The normal and mixed-projection tests below will also apply internally when the simple sites change.

Definition 11.1 (Pure marker lattice). For a finite set T of t markers, let F_T have labels

$$(I, V), \quad I \subseteq T, \quad V \leq S_{T \setminus I}.$$

Choose a set Z disjoint from T , with $|Z| = t + 1$ or larger, and interpret this label as the subgroup

$$S_{Z \cup I} \times V \leq S_{Z \cup T}.$$

Order, meet, and join are subgroup containment, intersection, and generation in this model. The markers in I are called absorbed. We write

$$X_I = (I, 1), \quad x_i = X_{\{i\}}, \quad M_D = X_{T \setminus D}, \quad C_D = (T \setminus D, S_D) \quad (D \neq \emptyset).$$

The bottom and top are $X_\emptyset$ and X_T . We also write F_t when only the cardinality matters.

Lemma 11.2 (Elementary marker properties). *The labels in Definition 11.1 are closed under meet and join. The coatoms are exactly the C_D , $\emptyset \neq D \subseteq T$. The axis filter $[X_I, 1]$ is $F_{T \setminus I}$, and the axis ideal $[0, X_I]$ is F_I . Moreover*

$$\text{ht}(F_t) = O(t \log \log(t + 3)).$$

Proof. The independent symmetric group on the absorbed part makes the intersection again a group with independent full symmetric action on its new absorbed part. In a join, that part expands exactly along the generated orbits meeting it: symmetric transpositions there and their conjugates give the full symmetric group on the expanded part, independently of the induced group on the remaining markers. Thus both operations stay within the indicated labels. These rules depend only on the marker labels (I, V) , so the resulting lattice is independent of the choice of Z with $|Z| > t$.

If a proper label has unabsorbed set D , it lies below C_D . The latter is the full stabilizer of the partition into $Z \cup (T \setminus D)$ and D . These parts have unequal sizes, the first larger than $t \geq |D|$; adjoining any element failing to preserve them generates the full symmetric group by the same absorption argument. Hence C_D is a coatom, proving the classification. The descriptions of the axis filter and ideal follow directly by treating $Z \cup I$ as the fixed absorbed set, or by restricting to $Z \cup I$, respectively.

For the height use $|Z| = t + 1$, embedding F_t in the subgroup lattice of S_{2t+1} . Every strict step in a subgroup chain consumes at least one prime factor, with multiplicity, of $(2t + 1)!$. Writing $n = 2t + 1$, the total number of such factors is

$$\sum_{p \leq n} \sum_{a \geq 1} \left\lfloor \frac{n}{p^a} \right\rfloor \leq n \sum_{p \leq n} \frac{1}{p-1} = O(n \log \log(n + 2)),$$

by the reciprocal-prime bound (the difference between $1/(p-1)$ and $1/p$ is summable). This proves the estimate. $\square$

Lemma 11.3 (Marker normal test). *For sufficiently large t , every exact finite group interval isomorphic to F_T or F_T^{op} has property $(*)$.*

Proof. Let $\omega = (I, V)$ label the base times a normal subgroup. As in (10.1), it and all its projections to subintervals satisfy the left-modular law in either orientation. Indeed in actual subgroup order, with base H_0 , normal $N \trianglelefteq G_0$, and $H_0 \leq D \leq E \leq G_0$, the identity is

$$(DN) \cap E = D(N \cap E) = D \vee ((H_0 N) \cap E).$$

Projection to $[D, E]$ gives $D(N \cap E)$, with $N \cap E \leq E$. Self-duality of the left-modular identity gives the reversed statement as well. In particular the configuration

$$\omega \wedge b \leq a < b \leq \omega \vee a \tag{11.1}$$

is impossible, also after any projection.

Suppose I is nonempty and proper, with $|I| \geq 2$. Choose four positions P_4 in cyclic order such that membership in I is nonconstant, occurs in both parity classes, and is not invariant under rotation by one. Such a choice is possible for large t : two chosen positions of I can be adjacent, and at least one of the other positions lies outside I ; if the complement has only one point, use three from I . For the rotation s put

$$a = (T \setminus P_4, \langle s^2 \rangle), \quad b = (T \setminus P_4, \langle s \rangle).$$

The I -membership condition forces $\omega \wedge b \leq a$. Each parity orbit of s^2 meets the absorbed part of ω , so $\omega \vee a = 1$. Thus (11.1) holds.

If $I = \{j\}$ and V moves some marker k , choose P_4 so that k has opposite parity from j and its image under some element of V lies outside P_4 . That image is already absorbed in a ; the two parity classes are therefore absorbed in $\omega \vee a$. Again the same rotation gives (11.1). If $V = 1$, use three positions containing j , with $C_3 < S_3$ in place of the two rotation groups and absorb the outside. The join with the C_3 absorbs all three positions, while the meet with the S_3 lies in C_3 . Thus nonempty proper I is impossible in all cases.

Now let $I = \emptyset$ and suppose V moves j . Project above x_j . In the marker description of that axis filter, $\omega \vee x_j$ has nonempty absorbed set because the V -orbit of j contains another point. The preceding test makes it the top. Consequently V is transitive on T . Every marker atom x_j complements ω .

First take the forward orientation. Complementation of a base-normal multiple gives the interval isomorphisms

$$[0, x_j] \longrightarrow [\omega, 1], \quad u \longmapsto u \vee \omega.$$

As x_j is an atom, ω is a coatom; since its absorbed set is empty, Lemma 11.2 gives $\omega = C_T$. Factor by the base core. Choose a minimal missing normal subgroup inside the putative normal subgroup; it still has a nonbottom proper normal-multiple label, and the same argument identifies that label with C_T . Write this minimal normal as N and the base as H . At a transposition label

$$0 < d = (\emptyset, \langle(i\ k)\rangle) < C_T,$$

the intersection $N \cap K_d$ is a nontrivial proper subgroup of N , and $H(N \cap K_d) = K_d$. For every j the marker operations give

$$C_T \wedge (d \vee x_j) = d.$$

Because N is normal in the whole upper endpoint and lies in K_{C_T} , this equality shows that K_{x_j} normalizes $N \cap K_d$. The groups K_{x_j} generate the upper endpoint. Thus $N \cap K_d$ is a nontrivial proper normal subgroup of N normal in that endpoint, contradicting minimality.

In the reverse orientation the normal projection instead yields isomorphisms, in the displayed F_T order,

$$[x_j, 1] \longrightarrow [0, \omega], \quad u \longmapsto u \wedge \omega.$$

For $k \neq j$ the label $X_{\{j,k\}}$ is strictly above x_j ; therefore $X_{\{j,k\}} \wedge \omega > 0$. This meet is nontrivial only if V contains the transposition $(j\ k)$. Hence $V = S_T$ and $\omega = C_T$. But a supported three-cycle group containing j cannot be in the image of the indicated isomorphism. Any label above x_j whose meet with C_T contains that cycle must absorb all three positions; its meet then contains the full S_3 , rather than just the cycle group. This contradicts surjectivity.

We have proved the normal endpoint assertion. The lattice F_T is nonmodular for large t : it contains the full subgroup lattice of S_T , or one may use the rotation obstruction just exhibited. In the faithful base-coset action after factoring the core, every minimal normal subgroup is transitive because it supplements the base. An abelian minimal normal identifies the interval with an invariant-subspace lattice and makes it modular. Two distinct nonabelian minimal normals commute; supplementation by one makes the base induce all inner actions on the other, so intersections there are normal subproducts of simple factors, again forming a modular lattice. These contradictions give the unique nonabelian monolith required by property (*). $\square$

Lemma 11.4 (Exclusion of a mixed marker base). *For sufficiently large t , suppose that*

$$F_T \cong [P, S^b]^\Delta$$

in forward order, with S nonabelian simple and Δ transitive on its factors. If c denotes coordinate-product closure, then

$$P < c(P) < S^b$$

is impossible.

Proof. Assume the mixed case and let $\omega = (I, V)$ label $c(P)$. The projections of P are nontrivial and proper at every site, by factor transitivity. A label joining with ω to the top must be subdirect: adjoining $c(P)$ does not change its site projections. For two distinct comparable subdirect labels, their intersections with ω are distinct. Indeed their tied coordinates and restricted twisting maps are realized using the nontrivial full projections of P to the factors of $c(P)$; a strict refinement of ties gives a strict change of the intersection. Conversely, comparable subdirect groups with unchanged ties and twists coincide.

Apply the rotations in the proof of [Lemma 11.3](#). They give $\omega \wedge b \leq a < b$ with both a and b joining ω to the top. Hence both are subdirect, whereas

$$a \wedge \omega = b \wedge \omega,$$

contradicting the preceding strictness. This eliminates nonempty proper I , including the singleton cases of that proof. The top absorbed set is impossible because ω is proper.

If $I = \emptyset$ and V moves j , consider $c(x_j)$. It contains $\omega \vee x_j$. In the axis filter above x_j , apply the nonempty-absorption test just used if the product closure is still a proper mixed label. If x_j itself is already a product, its closure equals it; this cannot happen because it contains $\omega \vee x_j > x_j$. Thus $c(x_j)$ must be the top. So x_j is subdirect. Its join with ω is then the top: ω has independent nontrivial supports, whose conjugates under a subdirect group generate each simple site by simplicity. In parameter terms this makes V transitive. All x_j are therefore subdirect and have $x_j \wedge \omega = 0$. Strictness for the comparable subdirect labels $x_j < X_{\{j,k\}}$ gives $X_{\{j,k\}} \wedge \omega > 0$; as before, V contains every transposition, and so $V = S_T$.

Each meet $x_j \wedge \omega = 0$ is the same subgroup P . Write $c(P) = \prod_o T_o$ over the simple sites, where every T_o is nontrivial. On a tied pair of sites, the two-coordinate projection of P is the graph of the identifying isomorphism restricted to T_o ; on an untied pair it is the whole product. Thus P determines the ties and the restrictions of their twisting maps to the T_o , although it need not determine the full maps on S . Every invariant overgroup of a subdirect label is obtained by an operator-invariant refinement of its ties, retaining the twists on the remaining parts. Its intersection with $c(P)$ depends only on that refinement and the restricted twists. Consequently meeting with ω has the same image on every filter above x_j .

This is false for $\omega = C_T$. Fix a triple of markers. If j lies outside it, joining its supported three-cycle label with x_j and then meeting C_T recovers exactly that three-cycle label. If j lies inside the triple, any label above x_j whose meet with C_T contains the cycle must absorb all three positions; its meet then contains the full S_3 . It cannot yield just the three-cycle subgroup. A triple and an outside marker give the contradiction. $\square$

Lemma 11.5 (Ordinary site interval for product markers). *Suppose $F_T = [P, S^b]^\Delta$ exactly in forward order, with Δ transitive on the simple factors and P a coordinate product satisfying $P \neq 1$ or $b = 1$. For sufficiently large t , all coatoms and all axis labels are products. At one site, let Q be the projection of P and let $\Lambda \leq \text{Aut}(S)$ be the image of the site stabilizer. Then $\Lambda \cap S \leq Q$, and the product labels form the ordinary interval*

$$[H, S\Lambda], \quad H = \Lambda Q.$$

This correspondence preserves meets and joins in the full invariant interval. We denote its subgroup at a product label u by K_u .

Proof. A proper subgroup projecting fully at every simple site cannot contain the independent nontrivial base projections: their conjugates normally generate the full simple sites. At $b = 1$ this is automatic. Hence each coatom has proper product closure and equals that closure by maximality. For a proper coatom let T_1 be its projection at the fixed site. It is nontrivial and proper. The group $T_1(\Lambda \cap S)$ is still invariant and transports to every site. It is proper, since equality to S would imply $T_1 \trianglelefteq S$. Maximality consequently gives $T_1(\Lambda \cap S) = T_1$. The singleton-omission coatoms $C_{\{i\}}$ intersect to the base, so their site projections give $\Lambda \cap S \leq Q$. The same intersections over $i \notin I$ give X_I ; thus every axis label is a product.

An invariant site projection containing Q transports uniquely to a Δ -invariant coordinate product. As $\Lambda \cap S \leq Q$, these projections are exactly the intersections with S of the overgroups of H in SA . An overgroup is recovered by multiplying its intersection with S by H . Finally, coordinate products are closed under intersection and generation in the original simple power, proving the asserted compatibility. $\square$

Theorem 11.6 (Forward marker theorem). *For every nonnegative integer m , there are integers $a(m)$ and $t_0(m)$ such that the following holds whenever $|T| \geq t_0(m)$. Every exact forward realization*

$$F_T = [P, S^b]^\Delta$$

with S nonabelian simple, Δ transitive on the factors, and invariant coordinate product $P \neq 1$ (or $b = 1$), has the following configuration in the ordinary site interval of Lemma 11.5. There is $A \subset T$ with $|A| \leq a(m)$ such that K_{M_A} has an actual high-degree alternating simple normal subgroup $B_A \leq S$ satisfying

$$HB_A = K_{M_A}, \quad C_{K_{M_A}}(B_A) \leq H, \quad C_{K_{M_A}}(B_A) \text{ is soluble.}$$

In the almost simple quotient $K_{M_A}/C_{K_{M_A}}(B_A)$, at least m distinct labels $M_{A \cup \{i\}}$, with fresh indices $i \in T \setminus A$, are full stabilizers of individual points in the natural alternating action. The constants $a(m)$ and $t_0(m)$ are independent of S, b, Δ, P and the realization; the high-degree threshold is the fixed one used in the structural exclusions.

The proof of Theorem 11.6 occupies the rest of this section and Section 12. Its initial site reduction is Lemma 11.5. The normal test and mixed-base exclusion apply whenever new simple sites occur internally. The bounded-type step uses exactly the sufficiently-general bounded-complexity input established in Section 8; the high-type steps retain the protected root and tensor conditions of Section 9.

11.1 Pure links and simultaneous pruning

We prove Theorem 11.6 by successive exclusions, always stopping if the required point stabilizers have already appeared. The present subsection records the transfer and selection facts that allow every exclusion to be made simultaneously below each bounded path prefix. All labels refer to the original marker set of Definition 11.1; ordinary labels are those represented by products at the original simple sites.

Lemma 11.7 (Exclusion of bounded types). *In the product configuration of Theorem 11.6, the ambient simple group has unbounded order and unbounded classical rank or alternating degree as the*

required marker reservoir grows. More precisely, any prescribed bounds on order and on Lie rank can be excluded by choosing the reservoir sufficiently large. This assertion also holds in every exact pure-marker filter obtained by the changes of simple sites below.

Proof. Bounded order is immediately impossible by the length of an axis chain. In bounded Lie rank use the argument of [Theorem 8.2](#), [Lemma 8.4](#), and [Proposition 8.6](#). For a prescribed sufficiently large constant k test M_U , $|U| = k$. There must be some for which all ordinary labels above it admit the sandwich on intersection with S : otherwise pick many disjoint such sets and take proper bounded-equation stabilizer witnesses as there. Each has omitted (nonabsorbed) set nonempty inside its tested set, hence intersections strictly descend (also ordinary products, strict on intersection with S). The linear spaces of equations cannot give such long descent. Now residual lengths under M_U are arbitrarily large and the same height and order comparison for the sandwich maps bounds the number of ordinary labels above it by $C2^h$, $h = O(k \log \log(k+3))$. But all partitions of U occur via intersections of cuts C_D (individual symmetric group on every part), giving at least $\exp((1/2 + o(1))k \log k)$ for example by pairings. This is impossible. In particular we can require arbitrarily high thresholds on the remaining types here or at any use with a large enough marker set. □

Convention 11.8 (Relative labels at an ancestor). At a path prefix A , suppose that we have the prepared-path data of [Definition 9.1](#): an actual simple subgroup

$$B_A \trianglelefteq K_{M_A}, \quad B_A \leq S, \quad HB_A = K_{M_A},$$

whose centralizer in K_{M_A} is soluble and contained in H . Initially $A = \emptyset$. We work in its almost simple quotient, and write $T' = T \setminus A$ for the relative marker set. Thus the relative top is M_A , even when a displayed relative label uses only T' . A proper full stabilizer containing a tested axis with a bounded number of further omissions has a product label

$$r_D = (T' \setminus D, V_D), \quad g_D = (\emptyset, V_D) = r_D \wedge X_D, \quad \emptyset \neq D \subseteq T', \quad V_D \leq S_D. \quad (11.2)$$

Here D is a subset of those further omissions, and V_D fixes every position outside D in the expression for g_D . A normal subgroup is *retained* at this link if it is contained in K_{g_D} ; otherwise it is *missing*.

We will use exact pure-marker links before passing to the ordinary site interval. These are different intervals: the ordinary one corresponds exactly to the coordinate-product labels in the full formal link, and need not contain every formal label. Normal subgroups can nevertheless be tested in the full link by transporting them across the original simple sites. We first prove this normal-subgroup transfer, then use it to find independently supported groups inside intersections.

Lemma 11.9 (Normal transfer on pure links). *For the labels in [Equation \(11.2\)](#), the full formal interval $[g_D, r_D]$ is $F_{T' \setminus D}$. If this residual set is sufficiently large, its ordinary group interval $[K_{g_D}, K_{r_D}]$ has the normal property of [Lemma 11.3](#), including its nonabelian monolith modulo the base core. For pairwise disjoint supports D_i , put*

$$J = \bigwedge_i r_{D_i}, \quad g_* = \bigvee_i g_{D_i}, \quad D_* = \bigcup_i D_i.$$

The same conclusion holds for $[K_{g_}, K_J]$, and for distinct i, j one has $g_{D_i} \leq r_{D_j}$ and $r_{D_i} \vee r_{D_j} = M_A$.*

Proof. The full formal link $[g_D, r_D]$ is $F_{T' \setminus D}$: every intermediate increase keeps V_D independent. For disjoint supports the label calculation gives $g_{D_i} \leq r_{D_j}$ and $r_{D_i} \vee r_{D_j} = M_A$. The link $[g_*, J]$ is likewise pure on $T' \setminus D_*$, with independent side group $\prod_i V_{D_i}$. We prove the normal assertion for either pair of endpoints, denoted by g, r .

Work before any ancestor-centralizer quotient, and let S denote the original simple site. Put

$$L = K_g, \quad R = K_r, \quad Q = L \cap S, \quad Z = R \cap S.$$

Let P_g, P_r be the corresponding products at the original sites, with fixed-site projections Q, Z . They have these projections because $\Lambda \cap S$ lies in the original base. With Δ replaced by its finite acting image if necessary, the full formal link has the exact ordinary realization

$$[P_g \rtimes \Delta, P_r \rtimes \Delta].$$

Indeed its intermediates all contain Δ , and hence correspond to precisely the Δ -invariant subgroups between the two products. The normal test applies to this full interval.

First let $N \trianglelefteq R$ lie in Z . Since $\Lambda \leq L \leq R$, its transport to the other sites is well-defined. The resulting product is normal in $P_r \rtimes \Delta$. By [Lemma 11.3](#), either $N \leq Q$ or $QN = Z$; in the second case $LN = R$. Thus every missing normal inside the original site supplements the ordinary lower endpoint.

Now let $N \trianglelefteq R$ satisfy $N \cap Z \leq L$. Then $[N, Z] \leq Q$, so N normalizes the intersection with S of each singleton-omission axis coatom of the link. These coatoms are products, are strictly intermediate, and have meet L . If N escaped one of them, adjoining it to that coatom would generate R ; the coatom's inner intersection would therefore be normal in R . It strictly contains Q , but its product with L is only the proper coatom, contradicting the preceding paragraph. Hence $N \leq L$. Together the two paragraphs prove the normal endpoint assertion for every normal subgroup of R .

Finally put $C = \text{core}_R(L)$. Two missing normals commuting modulo C , or one missing normal abelian modulo C , would transport their intersections with Z to missing normal products in the exact semidirect interval. Their commutators lie in the transport of $C \cap Z$, a normal product contained in its base. This contradicts the unique nonabelian monolith supplied there by [Lemma 11.3](#). The ordinary interval therefore has property $(*)$. Quotienting an ancestor centralizer preserves it, since that kernel already lies in L . □

The lower endpoint for retention is thus K_{g_D} on the link indexed by D , and K_{g_*} on a simultaneous intersection link. We keep these endpoints when applying the subsequent normal-product arguments.

Lemma 11.10 (The disjoint-support join restriction). *For disjoint nonempty D_1, D_2 and $J = r_{D_1} \wedge r_{D_2}$, every formal label u satisfies*

$$J \leq u < r_{D_1} \implies u \vee r_{D_2} < M_A. \quad (11.3)$$

Proof. Indeed, all positions outside $D_1 \cup D_2$ are already absorbed at u . The part of D_2 still not absorbed there stays separate also from D_1 since $u \leq r_{D_1}$, and is preserved by V_{D_2} since $J \leq u$; if this part were empty one would already have r_{D_1} . □

Lemma 11.11 (Independent support in intersections). *In the setting of [Lemma 11.9](#), suppose that $Y \trianglelefteq K_{r_D}$, with $Y \leq B_A$, and that a retained normal subgroup $Z \leq Y \cap K_{g_D}$ makes Y/Z a*

missing power of nonabelian simple groups. Assume that K_{g_D} acts transitively on its simple sites with a bounded number of orbits on ordered pairs. At a sufficiently long residual link, singleton axis intersections with this power are products and their simple sites have arbitrarily high type. Moreover, in $E = K_J$, the product of the individual simple-site intersections of $(E \cap Y)/Z$ is E -normal and missing relative to K_{g_*} . It has nonsoluble factors and its image contains the nonabelian monolith modulo the base core of E . If $J < r_D$, these singly supported factors are proper.

Proof. There are two sets of coordinates in this argument: the original b simple sites, and the simple factors of Y/Z inside each of them. We first obtain an exact interval at the new sites, then show that its axis groups contain independent supports.

Lift Y, Z inside the actual $B_A \leq S$, before any ancestor quotient, and transport them across the original sites. Since $K_{g_D}Y = K_{r_D}$, we also have

$$(K_{g_D} \cap S)Y = K_{r_D} \cap S.$$

Thus the transported product of copies of Y supplements the full lower endpoint, while the transported product of copies of Z is retained. Every intermediate subgroup is its lower endpoint times its intersection with that normal product. After factoring the retained product, $[g_D, r_D]$ is therefore an exact invariant-subgroup interval in $(Y/Z)^b$. The operators are the original action together with conjugation by the label at g_D ; they act transitively on the new simple sites.

Consider a singleton axis $g_D \vee X_{\{i\}}$, with i outside the used supports. Its intersection with the new power remains a product across the original b sites. It has nontrivial projections at the new sites, by strict increase from the lower endpoint and transitivity. These projections cannot be full. Otherwise every subgroup on a long axis chain above it would be subdirect. Strict axis increases give strict intersection increases by supplementation; inside an original site these would require strict refinements of K_{g_D} -invariant tie partitions. The bounded number of orbits on ordered pairs bounds the length of such a chain, as in Lemma 6.7, whereas the residual marker set is arbitrarily large.

The filter above this singleton axis is again an exact pure-marker interval. Its bottom has nontrivial proper projections to the new sites, so Lemma 11.4 makes it a coordinate product there. Lemma 11.5 then makes all subsequent axes products at those sites as well. Applying Lemma 11.7 to this exact filter forces its new site simple to have arbitrarily high type. This uses the bounded-type exclusion, not the conclusion of the forward theorem being proved.

Now put $E = K_J$. The axis

$$g_D \vee X_{T' \setminus D_*} \leq J$$

supplies independent new-site supports in E , modulo Z . Inside one original site take the product of all individual simple-factor intersections of $(E \cap Y)/Z$. This product is E -normal. It is missing relative to K_{g_*}/Z : the axis intersection just obtained is a product contained in it, and supplementation recovers that axis label by multiplying its lift by K_{g_D} . That axis is not below g_* .

By Lemma 11.9 on $[K_{g_*}, K_J]$, the supported normal product has image containing the nonabelian monolith modulo the base core. It is nonsoluble, and transitivity makes every supported factor nonsoluble. If $J < r_D$, each factor is proper: a full one would, by the transitive K_{g_D} -action and supplementation, give $E = K_{r_D}$. This proves precisely the supported-product conclusion needed below; it does not identify that product with all of $(E \cap Y)/Z$. In direct-system applications the block centers are retained, so lifting these supports with their centers gives the required actual block-supported matrices.

□

Remark 11.12. When reasoning in the almost simple quotient at A we generally keep ordinary subgroup labels (including H) for their images as well. Overgroups there containing the base

correspond without change of order to ordinary labels below K_{M_A} , since the kernel was already in the base. The image of B_A still identifies with the *actual* B_A by injectivity. Thus even if normal-product transport is used while reasoning in this quotient, the inner groups transported can be taken back inside that actual simple group first. Ordinary labels throughout are products at the original sites; their being a sublattice, rather than necessarily the full displayed label lattice, matters for cover/size arguments. Within the ideal every relative cut C_D is product (global cut intersected with M_A); in particular a maximal ordinary proper label has to be a cut, by containment.

Lemma 11.13 (When a product cover is a full cover). *Let Δ act on a finite direct product by coordinate-permuting automorphisms. Let $X = \prod_i X_i < Y = \prod_i Y_i$ be invariant coordinate products, with no invariant coordinate product strictly between them. If an invariant subgroup U satisfies $X < U < Y$, then $X_i \trianglelefteq Y_i$ at every site. Consequently, nonnormality at one site makes $[X, Y]^\Delta$ a cover in the full invariant-subgroup interval as well.*

Proof. The coordinate closure of U is Y . Conjugating the independent copies of X_i inside U , whose projections are all of Y_i , therefore gives

$$N = \prod_i \langle X_i^{Y_i} \rangle \leq U.$$

This is an invariant coordinate product with $X \leq N < Y$, so the product-cover hypothesis forces $N = X$. Thus every X_i is normal in Y_i . $\square$

We apply this lemma before taking ancestor-centralizer quotients. An ordinary cover is a cover among the original-site product labels. If a formal intermediate exists, the lemma forces normality of the lower site projection in the upper one. This normality persists on passing to an ancestor quotient and intersecting with an actual supported subgroup there. The concrete nonnormality tests below can therefore detect a formal intermediate even when that intermediate is not itself a product label.

Lemma 11.14 (The three cuts above a disjoint meet). *If D, D' are nonempty and disjoint, the only coatoms above $C_D \wedge C_{D'}$ in the relative marker lattice are*

$$C_D, \quad C_{D'}, \quad C_{D \cup D'}.$$

Each of the first two covers the meet, whereas the interval up to the third is the Young subgroup interval $[S_D \times S_{D'}, S_{D \cup D'}]$. These three cuts and the first two cover relations persist in the ordinary product sublattice.

Proof. In particular, in the relative label lattice, the meet of cuts on *disjoint nonempty* D, D' absorbs everything outside $D \cup D'$, with side group $S_D \times S_{D'}$. The only nonempty sets available for coatom cuts above that meet are $D, D', D \cup D'$ (unions of the unabsorbed orbits). The first two coatoms cover the meet, since inside one such cut one would have to absorb from the other set, hence absorb that whole set, for any strict increase. Up to the union cut instead we have the full Young subgroup interval in $S_{D \cup D'}$. These are formal statements before the product selection (and within the ideal at M_A); in particular all three coatoms persist, and formal covers between products are still covers. In the other direction when needing an intermediate such as the formal wreath label, which need not itself be product, we will argue using product closure rather than just assuming persistence. $\square$

Lemma 11.15 (Finite selection simultaneously at all ancestors). *For the successive exclusions below, fix a bound on path-prefix size and a finite sequence of rounds, each testing boundedly many omitted positions and finitely many structure or comparison predicates. A sufficiently large marker reservoir can be thinned so that, conditionally on absence of the point configuration, each round applies simultaneously at every eligible prefix carrying the simple group of [Convention 11.8](#). At any such fixed prefix, a bad witness with omitted support of size k has replicas on every k -subset of a later reservoir. Any prescribed bounded unions of these replicas retain all properties established in preceding rounds. The required reservoir sizes depend only on these finite requirements, not on group dimensions or field sizes.*

Proof. Apply finite Ramsey homogenization to the bounded omission patterns. Concretely color bounded-size ordered index subsets by all the tested label predicates (Boolean, also subdivided into the boundedly many indicated cases) relative to their prefixes A whenever those have the required path-node normal simple in the ordinary product interval. It is canonical if it exists, by its soluble centralizer. Include type predicates and the bounded comparisons across subsets as used for selection. From a sufficiently large homogeneous set, for fixed A with fresh indices after it, any bad label with small nonempty omitted set after A thus gives replicas with *every* k -set of later fresh positions ($k = |D|$), including as many disjoint ones as needed. Choices of witness structures can also be further finitely colored and thinned. The exclusions are used successively: earlier ones for larger bounded sets as needed to make the intersections (which contain the axes omitting the support unions) common previous-good labels, just as properties there passed upwards from anchors. Thus they apply simultaneously relative to all indicated high-type ancestors. All bounds for combined supports in each round need only be finite in terms of subsequent round requirements, and never involve the ambient group dimensions or fields.

More explicitly one may use successively shorter *initial segments* of a homogeneous reservoir as rounds proceed. Assert each exclusion (conditional on absence of the point configuration) for all prefixes A of the allowed path sizes with the stated high simple data, and omitted sets in that round's segment, A before the further omissions in index order. If a bad full stabilizer contains a tested axis label, its exact omitted support is a nonempty subset of the tested further omissions. Replicate this support at fixed A using the spared tail of the preceding segment; earlier-good properties are then available for each specified bounded union of the replicas by the result of preceding rounds on that segment. Thus choose earlier omission bounds and segment lengths large enough *backwards* (e.g. include combined supports of the finitely many simultaneous intersections needed in a later round, and allow extra finite Ramsey thinning there). Predicate colors on ordered subsets can include all splits/sublist choices at the finitely bounded sizes, recording actual path-simple existence even before descending (B_A is unique by its soluble centralizer), needed same-characteristic/type tests, and existence of a witness with exactly those support positions in the relevant categories. Not the unbounded field sizes, group orders, etc., only finitely many case/comparison predicates are needed. In a contradiction at a fixed prefix one can further choose witnesses and color them by finitely many tests, without losing earlier-good properties. All unions and links still refer to the original T ; in particular the residual marker counts for the normal/missing-site tests remain arbitrarily large. $\square$

11.2 Alternating ancestors

Throughout this subsection B_A is alternating of sufficiently large degree. The homogeneous families, disjoint replicas, and inherited properties are supplied by [Lemma 11.15](#). A previous-good intersection means one to which the conclusions of all preceding pruning rounds apply.

Proposition 11.16 (Subset stabilizers produce points). *If a tested descendant is contained in a proper subset stabilizer, then, after the finite selections of Lemma 11.15, there are arbitrarily many distinct relative labels $M_{\{i\}}$ that are full stabilizers of individual natural points. In absolute notation these are $M_{A \cup \{i\}}$. Consequently, in the absence of the conclusion of Theorem 11.6, the required descendants and their bounded intersections are transitive.*

Proof. For subset stabilizers K_{r_D} , among the two supported alternating groups precisely one side is missing relative to K_{g_D} by the normal property of Lemma 11.9 (soluble quotient over both). Use the retained side's point subset U_D ; K_{g_D} acts transitively there by supplementation with the other side, even for small size. Thus retained point sets for disjoint supports are disjoint, by mutual containment of each smaller label in the other's stabilizer and distinctness. In a homogeneous k -set family we can take their sizes each below a small fixed fraction of the whole (color, using arbitrarily many disjoint replicas). Their stabilizers are maximal: an increase would give transitivity and overlapping conjugates of the large complementary supported alternating. So $r_D = C_D$ (relative labels; all relative coatoms are products by intersecting global cuts with M_A). The same maximality applies to the subset union for two disjoint k -sets. The union label contains $C_D \wedge C_{D'}$, and therefore equals $C_{D \cup D'}$, since the only coatoms above that meet are these three, and the union stabilizer moves the individual subsets (correct parity outside). Consequently the union itself is independent of the split by its small size and the stabilizer orbits.

Decomposing the retained subsets.

It follows that U_D splits into elementary disjoint subsets indexed by the individual $i \in D$, the same in every use. Indeed for any point the indicator p_D has $p_D + p_{D'}$ fixed by the union of disjoint index sets. Thus $p_{F \cup i} - p_{F \cup j}$ on replacing one index is independent of the other $k - 1$ indices F , by comparing with a common disjoint auxiliary $(k - 1)$ -set. For $k \geq 2$ we obtain a sum of individual real weights plus a constant: the replacement differences obey the triangle identity by using the same F for three positions, so are differences of weights, and replacements connect all the k -sets. Weights differ only by zero or one up to sign; with our ample reservoir, two or more each of a high and a low weight would produce a difference of two between indicator sums. So either all indicators vanish (a constant one violates disjointness) or just one position is exceptional and membership is exactly when it occurs (the reverse choice again violates disjointness). For $k = 1$ use disjointness directly. At most one elementary subset can be empty by label distinctness. For any nonempty one its stabilizer label is $M_{\{i\}}$, since it contains $C_D \wedge C_{D'}$ for pairs meeting just in i ; two such meets with remaining positions disjoint already join to $M_{\{i\}}$. Indeed each meet fixes position i and absorbs all outside $D \cup D'$; use four sets all meeting pairwise just at i . For $k = 1$ we already have the label $C_{\{i\}} = M_{\{i\}}$ itself.

Equal sizes and singletons.

Unions of two or four of these nonempty elementary sets again have the corresponding cut labels by the coatom-meet argument. For disjoint pairs P_1, P_2 , the full interval from the two cuts' meet J_0 to the union cut F_0 is that from $S_2 \times S_2$ to S_4 , with a proper wreath intermediate. If actual union sizes for the two differed, the ordinary interval would be a cover (project the full stabilizers to the symmetric group on the combined set, using outside parity correction). Apply Lemma 11.13 to the product endpoints in the original power. The formal wreath intermediate would force $K_{J_0} \cap S \leq K_{F_0} \cap S$ before the ancestor quotient. Passing to that quotient and intersecting with the actual B_A gives a contradiction: on the combined set the upper group induces the full symmetric group by outside parity correction, whereas the lower group induces exactly its two-part stabilizer. For unequal positive part sizes that stabilizer is maximal proper in the full symmetric group (an increase is transitive, merging overlapping larger parts via conjugates of their supported symmetric group), not normal. Thus pair-sums of sizes agree and all nonempty elementary sizes

agree (compare pairs sharing one index against a common disjoint pair). For two elementary sets the analogous formal interval is just a cover, so their equal size must be one by block-swapping with parity correction. This gives the desired point labels. Henceforth we can use transitivity after this exclusion if success was not already obtained. $\square$

Lemma 11.17 (Exclusion of imprimitivity). *Assume the transitivity conclusion of [Proposition 11.16](#) for all bounded unions required in this round. A homogeneous family of proper full stabilizers of nontrivial equal-block partitions cannot occur at the tested descendants.*

Proof. For full stabilizers of nontrivial equal-block partitions (the imprimitivity test) let the sizes be a per block, b' blocks. If $a \geq 3$ with $A_a^{b'}$ retained, the supported primitive groups force refinement into blocks of another system of the pattern (discrete restriction to a block is impossible by moving a point while fixing its equivalent outside neighbor), hence equality mutually. For $a = 2$ use instead retained even simultaneous flips of two blocks, precluding any other matching of this kind. Otherwise $a \geq 5$ and the alternating base product is missing. Here the full stabilizer is maximal by the supported A_a 's: an increase gives overlapping conjugate blocks, thus full alternating on merged components, and its primitive top on blocks forces total merging. Indeed supported alternating groups of these sizes on overlapping sets generate alternating on the union (conjugate three-cycles through a common point; even with just one common point this brings in three-cycles using one point of the other set and two of one's own). The full top symmetric action is available, with internal parity correction. Take two disjoint failures, giving coatoms $C_D, C_{D'}$ and a meet covered in each. The two block systems have common cell size h on occupied incidences by prior transitivity. They cannot coarsen one another by primitive top action, which is realized inside the intersection for both by supplementation relative to the g labels. If $h \geq 2$, the cell-refinement stabilizer intersects a row label properly (by its supported primitive block groups), strictly above the intersection because one can swap two cells within a row while fixing points elsewhere, with internal parity correction, breaking the columns. This contradicts covering. If $h = 1$, the intersection embeds subdirectly in $S_{b'} \times S_{b''}$ via the two top actions, both block counts at least five by the touching block sizes. The kernel of projection to the first top action must still supply all even-total base sign patterns there (supplementation with $A_a^{b'}$, which acts trivially on that action's blocks and signs). But this kernel injects as a normal subgroup of $S_{b''}$; its abelianization has size at most two, versus its required homomorphic sign image of size at least $2^{b'-1}$. $\square$

Proposition 11.18 (Exclusion of affine and Cartesian structures). *After [Lemma 11.17](#), the tested descendants are primitive. They cannot lie in full normalizers of elementary abelian regular groups or in proper Cartesian-system stabilizers.*

Proof. Using then common primitive previous-good labels E , full normalizers of elementary abelian regular groups are excluded by [Proposition 7.4](#): the regular groups are retained and hence mutually included in E , where each is minimal normal; distinct such groups commute, whereas an abelian regular group is self-centralizing. For Cartesian stabilizers, [Lemma 7.5](#) applies with mutual inclusion in intersections for disjoint supports; small slot sizes 2, 3, 4 are likewise precluded by the regular-group test. In the missing power case for two disjoint supports, the supported normal products in E are supplied by [Lemma 11.11](#), so the tuple commutator and point-stabilizer factorization in [Lemma 7.7](#) give a common Cartesian refinement with at most one nontrivial fine slot per pair of slot positions. Take E to be the two-label intersection. Its primitivity gives transitivity on occupied fine slots, say each of set size $h \geq 2$. The top action at each of the two original systems inside E

is transitive containing the alternating group by supplementation, hence primitive, so grouping fine slots into rows and columns splits each slot position on both sides into at least two fine ones (refinement would be improper by that primitivity). Both original slot set sizes are large by the bounded-type consequence for the missing powers.

Let u be the intersection of the full row-system stabilizer with the fine-slot stabilizer. This is a proper decrease by the supported natural alternating action on each row slot, and it contains E . It permits interchanging two fine slots in one row, leaving the others in place: if the parity of the interchange needs correction then h must be odd (at least three fine slots), so use an internal symbol transposition. Together with the full column-system stabilizer this generates the ambient group. Here are details of the latter assertion. Local alternating gates on two overlapping sets of Cartesian directions each of product size at least ten generate the alternating gates on their union. Split into differences and shared part of alphabet sizes a, c and b , all at least two (proper overlap). On the union the generated group is two-transitive: ordered distinct pairs can be changed to any one fixed pair distinct in all three positions using two-transitivity on each of the two touching position-sets (first obtain a difference in the shared part on a position-set with some difference, then set the first and shared values as desired, then the shared and last). For $b \geq 3$ use three-cycle gates on the two position-sets whose lifted supports intersect in one tuple there (one shared value appearing once in each cycle and disjoint other shared-part values). Their commutator supplies a three-cycle, sufficient by primitivity. For $b = 2$, both $a, c \geq 5$. Flip the shared bit on two values of the first alphabet, and use arbitrary A_c conditional on bit one. Commuting these and commuting again with conditional A_c gives A_c conditional on bit one and membership in the chosen two values (by perfection). Indeed the first commutation applies an element of A_c on one bit value and its inverse on the other just for those two first-alphabet values; intersect the condition with bit one by commutators again. Two such two-value choices intersecting once and commuting their conditional actions likewise give supported A_c on a single fiber, again sufficient (a three-cycle there, then Jordan). All gates obtained act independently of unused coordinates, meaning the same union permutation repeated on all outside tuples. Now we have the column gates and those on the conjugate columns by the interchange. Indeed an altered column and its original overlap properly, and merging them extends the latter by a slot from a second column, next allowing merger with that entire column by the same observation. Align actual permutations of the columns preserving fine coordinates up to a permutation (uniform fine-slot alphabets and common fine-slot count per column); even column permutations so aligned are in the socle as well. Thus the merged directions connect all columns by primitivity (for two columns the one interchange already connects). The gate observation yields the whole alternating group, hence full join (the base already supplements it). This contradicts $u \vee r_{D_2} < M_A$. $\square$

Lemma 11.19 (Exclusion of primitive diagonal structures). *After the preceding alternating rounds, primitive diagonal structures are excluded as well. Hence the required descendants of an alternating ancestor are primitive almost simple groups. The transitivity conclusions continue to hold relative to all such ancestors.*

Proof. By Lemma 7.9, every remaining primitive group that is not almost simple normalizes a transitive power $Y = B^s$, $s \geq 2$, with a full possibly twisted diagonal point stabilizer and natural degree $|B|^{s-1}$. This includes two opposed regular simple factors.

First consider retention. A retained power with transitive factor action is a minimal normal subgroup of the common primitive intersection E , with trivial centralizer because its diagonal point stabilizer is self-normalizing. Distinct retained powers would therefore coincide. If the factor action is intransitive, each factor-orbit product is transitive; in the diagonal action a transitive subproduct

uses at least $s - 1$ factors. Thus the only possibility is $s = 2$, the two opposed regular simple factors. A retained regular factor is minimal normal in E , and its centralizer supplies its opposite regular group. Distinct minimal normals commute and consequently determine the same opposed pair. These are exactly the retained alternatives in [Proposition 7.10](#), and exclude distinct disjoint-support witnesses.

It remains to consider a missing power with transitive factor action. Take disjoint labels and their proper previous-good intersection E . By [Lemma 11.11](#), E has a missing normal product $R = \prod_{a=1}^s R_a$ of singly supported proper groups, with nonsoluble factors. This assertion does not require $R = E \cap Y$. Choose a minimal E -normal subgroup $P' = C^u$ inside R . It is nonabelian by the affine exclusion, and transitive by primitivity of E . Every factor of P' lies in one R_a : the groups $[P', R_a] \leq P' \cap R_a$ are normal in P' , distinct slots commute, and these commutators generate P' , since $P' \leq R$ is perfect. The assignment is E -equivariant. Supplementation relative to g_D makes E transitive on the s slots, so there is a common positive number h of factors over each slot and $u = sh$.

The primitive-group argument of [Lemma 7.9](#), applied to E , says that the point stabilizer $Q = P'_x$ is a full diagonal. If $h > 1$, the product of the projections of Q to the s assigned batches is a proper intermediate subgroup between Q and P' , normalized by E_x , contradicting primitivity. If $h = 1$, equality of point degrees gives

$$|C|^{s-1} = |B|^{s-1}.$$

But a factor C embeds in a proper subgroup $R_a < B$, an impossibility. Thus the diagonal case is excluded, and the required descendants are primitive almost simple. These conclusions apply simultaneously at the eligible ancestors by [Lemma 11.15](#). □

11.3 Classical subspaces and projector stabilizers

We use the natural covering groups, forms, collineations, and anti-actions of [Section 6](#). All dimensions required to be large can be made so by [Lemma 11.7](#).

Lemma 11.20 (Exclusion of parabolic stabilizers). *For a high classical ancestor, the tested descendants cannot preserve a proper linear subspace in the absence of anti-actions, a proper signed flag in their presence, or a proper singular subspace in the isometry cases. The same assertion holds for the bounded intersections needed in subsequent rounds.*

Proof. Soluble unipotent radicals are retained, thus each lies in all other labels on a disjoint family. For linear subspaces (no anti-actions), signed flags (linear with anti-actions), or singular subspaces in the isometry types, the shear arguments of [Propositions 6.4](#) and [6.5](#) give a chain (using the smaller space of each signed flag, the other determined in reverse by an anti-action of the base), up to the bounded singular-space terminal exception. Pick two in a many-member disjoint family there (below terminal dimensions with ambiguities in orthogonal type, e.g. below Witt index minus one in split type D). Their full stabilizers and intersection meet the ancestor socle in the corresponding parabolics of compatible flags. By the standard parabolic overgroup theorem for the natural BN -pair in the simple Lie-type group (overgroups of a parabolic are parabolic, with the standard-parabolic subset order; in the classical interpretation use the indicated subspace or singular flags, away from the split type D terminal fork here), the labels are maximal and their meet has only the two maximal increases to coatoms: overgroups intersect the socle in parabolics obtained by dropping flag terms, determining the overgroup since it contains the supplementing H . In the signed case terms must be kept in duality-stable pairs (possibly a single middle term). This contradicts the three coatoms above the meet of two coatoms with nonempty disjoint supports.

□

Lemma 11.21 (Nonsingular radical lines in characteristic two). *In the orthogonal case in even characteristic, a full stabilizer of a nonsingular radical line is maximal. Such line stabilizers cannot occur in a homogeneous family of the tested descendants after Lemma 11.20.*

Proof. Indeed write the hyperplane of orthogonality to a line representative u as $u^\perp$; the simple-group stabilizer contains products of two nonsingular symmetries with centers therein. An increase gives another such hyperplane $w^\perp$; there is a common nonsingular partner for bridging the products (high dimension). Odd and even words of symmetries using these two hyperplanes then give by conjugacy all centers, so even products giving the simple group are contained. Indeed arbitrary even words from the two center sets together are available by inserting a common partner for mixed products; once all centers are obtained as symmetries by words (necessarily of odd length by Dickson parity), their pair products thus belong and give a nontrivial normal generating set projectively by simplicity. In detail, for nonsingular v outside both one can choose nonsingular $z \in u^\perp$ whose symmetry takes v into $w^\perp$, by choosing both polar pairings $b(z, v), b(z, w)$ nonzero there, then ensuring $Q(z) = b(z, v)b(z, w)/b(v, w)$ for the quadratic form Q . One can adjust this value arbitrarily with a large nondegenerate subspace orthogonal to the indicated vectors. Neither functional vanishes identically since the lines are distinct and $v \notin u^\perp$. This proves the generation assertion, also sufficient in the ambient quotient since the base supplements its socle. For two disjoint failures the pair span P must be nondegenerate, else its singular line would be preserved and has been excluded on the combined support. The pair intersection then agrees with the full intersection of either line stabilizer with the plane stabilizer (by covering). Indeed the latter cuts it properly, e.g. products of nonsingular symmetries in $u^\perp$ move the plane (choose one center pairing nontrivially with w and outside the plane, correcting by one fixing the plane). But this full intersection on the u side includes the symmetry at u corrected outside P , failing to preserve w . This gives a contradiction.

□

Proposition 11.22 (Maximality of an unbalanced ordered splitting). *Let $V = W \oplus U$, with $0 < \dim U < \dim W$, be a nondegenerate splitting in an isometry type, or an image–kernel splitting in linear type with anti-actions. Let M be a group of the corresponding classical actions containing the natural special group, and let K be the full ordered splitting stabilizer, equivalently the idempotent stabilizer for the ring action. Assume that K supplements the special group in M , that its collineations act semilinearly irreducibly on U , and that the supported special group on W is available, as is that on U when $\dim U$ is large. In the linear case assume also that K contains an anti-action fixing the projector, hence carrying its image–kernel data into the dual annihilator data. For sufficiently large $\dim W$, the subgroup K is maximal in M .*

Proof. Let an overgroup properly contain K . We show that the transported copies of the supported special group on W generate the whole ambient special group. First we find large summands close to W whose vectors span V in boundedly many summands. We then collapse those sums and recover the global elementary generators.

Throughout, supported groups use their natural matrices, equal to the identity off the indicated summand; their projective images are the groups in M . Transporting a splitting transports these groups with it. In linear type an anti-action transports the special linear support by inverse transpose, and exchanges the corresponding primal and dual data.

Near splittings and bounded sums.

Any increase contains transported copies of the supported special on W at the largest summand of any moved projector. This copy cannot normalize the original projector: by irreducibility on its support and dimensions it would first force its image space to agree with W , then its fixed complement to be U . The high classical special groups here are generated by matrices of bounded displacement rank (ordinary elementary/root generators, also obtained by Witt reduction). For instance take all inner conjugates of a noncentral elementary generator (in an isometry type one can use a special linear shear on two totally singular basis directions extended dually and identically off the associated two Witt pairs); they suffice by projective simplicity and perfection (generation modulo the special group's center suffices by perfection). Thus a near copy of the splitting with moved projector is attained, with new largest space W' differing from W by bounded codimension of intersection. Use the anti-action if necessary in linear type to get movement of the *image* (also to achieve kernel movement). Use these and their K -transports as near splittings (transports by anti-actions of these bounded-rank displacements remain near). Every vector is a sum of boundedly many from these large spaces. Indeed projections span U by semilinear irreducibility. If U is bounded-dimensional this suffices. Otherwise transport a nonzero such projection by the available classical actions to get bounded-sum generation there. In the linear or symplectic case use transitivity, and for a norm constraint use high Witt dimension: every vector v is a sum of boundedly many vectors of a given nonzero vector's type (norm, including isotropic nonzero). For instance prescribe y and $v - y$ both of the given norm by first imposing the corresponding polar pairing with $v \neq 0$ (a trace in hermitian type), then adjusting the norm in a large orthogonal subspace, avoiding zero vectors. Here we use the usual transitivity on nonzero vectors of given norm also under the high special by Witt extension with determinant, spinor or Dickson corrections: a large nondegenerate subspace orthogonal to the vector to be fixed supplies arbitrary such coset data (in unitary type norm-one determinants; in orthogonal type reflections/symmetries at nonsingular vectors, both norm-square classes available if needed). Likewise after prescribing the pairing to get two summands of the given norm, orthogonal adjustment can be made perpendicular also to a first solution of the pairing, using a large nondegenerate subspace representing every norm many times. Use W itself for the unprojected correction. In the linear setting dual generation holds also: under transport by a fixed anti-action at the original splitting, large-space images for near projectors correspond by a semilinear primal-to-dual identification to annihilators of the transported kernels. The family of these initial near splittings can be taken closed under that transport in both directions (bounded defect on both images and kernels). Thus these annihilators span the dual, with the same bounded-sum conclusion there; in isometry type use selfadjointness and primal spanning. In particular the common kernel of the available near projectors to the large spaces is zero.

Collapsing bounded sums.

One can collapse such bounded sums to one summand from some conjugate splitting, still within bounded codimension of W . Write W_i for summand spaces used and p_i for their projectors, with bounded pairwise defect, so intersections also contain large nondegenerate spaces when relevant. If $s = w_i + w_j$, $w_i \in W_i, w_j \in W_j$, has $p_i(s) \neq 0$, choose $d \in W_i \cap W_j$ so $p_i(w_j + d)$ is nonzero of matching type/norm as $p_i(s)$. Use largeness of the intersection to solve the norm condition, with nonzero choices, and the special-group orbit transitivity with corrections established above. The complementary components agree:

$$(1 - p_i)(w_j + d) = (1 - p_i)w_j = (1 - p_i)s.$$

The supported group on W_i can therefore send $w_j + d$ to s , mapping W_j to another near splitting. For sums of three or more, if all pair-collapse projections vanish, perturb two summands oppositely by an intersection vector to make another pair collapse. For example perturb the first and second

by a common nonzero vector with opposite signs; the first projection for the first-plus-third sum is now nonzero. Defects stay boundedly large at most because only boundedly many steps are needed. More precisely, write $\delta(E, F) = \dim E - \dim(E \cap F)$ for these equal-dimensional spaces. A transport g supported on W_i fixes its projector, so

$$\delta(gW_j, W_i) = \delta(W_j, W_i), \quad \delta(gW_j, W) \leq 3d_0$$

when both current defects from W are at most d_0 . The same calculation holds on the dual supported spaces in linear type, since g fixes that part of its splitting as well. Thus we maintain bounded defects for both images and kernels. Large nondegenerate parts of intersections in these steps follow since intersection with a near space has bounded codimension in the nondegenerate summand (and we may still take boundedly many orthogonality constraints). Finally if a nonzero pair sum lies in both projection kernels, take a fresh near W_3 with nonzero projection of the sum. For a nonzero summand w_i , choose in $W_i \cap W_3$ a nonzero vector of its type and transport it to w_i by the supported special group on W_i . This transport fixes the sum, since $p_i(s) = 0$. Its image of the W_3 -splitting therefore covers w_i and still has nonzero projection of s . Use that splitting to collapse the pair. The same construction holds dually in linear type, with both defect bounds maintained as above.

Global elementary generators.

Thus all centers are available near W . Supported special transvections now generate globally in unitary and symplectic type. In orthogonal type use products of reflections/symmetries of same norm, bridging by a common partner in the large near-space intersections; such products generate (in particular supply global nontrivial normal generators). In linear type for v, f covered respectively in some W_i and some dual supported W_j^* , $f(v) = 0$, take a commutator of supported transvections with tensors $v \otimes h$, $z \otimes f$, with $f(z) = h(v) = 0$, $h(z) \neq 0$, h supported in the first dual space and $z \in W_j$. These choices exist by the large intersections relative to the first projection kernel (one can take z in $W_j \cap W_i \cap \ker f$ independent of v , then choose h on W_i ; its scaling gives arbitrary transvection coefficient): in the linear argument keep bounded intersection defects simultaneously on images in primal and dual. This holds for the initial near splittings by bounded displacement (also of the contragredients); when a supported group transports another near splitting in a collapse, it keeps its own splitting fixed, so both defects still grow only boundedly. This gives the required full transvections. This proves maximality. $\square$

Lemma 11.23 (Orthogonality of the retained projectors). *After Lemmas 11.20 and 11.21, consider full stabilizers of proper nondegenerate subspaces in isometry type, or of proper nontrivial individual projectors in linear type with anti-actions. Let e_D be the projector onto the side whose supported special group is retained relative to g_D . After finite selection, witnesses on disjoint omitted supports satisfy $e_D e_{D'} = e_{D'} e_D = 0$. For the small orthogonal exceptions the nondegenerate-line round is performed first.*

Proof. Now take isometry stabilizers of proper nondegenerate subspaces, or linear-with-anti individual proper nontrivial projector stabilizers. Designate by U_D the side with supported special retained by the normal property of Lemma 11.9 over g_D ; exactly one side is missing. Write e_D for the projector to U_D . For a pattern with large retained dimensions, disjoint supports then have mutually orthogonal projectors as in Propositions 6.4 and 6.5, by supported irreducibility and mutual preservation. For a pattern with bounded retained dimension (take it constant), the full stabilizers used are maximal by Proposition 11.22. We explain small exceptions/ordering in isometry type: the full individual isometry group on U_D is available in projection with corrections on the large complement. It acts irreducibly except possibly split orthogonal planes over two and three. Over two there is a forbidden nonsingular radical line. Over three there are two nonsingular axes

individually invariant unless interchanged by a similarity giving semilinear irreducibility. Thus first run the nondegenerate line exclusion, then these other dimensions with invariant-line cases forbidden on the combined supports. There is no dependence of the line test itself on excluding these planes.

For two disjoint bounded-dimensional occurrences the sum of the small spaces is again complemented, by its orthogonal space or by the kernel intersection, via the prior singular/signed flag exclusions (also nonsingular radical lines when relevant). In linear type an anti-element of H fixing both projectors exchanges the sum with the kernel intersection, as do signed anti-actions of the pair group; a nonzero intersection gives a signed flag. Also images determine kernels in this case. The sum projector stabilizer cuts each individual stabilizer properly (large complementary supported group) and thus its full intersection with either equals the pair group by covering. Consequently all linear or isometric maps on one small summand, identity on the rest of the sum (split it using that summand's prescribed complement) and corrected outside, preserve the other small space. The outside here is in both projector kernels, giving a large space for arbitrary special-group coset corrections as above. Distinct small spaces (complements determined and dimensions taken constant) cannot contain each other; this also explains strict cutting by the sum stabilizer, since a projection of the sum to an individual large complement is then nonzero proper, not preserved by the full supported large action. Taking differences on the other small space under the indicated maps thus forces zero projection onto the chosen summand whenever its individual linear/isometry group acts irreducibly and nontrivially: a nonzero projection generates the irreducible space already by its orbit differences. Hence mutual orthogonality outside the indicated small exceptions. For the split orthogonal plane over three with interchange, the isometries here together with the intersection label interchanging the nonsingular axes by earlier line exclusion suffice (the minus sign supported there on the sum splits the other small space along the projection, and the joint action is irreducible on the summand by the interchange together with its isometries). For a binary linear line with anti-action, use the bilinear pairing giving a fixed anti-element: each line and kernel are mutual annihilators for this pairing, with self-value 1 at the line representative. On two distinct such representatives the two cross-values agree, since left and right orthogonality to each line coincide; value 1 off-diagonal would violate the complementation of the sum. This likewise gives projector orthogonality. $\square$

The orthogonal projectors admit an additive decomposition over the individual marker positions. We record the algebraic step separately because it works in every characteristic.

Lemma 11.24 (The projector fold). *Let e_D be a family of idempotents in one associative algebra, indexed by the k -subsets of a sufficiently large finite reservoir. Suppose that idempotents on disjoint sets annihilate one another on both sides, and that $e_D + e_{D'}$, for disjoint D, D' , depends only on $D \cup D'$. Then there are pairwise annihilating idempotents h_i such that*

$$e_D = \sum_{i \in D} h_i.$$

If the e_D are selfadjoint, so are the h_i .

Proof. For $k \geq 2$, $\Delta_{ij} = e_{F \cup i} - e_{F \cup j}$ is independent of F as in the subset argument. Disjoint-pair differences annihilate each other. Also e_D acts as identity on Δ_{ij} on both sides when $i, j \in D$, and as zero when $i, j \notin D$. The latter follows using a disjoint witness for the difference, the former by adjoining a spare disjoint k -set to D (the union projector supports a witness within the union). Then $h_i = \Delta_{ij} \Delta_{il}$ is independent of distinct auxiliary j, l , including order, by disjoint annihilation. For example, with a fourth index a , the triangle identities give

$$\Delta_{ij} \Delta_{il} = \Delta_{ij} \Delta_{ia} = \Delta_{il} \Delta_{ia} = \Delta_{il} \Delta_{ij}.$$

It meets each e_D by multiplication either as identity-action ($i \in D$, use a partner in D) or zero. To obtain $\Delta_{ij} = h_i - h_j$, choose D containing i, j and let D' replace them by two fresh positions a, b . Then

$$\Delta_{ij} = \Delta_{ij}(e_D - e_{D'}) = \Delta_{ij}(\Delta_{ia} + \Delta_{jb}) = h_i - h_j.$$

No division by the characteristic or by k occurs. Thus distinct h_i 's annihilate each other (separate by an e_D), and $h_i = (h_i - h_j)(h_i - h_l) = h_i^2$. They are orthogonal idempotents, selfadjoint when relevant, and the remainder in $e_D - \sum_{i \in D} h_i$ is constant, necessarily zero by disjointness and idempotence (each such remainder is itself an idempotent supported by e_D on both sides). For $k = 1$ use the projectors directly. □

Proposition 11.25 (Exclusion of proper projectors). *Under the preceding classical rounds, the tested descendants cannot lie in any of the proper stabilizers considered in Lemma 11.23. The remaining descendants and the required bounded intersections therefore have irreducible anchors; in linear type their collineation parts are also irreducible.*

Proof. In the resulting homogeneous k -set patterns choose dimensions small relative to total by orthogonality on disjoint supports. The stabilizers and disjoint-pair sum stabilizers are maximal by Proposition 11.22, and the sum label is $C_{D \cup D'}$ as in the subset argument (it moves each summand). For clarity the aggregate here has the requisite irreducible collineation action: split planes over two do not arise as sums, and over three in the line test choose both norms of the same class by coloring, avoiding the split plane. Thus the aggregate projector $e_D + e_{D'}$ depends only on the union; it is recoverable from its stabilizer, e.g. the supported large complementary special forces uniqueness among projectors with small image.

Apply Lemma 11.24 to write the projectors as sums of the h_i . In this marker configuration, at most one h_i vanishes, and each nonzero h_i has full stabilizer label $M_{\{i\}}$. For $k = 1$, the full stabilizers are already the singleton cuts $C_{\{i\}} = M_{\{i\}}$. For $k \geq 2$, two vanishing h_i 's would identify distinct cut projectors by replacing one index with the other. For a nonzero h_i , choose D, E meeting just at i ; then $h_i = e_D e_E$. Thus the paired cut-meet argument of Proposition 11.16 puts $M_{\{i\}}$ in its full stabilizer. That stabilizer is proper because h_i is a nonzero proper projector, with nonzero disjoint witnesses remaining. Since $M_{\{i\}}$ is a coatom, it is the full stabilizer.

For nonzero positions, in large fixed sizes $l, 2l$ with ample further spares, all sum-projector stabilizers on those sets P are again maximal (keep dimensions small relative to the ambient by thinning), hence are $C_{B(P)}$ for some nonempty $B(P) \subseteq P$, and distinct by projector uniqueness. Homogeneity/counting forces $B(P) = P$ on a thinned reservoir (homogenize the decreasing-size test; injectivity rules out strict decrease everywhere for that size). For disjoint l -sets, if the sum dimensions differ, their meet to the combined cut is an ordinary cover, by Proposition 11.22 applied within the combined aggregate (both pieces now sufficiently large). Indeed projection to ring actions on that aggregate identifies the meet as full ordered splitting stabilizer, with anti-action still present in the signed linear case. These applications have the supplement hypothesis: the ancestor outer actions are supplied by the common base, and restrictions of linear/special isometries from the inner ancestor are covered modulo the aggregate special group already by the splitting stabilizer, using the two large pieces for arbitrary determinant etc. (orthogonal determinant and spinor or Dickson corrections by Witt groups) and the large outside space for compensation. The formal wreath label is strictly between the two-cut meet and the combined cut. If these product endpoints formed an ordinary cover, Lemma 11.13, applied before the ancestor quotient, would force normality at each original site. This is impossible on intersecting with the actual supported high special group of the combined aggregate: its projective action is simple, and the lower intersection is a proper

nontrivial splitting stabilizer. The ancestor centralizer kernel is already in the lower endpoint, so this is also a valid nonnormality test in the quotient. Hence dimensions agree for disjoint l -sets, thus all individual dimensions agree (compare a pair of replacements to one common disjoint set). Align the individual summand forms themselves by finite coloring. Now on an l -set the sum-cut interval from the axis has at least the whole subgroup lattice of S_l just up to the stabilizer allowing permutations of the individual projectors (kernel the axis label, and align permutations with outside correction). The sum stabilizer exceeds this by its large full classical action, which cannot preserve that uniform direct decomposition (same orders as in the earlier direct-system pruning). Indeed the formal interval $[M_P, C_P]$ is exactly the subgroup lattice of S_P , with $|P| = l$. Its ordinary product sublattice cannot contain more labels. Yet the permutation stabilizer just described already supplies one ordinary label for every subgroup of S_l , and the strictly larger full sum stabilizer supplies an additional label. This is the required contradiction. Thus these failures too are excluded. As before we get prior irreducible anchors (collineation part also in linear type).

□

11.4 Direct systems

Use the direct systems and supported groups of [Lemmas 6.8, 6.9](#) and [6.11](#), now with disjoint label supports D_i , $K_i = K_{r_{D_i}}$ (as needed also for the following categories). Retention in the smaller g_{D_i} labels places a group inside *every* label K_j here. A previous-good anchor can be taken to be the intersection or a large axis group below it, omitting just the joint support (all relative to the ancestor). Thus we have transitivity and semilinear irreducibility on blocks by the collineation part of the intersection whenever using them.

Lemma 11.26 (Coincidence of exceptional matchings). *In the retained line-system and exceptional plane-system cases of classical pruning, a family of disjoint-support witnesses forces all the exceptional matchings to coincide. A fixed previous-good intersection then allows only boundedly many further systems. Thus these cases are excluded by taking sufficiently many replicas.*

Proof. In the line-system argument of [Lemma 6.8](#), the diagonal groups are still mutually included by retention, so it remains to treat its matching cases. All matchings relative to the first system would have to agree: the signs of one of the other systems already interchange first-system lines on a matched pair, simultaneously on a second pair if needed (choose two flips of equal norm class for determinant/spinor correction). There are plenty of possible second pairs by transitivity and at most two norm types (the two sign-orbit lines on a pair are of the same norm class); thus another invariant matching must pair these same two first-system lines (otherwise choose the extra flip away from a proposed different partner to keep that partner fixed). The choices on matched lines are unique as before. For the exceptional plane matchings of [Lemma 6.9](#), an individual retained plane sign likewise interchanges the two old planes, forcing matching coincidence. The common intersection of all the labels is transitive on the matching edges, so for that fixed intersection group bounded choices on one edge bound the number of further systems, impossible. Here and below common intersections for any bounded family can be required previous-good by earlier rounds at sufficiently increased omission bounds.

□

Lemma 11.27 (Support and compatibility for direct systems). *In the remaining direct-system cases, two disjoint-support witnesses have the supported block groups needed for the displacement-rank compatibility argument. The block groups can be chosen as actual block-supported matrices, with retained centers included. Retained systems are excluded by comparable refinements and the uniform order bound; a missing simple-site orbit has large block dimension and bounded pair-orbit count.*

Proof. In the remaining direct-system cases the supported groups of order greater than two for compatibility are available exactly as above: use the full retained groups, or, for a missing simple-site orbit product, [Lemma 11.11](#) with the previous-good intersection. (If two site orbits, can likewise use full supports from a retained one.) Centers included with these lifts are retained. In fact any missing simple orbit here forces large block dimension by the bounded-type consequence, with the action on sites of bounded pair rank by supplementation relative to g_{D_i} ; in particular bounded rank including the possible split four-space ambiguity cannot then occur. Block pair-orbits are bounded by the alternating actions on at most two isometry classes (one class in linear type), and the simple sites have fibers of size at most two over blocks. In applying the missing-product argument here the singly supported projections modulo centers in the intersection actually yield block-supported matrices projectively there: use representatives on that block in the supported choice of groups, since the center kernel needed when lifting already lies in the intersection. We can take all available such block matrices there, so the intersection's semilinear collineation stabilizer of a block normalizes the corresponding supported group. The block module for the supported group again is semisimple without trivial constituents by semilinear irreducibility for that stabilizer (use the module socle and the fixed group of vectors, as before) and nontriviality of the block action. (For a missing product the singly supported subgroups are even nonsoluble.) The displacement-rank proof of [Lemma 6.11](#) now applies: fixation of all the other system's blocks forces splitting by projections using orbit differences there; otherwise movement both ways forces equal block dimensions, and forces any moving block matrix to interchange a single pair of those other blocks as an involution, identity elsewhere there and with displacement of full rank on its own support, as in [Lemma 6.11](#). Its retained-case proof also applies: full retained actions force comparable refinements, whose height is absolutely bounded within the pattern by the uniform order obstruction on coarse blocks. $\square$

Lemma 11.28 (Merging compatible large summands). *Let W_1, W_2 be compatible summands obtained by grouping blocks of one common direct splitting, orthogonal in the isometry cases, with the corresponding complements. If $W_1 \cap W_2$ is sufficiently large and nondegenerate when relevant, their two supported special groups generate the supported special group on $W_1 + W_2$.*

Proof. Here the summands are direct groups of blocks of a common splitting, orthogonal in isometry cases, with the corresponding complements. Use W_1 of dimension at least $\dim W_2$. Every vector in the sum lies in some conjugate of W_1 by the supported group of W_2 : for a nonzero W_2 -projection attain it by transporting a vector of the intersection (matching norm using its high rank), and the remaining component already lies in W_1 . Thus we get the full symplectic or unitary transvection generation on the sum; likewise products of orthogonal symmetries/reflections of matching norm, using W_2 and partners in its large intersections with these conjugates to bridge any two centers. These elements give noncentral normal generation of the special. For linear type we also have dual coverage; as in the projector argument get a transvection tensor $v \otimes f$, $f(v) = 0$, by commutators from two supported spaces of dimension $\dim W_1$, sufficient because twice that dimension exceeds the sum dimension by a large amount. All assertions can use the natural covering matrices (the elementary generators obtained give the perfect special group), with the indicated support. $\square$

Proposition 11.29 (Exclusion of the remaining direct systems). *After the preceding classical rounds, a homogeneous family of proper full direct-system stabilizers cannot occur at the tested descendants.*

Proof. Treat missing systems with two blocks first. More generally once these are excluded, two-class block systems are impossible by coarsening, and the top action on blocks, induced inside

intersections by supplementation, is primitive (two blocks, or at least the alternating permutations with transitivity). Thus two disjoint-support instances give incomparable compatible systems and each block splits into at least two cells, of uniform dimension by transitivity. In the general round all those cells must in turn be mutually isometric when appropriate (else a two-class coarsening). In the two-block round there is a full 2×2 grid of spaces, and for some choice of the pair of instances we can interchange two same-row or same-column cells isometrically. Indeed if this fails the form must be orthogonal in odd characteristic with odd cell dimension by similarity/transitivity. Take three failures, choosing their common intersection to be previous-good as allowed by [Lemma 11.15](#). Its collineation irreducibility makes the occupied triple cells one orbit. Since its permutation action lies in S_2^3 , this orbit has size 4 or 8, and it projects onto all four cells for every pair. Size 8 violates odd pair-cell dimension by summing two equal triple-cell dimensions. For size 4, the third two-block system is the unordered diagonal partition of the four cells of the first pair (there is one third position at each occupied pair, with both third positions represented in each row and column), so the whole intersection of that pair of stabilizers preserves it, impossible by disjoint label supports.

Use a resolved pair, designated row and column, and let u be the intersection of the full row-system stabilizer with the stabilizer of the simultaneous refinement. It contains the joint intersection and is strictly smaller than the row label (uniform finer decomposition on each high-dimensional block). It permits a swap of two fine cells in a row, leaving the others in place, or a row three-cycle of cells. Indeed we have isometric choices as just described, and determinant, spinor or Dickson corrections inside the cells when of large dimension. For small cell dimension here the row block dimension is large (absolute thresholds can be made large enough to ensure at least three cells) and all cells are equivalent, so take a three-cycle via commutators of aligned permutations. Such collineations come from the ancestor socle.

We now apply [Lemma 11.28](#) to the original and altered column spaces.

A column space and its altered version lose/gain one cell by the within-row permutation and have large intersection (at least half a block); their supported groups thus merge, extending the column X by a cell from another column Y . If that cell is large we merge with Y immediately. Otherwise use the column label to transport within Y (alternating fine-cell permutations there while preserving X) and merge extensions along X until acquiring $X + Y$. The resulting merged column pairs propagate to a connected graph by block primitivity, and give full generation by repeated mergers with at least a column in common. This contradicts the formal disjoint-support rule $u \vee r_{D_{\text{col}}} < M_A$, and finishes direct systems. □

11.5 Large tensor factors and representation bounds

We supply maximality details for the forward field and ordered-factor tests, to avoid assumptions about arbitrary structure stabilizers being maximal. Use ambient notation $S \leq M \leq \text{Aut}(S)$ in the almost simple quotient at an ancestor, natural space V of dimension v , $\mathcal{A} = \text{End}_F(V)$. We only use full stabilizers in groups supplementing S already by the common base. Write Y below for the indicated supported projective special group(s), using their perfect matrix lifts (center need not disappear in the projective field and paired constructions).

Lemma 11.30 (Large-rank projective degree bounds). *For a sufficiently high-rank classical simple group of natural dimension m , with the type and characteristic conventions of [Convention 9.2](#), a nontrivial irreducible projective representation in defining characteristic has degree at least m . Unless it is natural up to Frobenius and duality, its degree is at least*

$$\frac{m^2}{2} - O(m).$$

A highest weight with l nonzero digit places has degree at least m^l . For field parameter P , every nontrivial cross-characteristic projective representation has degree at least P^{cm} , for an absolute $c > 0$; the same exponential lower bound holds for permutation degree. These lower bounds on nontrivial contributions apply to perfect central lifts and quotients. Finally, a high-degree alternating simple group acting projectively nontrivially in dimension v has natural alternating degree $O(v)$.

Proof. In defining characteristic, for classical simple groups of sufficiently high rank, with natural dimension m , the minimum nontrivial irreducible projective degree is at least m ; except for the natural module up to Frobenius/duality, it is at least $m^2/2 - O(m)$. To see this use the restricted weights on the universal cover and the Steinberg theory recalled in [Convention 9.2](#), using Jantzen [19], Steinberg [45]. For a restricted digit in A the permutation orbit already gives the latter bound except at a single endpoint only; at that endpoint with coefficient greater than one, simple-root lowering (coefficient less than p) and the resulting orbit give it as well. In B,C,D use signed coordinate permutations (even signs in D); at least two nonzero coordinates give the larger bound in high rank (including spin weights), and a single first-coordinate support with coefficient greater than one works again by lowering. Use B only in odd characteristic. Here the single first-node coefficient one is the irreducible natural module, of degree m also in odd-dimensional B. For $k \geq 2$ nonzero coordinates out of e in a signed word the orbit has size at least $2^k \binom{e}{k}$, allowing a loss of a factor two just at $k = e$ with even-sign restriction; the stated quadratic bound follows (including at $k = 2$). These arguments also give the bound by m^l for l nonzero digit places. Cross-characteristic nontrivial projective degree is at least P^{cm} for field parameter P and absolute $c > 0$, by the argument of [Lemma 9.5](#); in particular the same exponential bound works for permutation degree. These bounds on nontrivial contributions work also upon restriction to perfect central lifts/quotients, pulling to the universal covers: lift projective actions there and take a nontrivial composition factor (there is one by perfection if the projective action is nontrivial). The center acts scalarly on irreducibles. In particular minimum permutation degree can be tested this way by a permutation representation in another characteristic (the high perfect central lifts themselves then have no nontrivial small permutation image). We use the classification and covering-group structure in [Convention 9.2](#). An alternating simple of high degree acting projectively nontrivially in degree v has alternating degree $O(v)$, by the following elementary argument. Write its degree as n , choose $\ell \in \{3, 5\}$ different from the representation characteristic, and take $r = \lfloor (n - 2)/\ell \rfloor$ disjoint ℓ -cycles. They generate $E \cong C_\ell^r$, leaving two spare letters. The normalizer in A_n induces every permutation of these generators: a permutation of the ℓ -letter blocks has its parity corrected, when necessary, on the two spare letters. For matrix lifts of the projective action of E , scalar commutators define an alternating bicharacter $b : E \times E \rightarrow \mu_\ell$. Conjugation preserves b . Interchanging any two generators therefore makes their value equal its inverse; because ℓ is odd, that value is one. The lifts commute. Rescaling them makes their ℓ -th powers one, and, since ℓ is prime to the characteristic, they are simultaneously diagonalizable. The projective action of A_n is faithful by simplicity, so its restriction embeds E in the ℓ -torsion of the diagonal torus modulo scalars. Consequently $r \leq v - 1$, and $n \leq 5v + 1$. $\square$

Lemma 11.31 (Coefficient twists on finite covering groups). *Let a classical finite covering group be defined by a standard pinned graph-field map of height f_0 . Coefficient twisting a defining-characteristic irreducible representation by p^j is equivalent, after restriction, to twisting its rational highest-weight lift by the source split p^j -Frobenius. Thus the nonzero digit positions are shifted by j modulo f_0 , with the diagram action at each wrap. The same comparison may be made on composition factors. Equivalence of fixed projective actions on a perfect covering group implies linear equivalence of their lifts.*

Proof. With a standard pinned graph/field map of height f_0 , coefficient p^j -twisting a defining-characteristic irreducible is equivalent on the finite cover to twisting its rational highest-weight lift by the split p^j -Frobenius of the source before restriction. Indeed in split pinned coordinates, transforming coefficients in the rational representation together with the coefficients in the argument (i.e. conjugating by coefficient Frobenius on points of target and source) still gives an equivalent rational irreducible by highest weight. The split map permutes the standard pinned finite fixed group; as in [Convention 9.2](#), digit positions thus move by j modulo f_0 , with diagram action upon wrapping. This reasoning up to equivalence suffices (also for testing composition factors); on a perfect covering group an equivalence of fixed projective actions lifts to linear equivalence because scalar discrepancies are characters. □

Proposition 11.32 (Maximality of an unequal ordered tensor factor). *Let S be a high-rank classical simple group on V , and let an ordered proper matrix tensor factor give natural dimensions $x < y$, both sufficiently large. In isometry type assume that the factor is selfadjoint. In characteristic two assume that the bilateral factor forms are alternating, and in orthogonal type use the canonical tensor quadratic form. Exclude symplectic type in characteristic two. Then the full stabilizer in S of the ordered factor is maximal.*

Proof. Let Y_1, Y_2 be the supported perfect projective special groups on the two spaces, with the alternating special groups (symplectic groups) as needed on factors. They lie in S . For an overgroup within S take a minimal nontrivial normal subgroup N there. Its lift algebra is semisimple by absolute irreducibility of the supporting action $Y_1 Y_2$, and has no multiple center components: the larger simple action fixes all projectors by its minimum permutation degree, then there are at most x of them (invariant spaces are modules over the larger matrix factor), so the smaller also fixes them. The center is then scalar by perfection and full spanning by Y_1, Y_2 .

Any proper nonscalar matrix factor fixed individually by Y_1, Y_2 gives precisely the original factorization. Indeed lift the projective actions on factor spaces and tensor by perfection; all are irreducible on the universal covers jointly by tensor irreducibility. The lower bounds x, y for any respective nontrivial contributions (and xy if both in one slot) force their allocation to distinct unique slots of a nontrivial split. Their matrix spans recover the factors. Here the separate lifts commute as needed (any residual scalar commutators against a perfect factor give a trivial character); joint irreducibles of the direct product over the closure are outer tensors. The supported matrices originally span the actual factors over F by natural absolute irreducibility, and a trivial projective action on other tensor directions means these matrices belong to the indicated one by commuting with the others. Even for a permuted tensor system, Y_2 on the larger space fixes all slots, and acts projectively on just one (two would cost at least y^2); Y_1 preserves that one and must fix the others since their product dimension is at most x . Thus a proper nonscalar lift algebra for N already forces ordered stabilization by the overgroup. If the algebra is full and N is a power of nonabelian simples, their lift algebras tensor to it by commuting as usual, so multiple components also force ordered stabilization (no swap of unequal degrees). The elementary abelian full-algebra case has the nondegenerate scalar commutator pairing, order v^2 and projective linear centralizer just itself. Its normalizer inside the simple group is far too small for $Y_1 Y_2$ (at most $v^{O(\log v)}$ by automorphisms of the elementary abelian group, while $v \leq y^2$).

Thus otherwise $N = B$ is a single simple with trivial projective linear centralizer and contains Y_1, Y_2 by Schreier. It must be high classical in the same characteristic: bounded Lie ranks are excluded by the minimum faithful projective/linear degree of the larger simple support in any characteristic (use also bounded-dimension adjoint realizations there); alternating type and cross-characteristic classical type are excluded by the exponential bounds on embedding the larger support,

together with the action of B in degree xy . In the alternating alternative the alternating degree would be at most $O(xy)$ by that action but at least exponential in y by restriction to Y_2 . In the other-characteristic high classical alternative the natural dimension there would already be at least exponential in y and its own cross-characteristic degree far exceeds xy . Its natural dimension w is at least $x + y$: restrict its projective natural action to the commuting supports, lift from universal covers of each (commuting linearly by perfection), and count composition factors with nontrivial contribution from each (a joint irreducible with both nontrivial is an external tensor over the closure, costing at least xy). Thus by the stronger degree bound, V is projectively the natural representation of B up to twist, $v = w$: we have absolute irreducibility by the full matrix span, and $w^2/2 - O(w) > xy$.

In this equality case B matches the ambient root series. In ambient A , the supporting representation is not self-dual even over the algebraic closure for the finite supporting special groups; in signed types the form symmetry and dimension determine the series (and type A natural is not self-dual). In characteristic two a symplectic natural B cannot preserve the orthogonal quadratic (see also the field-transfer transitivity test in [Section 12](#); over its natural finite field an invariant quadratic, even allowing coefficients in the closure, would have constant evaluation on nonzero vectors). Such finite-group preservation on perfect linear lifts would be exact. Now apply [Corollary 9.10](#): either $B = S$ or it comes from a proper root of the ambient Steinberg map, of field height strictly between zero and the ambient field height f , fixing the supports pointwise projectively. The latter is impossible by restricted-weight uniqueness: on a supporting group the digit is natural at field exponent zero (parameter p^f), not its twist through such a height even up to duality. Frobenius tests on projective representations here and below agree linearly on universal covers by perfection. This proves the claim. □

The field-transfer and exchanged-pair overgroup arguments use [Lemmas 11.30](#) and [11.31](#) and [Proposition 11.32](#). They complete the remaining classical exclusions in [Section 12](#).

12 Completion of the forward marker theorem

The preceding rounds have made the tested classical descendants irreducible and excluded their direct decompositions. We now remove the remaining field and tensor structures. The order matters: the first tensor argument excludes factorizations with two large factors but leaves a few possible retained plane factors. The field and exchanged-pair arguments then remove those planes, because each carries one of the structures tested there. Once these exclusions are complete, every nontrivial inner subgroup normalized by a previous-good label spans the full endomorphism algebra. This permits the final tensor-system exclusion and the construction of actual normal simple descendants.

Work relative to a fixed path ancestor, with almost simple quotient $S \leq M \leq \text{Aut}(S)$, and retain the ordinary subgroup labels K_r . Its natural space is V , its ground field is F , and $\mathcal{A} = \text{End}_F(V)$, with $v = \dim_F V$. The base already supplements S . The witness labels r_D, g_D , the cut labels C_D , and the axis labels M_D have the relative meaning fixed in [Convention 11.8](#). Every group tested below contains a previous-good axis label; its omission bound was enlarged before the present round as in [Lemma 11.15](#).

The pure-link normal test of [Lemma 11.9](#) applies with lower endpoint g_D , or g_* at a simultaneous intersection. Missing normal powers supply the independently supported subgroups of [Lemma 11.11](#); we do not require them to exhaust the intersection with that power. For disjoint witness supports, a proper increase above their meet must obey the join restriction in [Lemma 11.10](#). These are the interval properties used in the remaining exclusions.

All lower dimension bounds in this section are fixed absolute thresholds, chosen before the marker reservoir. The representation bounds and coefficient-twist convention are those of [Lemmas 11.30](#) and [11.31](#). After the structural exclusions, we select descendants having the additional root and tensor protections relative to every earlier member of a classical run. The absolute path bound will then force the desired point configuration.

12.1 Overgroups of field and paired stabilizers

We first establish the overgroup information needed by the field and paired tests. Under the following hypotheses, a prime-degree field stabilizer or an exchanged-pair stabilizer is maximal. For a field of two-prime degree, every maximal overgroup must preserve one of its two prime-degree subfields. Thus, when two disjoint witnesses have such a field stabilizer as their intersection, they cannot leave room for the third cut supplied by [Lemma 11.14](#).

Proposition 12.1 (Field and paired overgroups in the prepared configuration). *Use the transferred forms of [Section 6](#). Let $D \subseteq A$ be a field containing F , with $[D : F] = n$, where n is prime or a product of two distinct primes, and let $m = v/n$ be sufficiently large. In an isometry type assume that D is selfadjoint; in orthogonal characteristic two assume the irreducible normalization already obtained in the preceding rounds. Omit symplectic characteristic two with nontrivial adjoint on D .*

Alternatively, in an isometry type other than symplectic characteristic two, let $e, e^ = 1 - e$ be exchanged complementary idempotents, with totally singular images when a quadratic form is present, and let $m = v/2$ be sufficiently large. Suppose that the collineation action of each overgroup under consideration is semilinearly irreducible, as ensured here by the previous-good axis.*

Every proper overgroup in M of the full field stabilizer preserves a nontrivial subfield algebra of D over F . Every proper overgroup of the full paired stabilizer preserves that pair. Consequently a prime-degree field stabilizer and an exchanged-pair stabilizer are maximal; for a compositum of two prime degrees its maximal overgroups can only be the stabilizers of its prime-degree subfields.

Proof. In the field case the stabilizer includes the perfect transfer group Y over D , of large natural degree m . In the paired case use the paired perfect $\mathrm{SL}_m(F)$. Its centralizer algebra is $Fe + F(1 - e)$, by natural versus adjoint-dual representations; that for the field case is D .

The full stabilizer contains a supplement to S . To prove the assertion take a minimal subgroup $N \neq 1$ normal in the overgroup inside S . The lift algebra is semisimple, and no center components can be permuted nontrivially by Y , by its minimum permutation degree (at least exponential in an absolute positive constant times mf_Y , where p^{f_Y} is its field parameter with $mf_Y \geq v/2$; the number of center components is at most v). Thus its center lies in the displayed centralizer alternatives (perfection). Any enlargement from scalars already suffices: a larger center normalized by the overgroup then gives the indicated field (inside D) or exchanged pair. Here the high-degree centralizer facts use the transfer/paired representations as in the earlier pruning. Nor can there be a proper nonscalar central simple tensor factor, or multiple simple components of N tensoring to the full algebra: Y must individually stabilize such matrix factors in either setting (by order), acting projectively nontrivially on every factor space, as the centralizer algebras are commutative.

Indeed any such nontrivial projective representation of Y over F has degree greater than $\sqrt{v}$. For the paired group this follows from m ; in the field case lift on its universal covering type, with some nontrivial composition factor. The factors are stable as a multiset under twist of field height $h = \log_p |F|$. Indeed coefficient twisting the lifted representation through this height preserves the projective matrices (taken over F), hence gives the same representation by perfection on the universal cover, and in particular the same composition factors. If that constituent has orbit length d , its restricted highest weight has at least $f_Y/(dh)$ nonzero digits, since the twist circulates them

modulo field height f_Y of Y , where $f_Y/h \geq n/2$. Diagram twisting does not affect the support positions. This costs dimension at least $dm^{\lceil n/(2d) \rceil} \geq nm/2$, counting the orbit constituents with their Steinberg dimensions (use $m^{j-1} \geq j$ for $m \geq 2$, $j \geq 1$). Thus two tensor directions are already impossible.

The full-algebra elementary abelian normalizer inside S is again too small by orders to contain Y . Thus otherwise we have a full-spanning simple $B = N$, with $Y \leq B$ as before (centralizer inside S trivial). The order/degree exclusions again force B high classical of the same characteristic, since Y 's field-height times natural degree is at least a constant times $h\nu$. Let its natural dimension be $w \geq m$, field height f_B , and put $t = f_B/\gcd(f_B, h)$. The absolutely irreducible projective representation in V costs $v \geq w^t$ by Lemma 11.31. Orders compared with Y give

$$hvm \leq C f_B w^2$$

for an absolute constant C . With m large this forces $t = 1$ (otherwise $m \leq Ctw^{2-t}$), then $w^2 \geq vm/C$. Again the representation must be natural up to twist by the nonnatural lower bound of Lemma 11.30.

We detail series and root checks here. In ambient linear type the field support uses the natural linear type over D with its embedding-conjugates and is not self-dual. In unitary type we necessarily have odd n (involution extending the ground involution); the n embedding twists of the natural unitary type occupy distinct digit positions modulo nf , so again cannot match to form a self-dual module. For a pair in unitary type the natural and conjugate-dual of $\mathrm{SL}_m(F)$ at distinct digits likewise do not pair to a self-dual module. Signed types match by the previous tests; in the included symplectic characteristic two field case (trivial adjoint on D), Y is symplectic over D and cannot preserve a nonsingular quadratic even with extended coefficients. For example its lift is transitive on nonzero ground vectors, impossible for such a quadratic by evaluation there (also over the binary field, using polarization or a 3-space). The square-form preservation by a perfect group would be exact. Now all supporting constituent nonzero digits lie at multiples of the ambient f with f_Y a multiple of f (also in the indicated paired representations). No proper smaller positive height could fix the action projectively, even up to duality. To see the digit-position convention explicitly, the field-transfer action over the closure uses natural copies over the embeddings of D extending the ground embedding, hence twists spaced by h , with $f_Y = hn$ in the linear or trivial-adjoint case and $hn/2$ in the involutory case. In the latter, either $h = 2f$ in unitary type, or $h = f$ in bilateral type and the nontrivial involution fixes F , so n is even there. For a pair one uses $f_Y = h$ with natural and adjoint-dual (thus conjugate-dual in unitary type). Wrapping may diagram-twist but does not affect the positions modulo f . A fixing ambient root would intertwine the representation with a coefficient twist by its height (possibly with duality) as natural projective matrices, hence on perfect lifts as above, rotating the positions to nonmultiples of f . This rules out a proper root as before and proves the assertion. In particular prime field labels and exchanged-pair labels in these uses are maximal. For a two-prime compositum the only possible maximal overgroups correspond to prime subfields. $\square$

Lemma 12.2 (Canonical quadratics in symplectic characteristic two). *In the prepared symplectic configuration in characteristic two, replicated failures whose full stabilizers preserve canonical quadratics with the ambient polar form are impossible. This excludes ordered tensor stabilizers with alternating factor forms, nontrivial-adjoint field stabilizers, and exchanged pair stabilizers as soon as the common previous-good label is irreducible.*

Proof. For two disjoint failures whose stabilizers respect such quadratics (transforming as the bilinear form), irreducibility of a common previous-good label forces the same quadratic by square-difference,

yet the two labels join to the top, not preserving it. This applies to tensor stabilizers (factor forms then alternating), to fields with nontrivial adjoint by hermitian trace, and exchanged pairs by their canonical hyperbolic quadratic. One can exclude all such additional quadratic stabilizers as soon as previous irreducibility on the joint support is available, also for use when a later tensor label is known to force one of them (alternation of ordered factors by previous irreducible normalization; for a slot system transitivity will give alternation).

□

12.2 Ordered tensor factors

Proposition 12.3 (Exclusion of two large ordered factors). *At the present forward-test stage, a homogeneous family of failures preserving proper ordered matrix tensor factorizations cannot have both factor degrees above the fixed large-degree threshold. In isometry types the factors are selfadjoint and have the normalized factor forms.*

Proof. The factor-form arguments of [Section 6](#) apply. Take a homogeneous pattern on all k -sets with designated algebra A_D on the retained side (over g_D ; the supported special groups are commuting normals with soluble quotient). For disjoint sets these large supported groups normalize each other; nontrivial intersection would force equality of simple groups, hence algebras and labels. Thus they commute projectively and linearly on perfect matrix lifts, so the designated algebras commute and multiply as independent factors. By coloring one can keep all their degrees small relative to total, e.g. at most $v^{1/10}$, by using arbitrarily many disjoint sets. Now each such factor stabilizer, and the stabilizer of the product on a disjoint pair, is maximal by [Proposition 11.32](#) and supplementation of the socle by the base. Thus we have labels C_D and product label $C_{D \cup E}$ for disjoint D, E , by the three-coatom argument of [Section 11](#) (the aggregate supported group moves its constituent factors). The aggregate small-side algebra $A_D A_E$ depends only on the union: among such small sides it is unique for its full stabilizer, by the supporting-group identification of individually preserved tensor factors in the observation. By repartitioning two sets at a time, products for any larger disjoint family likewise depend only on the union.

Recovering elementary factors.

Consequently A_D tensors into elementary matrix factors T_i indexed by $i \in D$, independent of D (scalars allowed). Here are details for $k \geq 2$. At a fixed B take external mutually disjoint k -sets B_i for $i \in B$. Repartition using sets X_i each consisting of i and all but one entry of B_i , plus the final set of leftovers. Then $A_{X_i} \subseteq A_B A_{B_i}$; take its coefficient algebra $L_i(B)$ in A_B relative to this tensor expression (generate from all coefficients). These coefficient algebras commute with each other since the A_{X_i} commute and the external factors are independent. They jointly generate A_B , since the leftover-set factor lies entirely in the product of the A_{B_i} 's (repartitioning the auxiliary sets), and by repartition of the whole union including B , A_B now lies in the product of the coefficient-generated algebra with the independent external tensor factors. Thus they are central simple tensor factors (commuting generating algebras have scalar centers and no radicals). For another choice with disjoint auxiliaries they commute across choices at different positions by the same reasoning, so coincide at matched positions by centralizers. Compare arbitrary choices through spare disjoint auxiliaries. And for a different B still containing i , one can use the same X_i, B_i auxiliary choice, getting exactly the same coefficient algebra (take coefficients into the commutant of A_{B_i}). This proves the assertion; these factors are selfadjoint when needed (take coefficients also in the adjoint basis, since the algebras used are selfadjoint). They commute across arbitrary positions, hence are simultaneously tensor independent by central simplicity. At most one can be trivial by label distinctness, and each nonscalar T_i has stabilizer $M_{\{i\}}$ as usual, since it is $A_D \cap A_E$ for $D \cap E = \{i\}$.

(two such paired meets of cut labels, with remaining indices disjoint, join to $M_{\{i\}}$; the stabilizer is proper by the tensor order bound). For $k = 1$ this conclusion is immediate instead.

The aggregate supplement.

Take prescribed large fixed set sizes, on nontrivial positions with degrees again kept small enough relative to total by thinning. For all such sizes product-factor stabilizers on sets P are maximal (large unequal sides), hence $C_{B(P)}, \emptyset \neq B(P) \subseteq P$; they are distinct and thus $B(P) = P$ after homogenizing on a sufficiently large reservoir. Use sizes including $l, 2l$ for sufficiently large even l . For disjoint D, E of size l , $P = D \cup E$, write $J = C_D \wedge C_E, F_0 = C_P$. On the aggregate for P we have the full supported high special projectively, say R , normal in K_{F_0} and lying inside the actual ancestor simple. There is no loss from possible outer cosets: we claim

$$K_J R = K_{F_0}.$$

Indeed transport R as a normal product over initial sites at the product label F_0 , and join to M_P , giving the product label of $K_{M_P} R$ in the exact interval $[M_P, F_0] = \text{Sub}(S_{2l})$. Its meet with the alternating group there is either endpoint on $\text{Sub}(A_{2l})$, by the earlier exact normal test. If the meet is trivial, the projected normal-multiple label in $\text{Sub}(S_{2l})$ itself has order at most two, and thus preserves some l -set there (l even). This puts R inside the actual factor stabilizer by the cut labels, impossible on its high aggregate by the tensor order bound. So it contains the alternating group formally, giving the claim by joining with J .

Degree equality and the characteristic-two exception.

In orthogonal characteristic two use instead disjoint unequal large sizes also chosen as above, with their sum. Then $[J, F_0]$ is formally just a cover (Young subgroup with unequal blocks in the combined symmetric group; any movement of the split yields a crossing transposition by conjugation). But R includes the supported symplectic group on the alternating aggregate (tensor with a further alternating space, with canonical total quadratic). Within F_0 we have a proper intermediate between the split stabilizer J and the upper label by imposing the canonical tensor quadratic on the aggregate, with properness over the split by its supported high orthogonal group and tensor orders. Indeed K_J stabilizes both ordered pieces and acts by semilinear similarities on them, respecting the split's canonical aggregate quadratic (zero on pure split tensors) with the aggregate polar multiplier. Not all of the supported symplectic group respects it. This excludes the case directly.

In the remaining types use our equal-size index sets. If the two actual side degrees differ, [Proposition 11.32](#) makes $K_J \cap R$ maximal and nonnormal in R . This intersection is the full ordered stabilizer there: within R , fixing one piece also fixes its commutant piece on the aggregate. Since $K_J R = K_{F_0}$, intersections with R determine all ordinary intermediates, so $[K_J, K_{F_0}]$ would be a cover. Yet the formal interval has a proper wreath intermediate. Apply [Lemma 11.13](#) at the original sites before the ancestor quotient. It would make the lower site projections normal in the upper ones, and therefore make their intersection with the actual R normal as well, contradicting the preceding nonnormality. Thus all disjoint l -products of degrees agree. Comparing replacements with spare sets makes all elementary degrees agree; finite coloring aligns their form classes.

The final permutation count.

Now under F_0 the label allowing permutations of the chosen slots in P has kernel label M_P for that action. It induces all symmetric permutations. Indeed use the aligned tensor transpositions in the ambient socle, with corrections as follows. For common degree d the interchange has negative-eigenspace dimension $\binom{d}{2}v/d^2$ for sign purposes in odd characteristic, which is even when d is even by the many slots, and also when d is odd and the total dimension even. Linear/unitary determinant can always be corrected (by minus identity for odd total dimension). In orthogonal odd characteristic with even slots the tensor complement of the two has even dimension and square determinant by

additional even slots. The negative-space form determinant modulo squares for the involution is then square by tensoring, giving also spinor membership. For odd slots use isometries on the large complement of all the chosen slots in P for determinant/spinor corrections: tensoring with identity on an odd-dimensional symmetric factor still gives both independent invariants (diagonalize that factor; determinant unchanged and spinor unchanged up to a determinant-dependent scalar class). The full aggregate group exceeds this permutation stabilizer by orders. Thus above M_P there are more elements than the entire formal interval to F_0 , contradiction. $\square$

Lemma 12.4 (The remaining ordered-factor possibilities). *After Proposition 12.3, the missing side of an ordered-factor failure has large degree. Its retained side can only be a linear plane over $\mathbf{F}_2$, an odd-characteristic bilateral orthogonal plane, or a symplectic plane over $\mathbf{F}_2$.*

Proof. For remaining ordered-factor failures the missing side must still be of large dimension by the missing-site bound: the normal products of supported groups modulo centers have soluble remaining quotient and at most the two-slot ambiguity, so some missing simple orbit would otherwise have bounded type. (If there are two projective simple components on a side, use a missing transitive orbit thereof; supplementation over g_D gives the top action, here of bounded pair rank.) Thus it is the retained side that is now small. The component list and arguments of the earlier ordered-factor pruning then apply except for retained nuisance factors: a linear plane over two, an odd-characteristic bilinear orthogonal plane, or a symplectic plane over two. Indeed across a disjoint pattern all other retained components lie mutually in the stabilizer labels, with exactly the same normalization and spanning arguments giving commuting distinct component algebras (one or two per label); the cases giving forbidden direct systems there are already excluded. From arbitrarily many disjoint labels then take many distinct commuting slots. Label distinctness indeed requires many distinct slots since the designated slot set determines the designated algebra. The common intersection preserves each label's set of at most two slots (orbit sizes at most two even on the union), so preserves an ordered aggregate of many with still many unused, impossible by Proposition 12.3 on the combined support. This proves the partial exclusion needed. $\square$

12.3 Field algebras and paired projections

Lemma 12.5 (Transfer degree at full simultaneous stabilizers). *For the prime-field witnesses in this round, the transferred natural degree exceeds any prescribed absolute bound after the earlier missing-site tests have been enlarged. The same conclusion holds at the full simultaneous field or paired stabilizers used below, relative to their lower endpoint g_* . Prime-field labels are coatoms. Two disjoint witnesses whose fields commute cannot have a field compositum.*

Proof. For any prime field algebra label under consideration, the large transfer degree hypothesis of the stabilizer observation holds: the transferred special inside the ancestor simple gives, modulo its soluble center, nonabelian simple orbit product(s) with soluble remaining quotient, one of them necessarily missing by the pure-link normal test, hence of high type. To spell out this use of the standard orders/structures: if the module degree were bounded, total ground dimension large would force the extension and its parameter large; permanently soluble dimensions then give only a soluble stabilizer, and otherwise the natural perfect special modulo center gives the simple(s), with at most the split orthogonal four-space ambiguity. Semilinear field automorphisms, similitudes, extra scalars or anti-actions do not affect quotient solubility. We use the same argument for the transferred groups in simultaneous intersections below *when the intersection is the full simultaneous stabilizer as*

indicated, over g_* there. More explicitly, any center lost on passing from such a normal transferred perfect special group to its simple site(s) is soluble normal and retained; if the simple sites are not transitive take a missing orbit (by the normal test and solubility of the rest above the supported products), with boundedly many sites and the lower pure-link endpoint supplying the orbit action by supplementation. Thus the high-type conclusion of the missing-site argument applies; we may demand the transferred natural degree exceed any needed absolute bound, by the bounded-rank test there. Thus field labels are coatoms by [Proposition 12.1](#) and [Lemma 12.2](#). For two disjoint-support instances the intersection J is covered in each, with also a third coatom above it. In particular if the two fields commute with compositum a field, their intersection is the full normalizer of that field, of two-prime degree (equality would give identical labels); its large transfer and [Proposition 12.1](#) rule out the third coatom. For any such application in the symplectic even case with individual trivial adjoints, the compositum also has trivial adjoint. □

Proposition 12.6 (Exclusion of fields, pairs, and nuisance planes). *In the prepared forward-test configuration, the prime-field and exchanged-pair failures are impossible. The nuisance planes of [Lemma 12.4](#) are therefore impossible as well, completing the ordered-factor exclusion.*

Proof. Linear type.

In linear type the prime-field proof with retained projective tori from [Section 6](#) thus works: commutation gives a field by the direct-system exclusion; the only possible noncommuting exceptional pair gives a quaternion M_2 factor over three normalized by the intersection, now forbidden by [Lemma 12.4](#). We thereby also exclude the remaining linear ordered factors (plane over two gives a canonical quadratic field).

Fields with nontrivial adjoint in isometry types.

In isometry types first take field patterns with nontrivial involution restriction. For two disjoint occurrences use $R = F[D_i, D_j]$ as in the previous pruning, semisimple with either center field or two exchanged central components. We give the adaptation. If a larger center field gives an invariant central prime D' , then either K_i normalizes it (forcing $D' = D_i$ by the perfect transfer and centralizer), or the full simultaneous stabilizer of D_i, D' equals J by covering. In the latter, if $F[D_i, D']$ is a field its large transfer inside J centralizes D_j , giving commutation. If not, there must again be two exchanged components, from equal prime degrees, and the full intersection includes the paired special linear group over D_i on the components, normalized there (D' is likewise scalar over D_i on each, by the tensor of the two equal-degree fields, a product of copies of D_i). Its two spaces are dually totally isotropic and totally singular when relevant by previous irreducibility. The quotient over this paired type of group is soluble, so by the pure-link normal test and the missing-site bound its module degree on each component is again large. Indeed on a pair commuting with D_i use the transferred pairing over D_i , acting by a determinant-one matrix and its inverse adjoint. In the split-algebra use this also respects D' since on each component the two prime fields act by the same field of scalars. Quadratic restrictions with zero polar on the pair spaces in characteristic two are zero by previous-good irreducibility (else the kernel sum gives an invariant proper nonzero subspace), so pairing preservation suffices there. This special lies inside the ancestor simple by perfection in the nonsoluble large-parameter/dimension range, normalized in the simultaneous stabilizer which acts by paired semilinear similarities allowing exchange (thus soluble remaining quotient; dimension one itself would just give solubility). It too centralizes D_j , and its ground-field centralizer commutes with D_i , as in the earlier paired situation: in the now large degree the natural and adjoint-dual constituents over the embeddings are pairwise distinguished, including across the two sides. The same argument applies directly if the center of R has two exchanged components, using their pair

stabilizer which cannot contain K_i by the field centralizer; again its full intersection with K_i must be J .

For center F , $R \neq \mathcal{A}$ since otherwise the intersection has soluble image on the fields with soluble kernel, impossible. Thus R gives an ordered tensor, necessarily with retained nuisance plane side. If R itself is that plane algebra, it cannot contain two distinct fields of the nontrivial-adjoint kind: on an odd orthogonal plane such a field has the two singular lines as eigenspaces over the closure and is uniquely recovered, and over two the quadratic field in a plane matrix algebra is unique. If its commutant is the nuisance plane instead, K_i cannot preserve the tensor (large transfer and field centralizer), so the full intersection of K_i with this ordered tensor stabilizer is J . On the large factor we have the perfect transfer relative to D_i , with again large degree by the missing-site argument (similarity projection on the nuisance side is soluble), supported inside J . Here transfer is on that large factor space for its factor form; in the bilinear characteristic-two setting this is alternating, and the tensor with the nuisance-side identity preserves the canonical total quadratic in orthogonal type. Its centralizer in R is D_i , but would include D_j , impossible. In this transfer-on-a-factor use, the full simultaneous stabilizer acts on that matrix factor by semilinear similarities respecting D_i , so on its factor space one again has only a soluble remaining similarity quotient over the transferred special. On the nuisance plane factor the whole similarity projection is already soluble. Likewise for the simultaneous paired transfers, after fixing field scalars and the two summands, linear action on one summand and the pairing multiplier determine the action on the other. Thus the soluble-quotient tests indeed use the corresponding supported perfect transfer images inside the simultaneous group, without needing arbitrary individual similitudes there to embed unchanged in the ancestor simple.

Consequently any remaining pair commutes. Two distinct commuting prime fields of the same degree generate a split algebra; for homogeneous adjoint type its primitive component projectors are selfadjoint (relative identifications of the fields commute with their indicated adjoints), contrary to direct-system exclusion. Distinct prime degrees give the field compositum contradiction already proved. This finishes the first prime-field case.

Exchanged pairs and trivial-adjoint fields.

Next exclude exchanged pairs $e, e^* = 1 - e$ of idempotents; total singularity when necessary follows as before. Their labels likewise are coatoms by [Proposition 12.1](#). For the algebra generated by two projections from disjoint instances a larger center field would give a prime D' commuting with them and normalized by the intersection. It is not normalized by the first label (large paired group centralizer). The full intersection of the first pair stabilizer with $N_M(D')$ thus equals the pair-label intersection J . Again it includes a paired D' -linear special group of large degree by the same argument. Centralizing the other projection (perfection), it forces commutation of the two projections by distinct natural and adjoint-dual constituents. But distinct commuting exchanged pairs normalized by the intersection give all four cells (zero cells paired oppositely would instead force the same pair), whose sums on adjoint orbits give a forbidden nondegenerate direct split, just as before. A center with two exchanged components likewise would force that impossibility since its pair commutes with both. For center F , their generated algebra is $M_2(F)$ by the scalar $(e - f)^2$ argument for two idempotents e, f . This must have nuisance plane type by the ordered-factor exclusion. An odd orthogonal plane gives at most one exchanged pair, and the plane over two now violates the nontrivial-adjoint field exclusion. This finishes paired exclusions, hence nuisance planes too by the canonical pair or nontrivial-adjoint quadratic field there. Remaining trivial-adjoint prime fields now have the same two-label proof as above (center F immediately impossible since all such ordered factors are excluded).

□

12.4 Permuted tensor systems

We first record the generation calculation that will replace direct-space merging when the objects being combined are sets of tensor directions.

Lemma 12.7 (Merging compatible tensor directions). *Consider compatible sets of tensor directions AB and BC in the prepared classical configuration, with common aggregate degree $\dim B$ sufficiently large. Suppose their tensor-supported natural special groups are available. They generate a group containing the tensor-supported special on ABC . In orthogonal characteristic two, use the canonical orthogonal special on every subset with at least two slots, with product polar form and quadratic zero on fine simple tensors. Symplectic characteristic two has already been excluded by Lemma 12.2.*

Proof. On compatible sets of directions call the exclusive parts A, C and common part B , also denoting their aggregate spaces by these letters. Assume B has sufficiently large degree, and the two direction sets have their full tensor-supported special groups. In orthogonal ambient characteristic two we need only the canonical orthogonal specials there and on the desired union when multislotting (still no symplectic characteristic two ancestor here). Then these groups generate so as to contain the union special; nested cases need no argument, so take both exclusive parts nonempty. Work first on $A \otimes B_0 \otimes C$ where B_0 is a hyperbolic summand of B of defect at most two, or the entire B in linear type. In AB take transfers $1 + N - N^*$ with N directed between signed hyperbolic B -coordinates $s \rightarrow r$ from distinct pairs, and arbitrary A -matrix there. These are in the natural special: the adjoint arrow uses the opposite coordinates in exchanged order, so $X = N - N^*$ has square zero and $1 + X$ is a determinant-one isometry (in odd orthogonal type unipotent of odd order gives trivial spinor norm). For the canonical quadratic in characteristic two the displacement X has totally singular image in the two A -fibers at mutually orthogonal isotropic B -coordinates, by canonical zero on pure tensors; also $b(x, Xx) = 0$ by symmetry and adjunction for its product polar form b , and displacement rank is even. In BC take similarly $t \rightarrow s$ with arbitrary C -matrix L in the transfer arrow. Use three distinct pairs. Their commutator gives the paired transfer $t \rightarrow r$ with product arrow tensoring the two arbitrary matrices, by elementary matrix multiplication; the higher cross terms vanish. Thus we have all such arrows on $A \otimes C$ by linear spanning. In linear type simply use single unpaired arrows with the same calculation. In individual basis coordinates now, even short paired arrows between nonopposite (distinct-pair) coordinates within one B -pair region follow by commuting through a spare pair. These standard short transfers give the split special on $A \otimes B_0 \otimes C$ supported there: ordinary elementary/root generation (in odd symplectic type commutators give also the long roots with factor two, and in hyperbolic unitary type the commutators through an intermediate index to the opposite of the initial one give all trace-zero long parameters). In orthogonal characteristic two they give the split type D generators for the indicated quadratic, which suffices. These are assertions on natural matrices (or their projective images), with identity on the complementary summand inside ABC , and acting by tensor identity in all outside directions.

To see the generator step explicitly, the arbitrary $A \otimes C$ -arrows between distinct B -pairs are additive under transfer multiplication, and the commutator used to obtain them has just the composite arrow $N_1 N_2$ and its negative adjoint when the three pairs are distinct. Choose individual hyperbolic bases on the paired $A \otimes C$ fibers over B_0 , totally singular on each coordinate side (also for the indicated quadratic). Rank-one arrows between distinct big pairs give the elementary paired updates there, then any two distinct individual pairs can use an intermediate signed coordinate from a spare big pair. Thus all distinct-pair root groups are present. If a long root is needed, commutation through an intermediate index with final index opposite to the initial one yields on that last arrow the composite minus its adjoint (higher products still zero), i.e. $2c$ in the standard

odd symplectic coordinates or $c - \bar{c}$ in normalized hermitian coordinates, with arbitrary composite parameter c . This supplies all long parameters even in even-characteristic unitary type. Ordinary root generation of the split special here thus applies (hyperbolic orthogonal uses just type D arrows, including at characteristic two). In linear type the unpaired calculation gives all elementary matrices analogously.

If a small anisotropic part remains, take an isometric copy of it inside B_0 . Using also the perfect special on B (embedded in a given supported group), conjugate to the special supported instead with B_0 moved onto the orthogonal complement of that copy, by Witt transitivity with corrections (the large orthogonal complement supplies determinant/spinor data as in the subspace arguments). Here hyperbolic splitting of B uses its bilateral or hermitian pairing; in bilateral characteristic two it is alternating with no remaining part. A large enough hyperbolic B_0 represents an isometric subspace copy of the small anisotropic part in the other form cases (e.g. embed together with its negative in hyperbolic pairs). The supported embedding on B used for transport lies in a given special by tensoring with identity (form preservation and high-rank perfection). These two resulting large direct spaces within ABC are compatible with large nondegenerate intersection and full sum, so [Lemma 11.28](#) finishes the argument. This proves the tensor merging assertion.

In applying this repeatedly in orthogonal characteristic two, use on every multislot subset the quadratic with the product polar and zero on the fine simple tensors there. It is also zero for pure tensors across any two-part cut there (apply the canonical construction for that cut and compare on a tensor basis), so column groups or their altered versions supply the required specials. Aligned internal fine permutations carry these choices as required; when transporting whole merged sets of at least two columns the form is canonical on column tensors too, so arbitrary column similarities in those transports cause no problem.

□

Proposition 12.8 (Exclusion of permuted tensor systems). *After the preceding rounds, neither a full-algebra elementary abelian normalizer nor a nontrivial permuted tensor-system stabilizer can be a failure label in the forward test.*

Proof. Common tensor refinement.

We now have the same full-spanning property as in [Lemma 6.24](#) for a nontrivial inner subgroup normalized by a previous-good label. Thus the elementary abelian exclusion again works by scalar commutator pairing and inclusion of all retained elementary groups in the common intersection (normal there). Take then the tensor-system labels allowing permutations, with the very same reductions of exceptional small types as there. In the retained-product case we again have minimal normals in the common intersection recovering the slot systems. Otherwise take two disjoint occurrences, with common intersection E , and missing transitive products with rich top action (thus high slot dimensions by the missing-site bound). The supported normal products in E supplied by [Lemma 11.11](#) contain in their images the same base-core monolith (using the lower endpoint g_*). Hence the tuple-intersection commutator argument, followed by full spanning, gives a common matrix tensor refinement by intersections assigned to slot pairs. Concretely use $N_i = \prod_a R_{i,a} \trianglelefteq E$ of singly supported subgroups in the two missing products as supplied above (the projective tensor supports here have one nonabelian simple per slot). In the quotient by the core of K_{g_*} , assign each simple factor of the monolith W for each i to some $\bar{R}_{i,a}$ containing it, using $[W, \bar{R}_{i,a}]$ since $W \leq \bar{N}_i$. Then $\bar{N}_1 \cap \bar{N}_2$ still contains W by commutators. Just as before take intersections of the assigned R_{i,a_i} with $N_1 \cap N_2$, normal there and their images containing the assigned simple by commutators; once again taking their mutual commutators gives its containment in the image of $R_{1,a_1} \cap R_{2,a_2}$. Thus the tuple intersections generate a nontrivial E -normalized subgroup and jointly

span fully. Distinct tuples have matrix lifts commuting linearly by their distinct slot supports in some system. So their nonscalar lift algebras are commuting central simple tensor factors as before (one per occupied slot pair), recovering each coarse algebra by containment and fullness. Use the same selfadjoint/form observations as before. The fine slots have uniform degree $h \geq 2$ and forms aligned up to scalar by previous-good transitivity; each row or column contains at least two by block primitivity in E (supplementation supplies transitive top on each system including at least all alternating permutations). Indeed a singleton width would occur throughout that system by transitivity and the other system would then group it into blocks, hence coincide by primitivity and having at least two slots in both, impossible for disjoint supports. In orthogonal characteristic two the fine forms are alternating and the total quadratic is canonical, zero on fine simple tensors; this canonical quadratic is also the one for grouping by *any* bipartition of the fine slots (uniqueness from values on a tensor basis). Here [Lemma 12.2](#) excludes symplectic characteristic two.

A local alteration of columns.

Use u the full row-system label intersected with the fine-system stabilizer, a proper decrease (order on any high row slot) above E . Within u we can interchange two fine slots of a row or cycle three, leaving others in place as slots. Indeed use aligned permutations, commutators if at least three. For a transposition the determinant/spinor arguments given earlier work since there are at least four fine slots, using internal corrections on another slot for odd orthogonal degree (then h odd, at least three, repeated with odd multiplicity). In characteristic two orthogonal type the tensor swap has even displacement rank by the even h and at least four slots. Thus one column gets altered by a replacement, remaining in large overlap with its original version (common product degree at least the square root of the column degree).

Generation and the join contradiction.

By [Lemma 12.7](#), adjoining u to the column label merges a column with its altered version. If the gained fine direction within another column has large degree, merge immediately with that column; otherwise transport to its other slots by aligned within-column alternating permutations (its fine-slot count then at least three) and merge extensions via their large common column. As with direct systems conjugation under the column label then gives merged pairs on a connected graph of columns by primitivity; overlapping edges (and enlarging unions) merge with at least a large column in common, giving the full special projectively, thus the ancestor simple, contradicting [Lemma 11.10](#). Thus these last systems too are excluded. □

12.5 Grounded descendants and uniform thresholds

Proposition 12.9 (Grounding at bounded omissions). *After the prescribed rounds, every tested descendant has an actual nested normal simple supplement with soluble centralizer in the base. Relative to a high classical ancestor it is absolutely irreducible and avoids all the excluded tensor structures. Relative to an alternating ancestor it is primitive and transitive unless the desired point configuration has already been obtained. Its simple type can be required to exceed any fixed bounded order or bounded-rank threshold.*

The same actual normal simplicity and full spanning apply to a label above a previous-good axis and below the ancestor when it has the indicated nonabelian part: either a contained perfect simple group as used here, or its own pure link satisfying the normal test. In the first case that perfect group lies in the resulting normal simple.

Proof. First extract a normal simple inside the ancestor socle, using the structural conclusions just proved. In classical type this is the minimal-normal argument of [Theorem 6.30](#): elementary

abelian normals have been excluded, while multiple nonabelian simple factors would give a forbidden tensor system through their commuting lift algebras. The remaining actual normal simple spans the whole matrix algebra. Its centralizer in the ancestor almost simple group therefore injects into the soluble outer quotient. In alternating type the primitive, non-affine, non-Cartesian and non-diagonal conclusions give an almost simple group, as in [Theorem 7.11](#).

In either case the tested label is soluble over this simple, also after pulling back the ancestor's soluble centralizer. The simple lifts inside the actual ancestor socle, where the quotient map is faithful; hence successive simples really are nested subgroups. For an axis descendant the pure-link normal test has lower endpoint H . It puts the soluble centralizer in H and forces the simple to be missing: if it were retained, the base-core quotient would be soluble, contradicting its nonabelian monolith. Thus the axis label is supplemented by this actual simple.

For a general label above the previous-good axis, the same extraction applies when its own pure link or a contained perfect simple group supplies the nonabelian part. We need no supplementation assertion with the original base in the latter case. Solubility of the quotient over the extracted simple already forces every such contained perfect group into it. This is the conclusion used at the third coatom in the protection arguments.

Finally, at axis descendants the missing-site product test applies to their actual simple supplements inside the original S , and [Lemma 11.7](#) makes their types exceed any fixed threshold. Every link and intersection used here has bounded omissions; the earlier exclusion bounds were enlarged to make those intersections previous-good. □

Remark 12.10 (Choice of thresholds). In arranging these rounds the lower high-dimension requirements in the merging, large-factor and projection calculations are fixed thresholds, not functions of the field parameters. For instance the bounded-codimension grouping/collapse of near direct summands needs only an absolute number of transports (separate off bounded small-space dimensions first, and in the high smaller dimension use the full special orbit argument); choose the lower dimension cutoffs there accordingly. When forcing actual missing sites to be large, take the bounded-type threshold in their product tests also large enough for subsequent requirements such as large column dimensions/degrees (and hence large overlaps after moving one cell). At full simultaneous field or paired stabilizer intersections a bounded module dimension would force large field size in their transferred actions by the high total dimension; the resulting missing-site tests still have arbitrarily long pure links by the original residual marker count. Product transfers of actual simples at axis descendants likewise require only that many residual markers, not an a priori classification of all product labels at that simple. The counts of indices/replicas used together depend on fixed tests and the subsequent bounded path/omission requirements, not the sizes of those simple groups. Thus the enlargements backwards and homogeneity relative to path prefixes as at the start suffice here.

12.6 Selecting protected extensions

We now use the actual nested simples and the path-length bound of [Definition 9.1](#) and [Theorem 9.21](#). The protections to be imposed on each continuing high-classical run are a single tensor atom and absence of a fixing prime root, relative to every earlier member of that run.

Write $S_i = B_D$ for any earlier member of the run, including the current parent, and work above a large supply of fresh ordered positions. For a trial extension at the parent with omission set $W \supseteq D$ take increments of prescribed bounded sizes, all beyond previous positions. At trial nodes of matching (continuing) characteristic and classical type relative to D , we have a single cycle under

σ on tensor atoms as before, by the structure grounding. The orbit, label and counting arguments of [Lemma 9.7](#) and [Propositions 9.8](#) and [9.9](#) apply, only now heights up to K_{M_D} above K_{M_U} are bounded by

$$O(n \log \log(n + 3)), \quad n = |U \setminus D|.$$

Lemma 12.11 (Reduction to singleton protection failures). *With the bounded predicates homogenized relative to the allowed ordered prefixes, choose trial increments $a < b$ of matching continuation color and with b sufficiently large compared with $(|W| + a)^4$. If a protection failure survives at a continuing trial extension, then one obtains arbitrarily many coatom witnesses on distinct singleton positions, of the same protection kind, corresponding to distinct primes. All these witnesses are counted for one fixed later descendant B_U .*

Proof. Use preliminary homogeneity for a bounded number of comparisons: existence of the actual path socles and their types/continuation relative to prefixes, the structure exclusions, and whether a bad protection label for (D, L) exists for a specified ordered sublist support (and of which of the two kinds), suffices. Choose extension sizes $a < b$ from W having matching color for continuation from W , with b very large compared with $(|W| + a)^4$ (use three spaced sizes). If not continuing at a , use that next size with no new run protection. Otherwise take $W \subset L \subset U$ with these increments, both continuing. For a run ancestor D , any bad prime-coarsening or prime-root label for B_L (here B_L denotes the simple at M_L) occurs among those counted for B_U . Indeed $B_U \leq B_L$ by actual normal simplicity and soluble centralizer as before. Individually invariant tensor slots for B_L must group atoms for B_U , with just the unique prime congruence coarsening at that prime; and fixing roots still fix B_U . Their stabilizer/normalizer labels for B_L are proper below K_{M_D} and contain K_{M_L} . If one such bad case occurs, we can vary the *entire* list $L \setminus D$ (even the positions from $W \setminus D$) within $U \setminus D$, retaining by homogeneity the same test relative to D and ordered omitted support. These altered intermediate labels still have the indicated actual data, and their simple groups contain B_U by the same soluble-quotient argument (the altered M_L still contains M_U , even when no longer under M_W). Indeed we are comparing throughout relative to the fixed D , which still comes first in index order. If the nonempty omitted support of the bad label has size at least two this gives at least $(n/(|W| + a))^2$ different supports (vary $L \setminus D$ over all subsets of that size in $U \setminus D$; each specified support of size z can then occur with probability at most $(|L \setminus D|/n)^z$), violating the combined atom-prime and root count $O(1 + n \log \log(n + 3))$ against the stipulated spacing. If singleton it gives arbitrarily many coatom labels omitting distinct single positions x , relative to D , of the same kind, still from coarsenings or roots for this same B_U . A singleton omitted support leaves no choice of side group on it. In the tensor case there is just one possible such full stabilizer per prime in the count for B_U , and in the root case at most two (height and diagram uniqueness), so we still get arbitrarily many distinct primes. □

Remark 12.12 (The third coatom). Take two of the singleton coatom witnesses from [Lemma 12.11](#). The following common setup applies to the two possible kinds.

For two such coatom labels we also have a third distinct coatom over their intersection in the interval under M_D , namely the cut on their position pair. Write K for this third coatom group in the almost simple ancestor. It lies above K_{M_U} so has full spanning and structure avoidance there. Thus it has an absolutely irreducible normal simple $B \leq S_i$ containing any perfect subgroup in $K \cap S_i$ by Schreier and the trivial inner centralizer in S_i . We will use these assertions when K contains the indicated high perfect group; they follow by [Proposition 12.9](#). Write v for the ancestor natural dimension.

Lemma 12.13 (Singleton root failures). *The singleton coatom witnesses in Lemma 12.11 cannot be normalizers of fixing prime roots.*

Proof. In the root case take distinct primes r, s . Their roots fixing B_U combine to Θ with $\Theta^{rs} = \Phi_i$ and the two given roots as powers as in Proposition 9.9. Now $Y = S_i(\Theta) = (G_i^\Theta)'$ lies in the intersection. Its presence forces B high classical in the same characteristic by Lemma 11.30, and the natural dimensions compared both ways force B to have dimension v and act naturally up to twist in the ancestor. The root series agrees by the same form checks using the natural Y . In particular if the ancestor here is symplectic in characteristic two, its natural subfield Y from Θ cannot preserve a polar-nondegenerate quadratic even over the closure, excluding orthogonal B acting naturally there. In natural subfield coordinates its symplectic matrices on the finite cover act transitively on nonzero finite vectors (invariance of a putative quadratic would be exact by perfection); a constant quadratic value on those vectors forces zero polar form (for the binary field use three independent vectors and polar bilinearity). In orthogonal ambient characteristic two the preservation test instead excludes a natural symplectic B itself the same way. Hence $B = S_i(\Psi)$ with $\Phi_i = \Psi^e$, $e > 1$. Applying Corollary 9.10 also to Ψ fixing Y gives $\Psi = \Theta^h$, $eh = rs$. Lifts normalizing B commute with Ψ by uniqueness with given outer part. Thus K already normalizes the simple of one of the two given roots, impossible. $\square$

12.7 Natural recognition in the tensor-coarsening configuration

The root case supplied a subgroup already acting naturally on the ancestor module. For tensor coarsenings the subgroup in the two-cut intersection instead acts as a tensor product of spaced natural twists. We must prove that the normal simple group in the third coatom still acts naturally on that module. The following argument uses this particular tensor configuration throughout.

Choose two distinct odd primes r, s , leaving many unused primes at least as large. Put $d = rs$, $m = v^{1/d}$; thus m is very large even compared to d^4 by the remaining prime factors in the atom cycle (the B_U -atom degree is at least two, raised to the cycle length divided by d). In particular throughout the following bounds we can impose sufficiently large absolute constants and $m \gg d^4$, by using a sufficiently large fixed number of distinct labels. In their intersection is the high derived fixed group Y on the d -slot coarsening, as in Proposition 9.8. On the ancestor module over the closure it acts projectively irreducibly with lift tensoring d natural modules of dimension m at twists spaced by the ancestor field height f , allowing duals. Indeed the cyclically embedded fixed group projects on one slot with parameter p^{df} , and the remaining slot actions are its successive coefficient twists by the ancestor map, as described there. Restricted weights have these d distinct natural (or dual) digits, hence the individual d slots are exactly its tensor atoms. If the ancestor is symplectic in characteristic two, the two coarsenings already give canonical quadratics (polar the given form, zero on their respective simple tensors). These are coefficient-ground rational by Frobenius invariance, and the respective labels respect them up to the semilinear similarity action: maps preserving a system of matrix tensor factors permute pure tensor lines (apply factor identifications to the induced semilinear matrix-algebra maps; an intertwiner after that is scalar). The quadratics are equal by square difference and absolute irreducibility under B_U , using perfect lifts for exact invariance. Their joining labels cannot then generate the ancestor (the natural symplectic group here does not preserve such a quadratic). We exclude that case here.

Thus, for the remaining argument, $d = rs$ with distinct odd primes, $v = m^d$, and m is larger than a sufficiently large absolute multiple of d^4 . The group $Y \leq B \leq S_i$ has parameter p^{df} , and its action on V has precisely d natural or dual-natural atoms at positions $0, f, \dots, (d-1)f$. The notation B still denotes the normal simple in the third coatom group K , supplied by Proposition 12.9.

Both Y and B act irreducibly on V . We first show that B is high classical in the same characteristic, and bound the support of the highest weight for its action on V . Those bounds then prove irreducibility of a different module: the natural module of B , restricted to Y . This permits us to lift the inclusion to algebraic groups and compose rational highest-weight representations. The ensuing universal cuts and weight-multiplicity comparisons force the action on V itself to be natural.

Lemma 12.14 (A single sparse restricted weight). *In this configuration, B is high classical in the same characteristic, with natural dimension $w \geq m$. Its action on V has highest weight $p^x \lambda$, where $\lambda \neq 0$ is p -restricted. Every active type- A node of λ defines a cut whose smaller side has fewer than $2d$ entries. In a signed type, its dominant word has fewer than $2d$ nonzero coordinates and has no spin support.*

Proof. Again B contains Y and is high classical in the same characteristic ($v = m^d$ is small compared with cross-characteristic/permutation bounds for Y). Its natural dimension w is at least m , by the exact minimum-degree bound in Lemma 11.30. Its own tensor atoms on the ancestor module are again one σ -orbit by tensor-system avoidance. They group those of Y , so if nontrivially split, K already lies below one of the two given coarsening labels, impossible: a prime congruence coarsening here would have to come from the d -cycle of Y -atoms, necessarily at r or s and coinciding with the corresponding system used above. Thus its lift there has highest weight $p^x \lambda$ with $\lambda \neq 0$ p -restricted. At every active node of λ in A the cut size or its complement has size less than $2d$ (Weyl orbit with binomial lower bound versus m^d). In signed types there are fewer than $2d$ nonzero coordinates of the dominant word, and no spin supports. Indeed with rank e and k nonzero coordinates the orbit has size at least $2^{k-1} \binom{e}{k}$, and use $e \geq (m-1)/2$, $m \gg d^4$: on $2d \leq k \leq e$ log-concavity puts the minimum of the bound at one of the endpoints, both too large. Likewise $\binom{w}{2d} > m^d$ in A . $\square$

Lemma 12.15 (Irreducibility on the natural module). *Under the preceding hypotheses, the projective natural B -module remains irreducible on the finite group Y over the algebraic closure.*

Proof. Suppose the projective natural module of B is reducible on the finite group Y over the closure. In linear type this puts Y in a proper parabolic. In a form type an invariant subspace either gives such a parabolic, gives a proper nondegenerate splitting, or has a nonsingular radical line in the characteristic-two orthogonal case. We treat the parabolic case first, then the nondegenerate splittings with two positive ranks, and finally the line cases.

A proper parabolic preserves the filtration of every nontrivial rational highest-weight module defined by a noncentral cocharacter (on the covering group if needed). That filtration is nontrivial: the projective kernel cannot contain the cocharacter torus, so distinct cocharacter weights occur. Parabolic containment would therefore contradict irreducibility of $L_B(p^x \lambda)$ on Y . For an invariant degenerate subspace in a form type, its radical gives singular or isotropic containment; in characteristic-two orthogonal type take the kernel of the quadratic form on the radical, leaving only a nonsingular radical line if that kernel is zero. It remains to exclude the nondegenerate and line cases. We may suppress the overall weight scale p^x in the following comparisons: both the extremal restrictions and the multiplicities scale by the same factor, by Steinberg's theorem.

Use the connected part of each indicated stabilizer on points, with covering groups: the projective finite source is perfect; component quotients for nondegenerate splittings here are soluble (independent isometries on complementary summands, each connected up to at most orthogonal component data; one can work on exact form-preserving lifts). Thus a connected covering subgroup above it would also have to act irreducibly. One can take the reduced identity component over the

indicated connected stabilizer using the ambient central covering: it surjects onto that stabilizer (finite surjection with finitely many components and connected irreducible target), by a central isogeny, and projective invariance on its action already tests linear subspace invariance. Ordinary nondegenerate splittings use the product of the two connected classical groups on points (up to central quotients/covers, which do not change their Weyl actions), including an orthogonal two-space torus if needed. For a nondegenerate splitting with both ranks positive, maximal tori of the two parts use all ambient coordinates, except possibly one unused in two odd parts. Among restrictions of ambient orbit weights of λ , those with zero on any missing coordinate have full maximal Euclidean norm (coordinate projection and highest-weight convexity), yet are not transitive under the two-part Weyl group, since the sparse nonempty support and spare zeros can be differently allocated. For instance one block has room (in torus rank) for the whole support, and then a nonzero coordinate can also be reassigned to the other. This contradicts irreducibility even on the corresponding connected group, whose simple has all maximal-norm weights in one orbit by the same convexity. In these coordinates one takes hyperbolic pairs on each part separately; if both are odd orthogonal (odd characteristic), combine the two leftover directions as one ambient pair over the closure, unused on restriction. The Euclidean norm in the surviving torus coordinates is Weyl-invariant for the product too, and pulling its torus back isogenously does not change this real coordinate calculation.

Line stabilizers.

For a line in odd orthogonal ambient type B (odd characteristic), restriction of tori to D uses the same coordinates, with even-sum root lattice. If k is the last positive position in λ , sparsity gives $\lambda_k = a$ with $0 < a < p$. Short root lowering gives $\lambda - \epsilon_k$ as a weight (pairing $2a$ nonzero in the odd characteristic), already impossible by parity (the difference after scaling is still odd-sum). For a line in type D, including the nonsingular line in characteristic two, use the standard B_{e-1} stabilizer embedding, dropping the last torus coordinate ϵ_e . Here is some detail for characteristic two. In a split hyperbolic basis z_i, t_i take the nonsingular vector $u = z_e + t_e$. On points the connected special isometry group's line stabilizer fixes u and injects on $u^\perp / \langle u \rangle$ into the symplectic group: kernel elements must fix the whole hyperplane by the quadratic, and then only the nonsingular symmetry (odd displacement rank) could remain besides identity, outside the special group. A connected subgroup R fixing u has full image there: take the torus on the first $e - 1$ hyperbolic pairs, their orthogonal root groups, and for each $i < e$ combine the two end-pair root groups of roots $\epsilon_i - \epsilon_e, \epsilon_i + \epsilon_e$ with identical parameter a_0 (and similarly in the opposite orientation). Their product sends z_e, t_e to $z_e + a_0 z_i, t_e + a_0 z_i$ respectively and t_i to $t_i + a_0 u + a_0^2 z_i$, fixing the rest. This gives the long symplectic root groups on the quotient by squaring (surjective on points); all other required roots come from the orthogonal ones already used. Take the connected subgroup generated (with Zariski closure). Thus R has the whole stabilizer's points and dimension of that symplectic group, and is semisimple (a connected solvable normal would have trivial image). Relative to the indicated maximal torus the generators just given have roots $\pm \epsilon_i, \pm \epsilon_i \pm \epsilon_j$ on $e - 1$ coordinates (nonzero differentials also for the combined roots), exhausting the dimension. The displayed torus is maximal by the finite kernel and rank on the symplectic quotient. So this is type B_{e-1} with the claimed coordinate restriction. We can use its simply connected cover lifting into the ambient cover (central isogeny lifting as in the path proof). In odd characteristic this is the ordinary odd-dimensional special orthogonal stabilizer on the complement up to components/centers.

For the line comparisons these character coordinates on restrictions can be used on the simply connected covers: they come from the vector weights on the indicated surviving pairs. In particular for the special stabilizer just described the characters $\epsilon_i, i < e$, on its torus pull back as the short-root coordinates of type B_{e-1} , and ϵ_e becomes zero even on the cover (its vector pair is fixed by this torus). Lifting from a simply connected covering source through an ambient central covering agrees projectively and hence for these linear vector matrices by absence of scalar characters.

Again let k be the last positive position and put $a = \lambda_k$, so $k \ll e$ and $0 < a < p$. Root lowering with pairing a gives two distinct ambient weights. On the line stabilizer they have the same restriction, since ϵ_e restricts to zero:

$$\begin{aligned} p^x(\lambda - \epsilon_k + \epsilon_e) &\longmapsto p^x(\lambda' - \epsilon_k), \\ p^x(\lambda - \epsilon_k - \epsilon_e) &\longmapsto p^x(\lambda' - \epsilon_k). \end{aligned}$$

Here λ' is the restricted highest weight; extend a positive-chamber cocharacter by zero in the last coordinate to verify that it is highest. The two ambient weight spaces are independent, so this restricted weight has multiplicity at least two. We show that an irreducible module of highest weight $p^x\lambda'$ has multiplicity at most one there.

After removing the scale, the relevant top Levi-face module has highest weight $a\omega_1$ in tail type B, since the difference down by ϵ_k uses just those tail nodes. The simple $L(a\omega_1)$ is a composition factor of the degree- a symmetric power of the natural B Weyl module, also in characteristic two. The ordinary Weyl character of that natural module has precisely the signed coordinate pairs and one zero, independently of characteristic. Its symmetric power has a unique top weight, giving the stated constituent, and multiplicity exactly one at $(a-1)\epsilon_1$: one zero-weight term and $a-1$ copies of the first coordinate are necessary. This bounds the multiplicity in the simple constituent by one and proves the contradiction. $\square$

Lemma 12.16 (Rational lifting and the composed digits). *The restriction in Lemma 12.15 is the restriction of a parameter-restricted rational irreducible of the simply connected algebraic group of Y , of highest weight η . It gives a morphism into the simply connected algebraic group of B , preserving the required forms. The rational composition with $L_B(p^x\lambda)$ is irreducible. After removing the common scale p^x , its highest weight μ has a single natural or dual-natural coefficient one at each nonzero digit.*

Proof. This natural restriction lifts as a restricted irreducible for the parameter of Y , with rational highest weight η . It gives a morphism from the simply connected algebraic group of Y into that of B , as in Proposition 9.15. The invariant bilateral form, when relevant, is rationally invariant by self-duality and uniqueness. For the quadratic in characteristic two, if η has multiple digits, their self-dual factors have alternating forms and hence supply the canonical quadratic, equal to the given one by uniqueness. For a single digit, untwist to a p -restricted weight using the standard pinned choices; split Frobenius permutes the finite points. The cocycle for the finite-invariant quadratic then takes values in the square module of the dual, a nontrivial P_Y -restricted simple module because $P_Y = p^{df} > 2$. By Lemma 9.13 its rational cohomology class is zero. Thus the cocycle is a coboundary, and that coboundary is zero because this module has no finite-fixed vectors. This proves rational invariance.

The composed rational module is irreducible by finite irreducibility: on the finite covering points the lift into the ambient simply connected group of B differs at most centrally from lifts of the original finite projective inclusion (same projective natural matrices by construction). Thus on $L_B(p^x\lambda)$ one still obtains projectively the embedding action just specified with its d atoms. Its highest weight, after removing the scale p^x , say μ , has only single natural (or dual-natural) coefficients one at its nonzero digits. Indeed every rational digit factor restricted after twist to the finite group is one digit there and by individual tensor invariance must be an aggregate of the d twisted natural atoms, thus exactly one by restricted-weight uniqueness (an aggregate of two or more has natural/dual digits at that many distinct positions of the finite restricted tuple). Removing the scale is integral by torus restriction. $\square$

Lemma 12.17 (Small cuts force endpoint support). *In this same configuration, λ is supported only at the two endpoints in type A, or at the first node in a signed type.*

Proof. Here is a cut argument to finish. Restrict natural torus weights of B to get the multiset Ω of the module with highest η . As in Lemma 9.16, one fixed Weyl translate of λ maximizes throughout the positive chamber for Y , simultaneously on each active fundamental summand. The restriction of each such translated fundamental weight is itself dominant (its orbit maximum, also against Y -Weyl translates). Indeed Weyl actions on the image torus extend on it via ambient Weyl elements: after transport of the ambient maximal torus by an image normalizer element, conjugate it back in the centralizer of the image torus. Recall the simultaneous maximum exists since at an interior Y -cocharacter a maximizing orbit weight of the upper module must actually restrict to its unique highest on composition (weight convexity), hence maximizes on the whole positive chamber. Each active fundamental summand is then maximal there since for each cocharacter there is a common ambient Weyl element maximizing all summands. Since their positive integral sum is μ , each has support only at endpoints for A-type Y , or the first node for signed Y . Each corresponds to a strict upper cut in Ω as in that result: all selected elements dominate all unselected and there are no identical vectors across the boundary, by highest multiplicity one (in signed type all active nodes here are early with spare zero coordinates). The sum of the selected weights is that fundamental restriction. Take the small side of size less than $2d$; for a co-small A-cut this means the negatives of the unselected values, still with that same sum and strict dominance in the dual multiset instead. More explicitly an exchange across a strict coefficient boundary compares natural weights against each other throughout the Y -chamber. Equal-vector ties across it would produce two distinct ambient orbit weights mapping to the highest vector weight, contrary to multiplicity one. In signed type one similarly flips nonzero coefficient coordinates in checking the maximum (compensating with zero-coefficient coordinates in D), so the positive directed choices lie nonzero in the positive cone and one can make the same strict comparisons in the full natural multiset.

The middle-coordinate Weyl group in type A for Y (permuting positions other than first and last), or signed Weyl group away from the first coordinate, fixes this sum, thus preserves the selected multiset by strict domination across the cut. By its small size, every selected weight must *individually* be fixed (nontrivial orbits cost at least $m/3$). (A permuted selection with the same sum still attains the maximal Y -chamber interior pairing, hence is the same multiset.) This includes the highest η or its dual, thus supported also only at the indicated endpoints or first node as appropriate. Its active simple reflections are then outside the cut, not fixed individually there. Dominating those forces each selected weight's difference down from highest to be supported only at the active node of any such reflection, which by individual fixedness allows no difference at all: the corresponding simple root (endpoint, or first in signed type) is not fixed by the subgroup either. Thus the cut takes just the highest line. We conclude λ has only endpoint coefficients in A, or just the first coefficient in signed types. □

Lemma 12.18 (Multiplicity forces the natural representation). *In the tensor-coarsening configuration above, λ is a natural weight, up to duality. Consequently $w = v$, and B acts naturally on V , up to Frobenius twist.*

Proof. It cannot have both endpoint coefficients $a, b > 0$ (ambient for B then A), giving $\mu = a\eta + b\eta^*$. At the lowest nonzero digit exponent y of η , if Y is type A with just one active endpoint at that digit, μ has both active there, impossible. In fact the two coefficients at that position are nonzero modulo p by $0 < a, b < p$ and the nonzero lone endpoint digit, with no carry into that position from below. Otherwise choose torus coordinates at the full maximum with first two values $\eta, \eta - p^y\alpha_1$, last

two values $-\eta^*, -\eta^* + p^y\alpha_1$ (last highest-coordinate minimum first in this listing). These distinct weights occur by digit activity (also for signed type with $\eta^* = \eta$), and middle coefficients of λ are constant. Here $\eta + \eta^*$ is not proportional to α_1 by the support already found in high rank, so the four entries are available distinctly at the maximum, rearranging the middle coordinates. Lowering at each of the two endpoint nodes of λ gives two distinct weights colliding at $\mu - p^y\alpha_1$, up to the overall scale, impossible by simple-root multiplicity one bounds in the simple composed module.

Thus up to duality $\lambda = z\omega_1$, $\mu = z\eta$. If $1 < z < p$, a coefficient one at any active node of the lowest η -digit would already contradict the required digit shape of μ . For coefficient at least two there we have three distinct nonzero nonopposite weights $\eta, \eta - p^y\alpha, \eta - 2p^y\alpha$ for that simple root (again the endpoint/first-node shape in high rank ensures nonproportionality to the root). Assign these to the first three natural coordinates of B at the maximum (in signed types choose signs using also spare zero-coefficient coordinates). The upper A_2 -face for λ gives the two symmetric-power weights lowering twice (one step to the third coordinate, or twice using the second) colliding again along just α . This forces coefficient one for λ .

All the collisions just used are comparisons in the rational composed simple module. With scales displayed, the first pair of distinct upper weights restricts to the same weight

$$p^x\mu - p^{x+y}\alpha_1,$$

and the symmetric-power pair in the last case restricts to

$$p^x\mu - 2p^{x+y}\alpha.$$

Each lies below the highest weight in a single simple-root direction, where the divided-power PBW bound gives multiplicity at most one, as in [Lemma 9.16](#). Hence the only remaining coefficient is one. $\square$

Proposition 12.19 (Singleton tensor failures). *The singleton coatom witnesses in [Lemma 12.11](#) cannot be prime tensor-coarsening stabilizers.*

Proof. By [Lemma 12.18](#), $w = v$ and B acts naturally up to twist. Its series matches the ancestor by the previous tests: in type A the finite Y digit tuple is non-self-dual, and the symplectic characteristic-two ancestor case has been removed. The third coatom is proper and contains the supplementing base, so $B < S_i$. By [Corollary 9.10](#), a proper root of smaller height fixes B , hence Y .

Fixedness identifies the projective Y -action with its coefficient twist through a height $0 < g < f$, possibly with duality, using the matrix realization of the ambient point maps. These perfect lifts agree linearly. But the support of the finite restricted tuple is

$$\{0, f, \dots, (d-1)f\} \subseteq \mathbf{Z}/(df)\mathbf{Z}.$$

A shift through $0 < g < f$ moves every one of these positions to a nonmultiple of f . A diagram action on wrapping may exchange a natural weight with its dual, but cannot change which digit positions are nonzero. Thus no such smaller-height root fixes Y , giving the required contradiction. $\square$

12.8 Conclusion of the forward test

Proposition 12.20 (Completion of the forward marker argument). *The forward marker assertion of [Theorem 11.6](#) holds. In the prepared configuration, the assumption of no point success at all the prescribed bounded prefixes is impossible once the initial marker set is sufficiently large.*

Proof. Fix the requested number m of point labels. Choose the high-type thresholds first, and let D_0 be the resulting absolute bound in [Theorem 9.21](#). Reserve more than D_0 path nodes. Choose the trial omission sizes successively, with the spacing required by [Lemma 12.11](#); each is a finite function of the bounds already chosen. This fixes a bound on every path-prefix size. The earlier exclusion rounds can now have their further-omission bounds enlarged backwards to cover every bounded union used by a later round. All such comparisons are relative to one fixed ancestor, as in [Lemma 11.15](#).

Finally choose the initial marker reservoir large enough for those finite Ramsey requirements, for m point labels whenever [Proposition 11.16](#) succeeds, and for all residual pure links and missing-site tests. A terminal spare reservoir, and spaced prefixes when needed, provide the fresh indices throughout. Only the bounded case and comparison predicates already specified are colored; these choices do not depend on group dimensions or field sizes.

Start with the actual simple S in the ordinary site interval. At each subsequent bounded prefix the construction either supplies the required m point stabilizers, or continues with an actual nested simple supplement by [Proposition 12.9](#). In the second case the socles satisfy the transitivity and full-spanning properties relative to every ancestor, and [Lemma 11.7](#) excludes all bounded types. A continuing classical run receives both protections by [Lemmas 12.11](#) and [12.13](#) and [Proposition 12.19](#); an alternating or otherwise noncontinuing step needs no new run protection. Thus absence of point success at the bounded prefixes would give more than D_0 actual simple nodes satisfying all the group-theoretic hypotheses of [Theorem 9.21](#), a contradiction. The construction must stop with point success at one of those prefixes. Its prefix bound and reservoir threshold depend only on m and the fixed choices, proving the asserted uniformity. □

13 Weighted templates and simultaneous size readings

A weighted template starts with disjoint finite permutation sets whose cardinalities satisfy selected numerical relations. Its defining subgroup interval is itself finitely representable. Our task is to recover those relations from an arbitrary finite representation of a nontrivial lattice image of the template. The recovered cardinalities may differ from the defining ones. What must survive is their arithmetic, read simultaneously from the parts of one partition. The next section will bound the number of lattice elements independently of the defining cardinalities.

We first pass to an ordinary subgroup interval, allowing either orientation ([Proposition 13.6](#)). Normal and projection tests then produce a forward interval in a nonabelian simple power, with nontrivial base projections and product subgroups at its distinguished cut coatoms ([Theorem 13.19](#)). Applying the forward marker theorem to a pure-marker filter places the whole remaining template in an actual alternating power ([Lemma 13.20](#)). The cut intersections recover a single partition of its permutation domain ([Proposition 13.23](#)); equal totals and squares are read from that partition. Only boundedly many markers are deleted, and the order of each simple site remains at most the original ordinary top-group order.

Literal lattice calculations throughout use the original parameter order, even when actual subgroup containment has the reverse order. The distinction between subgroups in the simple power and their ordinary one-site counterparts is fixed in [Conventions 13.12](#) and [13.21](#).

13.1 The template and its nontrivial images

Definition 13.1 (Units and stock). We use disjoint transitive permutation sets called units; their index set is denoted by T , and symmetric-group notation on a set of indices (when talking about

permutations of letters) uses the entire union of their units. There are singleton markers. On the units we take independent groups with product Y , of these kinds:

- full symmetric groups, including on the markers;
- primary squares of side x , with group $A_x \times A_x$ slotwise, where $8 \mid x$, $x \geq 16$;
- auxiliary units of shape $(x/4) \times 2 \times 2$ (same side-parameter restrictions), with *full symmetric groups slotwise*.

For each primary side there is its auxiliary type (total size x), and for each axis length used there are full reference units of that size. Units are supplied in cohorts of identical type and size. Also, for each non-marker cohort so required (or basic extra full cohort used), include a full offset cohort of size per unit one greater than its total size per unit. These extra offsets themselves need no new offsets. In what follows every cohort, including markers, can be required to have at least a sufficiently large number of members. For the group-structural tests this lower bound can be *absolute*, independent of sizes and number of cohorts. We explain the finite losses involved along with the arguments. Assume there are nonmarker units in this use.

Definition 13.2 (Weighted template). Adjoin a foundation set Z larger than all the letters together. Use the ordinary interval

$$F = [S_Z \times Y, S_{Z \sqcup (\cup T)}].$$

(Here $\cup T$ denotes the union of the units.) The notation for its labels is

$$(I, U) = S_{Z \sqcup (\cup I)} \times U, \quad I \subseteq T, \quad Y_{T \setminus I} \leq U \leq S_{T \setminus I}.$$

The set I is the *footprint*: its units have been absorbed into the foundation orbit.

Lemma 13.3 (Labels, cuts, and restrictions). *The labels in Definition 13.2 are exactly the overgroups in F . Its coatoms are the labels C_D below. Absorption in a join follows residual orbits. Axis filters and axis ideals again give the weighted templates on the units that remain.*

Proof. In any overgroup the conjugates of the supported S_Z generate the full symmetric group supported on the orbit of Z , by overlap (its size before expansion already exceeds half the ambient degree). That orbit absorbs entire units. Absorption in joins thus follows orbits; in particular the join of the x_j 's below will be top. We shall use the following labels:

$$\begin{aligned} X_D &= (D, Y_{T \setminus D}), & x_j &= X_{\{j\}}, & M_D &= X_{T \setminus D}, \\ C_D &= (T \setminus D, S_D) \quad (D \neq \emptyset), & N &= C_T, & J^* &= (\emptyset, \prod_{j \in T} S_{\{j\}}). \end{aligned}$$

The C_D 's are precisely the coatoms (here and below test by absorption, or by the unequal pieces of each cut with the large foundation). The filter above X_D and the ideal below M_D have the templates on the remaining units (in the filter the foundation has grown); inside M_D the independent unchanged Y_D factors out. $\square$

Convention 13.4 (Images and literal lifts). We will work even with an arbitrary nontrivial lattice image $\bar{F}$ of F (surjective lattice homomorphism); often the notation for labels refers also to their images. We call the ones before the map literal labels. Intervals between comparable literal labels map onto the corresponding intervals (join with the lower then meet with the upper to put a lift there); all meets and joins may be tested by first doing the literal operation and then mapping.

Lemma 13.5 (Footprints and faithful pure links). *In every nontrivial lattice image $\bar{F}$, the footprint is well defined. All cut coatoms survive distinctly and are all the coatoms of the image. Residual orbit discrepancies are detected by absorption. The pure marker links, and canonical permutation intervals on copies of any fixed package of units, are faithful.*

Proof. The footprint I is unambiguous in the image. In fact a collapse with changed footprint gives a comparable one (use meets), and for j gained it then identifies $N \vee x_j = 1$ with N , by meeting first with x_j . Indeed x_j is under the upper label, while its intersection with the lower is only residual (empty footprint). Thus the identification kills every x_R for full units R by intersection with N . For a grid unit O , in each direction take a label preserving a partition of $O \cup R$ into the parallel lines of that direction plus a reference block R of the matching size. Use full independent bases, *alternating top* on the line blocks including R , and unchanged Y elsewhere, without absorption. This contains the bottom: on parallel lines the grid groups induce even permutations (for the auxiliary grids both other lengths are even). It joins with x_R to absorb O . Intersecting all these labels with x_O gives bottom: preserving each parallelism forces slotwise permutations and on the primary squares the two alternating conditions enforce both prescribed slot groups. Thus all x_O are killed as well, impossible. In particular the cut coatoms survive, distinctly with no new ones, and orbit discrepancies of residual labels can be detected by joining with absorptions. Here adjoining x_j to a label for an unabsorbed j absorbs exactly its residual orbit in addition to the old footprint. On just markers, the resulting pure lattice, in particular in the filter absorbing everything else, is exact: for a strict change of residual group (on singleton positions) intersect down to a cyclic group generated by a discrepancy element, holding the footprint fixed. A proper subgroup there has a smaller orbit on some cycle (an index divisor hits the cycle length), detected by adjoining the absorption of one singleton. The same test works inside a permutation label acting canonically on copies of a package of units modulo the prescribed internal bottom actions. $\square$

Proposition 13.6 (From algebras to ordinary intervals). *Every nilpotent join-preserving deflation of $\bar{F}$ is zero. If some nontrivial image of F is the congruence lattice of a finite algebra of size m , there is an ordinary finite-group interval representing the dual of a nontrivial image of F , whose top group has order at most $60^m m!$. The full lattice F itself has a finite algebra representation.*

Proof. Let f be a nilpotent join-preserving deflation. It kills every atom: otherwise it fixes that atom, contrary to nilpotence. In particular, $f(x_R) = 0$ for every full unit. If A is any of the parallel-line labels used in Lemma 13.5, then $x_O \leq A \vee x_R$, so

$$f(x_O) \leq f(A \vee x_R) = f(A) \leq A.$$

Also $f(x_O) \leq x_O$. Intersecting over the directions therefore gives $f(x_O) = 0$. Since all the x_j 's join to the top, $f = 0$. The idempotent-restriction and unary-nonunit argument of Proposition 4.8 applies starting with a least carrier representation among all nontrivial images of F . Idempotent restriction to more than one point would still represent such an image since it gives a surjective lattice map; nonunits thus yield the nilpotent join deflation as there. Consequently this minimal representation gives an ordinary finite group interval for the *dual* of a nontrivial $\bar{F}$, by the diagonal construction in Equation (4.3). Its top order is at most $60^m m!$ if the representing size was m , taking the simple group A_5 . Also F itself is representable (the transitive coset action on its ordinary interval). $\square$

13.2 Normals: absorption, parallelism, and signs

We next show that a normal subgroup cannot produce an intermediate label between the base and the top. The tests successively exclude absorption of units, movement between units, disruption of

their parallel partitions, and finally the remaining primary-square signs. All calculations are made with literal labels and then passed to the image.

Convention 13.7 (Normal endpoint property). For an ordinary interval $[H, G]$, property $(*)$ means that every normal subgroup of G is either contained in H or supplemented by H , and that modulo $\text{core}_G(H)$ there is a unique minimal normal subgroup, a power of a nonabelian simple group, supplemented by the base. In the next three results, $[H, G]$ represents $\bar{F}$ in either orientation, and e is the label of HL , with $L \trianglelefteq G$. All inequalities between labels use the parameter order.

Lemma 13.8 (Projection and intersection tests for normals). *A normal-multiple label cannot satisfy*

$$e \wedge b \leq a < b \leq e \vee a.$$

Its axis projections are again normal-multiple labels. If L is minimal normal and nontrivial, a nontrivial proper intersection $L \cap G_b$ cannot be normalized at a family of labels whose actual groups generate G . The equations in the proof test precisely the normalizations needed below.

Proof. Write e for (the label of) a base times a normal, always reasoning with labels in the forward $\bar{F}$ order. Dedekind's law makes it left-modular: in particular $e \wedge b \leq a < b \leq e \vee a$ is impossible in either orientation. Projections to axis filters (by joining in the label order) again come from normals: in actual order enlarging the bottom to K projects HL_1 to KL_1 , and shrinking the top to K projects it to $HL_1 \cap K = H(L_1 \cap K)$, for $L_1 \trianglelefteq G$. Axis intervals used after deleting a bounded number of units are still nontrivial images on the corresponding templates, by interval surjectivity and footprint detection. For the normal tests below we suppose that e is a proper nonendpoint.

Here is also the intersection test used repeatedly below in the normal argument. Write G_b for the actual ordinary overgroup at label b , and $G_e = HL$ for the actual normal multiple, $L \trianglelefteq G$ (similarly after any indicated base-core passage). If $H < G_b < HL$, then $I_b = L \cap G_b$ is nontrivial proper in L , by $G_b = HI_b$. It is normalized at every s for which $\langle G_b, G_s \rangle \cap HL = G_b$, since then it is just $L \cap \langle G_b, G_s \rangle$. The sufficient equation in parameter order is thus $e \wedge (b \vee s) = b$ forward, $e \vee (b \wedge s) = b$ reverse. If L was taken minimal normal nontrivial and the indicated G_s 's generate G (label join forward, label meet reverse), this normalization is impossible. $\square$

Lemma 13.9 (A proper normal label is internal). *If e is a proper nonendpoint normal-multiple label, then $I(e) = \emptyset$ and $e \leq J^*$. The conclusion remains available after any of the bounded axis restrictions used in the proof, and when all unit actions are replaced by full symmetric groups.*

Proof. Absorption and the offset comparisons. Take four disjoint equal-size unions of units in a cyclic order. Absorb the outside and allow independent full symmetric groups on the four blocks. Adjoining the square of their rotation gives a label a , and adjoining the whole rotation gives $b > a$, with strictness detected by orbits. Suppose the four counts of absorbed letters from $I(e)$ are nonconstant, with a positive count in each parity class of positions. An odd rotation cannot preserve these counts, so $e \wedge b \leq a$. But the square rotation connects each parity pair, both of which meet the absorbed part of e ; hence $e \vee a = 1$. This is forbidden.

In any cohort, therefore, either every unit is absorbed or at most one is: otherwise put two absorbed copies in opposite parity positions and a nonabsorbed copy in one of the remaining positions. Each sufficiently numerous cohort consequently has a majority absorption bit. Compare a typical size- s unit together with a typical marker against a typical size- $s + 1$ offset unit. Write their majority bits as ϵ, μ, η , respectively. The four-block test, using both block types and repeating a positive one in both parity classes if their counts differ, forces

$$s\epsilon + \mu = (s + 1)\eta.$$

For bits in $\{0, 1\}$, this implies $\epsilon = \mu = \eta$. Every required cohort and its offset therefore have the marker cohort's bit. If that bit were one, the preceding cohort dichotomy would make every unit absorbed, contrary to $e < 1$. All majority bits are zero.

Finally, two absorbed units from different cohorts are impossible. Use four blocks each containing one copy from each of those cohorts, and put the two absorbed copies in different parity classes, choosing the other copies nonabsorbed. The same test applies. Thus $|I(e)| \leq 1$. These arguments use only sizes, offsets and boundedly many copies, so they also hold with full unit actions and after the indicated bounded axis deletions.

Excluding a singleton footprint. If $I = \{O\}$ and the residual group moves between other units, shift above x_j at the source of such a move. At least two absorptions remain (O and another unit hit from j), so the shift is top by the count just proved (the same count in a filter deleting a unit), meaning the residual group is transitive on the complement. On four copies including O the square test then gives total absorption on join anyway (we reach another unit besides O , then its whole transitive complement orbit), impossible. If instead the residual group preserves units separately, use three copies including O with rotation group $A_3 < S_3$ on the blocks, outside absorbed. This increase is strict (meet with the version allowing just one transposition on blocks, then detect orbits); intersection with e cannot move the blocks while join absorbs them. So $I = \emptyset$.

Excluding residual mixing. By shifting once at a mixed unit and applying the same conclusions, mixing units now forces transitivity on all the residual letters (there is an extra absorption in that shifted label, hence it is the filter's top). These nested temporary shifts delete only boundedly many units before doing the counts.

For such a transitive e , every marker x_i complements it. Normal complementation gives isomorphisms in actual subgroup order (when K complements HL , $L \trianglelefteq G$, project K to G/L to identify $[H, K]$ with $[HL, G]$). Forward this forces $e = N$ by marker atomicity (it makes e a residual coatom). Replace the base by the label J^* , i.e. switch to all full unit groups, with the same transitive projected label. In the new base-core quotient take a minimal normal inside the normal giving that label; it is still missing and proper, so still transitive by what was proved thus far (nonmixing would now mean bottom), hence still maps to N . For a residual marker transposition d over the new base, we have $0 < d < N$, $N \wedge (d \vee x_j) = d$ for every unit, using notation relative to the full-unit base. Thus the intersection of the minimal normal with the subgroup labeled by d is nontrivial proper and normalized by all labels x_j which generate the top, impossible. Indeed inside an actual base times that normal, all overgroups of the base factor as the base times their own intersections with the normal; and here enlarging d by any x_j leaves its intersection unchanged.

In reverse orientation the complementation identifies $[x_i, 1]$ with $[0, e]$ by meeting with e in $\bar{F}$. Thus $X_{\{i,k\}} \wedge e > 0$ on distinct markers, forcing every marker transposition label beneath e (the literal residual action in that intersection only has the two singleton markers available over bottom). A supported marker three-cycle group including i , which lies below e by these transpositions, then cannot be in the indicated meet image: joining it with x_i absorbs all three and they admit transpositions below e . These small permutation labels over bottom are exact by orbit detection. So in either orientation every proper nonendpoint normal label lies under J^* . $\square$

Lemma 13.10 (Parallel restrictions on an internal normal). *After quotienting by the base core, suppose that a minimal nontrivial normal subgroup gives a proper nonendpoint label e . Let B be the intersection of J^* with all full symmetric parallel-block labels. Then $e \leq B$. This case is impossible in reverse orientation. In forward orientation $[0, B]$ is the exact lattice of subspaces of the sign space having two coordinates for each primary square.*

Proof. Pass now to the original base-core quotient and, if a proper nonendpoint occurs, take the normal to be minimal nontrivial therein (within the purported normal), written L . This still gives

a proper nonendpoint internal e (actual base times L); internal representatives can be used since $e \leq J^*$.

For a parallel direction on O , write $A < W$ for the alternating-top parallel label used earlier and its full symmetric-top version, now allowing full internal groups on other units. This increase survives: adjoin a transposition d on two spare markers to both, then intersect with a graph label coupling this swap to odd line-block action, giving different marker orbits, detectable by absorption. Suppose $e \not\leq W$. Some element of an internal representative then disrupts the partition; conjugating supported full bases and using primitive alternating top generates independent full action there (a cross-transposition in addition to the within-block symmetric groups suffices), so $e \vee A \geq W$. Thus $D = e \wedge W \not\leq A$ by left-modularity. Forward, $0 < D < e$ in this situation, and

$$e \wedge (x_k \vee D) = D \quad (k \neq O),$$

since x_k , like the literal internal intersection giving D , preserves O and its indicated lines; on meeting back with the internal representative one thus stays under W . Hence the nontrivial proper intersection of L at D is normalized both at W (where the intersection is the same) and all these x_k (use $x_k \vee D$); together they absorb everything, impossible by minimal normality.

In reverse orientation in fact no internal element of a representative for e can preserve the direction's lines on O and act oddly on them. Otherwise set $K = e \vee d$ with d just a spare marker swap over bottom. This is strictly above e (orbit test) but residual. Take the graph label P' coupling this swap to odd line-top on O , otherwise allowing internal actions preserving units and this parallelism. We have $d \wedge P' = 0$, and $e \vee (K \wedge P') = K$ using the presumed odd element times the swap. Thus the intersection of L at K , nontrivial proper by $e < K < 1$, is normalized at d (by $K = e \vee d$) and at P' (enlarging actually, i.e. using $K \wedge P'$, leaves the intersection unchanged). These generate actual top by $d \wedge P' = 0$. This too is impossible; it rules out in particular $D \not\leq A$ obtained above under a disruption (an odd top element in the internal intersection with W fixes the reference block, hence acts oddly on the lines of O).

It follows in both cases that $e \leq B$, the intersection of J^* with all the W 's. This is slotwise full on the grids and differs from bottom only by primary square sign bits. Reverse has now been excluded (meet to place the representative under literal B , then use the odd-line test).

In the forward case $0 < e \leq B$, so the image of the literal sign-space interval is nontrivial. The subspace lattice is simple: a nontrivial congruence collapses an atom by atomisticity, and every two distinct atoms lie in a plane, where such a collapse propagates to all atoms. Hence this interval is exact. $\square$

Proposition 13.11 (Normal endpoint property for weighted images). *For sufficiently large absolute cohort stock, every ordinary interval representing a nontrivial weighted image, in either orientation, satisfies (*). Its base acts transitively on the simple factors of its monolith modulo the base core. The same assertion holds for the bounded axis deletions used here, provided the remaining stock is sufficient.*

Proof. By Lemmas 13.9 and 13.10, only the forward sign-space case remains. Take a minimal missing normal inside any purported proper normal multiple after passing to the base-core quotient.

Forward, $[0, B]$ must now be an exact subspace lattice over $\mathbf{F}_2$, since that literal interval is simple and its image is nontrivial ($0 < e \leq B$). Here the slotwise groups over literal bottom have quotient given by the sign space with two bits per primary square. Subspace lattices here are simple, e.g. by atomisticity and propagation of any atom collapse across the planes. For an active bit of the sign space of e , meeting with its parallel alternating-top test A gives its kernel D' . If this is nonzero, the same formula $e \wedge (x_k \vee D') = D'$, $k \neq O$, and normalization at A give the earlier forward

contradiction. Thus e has sign dimension one with vector $v \neq 0$. Swapping the two coordinates of any primary square by τ must fix v : otherwise use as a the label adjoining τ to bottom and as b this label extended by the sign line $\langle v + \tau v \rangle$; $e \wedge b \leq a < b \leq e \vee a$, strictness tested under B . Hence at an active square both bits occur together.

Choose an active square and use its x lines in one direction, together with its size- x auxiliary unit as one additional block. Let a have independent alternating groups inside these blocks and the alternating group on their $x + 1$ positions. Match coordinates along the lines and choose any matching with the auxiliary block. Outside this union allow full internal groups, without absorption. The prescribed bottom lies in a . Let b instead allow all base sign vectors of even total, with the same alternating top. Then $a < b$: the base sign which is odd on every original line and even on the extra block lies in b , and its nontriviality is detected by meeting with the exact sign interval $[0, B]$.

Write α for the line-position sign and β for the repeated within-line sign of the chosen square. The preceding swap test gives $\alpha(v) = \beta(v) = 1$. An element of $e \wedge b$ has $\alpha = 0$, since b has alternating top. As e is the single sign line $\langle v \rangle$, its sign component must therefore vanish, including β . The auxiliary bottom action is also even on its letters: its first-slot permutations repeat four times, and each binary-slot swap repeats an even number of times. Consequently $e \wedge b \leq a$.

On the other hand, a representative of the nonzero sign of e has base-sign vector

$$u = (1, \dots, 1, 0)$$

on these $x + 1$ blocks, and may also permute the original lines. Conjugate the supported matched alternating top by this representative, then remove the top permutations using a . Modulo the alternating bases this produces $u - hu$ for every even block permutation h . These differences span the augmentation space: moving the unique zero gives every pair of coordinate positions. The conjugations remain supported on this union regardless of the representative's actions on other units. Hence $e \vee a \geq b$, contradicting the normal-multiple test. This proves the normal endpoint property.

The lattice is nonmodular (the exact pure marker link includes the subgroup lattice of S_4 , already nonmodular: with a 4-cycle r and a dihedral reflection s inverting it, $\langle rs \rangle$ has trivial meet with $\langle s, r^2 \rangle$ but joins with $\langle s \rangle$ above it). For completeness, in the faithful base-coset action every minimal normal is transitive by the endpoint property. An abelian minimal normal would identify the interval with an invariant-subgroup interval in an abelian group, which is modular. Two distinct minimal normals commute; supplementation by one then makes the base induce all inner actions on the other. Intersections in the latter are invariant normal subproducts, again giving a modular interval. Thus there is a unique nonabelian monolith modulo the base core, and it is supplemented. The base is transitive on its simple factors, because the full group is and the monolith acts internally on each factor.

The stock bounds thus far only spent boundedly many copies for parallel tests, rotations, and nested shifts deleting at most a bounded number of units. $\square$

13.3 Product closures in both orientations

Convention 13.12 (Actual groups and product closure). After passing to the base-core quotient we thus have $G = H\mathcal{W}$, $\mathcal{W} = S^s$ a minimal normal nonabelian power, and the labels are exact H -invariant subgroups K_b in $[P, \mathcal{W}]$, where $P = H \cap \mathcal{W}$, transitive operator action on the sites. Thus K_b is written on the simple-power side (actual subgroups over the interval base correspond by intersection with $\mathcal{W}$), not a formal permutation label. In reverse, label joins mean actual intersections, and label bottom means actual $\mathcal{W}$. We use c for coordinate product closure, and call these its product labels when closed. Recall the extension description in [Lemma 4.10](#) of subdirect labels (site stabilizer $D_0 \leq H$, action image $\Lambda \leq \text{Aut}(S)$, extensions to overgroups of D_0 in H);

above a given subdirect group in actual order all are just restrictions on its domain interval, by reverse correspondence. Also, intersections of comparable distinct subdirect actual groups with a fixed invariant product must differ if nontrivial: the restrictions are independent across the tie parts, with nontrivial projections at every site by operator transitivity, so a strict tie change distinguishes them. And a subdirect joined (in actual order) with a nontrivial invariant product gives the whole power, by conjugating independently supported elements and simplicity. These facts and the projection arguments below can use the same transitive operator action on restricting a filter of actual groups (or add the enlarged base to the operators by inner actions).

Proposition 13.13 (Forward proper projections). *In a forward realization $[P, \mathcal{W}]^H$, suppose $P > 1$ and $c(P) < \mathcal{W}$. Every cut coatom is a product label. If $P < c(P)$, its closure label has empty footprint and lies below J^* .*

Proof. Forward, suppose first $P < c(P) < \mathcal{W}$, and put $K_p = c(P)$. Joining a label with p to top forces subdirectness, which also suffices since K_p contains nontrivial independent support. Thus a test $p \wedge b \leq a < b$, $p \vee a = 1$, is impossible by strictness on intersections (both K_a, K_b then subdirect, intersections containing $P > 1$). Exactly the same four-block rotation test gives $|I(p)| \leq 1$. In the singleton case $I = \{O\}$, if any other copy of O is moved in the residual action to touch a different unit, place it opposite in parity from O in the four-copy test, with destination outside the chosen four. The join again absorbs everything. If none are thus moved use the three-copy test instead. Hence $I = \emptyset$.

Shift above x_j ; if its closure is not full the closure label has empty extra footprint there by the same tests (or trivially when x_j is already product). So if p mixes j with other units then x_j must project fully by monotonicity of closure, and $x_j \vee p = 1$, forcing transitivity of the residual group. In that case all marker x_i 's are subdirect and their intersections with p are bottom. Meeting also with $X_{\{i,k\}} > x_i$ forces all marker transposition labels beneath p by strictness. The K_{x_i} 's have identical ties and identical twists restricted to the projections of P , by the common intersection P with K_p . Therefore the images of all the filters above x_i under meeting with p must agree (ties may be split by any invariant refinement, retaining the twists, then taking the product of the correspondingly restricted diagonals on intersection with K_p). Yet a supported marker three-cycle group d' lies beneath p by the transpositions, and occurs in the meet image for i outside its three positions, as $p \wedge (x_i \vee d')$ (p is residual, i is singleton). It cannot occur for i inside by the same absorption contradiction as before. Thus $p \leq J^*$. Now every cut coatom is product, since subdirectness there would join to top with p . The same cut conclusion holds immediately if $P = c(P) \neq 1$.

To elaborate on the common meet-image test in that argument, put $\mathcal{T} = c(P) = \prod T_o$ with the index here ranging over sites. A subdirect K_{x_i} meeting $\mathcal{T}$ in P restricts on each tie part to the full twisted diagonal of the corresponding T_o 's in that intersection: the projection there is exactly T_o since it comes from P . These are independent across parts. Thus whether two sites are tied, and if so the restriction of their identifying isomorphism to T_o , can be read from the two-site projection of P (graph of an isomorphism versus a whole product; $T_o \neq 1$). Above K_{x_i} one may split by every operator-invariant refinement, preserving its twists within the remaining ties; this is again invariant by equivariance of the original ties, and these are all the invariant overgroups in the power here. Their intersections with $\mathcal{T}$ depend only on that same restricted-twist data and the refinement. This proves the claimed agreement even if the full twists were not specified by P . In marker notation with i outside the triple one can bound $p \wedge (x_i \vee d')$ also by $N \wedge (x_i \vee d') = d'$ calculated literally, and it contains d' in the image. $\square$

Definition 13.14 (Product success). We say that *product success* holds when a forward realization in a nonabelian simple power has nontrivial projections of its base intersection at every site and

every cut coatom is a product subgroup. We allow passage to this configuration by deleting one marker in an axis filter.

We next establish in *reverse* orientation that for $P > 1$ we must have $c(P) = \mathcal{W}$, and for $P = 1$ all nonbottom actual labels are subdirect. The $P > 1$ argument only needs an invariant interval with those base and action conditions.

Lemma 13.15 (Subdirect copy actions in reverse orientation). *In a reverse invariant realization with $P > 1$, let z canonically permute k identical nonempty packages of units, acting as the prescribed bottom inside each package. For a sufficiently large absolute k , K_z is subdirect. The same holds for its canonical alternating-copy label once that label is sufficiently large.*

Proof. Take disjoint identical nonempty packages of units and a residual label $z = Y \rtimes S_k$ permuting k copies canonically (fix coordinated matchings of corresponding units respecting their types and actions). Foundation actions are implicit in residual notation; by a canonical subgroup label here we mean the inverse image in z of a subgroup of S_k . The interval $[0, z]$ in label order is exact $\text{Sub}(S_k)$, by the cyclic discrepancy and orbit test on these copies. We claim K_z subdirect for sufficiently large absolute k , here when $P > 1$. Otherwise its closure corresponds to a nonidentity subgroup $U \leq S_k$. Labels indexed by subgroups of U generated by prime order elements are product, since the corresponding atoms (actual coatoms above the nontrivial product K_U) must be so. Hence U cannot contain a large supported A_v : that subgroup is prime-generated and would give a product base $K_{A_v} \neq 1$ for an exact reverse $\text{Sub}(A_v)$ inside this invariant interval, contrary to the reverse alternating exclusion of [Theorem 10.1](#).

If $a < c_0$ in $\text{Sub}(S_k)$ and $c_0 \cap U = 1$, their labels are subdirect (by monotonicity, the label of each closure is below U and below c_0), so $\langle a, U \rangle < \langle c_0, U \rangle$ by intersection strictness with K_U . Thus any transposition moved by conjugation by $u \in U$, together with its conjugate, generates a group meeting U nontrivially. We get some prime-order $h \in U$ supported on at most four letters. Match a nontrivial orbit of h to a fresh ordered set of the same size (2 or 3) and exchange them simultaneously pairwise by an involution. By the very same strict-join test its group together with its h -conjugate (dihedral, two such exchanges) meets U nontrivially. Any nonidentity element there rotates the two sets nontrivially or exchanges them in matched fashion. By finite Ramsey homogenize a large reservoir of fresh points so a constant such pattern on increasing tuples lies in U . Comparing two tuples differing at just the last entry y, y' gives a supported three-cycle by taking quotients: the two permutations differ by conjugation by $(y y')$ and the first (say w) fixes y' and sends y to a common anchor (possibly of the original orbit) distinct from y . Thus one quotient is $(w(y) y')(y y')$. This anchor is fixed as a choice of position when the prefix is fixed and the last entry varies. Varying these cycles gives a large supported alternating group, contradiction. All these Ramsey requirements are absolute (orbit lengths two or three, bounded many permutation patterns, with sufficiently many points for the absolute reverse alternating exclusion). This proves the symmetric-copy claim. Its alternating-copy subgroup also has a subdirect actual group: in reverse order that group contains the subdirect group just obtained. $\square$

Lemma 13.16 (Small labels are subdirect in the reverse proper-closure case). *Suppose $P > 1$ in reverse orientation and $K_p = c(P) < \mathcal{W}$. Then $I(p) = \emptyset$, and every label supported on any fixed bounded number of units is subdirect. For the subsequent tests, a support bound of ten is enough. The required cohort stock is absolute.*

Proof. Suppose now $K_p = c(P) < \mathcal{W}$. We have $I(p) = \emptyset$: for any absorbed O use such a canonical permutation label on many copies including it; its alternating and symmetric versions are distinct subdirect labels, yet have the same joins with p (all copies are absorbed), impossible.

Call a label small here when supported on a bounded number of units (literally below their X -label); we need only fixed bounds, say up to ten. We show that every such label has a subdirect actual group. Otherwise an atom e_i (actual coatom) beneath its nonzero closure label in parameter order must be product, and $e_i \leq p$: the actual coatom contains the nontrivial product closure, so cannot itself be subdirect, hence is closed by maximality. It is residual and lifts to a literal label supported on the package in question at empty footprint. Take there a literal cover $B_0 < C_0$ of groups giving the images bottom and e_i (groups written on just the package). Choose many disjoint clones and write e_j for the analogous images of C_0 ; they too are distinct atoms and the lowers collapse. Indeed image congruences within residual package intervals transfer both ways by canonical swaps: join with the swap label over Y , then meet with the full residual symmetric label on the other package (only Y elsewhere). Literally this copies a supported group from one package to the other, by independent support and conjugation. Thus both collapses and noncollapses there transfer, and each supported interval maps onto its image interval. Distinctness of e_j 's follows also by their bottom meets.

Transporting product atoms. Every e_j must likewise be product. Indeed a coatom if not product is subdirect. If subdirect here for $j \neq i$, it contains (in actual order) both intersections of K_{e_i} with the actual groups at the canonical labels of $\langle(ij)\rangle$ and $\langle(ijl)\rangle$ on copies. This is because on the parameter side, joining e_i with either permutation label contains e_j by conjugating the literal C_0 . Thus its ties refine both sets of permutation-label ties, since the indicated intersections with the product K_{e_i} project nontrivially and independently across each set of parts. But their common refinement is discrete: the two canonical labels are subdirect with a common subdirect lower group K_z , hence consistent twists, and generate actual top (the canonical parameter subgroups meet trivially); any sites tied in both would keep that common twist on joining. This contradicts proper subdirectness of K_{e_j} . Being product and above P as an actual subgroup, each is above $c(P)$, i.e. $e_j \leq p$.

The faithful product interval. Use k large such packages. Between the products of the literal covers the interval is Boolean if B_0 is not normal in C_0 (any full projection conjugates the supported B_0 to generate C_0 there); otherwise a subspace lattice over $\mathbf{F}_\ell$ for the prime order of C_0/B_0 . This interval maps faithfully to $[0, \bigvee e_j]$ by survival or simplicity, and all entries are product (the join of the e_j 's lies under p , hence all atoms under it too must be coatom products in actual order; now use atomisticity). Take k coprime to ℓ in the latter case (one of two consecutive large counts suffices). Let E be the augmentation hyperplane in that case, the whole join in the Boolean case. It is nonzero and no atom beneath it is fixed by the canonical alternating copy action (label a over Y), using the faithful literal product interval. For the hyperplane an invariant line would have a fixed nonzero vector by perfection of A_k , hence constant on the copies, impossible by $\ell \nmid k$.

The extension image. For the subdirect K_a , write D_a for its extension domain with map β to $\text{Aut}(S)$. By restrictions $[D_0, D_a]$ realizes exactly $\text{Sub}(A_k)$, so the kernel of outer action must be supplemented by D_0 (the normal test in [Section 10](#); if retained, the base-core quotient would be soluble by Schreier, impossible there). Thus $\beta(D_a) = (\beta(D_a) \cap S)\Lambda$, identifying S with inners. Also

$$(a \vee E) \wedge p = E$$

by factoring the literal join as a set product (the lift of E is normalized by the copy action) and using a representative of p above that lift, since $a \wedge p = 0$. Here $a \wedge p = 0$ comes from joining a subdirect with a nontrivial product in actual order, and literally the set product intersected with that representative of p is the lift of E times the intersection with the lift of a . Hence the product closure of $K_a \cap K_E$ is K_E (the equality gives generation with $c(P)$, already under the closure by $P \leq K_a \cap K_E$). Writing $T_e = \text{proj}(K_e)$ at the site, this means $\beta(D_a)$ normalizes T_E .

Indeed intersection with a product has full projection onto it precisely when its factors are matched by the diagonal twists; by transport along the tie through the site this is precisely invariance of the site's factor under the extension image.

Here T_E is nontrivial and proper, by $P > 1$, site transitivity and the proper product $K_E < \mathcal{W}$. Its normalizer in S is proper by simplicity, and is invariant under Λ . Transporting it gives a product label $0 < f \leq E$. Choose an atom $h \leq f$, and put $B_a = \beta(D_a) \cap S$. The normalization just proved and the reversal of label order give

$$B_a \leq N_S(T_E) = T_f \leq T_h.$$

Also Λ normalizes T_h , as it does every invariant product projection. Thus $\beta(D_a) = B_a \Lambda$ normalizes T_h . The same intersection criterion now gives

$$c(K_a \cap K_h) = K_h.$$

The alternating copy action moves the literal lift of h to a different atom h' under E , so $h' \leq a \vee h$ in label order. Its actual group is a product and contains $K_a \cap K_h$; it must therefore contain the displayed product closure K_h . This contradicts the distinct actual coatoms $K_h, K_{h'}$, proving the small-label claim.

In this extension-image argument, $[D_0, D_a]$ really is the whole ordinary interval: all actual overgroups of the subdirect in the invariant interval are still subdirect and correspond exactly to restrictions of its extension map. For the normalization criterion at the tie through a site o , at to , $t \in D_a$, the product factor of K_E is the transport by t of T_E , whereas the entry matched by K_a to u at o is the transport of $\beta(t)^{-1}(u)$. Thus all $u \in T_E$ occur in the intersection exactly for stability under $\beta(D_a)$, and similarly for the product at h ; other tie parts transport. Finally the “different atom” test only conjugates a literal lift within the displayed Boolean/subspace interval, where alternating copy permutations preserve the lift of E and move the given atom's lift to a different atom there. The distinctness survives by faithfulness on this interval, and the conjugate is contained in the literal join with the copy label, so the conclusion in the image does not assume invariance of the entire quotient map under copy permutations. $\square$

Lemma 13.17 (Residual products are impossible in reverse orientation). *Suppose all labels on the bounded supports used below are subdirect in a reverse realization. There is no nontrivial proper product actual group whose parameter label p has empty footprint. This assertion applies both when $P > 1$ and when $P = 1$.*

Proof. For small residual $a < b$ we must have $a \vee p < b \vee p$ by intersection strictness (actual intersections nontrivial since joins are residual). A representative for p cannot move between units. Otherwise for an element sending a letter of j into $k \neq j$, choose a second letter outside k mapping outside k (available by sizes with copies). Use a supported full symmetric label on the one or two source units. Conjugation of the source-letter transposition in the join gives a supported transposition connecting k with somewhere else, adjoining which to the source label is a small strict increase by orbit detection at k , impossible.

Preserving parallelisms. On a square both parallelisms, and on an auxiliary grid both length-two parallelisms, must be preserved. Otherwise use the parallel partition plus matching reference block with full symmetric wreath action and only Y elsewhere as a , and the independent full symmetric action on that union as b . Again disruption generates b in the join with p ; strictness is detected by meeting with the other indicated parallel label of equal length using the *same* reference block, then testing whether the reference unit connects to the grid (it is the only common block of the two partitions).

Uniform auxiliary flips. Thus on an auxiliary unit $h_0 \times 2 \times 2$ a representative acts by component permutations on h_0 , and separate bit flips possibly per component. They must actually be uniform. If for instance the first flip varies, take a acting on the (component, second bit) positions by a full symmetric group and separately by constant first flip, with only Y elsewhere. Conjugation of its supported top permutations by the varying element produces (after removing top action) first-flip differences generating all even-sum flips on those positions: differences of the nonconstant flip vector with its translates include weight two and are supported within this unit regardless of actions on other units. Adjoining those to a strictly increases it. Indeed meet with simultaneous preservation of the two parallelisms: the first flips formerly constant induce with component permutations only even actions on the (component, first bit) lines (second-bit direction, h_0 even), but flipping just one component's first bit now occurs and acts oddly. As before distinguish by adjoining a spare marker swap then intersecting a parity graph label, which contains bottom. This violates strict joins again.

The remaining primary signs. We conclude $p \leq B$ of the earlier sign-space test, with exact $[0, B]$ since $p > 0$. On any primary slot with active sign take as a the alternating bases on those lines plus the auxiliary size- x block, with *symmetric* top, outside only Y . In the join with p , conjugation again adds independent augmentation base signs by discrepancy with the even auxiliary action, producing the strict increase b tested under B as before. This final contradiction (or triviality already if no signs) eliminates the proposed residual product. $\square$

Proposition 13.18 (Reverse projections and the trivial-base cases). *In reverse orientation, if $P > 1$, then $c(P) = \mathcal{W}$. If $P = 1$, every nonidentity actual label is subdirect and the site-action image Λ contains the inner automorphisms of S . In forward orientation with $P = 1$, either a marker axis filter gives product success, or the same subdirectness and inner-action conclusions hold.*

Proof. For $P > 1$ in reverse orientation, a proper closure would be residual by [Lemma 13.16](#) and would contradict [Lemma 13.17](#). This proves the first assertion.

If $P = 1$, apply the result just obtained in the reversed filter with parameter range $[0, M_{\{i\}}]$ for a spare marker i outside any tested small support. The actual base there is nontrivial with still the same transitive operators, so it projects fully by the reverse $P > 1$ result and the desired small labels are indeed subdirect. A proper nontrivial product group would have residual label p , since containing a singleton absorption in parameter order would put independent nontrivial support inside that small proper subdirect group. The residual argument rules it out, hence all nonbottom groups are subdirect as claimed. Here Λ must then contain inners (otherwise its proper nontrivial inner part, or the proper-invariant-subgroup result for soluble operators in [Section 4](#), yields a proper nontrivial invariant product).

Back forward, suppose $P = 1$. If some marker x_i does not project fully, its axis filter gives product success by the forward argument above (we may take the enlarged actual base as operators then). Otherwise the extension at each marker x_i has image containing inners by atomicity and the same proper-invariant-subgroup argument, now on a diagonal part: a nontrivial proper subgroup invariant there would transport and multiply to an invariant group strictly between $P = 1$ and K_{x_i} . Its domain interval via restriction realizes the reverse weighted test on one less marker. If Λ itself does not contain inners, the kernel of that extension map is retained by the normal test (supplementation would give image just Λ), hence this gives an ordinary interval within the image in $\text{Aut}(S)$, with base Λ , top containing S . Missing inners there must be supplemented, so this is a reverse invariant interval in one simple site with proper base $\Lambda \cap S$, impossible by the reverse result just proved: a nontrivial base cannot project fully here, and for trivial base having all nonbottom actual groups project fully leaves no interior labels. Thus Λ contains inners and again all nonbottom labels are subdirect. $\square$

13.4 Extension domains and the passage to product success

Theorem 13.19 (Product success with absolute stock). *There are absolute stock and marker-loss bounds with the following property. Any ordinary realization, in either orientation, of a nontrivial weighted image yields a forward realization of a nontrivial image of the weighted template with at most boundedly many markers deleted, in which product success holds. The stock bound is independent of all numerical unit sizes and of the number of cohorts. The simple group at a site has order at most the original top-group order. A larger prescribed fixed stock can be retained by increasing the initial absolute stock.*

Proof. Take an ordinary realization of minimum top order among both orientations of the same quotient; its order is at most the input order. If $c(P) = \mathcal{W}$, the ties of P are fixed and every actual label refines them. On the tie part through a chosen site these refinements are exactly the invariant equivalences under its part stabilizer in H . They give an ordinary realization of the opposite interval with top order at most $|H| < |G|$, as in the proof of [Item \(i\)](#). This contradicts minimality. Thus absent product success we have the artificial-top extension case in the opposite orientation, with base map containing inners.

The extension datum and its flat labels. To apply [Lemma 4.14](#), distinguish the new base from the old operator group. Write H_{op} for that operator group, $A_0 \leq H_{\text{op}}$ for its site stabilizer, and $\alpha : A_0 \rightarrow \text{Aut}(S)$ for its action on the site. The remaining case gives $\alpha(A_0) \geq \text{Inn}(S)$. Thus the extension datum in that lemma is

$$(K, H, B, \psi) = (H_{\text{op}}, A_0, S, \alpha).$$

Put $C_0 = \ker \alpha$ and $A_{\text{in}} = \alpha^{-1}(\text{Inn}(S))$; these are its base kernel and inner preimage.

Index the points of an affine four-space over $\mathbf{F}_3$ by distinct markers. For a nonempty flat X of dimension at most two, let Z_X be the extension label given by M_X in forward extension order and by X_X in reverse extension order. Write D_X and E_X for its domain and kernel. Restriction identifies the entire lower ideal of this extension with the ordinary interval $[A_0, D_X]$. In parameter order this is the axis ideal $[0, M_X]$, or the axis filter $[X_X, 1]$ with order reversed. It is a nontrivial weighted image deleting just the markers of X , so $D_X > A_0$ and its normal property $(*)$ is available.

Apply that property to $E_X \trianglelefteq D_X$. If its kernel were retained in A_0 , the whole preimage of $\text{Inn}(S)$ would lie in A_0 : an element of that preimage has the same image as some element of A_0 , and their quotient lies in E_X . This normal subgroup would lie in the base core, leaving a soluble quotient by Schreier's theorem, contrary to the nonabelian monolith in $(*)$. Hence the kernel supplements the base. Each flat extension is pure, with

$$D_X = A_0 E_X, \quad E_X \cap A_0 = C_0.$$

Only these flat labels need purity.

Proper and incompatible kernel joins. The literal map $D \mapsto X_D$ preserves Boolean joins and meets, and footprints preserve its strictness in the image. Consequently the Z_X 's reverse flat containment, and for $X \cap Y = \{i\}$ their join in extension order is the point label. Restricting the point extension to the generated domain gives

$$D_i = \langle D_X, D_Y \rangle = A_0 \langle E_X, E_Y \rangle, \quad E_i = \langle E_X, E_Y \rangle.$$

The kernel equality follows because the generated kernels lie in E_i , have the same intersection C_0 with A_0 , and already generate its domain together with A_0 .

Distinct point labels have incompatible join. For their generated kernel group E , the equality $E \cap A_0 = C_0$ would define a common pure extension on $A_0 E$, which is impossible. Hence $E \cap A_0 > C_0$.

It is normal in A_0 , and its nontrivial α -image contains the inner simple subgroup. Since $C_0 \leq E$, this gives $A_{\text{in}} \leq E$, exactly the incompatible-join condition of the lower-grid lemma.

Subdirect intersections in the point monoliths. Apply the projection conclusions to each ordinary point ideal $[A_0, D_i]$, without a minimality assumption. If they give product success, the theorem is already proved on the corresponding bounded marker restriction. Otherwise a nontrivial base intersection has full projection: this follows from the reverse projection result in reverse orientation and from the absence of forward product success in forward orientation. With trivial base intersection, every nonidentity label is subdirect by [Proposition 13.18](#), again allowing its marker-filter success alternative.

Thus the domains of lines and planes through i intersect the point monolith subdirectly modulo the point core. These intersections are strictly above the point-base intersection: each such domain is strictly above A_0 , by footprints, and supplementation recovers the domain from its intersection. The flat ideals have $(*)$, the proper kernel joins were just verified, and incompatible point joins contain A_{in} . All hypotheses of [Lemma 4.14](#) therefore hold, giving the required contradiction.

Stock and order bounds. Each test uses boundedly many units: four-block comparisons, packages of at most ten units, or the fixed affine grid. The Ramsey reservoir for the copy-action argument concerns package indices and bounded permutation patterns, independently of the numbers of letters inside a package. Cloning a bounded package therefore needs only an absolute number of copies from any one cohort, including the two consecutive copy counts used earlier. Tests on different cohorts reuse these stocks; they do not delete the tested units. The lasting deletions are the boundedly nested marker restrictions in the extension, atom-filter and point-ideal steps. Increasing the initial stock thus leaves any prescribed fixed number of spare copies.

Only core quotients, subgroups and domain intervals have been used; the extension domains lie in the preceding operator group. Hence every resulting simple site has order at most the initial minimal top-group order, itself at most the input order. The reverse projection step uses [Theorem 10.1](#). $\square$

13.5 A single alternating action for all readings

Lemma 13.20 (Restart in an actual alternating power). *From product success one obtains, after boundedly many further marker deletions, an actual power A_Ω^s with transitive operator action, base intersection $H \cap A_\Omega^s$, and product subgroups at every cut. The degree is high, and at least forty specified surviving marker cuts are genuine distinct alternating point stabilizers at each site. All previously product labels below the new top remain products.*

Proof. In the forward product success case, take the pure filter absorbing everything except markers (one can leave just a sufficiently large fixed substock unabsorbed and put many spare markers among the absorbed units). Its base K_{sat} inside $\mathcal{W}$ is product by intersection of marker point cuts, nontrivial, and the filter exact pure. Apply the forward marker theorem for, say, at least 40 distinct points. Thus at M_A for boundedly many omitted markers we have an actual high alternating simple B_A at one site in the ordinary product group ΛT_{M_A} ($T_b = \text{proj}(K_b)$), normal and supplemented by the pure base there. Here $\Lambda \cap S \leq T_{\text{sat}}$ as in that theorem; thus $B_A \leq T_{M_A}$, and multiplying its transported factors gives a normal group at actual HK_{M_A} , not contained in the pure base. In the whole ideal below M_A , removing these boundedly many markers, $(*)$ shows that already H supplements this normal group. Explicitly, normality of the local B_A under Λ makes its transport independent of the chosen element of H . The resulting actual subgroup $N = \prod B_A$ lies in the old power, is normalized by K_{M_A} and by H , and is not contained in HK_{sat} , hence not in H . The whole-ideal normal test gives $HN = HK_{M_A}$; the original transitive action of H on the sites permutes these very factors. Restart therefore with its power of alternating groups B_A^s , transitive

operator action, and base intersection $H \cap B_A^s$; labels that were products below M_A remain product by intersection. In particular all relative cuts are product (intersect original cuts with M_A). The specified genuine point stabilizers still hold by intersection with B_A . Below use notation on the remaining template (removing A) and this new power.

If success was obtained by an earlier axis-filter shift deleting a marker, take H to be the ordinary base of the resulting whole weighted template (thus enlarged at that prior shift). It can remain the operator group in testing the pure filter $[K_{\text{sat}}, \mathcal{W}]^H$; that filter's M_A is exactly the lift M_A in the whole template. In the transport just used, K_{M_A} itself is a product by the marker-cut intersections, Λ normalizes the local B_A , and T_{M_A} normalizes it as well. Its power is not in the pure saturated base since M_A strictly increases saturation and the local simple supplements it on ordinary product labels. After supplementation by H , intersection with this new power at each label is just the intersection of the old K_b with it (the old power contained the new one). For $M_{A \cup i}$ from the pure theorem, local intersection with B_A can be taken on the ordinary site label as well, since $\Lambda \cap S \leq T_{\text{sat}}$; the almost simple quotient's centralizer kernel was in that ordinary base. Thus it gives exactly the alternating point stabilizer inside the new site. This step uses the forward marker result and its high-type reductions in [Section 12](#). $\square$

Convention 13.21 (Ordinary site groups and groups in the power). Write $B_A = A_\Omega$; locally let Θ be the site action image, in S_Ω since the degree is high. The projection of the new base intersection lies in Θ , and all cut site groups are nontrivial proper normalized by Θ . They contain $\Theta \cap A_\Omega$: adjoining it there could not reach A_Ω by simplicity, hence could not increase by the product coatom property. We may thus look also at the ordinary cut groups multiplying by Θ , inside ΘA_Ω . This respects intersections (the cut site groups all contain $\Theta \cap A_\Omega$); covers between such products remain covers as ordinary site groups since any strict intermediate, intersected with A_Ω , would transport to a strict intermediate product.

In particular, throughout the readings K_b on this restarted template denotes an actual intersection inside the normal power (before multiplying by any ordinary base). We use the ordinary cut groups with Θ mainly to test permutation actions; when using a projection or a graph in the power keep the parity restrictions in the alternating groups. A proposed intermediate *product* obtained sitewise is a label if it contains the base intersection and is invariant (transport to the other sites is implicit); arbitrary intermediate labels need not themselves be products.

Lemma 13.22 (All cuts are full nonhalf subset stabilizers). *In the alternating realization of [Lemma 13.20](#), every cut coatom has site group the alternating intersection of a full subset stabilizer on Ω . The two parts of this subset cut have unequal sizes. The ordinary site groups obtained by multiplication by Θ respect all cut intersections and the indicated covers.*

Proof. Every cut coatom C_D (remaining indices) must give a full nonhalf subset stabilizer on Ω . Indeed use at least eight genuine distinct point markers all on one side of the formal cut, and up to six of them at a time as a prefix P_1 . The cut group intersected with this full pointwise stabilizer (axis M_{P_1} , by meeting the individual full point stabilizers) is maximal proper there including in ordinary site notation with Θ . In labels it is a relative cut coatom after deleting the prefix, and these are products; any ordinary intermediate would transport to a product. Also the intersection with a single such point stabilizer is covered by the cut itself, literally by the independent full groups on the two pieces in the template (strictness as well by footprints or orbit detection, and use the cover transfer between the indicated products as ordinary site groups). Thus if the ordinary cut group were transitive, it would be primitive by point stabilizer maximality. Here the tested genuine points are fixed individually by Θ (it normalizes their respective alternating point stabilizers), so prefix stabilizers indeed cut down inside a pointwise ambient alternating or symmetric group. Every

tested prefix stabilizer in it would have to be transitive on the remaining letters: otherwise by its maximality in the full pointwise ambient group it would be a full subset stabilizer there and contain a three-cycle fixing the prefix points, impossible in the original assumed transitive primitive cut group by Jordan's Theorem (its intersection with A_Ω is proper). Indeed on deleting the prefix the residual cut is still proper with a nonempty unabsorbed side by our spare markers; for a single chosen marker the upward cover back to the cut is just restoring one singleton to a full symmetric action on its piece. Strictness if that piece is unabsorbed follows by absorption on adjoining that marker. Taking successive nested prefixes gives at least six-fold transitivity, impossible here by the classification of multiply transitive groups. Thus the cut is intransitive and by maximality a full subset stabilizer (its alternating intersection is exactly the product site group); a half subset would extend properly to the transitive two-block stabilizer instead. The permutation-group inputs in this argument are Jordan's three-cycle Theorem and the classification of multiply transitive groups, in the high-degree range, as recorded in [Section 4](#); see also Jones [20, Theorem 1.1 and Section 2] for the prime-cycle statement and the exclusion of proper six-transitive groups. $\square$

Proposition 13.23 (Common alignment of all cuts). *There is a single bijection from the formal units together with the foundation position to the Θ -orbits on Ω , under which every cut and every intersection of cuts is the corresponding stabilizer of unions of those orbits. Write $u_j > 0$ for the size of the cell of unit j . The specified genuine point markers have size one.*

Proof. **Every orbit bipartition is a cut.** The preceding lemma assigns a bipartition into unions of Θ -orbits to every cut. Conversely, take any nontrivial such bipartition. Its individual-part stabilizer in A_Ω transports to an invariant product containing the base. A coatom above this product is a subset stabilizer by [Lemma 13.22](#). It must preserve the same bipartition, since the original stabilizer together with Θ is transitive on each of its two parts. Thus the product was already a coatom. This also excludes a half-sized bipartition. We have obtained a bijection between all formal cuts and all bipartitions of the orbit list.

Intersections recover the partition poset. At least one Θ -orbit is nonsingleton. Otherwise an even degree admits a half-sized bipartition. In odd degree, $\Theta = 1$ and the base intersection is trivial; a full cycle then transports to a proper transitive product, whereas every coatom above it would be an intransitive subset stabilizer, also impossible.

For any partition of the orbit list, intersect the cuts separating its parts. At a site this gives the full alternating intersection of the independent symmetric groups on their unions. Multiplying by Θ leaves exactly these unions as its orbits. The only possible parity exception would be a part of size two with every other part a singleton: its transposition has no outside parity correction. By the nonsingleton-orbit conclusion, that two-point part is already one Θ -orbit, so multiplication by Θ restores transitivity there too. Hence distinct partitions give distinct intersection labels, with their correct containment order.

On the template side, intersections of cuts likewise form the partition lattice on the units and the foundation position. A strict change of partition changes either the absorbed set or a residual orbit, so footprint and orbit detection preserve strictness in the image. The cut-intersection poset therefore identifies these two partition lattices.

Recovering individual cells. A partition lattice on sufficiently many positions has only position permutations as automorphisms: its merging atoms represent pairs, their joins detect intersecting pairs, and the large incident stars recover the individual positions. The poset identification thus gives one bijection of formal positions with actual orbit cells, compatible with every cut and every intersection. Denote the cell of unit j by Ω_j , and put $u_j = |\Omega_j| > 0$. Transport these cells to the other simple sites by the operator group. This is well-defined because the site stabilizer Θ preserves

each of its orbit cells; invariance of the cut labels gives the same formal alignment at every site. For a specified genuine point marker, the formal singleton cut is an actual one-point stabilizer. Its one-cell side must consequently correspond to that single point, so this marker has size one. $\square$

13.6 Equality and square readings in that alignment

The following two readings use the same alignment of [Proposition 13.23](#).

Proposition 13.24 (Equal totals and equal copies). *Let D, E be disjoint bounded collections of nonfoundation units with equal formal total size, with ample spare markers outside them. If at least one of D, E contains more than one unit, then*

$$\sum_{j \in D} u_j = \sum_{j \in E} u_j.$$

In a sufficiently numerous cohort, all copies have the same actual cell size. In particular, all surviving markers have size one.

Proof. Put $J = C_D \wedge C_E$ and $F' = C_{D \cup E}$. Equal formal totals supply a two-block wreath label W between them. It survives strictly above J : joining an absorption on one side detects that W connects the two sides. To distinguish W from F' , intersect with a cut fixing a proper nonempty subcollection of the multi-unit side. This prevents the wreath exchange, while the upper label still mixes letters across the two sides. Thus $J < W < F'$ in the image.

Suppose the actual totals differ. At one site write J_o and F_o for the projections of the product groups at J, F' . Projection onto the combined D, E -cells maps F_o onto their full symmetric group: spare units provide outside parity correction. Its kernel lies in J_o , whose image is the stabilizer of two unequal nonempty parts. That stabilizer is maximal and nonnormal, as is seen by conjugating its supported transpositions. Therefore $J_o < F_o$ is a cover and J_o is not normal in F_o .

The actual group K_W need not be a product. Its coordinate closure is nevertheless a product label between J and F' , so the site cover forces that closure to be $K_{F'}$. Now K_W contains the independently supported copies of J_o and projects onto F_o at every site. Conjugating those copies inside K_W therefore puts their independent normal closures in F_o inside K_W . Nonnormality and the cover give $\langle J_o^{F_o} \rangle = F_o$. Hence $K_{F'} \leq K_W$, contradicting $W < F'$. The actual totals must agree.

Apply this conclusion to disjoint pairs of copies in one cohort. Comparing two pairs which differ at one position against a common disjoint pair equalizes the individual cell sizes. This works for the marker cohort as well; its already identified genuine points make every surviving marker size one. $\square$

Theorem 13.25 (Reading a primary square). *For a primary square O and its full side reference R , let $r = u_R$ in the alignment of [Proposition 13.23](#). If r is even and $r \geq 8$, then $u_O = r^2$.*

Proof. We first construct the two actual parallel systems, then determine their common incidence cells and the number of blocks.

The two parallel systems. Put

$$M = M_{\{O, R\}}, \quad F' = C_{\{O, R\}}, \quad J = C_{\{O\}} \wedge C_{\{R\}}.$$

For each of the two formal parallel directions, let V_j be the full wreath label whose blocks are the parallel lines and the reference unit R , with everything outside absorbed. Each is maximal below F' : adjoining a permutation disrupting its blocks and conjugating the supported base transpositions generates the full symmetric group on their union. The label V_j mixes R with O , whereas $V_1 \wedge V_2 \leq J$,

since R is the only common block of the two formal partitions. Orbit detection preserves these distinctions in the image. In particular both V_j 's remain proper and maximal below F' .

First consider the product N_R of the independently supported alternating groups on the actual cells Ω_R at all sites. It is normal in the ordinary group at $C_{\{R\}}$. Since the formal reference unit is full, $C_{\{R\}} = M_{\{R\}}$; the ideal below it is a weighted image with just that reference unit deleted. Its normal property (*) applies. But N_R lies inside the separate-cell stabilizer at $J < C_{\{R\}}$, so it cannot supplement the base. It must lie in the base, and hence is available independently at every site in each K_{V_j} .

Fix a direction j . The projection of K_{V_j} , together with Θ , is transitive on $\Omega_O \sqcup \Omega_R$: Θ is transitive on each cell, and V_j mixes them. Suppose two translates of Ω_R overlap without coinciding. After conjugation one of these translates is Ω_R itself. The component of the overlap relation containing Ω_R then meets Ω_O and is Θ -invariant. Cell transitivity makes it the entire union. Alternating groups on intersecting supports of size at least eight generate the alternating group on their union. Conjugating the supported factors of N_R along the translates therefore gives, inside K_{V_j} , the independently supported alternating groups on $\Omega_O \sqcup \Omega_R$. Conjugation by a site projection is realized by an element of K_{V_j} ; conjugation by Θ uses operator invariance. Both preserve single-site support.

Let N be the product of these larger supported groups. It is normal in the ordinary group at F' , since the cell union is invariant. In the ordinary interval from the group at M to the group at F' , let e label the lower endpoint multiplied by N . We have

$$M \leq e \leq V_j, \quad e \not\leq J.$$

For the other direction k , set $a = J \wedge V_k$ and $b = V_k$. Then $a < b$ and $e \wedge b \leq a$.

The remaining join calculation takes place in the literal template. Lift e between M and V_j . Because its image is not below J , this lift moves the block R to some j -line. The literal label a contains the full supported symmetric groups on R and on every k -line. Conjugating its group on R supplies the full symmetric group on that j -line. Its crossings with the k -lines generate the full symmetric group on O , and the movement of R then generates the full group on $O \cup R$. Hence $e \vee a = F'$. This is the forbidden normal-multiple configuration, a contradiction.

Thus the translates of Ω_R are disjoint or equal and form an equivariant block partition at every site. Its full product stabilizer is proper, contains K_{V_j} , and lies below $K_{F'}$. Maximality forces equality with K_{V_j} . Removing its block Ω_R gives a partition of Ω_O into blocks of size r . Doing this for both directions yields two systems, each with $m' = u_O/r$ blocks.

The sign graph across all simple sites. Set $a_j = V_j \wedge J$ and $b_0 = a_1 \wedge a_2$. These are full product subgroups at the simultaneous structures. Let L be the projection of K_{b_0} at a site onto the actual O -cell alone. Outside parity correction makes this the full common stabilizer of its two systems. We show that $L \leq A_{u_O}$.

Choose two singleton markers outside O, R . From b_0 obtain b^- by fixing the markers individually and b^+ by fixing their pair setwise; from J obtain j^+ by fixing their pair setwise. Each restriction is an intersection with the corresponding cut labels. Formally the group on O at b_0 is $S_x \times S_x$ in its grid action, and every such permutation is even because x is even. The literal graph label g inside j^+ , which couples the marker swap to the sign on O , therefore satisfies

$$g \vee b^+ = j^+, \quad g \wedge b^+ = b^-. \quad (13.1)$$

It does not fix the markers individually: this survives the lattice map by orbit detection. This is a valid literal label before taking the image, since the bottom action on O is even; actions on the other parts remain independent.

Now use the actual subgroups in A_Ω^s . Project to the O -cell and the marker pair at every site, correcting parity on R . The three product subgroups K_{b-}, K_{b+}, K_{j+} have a full common kernel, say Q_0 , and their quotients are respectively

$$L^s \times 0, \quad L^s \times C_2^s, \quad S_{u_O}^s \times C_2^s.$$

The restrictions distinguishing them concern only O and the marker pair, so the same full kernel occurs in all three. By the meet equality, $Q_0 \leq K_{b-} \leq K_g$. Put $Q = K_g/Q_0$. The same equality gives

$$Q \cap (L^s \times C_2^s) = L^s \times 0.$$

In particular, the projection of Q to $S_{u_O}^s$ is injective and its image contains L^s . The join equality in Equation (13.1), taken with $L^s \times C_2^s$, forces that image to be all of $S_{u_O}^s$. Indeed, if the image is P , then $L^s \leq P$, so the projected join is $\langle P, L^s \rangle = P$; it is the entire product, not merely a subgroup surjecting to each site separately. Consequently Q is the graph of a homomorphism

$$f : S_{u_O}^s \longrightarrow C_2^s, \quad f(L^s) = 0.$$

No assumption is made that f acts separately at the different sites. It is nonzero: if it vanished, K_g would lie in the product restriction of K_{j+} fixing the two actual points individually, the intersection with their individual cut labels. The literal graph connects the two formal singleton units whereas that intersection does not, with the same outside absorption, contradicting orbit detection.

Every homomorphism to an abelian group factors through the sign abelianization of $S_{u_O}^s$. If L contained an odd permutation, then the signs of L^s would fill C_2^s , and $f(L^s) = 0$ would force $f = 0$. Hence L is even. Transport identifies the actual copies of L without changing parity.

The common incidence cells of the two systems on O now have at most one letter each: otherwise a transposition in one such cell would belong to L . Thus the letters of the O -cell are edges of a simple r -regular bipartite graph, with m' vertices in each part. The group Θ is transitive on edges, because it is transitive on the O -cell. In particular, $m' \geq r$.

The row chain and the column signs. The literal interval $[b_0, a_1]$ has exactly four elements. On the formal O -grid its lower endpoint contains the diagonal base ΔS_x and all pure row permutations S_x , while the upper endpoint is $S_x \wr S_x$. An intermediate is therefore determined by its base kernel. Intersecting this kernel with A_x^x gives a subdirect group containing ΔA_x . Its tie partition is invariant under all row permutations, so either all rows are tied or all are independent. In the first case the intersection is ΔA_x , whose normalizer in the base is ΔS_x , since A_x has trivial centralizer in S_x . In the second case the kernel contains A_x^x . Its quotient is an invariant sign space containing the constant vector. Taking differences under transpositions shows that the only possibilities are the constant line, augmentation, and the full sign space. Thus the four base kernels are

$$\Delta S_x < A_x^x \Delta S_x < (S_x^x)_{\text{even}} < S_x^x.$$

Here x is even, so the constant line lies in augmentation. The actions outside O , including on R , are independent and full at both endpoints. Passing to the lattice image leaves at most these four elements.

In the actual realization consider the following chain on the O -cell, where even refers to total base parity:

$$L < LA_r^{m'} < L(S_r^{m'})_{\text{even}} < LS_r^{m'} \leq S_r \wr S_{m'}.$$

All groups in this chain lift with the full common kernel; parity can be corrected freely outside the O -cell. They contain the base intersection and are Θ -invariant, because Θ preserves both systems

and the displayed base layers are canonical within the row system. Thus they are actual product labels in $[b_0, a_1]$.

The first increase is strict: a nontrivial three-cycle supported on one row cannot preserve the column partition, since the incidences are simple and every column has degree $r > 1$. For the next increase, examine the kernel of the row action of L . It can only permute columns with the same neighborhood, called twins. A permutation inside one twin class has the same parity contribution on every row of that neighborhood, and contributes zero on other rows. Its row-sign image therefore has rank at most the number of twin classes. Edge transitivity implies column transitivity, so all twin classes have the same size t . If there are no twins the rank is zero; otherwise $t \geq 2$, and it is at most

$$\frac{m'}{t} \leq \frac{m'}{2} < m' - 1.$$

Intersecting the displayed groups with the row base and comparing their sign images modulo $A_r^{m'}$ proves that the augmentation increase is strict. The final increase is strict as well: a row permutation has sign $(\text{sgn } \sigma)^r = 1$ on letters because r is even; since L is even, $L(S_r^{m'})_{\text{even}}$ is even, whereas $LS_r^{m'}$ contains an odd row-base element.

There are now four distinct labels in an interval which is an image of a four-element chain. They exhaust it, so $LS_r^{m'} = S_r \wr S_{m'}$. In particular the row action of L is $S_{m'}$. The orbit of any column neighborhood under this row action contains every r -subset of the rows, whereas there are only m' columns. Hence

$$\binom{m'}{r} \leq m'.$$

As $m' \geq r \geq 8$, this forces $r = m'$ or $r = m' - 1$. In the latter case the missing incidences form a perfect matching: every row and every column misses exactly one partner. Label each column by the row it misses. The letters then identify with ordered pairs (i, j) of distinct indices, and every element of L acts by the same permutation on both indices. The full row action $S_{m'}$ supplies an element inducing a transposition $(a\ b)$. On letters it exchanges (a, b) with (b, a) , and for each other index c it exchanges both (a, c) with (b, c) and (c, a) with (c, b) . This is a product of $1 + 2(m' - 2) = 2m' - 3$ transpositions, hence odd, contrary to $L \leq A_{u_O}$. Thus $m' = r$, and the O -cell has $m'r = r^2$ letters. $\square$

Figure 4 illustrates the incidence conclusion just proved.

Theorem 13.26 (Weighted realization and simultaneous numerical readings). *Fix any bounded support size for the comparisons and any desired fixed number of spare copies. There are absolute initial cohort-stock and marker-loss bounds, independent of the numerical unit sizes and the number of cohorts, with the following property. The bounds may depend on the fixed support and spare-copy requirements. An ordinary realization of a nontrivial weighted image, in either orientation and with top order M , gives a forward realization after at most that many marker deletions, with all cuts product in an actual alternating power. A single common cell alignment gives positive sizes $u_j \leq M$, equal on each cohort and one on all surviving markers. In this same alignment the equal-total reading of Proposition 13.24 and the square reading of Theorem 13.25 hold.*

If the input is a finite algebra of size m representing a nontrivial image, the same conclusion holds with $M = 60^m m!$, for a possibly different nontrivial image of the original template. The full template itself is finitely representable.

Proof. Apply Proposition 13.6, Theorem 13.19, and Lemma 13.20, then Propositions 13.23 and 13.24 and Theorem 13.25. All sizes read here are bounded by the order bound already given, since

the high alternating occurs inside an initial site. Stock bounds, including losses to A and marker requests, remain independent of the numerical sizes read. In particular $u_j \leq |\Omega| \leq |A_\Omega| \leq M$. All comparisons are made in this one realization; none requires a new choice of simple sites or a new alignment. The original stock can include the fixed losses and the fixed spare copies required at the end. $\square$

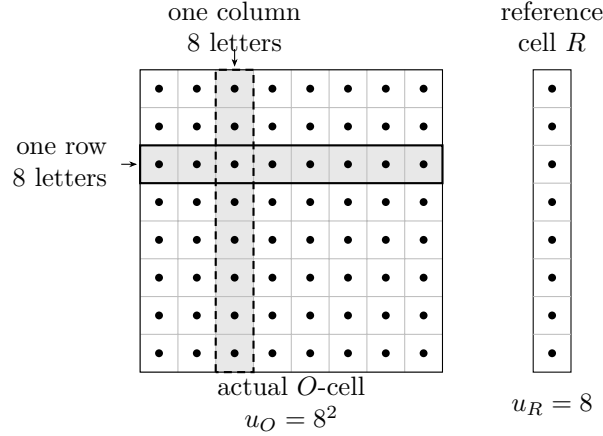


Figure 4: The established conclusion of [Theorem 13.25](#), illustrated for actual size $r = 8$ at one simple site. In the common cell alignment, the O -cell has r row blocks and r column blocks, each with r letters; each intersection contains one letter (a dot). The separate reference cell R contains r letters.

14 A uniform computable bound for weighted templates

The letter sizes in a weighted template can be arbitrarily large. We will bound its number of lattice elements using only the number of units. By [Lemma 13.3](#), it suffices to count permutation overgroups of the prescribed actions on each residual union of units. Such an overgroup may have several orbits. On each orbit we need both a bound for the number of transitive overgroups and a bound for their quotients by the normal closures of the prescribed actions. The latter bound controls how the different orbit actions can be coupled in one subgroup.

We begin with products of natural alternating actions, allowing one factor to act on several copies of its alphabet. Compression in bounded coordinate groups will let us lift a count from a block action to the full permutation action. We then include the private square actions present in the weighted templates and assemble the residual orbit actions. All bounds are computable functions of the stated parameters; none depends on the numerical weights of the units.

Definition 14.1 (Controls). A *rank-one control* on a finite set I is a group X that is a product of independent alternating groups A_{n_i} , with $n_i \geq 7$. Each factor acts naturally on one or more disjoint copies of its alphabet, using the same permutation on all its copies, and fixes all other letters. Additional fixed letters are allowed, as is the trivial group. Its *complexity* is the number of its orbits on I .

A *control with private squares* is the disjoint combination of a rank-one control and square orbits, each with its own two independent natural alternating factors. The factors of a square act on no

other orbit. The two alphabets of each square have the same size, at least seven. Its complexity again counts all orbits, including fixed letters.

A *trimmed control* fixes a specified bounded number of entries in each alphabet and retains the alternating action on the remaining entries. If fewer than seven entries remain, the whole remaining alphabet is declared fixed.

Fixing at most k entries per alphabet changes complexity by a computable function of k and the original complexity. Indeed, each natural copy splits into the fixed entries and at most one remaining natural orbit; an alphabet discarded at this step has size at most $k + 6$. A private square splits into products of these alphabet orbits, so it contributes at most $(k + 6)^2$ orbits after trimming. There is also a computable function $q_r(s)$ bounding the number of orbits on I^r for either kind of control of complexity at most s . To see this, specify the orbit containing each tuple entry and then the equality patterns of the alphabet entries in each factor. There are at most $2s$ factors and at most $2r$ alphabet entries to consider. Passing from the corresponding symmetric groups to alternating groups splits an orbit into at most 2^{2s} orbits. For example, with B_j the j th Bell number, one may use

$$q_r(s) = 2^{2s}(s + 1)^r B_{2r}^{2s}. \quad (14.1)$$

Consequently the number of X -invariant r -ary relations is at most $2^{q_r(s)}$. These estimates apply to equivariant images of invariant subsets as well: relations on such images pull back to relations on the original tuple sets.

Lemma 14.2 (Compression in a bounded base group). *Let X be a rank-one control of complexity at most s on I , and let D be a finite group of order at most a . There are computable bounds, depending only on (s, a) , for the following quantities:*

- (i) *the number of X -invariant subgroups $J \leq D^I$;*
- (ii) *the number of X -invariant right or left cosets of each such J ;*
- (iii) *the number of tuples needed to generate J under the action of X , each of which can be chosen fixed by a trimmed control of bounded complexity.*

Every invariant coset has a representative fixed by such a trimmed control. No commutativity assumption on D is required.

Proof. For one factor A_n , combine the coordinates in its repeated natural copies into an entry in a product $E = D^c$, where $c \leq s$ is the number of copies. Thus the coordinates on this alphabet form a list $(x_1, \dots, x_n)$ in E^n . Suppose first that x belongs to an invariant left coset xJ ; the case $x \in J$ is included. If σ is a three-cycle of alphabet entries, then $x^{-1}x^\sigma \in J$. This tuple is supported on the three moved entries, and an appropriate choice of σ gives any prescribed difference $x_i^{-1}x_j$ at one of them.

Let $z \in J$ be such a tuple, and let its value at a chosen entry be $t \in E$. Choose two fresh entries outside its support and conjugate z by a three-cycle that moves the chosen entry through those two positions, fixing all other support entries. If z' is the resulting tuple, then $z(z')^{-1}$ is supported on a pair and has values t, t^{-1} there. The other entries cancel coordinatewise. Since $n \geq 7$, the fresh entries exist; since A_n is two-transitive, such a pair can be moved to any ordered pair of distinct positions.

Keep position 2 as a reference. For each $i \neq 1, 2$, use a pair tuple in J with value $x_i^{-1}x_2$ at i and its inverse at position 1. Right multiplication changes entry i to x_2 and accumulates the correction at position 1. These are valid coordinatewise products even when E is nonabelian. After all corrections,

every entry except possibly the first is the common reference value. Do this independently on every alphabet. The resulting representative is constant beyond the first two positions of every alphabet and is fixed by the control fixing those positions.

Let $\mathcal{C}$ be the set of all tuples in D^I constant beyond the first two entries on each alphabet. The fixed coordinates and the three values per natural copy give, for example, $|\mathcal{C}| \leq a^{3s}$. Every pair multiplier used above is an X -translate of an element of $J \cap \mathcal{C}$. Therefore J is generated by the X -translates of $J \cap \mathcal{C}$. This bounds its generating list and the number of possible J 's, for instance by $2^{|\mathcal{C}|}$. Every invariant left coset meets $\mathcal{C}$, so there are at most $|\mathcal{C}|$ such cosets for fixed J . Inversion gives the corresponding assertion for right cosets. The required trimming bounds were established after [Definition 14.1](#). $\square$

Lemma 14.3 (Lifting across bounded fibers). *Let X be a rank-one control of complexity at most s on I , let D have bounded order, and let $U' \leq S_I$ contain X . Suppose U' is generated over X by a bounded list of elements g such that each $X \cap gXg^{-1}$ contains a rank-one control of bounded complexity. Then the number of subgroups*

$$V \leq D^I \rtimes U', \quad \text{proj}(V) = U', \quad (1, X) \leq V,$$

is bounded computably in s and these parameters. If $[U' : \langle X^{U'} \rangle]$ is bounded, then so is $[V : \langle (1, X)^V \rangle]$.

In the permutation action with bounded fibers, each such V is generated over its pure copy of X by boundedly many elements whose intersections with that pure control contain rank-one controls of bounded complexity on the expanded point set.

Proof. The base kernel $J = V \cap D^I$ is X -invariant, so its possibilities are bounded by [Lemma 14.2](#). For a prescribed generator g , write the fiber of the projection as (Jf, g) . If $h \in X \cap gXg^{-1}$, multiply a lift on the left by $(1, h)$ and on the right by the matching pure lift of $g^{-1}h^{-1}g$. The resulting element still lies above g , and its base entry is $h \cdot f$. Thus Jf is invariant under the specified intersection control. [Lemma 14.2](#), applied to that control, bounds its possible cosets and supplies a representative f fixed by a trimmed subcontrol. The kernel, these lifts, and $(1, X)$ generate V , proving the count.

Put $N = \langle (1, X)^V \rangle$. In V/N , all X -translates of a kernel generator have the same image. Hence the image of J is already generated by the images of a bounded list of tuples, with no translates required. A subgroup of D^I generated by k tuples has order at most

$$|D|^{k|D|^k}. \tag{14.2}$$

Indeed, each coordinate has a type in D^k , and coordinates of the same type evaluate every word in the generators identically. Projection to one coordinate of each type embeds the subgroup into at most $|D|^k$ copies of D . The quotient of V/N by the image of J is $U'/\langle X^{U'} \rangle$, which proves the index bound.

For the last assertion, let h belong to the trimmed intersection control fixing the chosen f . Conjugating its pure lift by (f, g) takes it back into pure X : the base term disappears because h fixes f , and $g^{-1}hg \in X$. The compressed kernel generators likewise commute with their trimmed pure controls. A natural action repeated over bounded fibers is still a rank-one control, with complexity multiplied by a bounded factor. These observations give all the asserted intersections on the expanded set and remain valid after a bounded number of such lifts. $\square$

We use the following consequence of the primitive order theorem with an explicit effective threshold. Maróti's Theorem [\[28, Theorem 1.1\]](#) says that a primitive group of degree d is either one

of the four Mathieu exceptions, has order at most $d^{1+\log_2 d}$, or lies between

$$A_\ell^e \quad \text{and} \quad S_\ell \wr S_e \quad \text{on degree} \quad d = \binom{\ell}{h}^e.$$

Replacing subsets by their complements permits $h \leq \ell/2$.

Lemma 14.4 (Large primitive parts). *For each positive integer s there is a computable threshold $H(s)$ with the following properties for $d > H(s)$. A primitive group of degree d and order at least $\lfloor \sqrt{d/s} \rfloor! / 2$ has one of the alternating-product actions*

$$(e, h) = (1, 1), \quad (1, 2), \quad (2, 1), \quad (14.3)$$

with ℓ arbitrarily large as the threshold is increased. Its socle is A_ℓ^e and has index at most eight. If its order is at least $\lfloor d/s \rfloor! / 2$, it contains A_d in the natural action. One may take

$$H(s) = 2^{2^{(s+100)}^{10}}. \quad (14.4)$$

Proof. Write $p = \log_2 d$. Outside the bounded Mathieu exceptions, the small-order alternative has logarithm at most $p(1+p)$. In an alternating-product action, $e \leq p$. Unless one of Equation (14.3) holds, $\ell \leq 4d^{1/3}$: if $e \geq 3$, use $\binom{\ell}{h} \geq \ell$; if $e = 2, h \geq 2$, use $\binom{\ell}{h} \geq \binom{\ell}{2}$; and if $e = 1, h \geq 3$, use $\binom{\ell}{h} \geq \binom{\ell}{3}$. Consequently the logarithm of $(\ell!)^e e!$ is at most $10p^2 d^{1/3}$. For the two additional possibilities (1, 2) and (2, 1), the same estimate with $d^{1/2}$ in place of $d^{1/3}$ is an upper bound.

Under Equation (14.4), we have $s < p$ and $2^{p/6} > 100p^4$. Compare the preceding upper bounds with

$$\log_2(v!/2) \geq v - 2, \quad v = \lfloor \sqrt{d/s} \rfloor \quad \text{or} \quad v = \lfloor d/s \rfloor,$$

respectively. The square-root factorial excludes the small-order alternative and all cases outside Equation (14.3); the linear factorial excludes the other two actions as well. The socles in the three remaining cases are respectively A_ℓ , A_ℓ , and A_ℓ^2 . Their full indicated overgroups have quotient orders two, two, and eight. In the last case primitivity forces transitivity on the two slots: fixing the slots would preserve the partition into lines in either direction. The centralizer of a natural alternating group, and of its action on two-subsets, is trivial; a commuting permutation is determined by the image of one point, whose stabilizer fixes no other point in these actions. In the product action, commuting with each independently supported factor likewise fixes each coordinate. Finally, the intersection of a normal subgroup with the alternating product is a subproduct of its simple factors, and slot transitivity forces every nontrivial such intersection to be the whole product. A normal subgroup with trivial intersection centralizes that product and is therefore trivial. This proves the assertions about the socle and unique minimal normal subgroup. $\square$

The next overgroup argument needs bijections between equal-sized parts that commute with one common subcontrol. For a rank-one control fixing each part setwise, its *profile* on a part records the number of fixed letters and the number of natural copies of each factor of that control. Equal profiles give equivariant bijections by matching copies of the same factor with their alphabet coordinates, and matching the fixed letters. Equal cardinalities alone need not give equal profiles; the following lemma supplies a suitable subcontrol.

Lemma 14.5 (Effective equal profiles). *Suppose a finite set is partitioned into at most s equal-sized parts B , each preserved by a rank-one control X . Write the size of B as*

$$b_B + \sum_i k_{Bi} n_i,$$

where b_B counts its fixed letters, k_{Bi} counts its natural copies of the factor A_{n_i} of X , and the number of factors and all nonnegative integers b_B, k_{Bi} are at most s . There is a rank-one subcontrol of X , of complexity bounded computably in s , that has identical profiles on all parts. If the common part size is at least two, the subcontrol may also be chosen with at least two fixed letters in every part.

Proof. The empty set is immediate, so suppose there is at least one part. Subtract the expression for one part from each other expression. This gives a system $An = c$ with bounded integer coefficients and bounded numbers of equations and variables. No equation prescribes the common total size. Every nonnegative integer solution admits a decomposition

$$n = u + \sum_{w \in \mathcal{H}} m_w w, \quad m_w \in \mathbb{Z}_{\geq 0}, \quad (14.5)$$

where u belongs to the finite set of coordinatewise minimal nonnegative solutions of $Au = c$, and $\mathcal{H}$ is the finite set of minimal nonzero nonnegative solutions of $Aw = 0$. The finiteness is Dickson's Lemma [9]. Given n , choose such a $u \leq n$; then $n - u$ is homogeneous. Unless it is zero, subtract a minimal nonzero homogeneous solution lying below it and repeat. The sum of its coordinates strictly decreases, proving the decomposition.

These finite lists can be found effectively. Enumerate solutions in larger and larger boxes and test whether their upward orthants cover all nonnegative solutions, using the first-order theory of addition and order [39]. The test has only fixed integer coefficients: for a proposed particular list it asks whether

$$\forall v \in \mathbb{Z}_{\geq 0}^r \quad (Av = c \implies \bigvee_u u \leq v),$$

and the homogeneous test adds $v \neq 0$. Dickson's Lemma guarantees termination; discarding nonminimal listed entries gives the desired lists. There are only boundedly many possible coefficient systems, so the lengths and coordinates of all these lists have computable uniform bounds.

For each original alphabet i , fix u_i entries and split the rest into w_i sets of size m_w for each $w \in \mathcal{H}$. For each w , use one common natural alternating factor A_{m_w} diagonally on all these sets, across all original alphabets and repeated copies. Thus the profiles on different parts refer to the same actual factors, even when two factors have equal degrees. If $m_w < 7$, fix those entries instead. The number of copies on part B for the new factor is $\sum_i k_{Bi} w_i$, independent of B because $Aw = 0$. The remaining fixed-letter count is also independent of B , by $Au = c$. This proves identical profiles and the complexity bound. Suppose the common part size is at least two. The number of fixed letters is already the same in every part. If it is less than two, some new factor A_m , with $m \geq 7$, acts on a part. By the identical profiles it has the same positive number of natural copies on every part. Fix two entries of its alphabet, simultaneously on all these copies. If $m \geq 9$, retain A_{m-2} on the remaining entries. If $m = 7$ or 8 , declare all entries of these copies fixed, as required by the trimming convention. This adds at most eight fixed letters per copy, and the number of copies is computably bounded by the preceding construction. The new profiles are still identical, and every part now has at least two fixed letters. If no factor acts, all letters were already fixed, so the same conclusion follows from the common part size. The trivial control is allowed; discarding the last active factor therefore causes no exception. This proves the fixed-letter assertion with computably bounded complexity. $\square$

Proposition 14.6 (Rank-one overgroups). *There are computable functions of s that bound, for every rank-one control X of complexity at most s on a finite nonempty set I ,*

- (i) *the number of transitive groups U with $X \leq U \leq S_I$ and the index $[U : \langle X^U \rangle]$;*

- (ii) *the length of a generating list for U over X , chosen so that every generator g has a rank-one control of computably bounded complexity inside $X \cap gXg^{-1}$.*

Proof. We prove a more precise assertion with a second parameter h bounding the number of strict steps in any chain of U -invariant equivalences. Every such equivalence is a union of X -orbits on I^2 , so initially $h \leq q_2(s)$ suffices. We induct on h , allowing the complexity to change by computable functions at a step. If $|I| = 1$, the assertion is immediate. Otherwise choose a minimal nondiscrete U -invariant partition $\mathcal{B}$; the universal partition is allowed. Its parts have a common size $d \geq 2$, and a part stabilizer acts primitively on its part. There are at most $2^{q_2(s)}$ possible partitions. Let I' be the set of parts.

A natural copy maps injectively to I' . The map from an X -natural copy to I' is either constant or injective, since a natural alternating action is primitive. If a copy for one factor maps injectively, all letters mapping to that moving orbit of parts come from injective copies of the same factor. Indeed a copy on which this factor is trivial cannot have an equivariant image in its nontrivial natural orbit. Each contributing copy supplies only one letter to a part, whence $d \leq s$.

Retain only the natural factors of degree greater than $d + 2$ to form a smaller control X_0 . Discarded factors have bounded degree, so X_0 has bounded complexity. It induces a rank-one control X'_0 on I' , whose complexity is no larger than that of X_0 . A part stabilizer in X_0 is a product of alternating groups or natural point stabilizers. Each nontrivial factor is simple and has order greater than $d!$, so its action on a fiber of size d is trivial. Label one fiber in each X_0 -orbit of parts and transport the labeling by X_0 . The trivial stabilizer action makes this well-defined. This labels X_0 as a pure control in a bounded-fiber wreath product, and the labeling can be fixed from X_0 and $\mathcal{B}$, independently of U .

The projected transitive group U' has equivalence-chain bound at most $h - 1$: lift a chain on I' and append the discrete partition of I at its fine end. Apply the induction hypothesis to (X'_0, U') , then [Lemma 14.3](#) with fiber group S_d . This bounds the choices of U , its index modulo the normal closure of X_0 , and its generators with the required intersection controls on I . They also prove the assertions for $X \geq X_0$: its normal closure is larger, generation over it needs no additional elements, and the intersection controls remain contained in $X \cap gXg^{-1}$.

Every natural copy maps constantly to I' . Now X fixes every part, so their number b is at most s . If $d \leq H(s)$, then $|I| \leq sH(s)$ is computably bounded, and finite subset counts bound every subgroup choice and generator list. The trivial subgroup is an admissible intersection control of bounded complexity on this bounded domain. Thus assume $d > H(s)$. Each part contains at most s natural copies or fixed letters, so some natural axis of degree at least d/s occurs in it. [Lemma 14.4](#) implies that its primitive part-stabilizer action contains A_d .

Let K be the kernel of the action on I' and put

$$T_0 = K \cap \prod_{B \in \mathcal{B}} A_B.$$

The projection of K to each part is normal in its primitive part-stabilizer group and contains the nontrivial axis action. It therefore contains A_B . The commutator subgroup of K lies in T_0 and still projects onto every A_B . Thus T_0 is a subdirect product of these nonabelian simple groups, hence a product of full twisted diagonals over a partition of the parts. It is normal in U .

For completeness, such a subdirect product is self-normalizing inside the direct product of its simple factors. On each diagonal class, an element normalizing the diagonal has coordinate conjugations that agree through all the tying isomorphisms. Its coordinates therefore differ from a member of that diagonal by elements of the centers, which are trivial. Consequently the whole

block-stabilizer normalizer satisfies

$$[N_{S_d S_b}(T_0) : T_0] \leq 2^b b!. \quad (14.6)$$

In high degree, automorphisms of A_d are induced uniquely by permutations of its natural alphabet. Record each twisted tie by matching the corresponding letters across its tied parts. The resulting relation on I^2 is X -invariant, since U normalizes T_0 . It determines the ties and their twists, so there are boundedly many choices of T_0 . For fixed T_0 , Equation (14.6) bounds all choices of U by subset counts in its quotient.

Here $1 < X \leq T_0$: X fixes each part and its natural alternating actions are even on it. Since U is transitive on parts, it is transitive on the diagonal simple factors of T_0 . A normal subgroup of their product is a subproduct. Thus $\langle X^U \rangle$ contains all of T_0 , giving the index bound in (i).

To obtain (ii), first generate T_0 by supported three-cycles in its diagonal factors. Such a cycle is specified by at most $3b$ letters, recording its simultaneous support in all tied parts. There are boundedly many X -orbits of these specifications by Equation (14.1). Take one representative from each orbit. Each commutes with the control trimmed to fix its support, and these representatives generate T_0 over X .

It remains to choose suitable representatives for U/T_0 . Write the size of each part as $b_B + \sum_i k_{B_i} n_i$, with fixed-letter counts and copy multiplicities for X . Their coefficients are bounded by s . Lemma 14.5 gives a subcontrol $X_1 \leq X$ of bounded complexity, with identical profiles and at least two fixed letters on every part. Fix a coset in U/T_0 . By multiplying its representative by T_0 , the map from one representative part in each diagonal class to its destination may be changed to any bijection of the prescribed parity. The maps on its other tied parts then correspond through the tying identifications, since the representative transports the diagonals. Those identifications are X -equivariant, because $X \leq T_0$. The identical X_1 -profiles allow the chosen bijections to be X_1 -equivariant; a swap of two fixed letters corrects their parity if necessary. The resulting coset representative centralizes X_1 , so $X_1 \leq X \cap gXg^{-1}$. There are boundedly many cosets by Equation (14.6).

This completes the induction. All coefficient lists used for profiles are drawn from bounded ranges, and all bounded-domain branches use finite subset counts. The only recursive call strictly decreases h , even when the complexity grows after small axes are discarded. Hence the bounds proved are computable functions of (s, h) , and substituting $h = q_2(s)$ proves the Proposition. $\square$

Proposition 14.7 (Overgroups of controls with private squares). *There are computable functions $B(s), C(s)$ such that every control X with private squares of complexity at most s on a finite nonempty set I has at most $B(s)$ transitive overgroups in S_I , and every such overgroup U satisfies*

$$[U : \langle X^U \rangle] \leq C(s).$$

The functions can be taken nondecreasing and at least one.

Proof. We count three successive choices: the block partition, the local primitive structures and their diagonal ties, and the overgroup in the resulting wreath product with bounded local groups. The first two choices will be encoded by invariant relations on tuples of bounded length.

Again choose a minimal nondiscrete U -block partition $\mathcal{B}$, with part size $d \geq 2$ and part set I' . The case $|I| = 1$ is immediate, and the choices of $\mathcal{B}$ are bounded by Equation (14.1).

The restriction of this partition to a square orbit is either constant, a partition into lines in one direction, or discrete. Indeed the point stabilizer in the product of its two alternating factors is the product of two maximal nonnormal natural point stabilizers. If an overgroup of that stabilizer projects larger in an axis, conjugating the independently present point stabilizer by a

mover generates that entire alternating factor independently. These are exactly the four resulting block possibilities. The discrete possibility cannot occur here: privacy means that no other X -orbit can map equivariantly onto the moving orbit of parts of this square, so a part meeting it would have size one, contrary to $d > 1$.

It follows that the image X' on I' is rank-one. Each moving part orbit is identified equivariantly with the natural alphabet of its contributing axis. Distinct such orbits for the same axis are diagonal natural copies; the factor is trivial on every other part orbit, and the active factors remain independent. There are at most s part orbits. Moreover, an injective rank-one copy on I' can only share its parts with copies of the same factor, one letter per copy. Other orbits, including all the private squares, have trivial action of that factor. Thus an injective rank-one copy still forces $d \leq s$.

If d is bounded in terms of s , every private square has side at most d , since one of its axis lines lies in a part. Discard its factors and declare its letters fixed. There are boundedly many such letters, and what remains is a rank-one control of bounded complexity on I . [Proposition 14.6](#) bounds the desired number and index, also for the original larger X .

Assume henceforth that $d > H(s)$, increasing the displayed threshold if needed only within a computable bound. In a part B , each original orbit contributes a whole orbit, an axis line, or a singleton. One contribution has size at least d/s . Therefore some axis of degree at least $\sqrt{d/s}$ acts nontrivially inside B . It lies in the kernel of the action on I' : square axes are private, and an injective rank-one occurrence of its factor anywhere would force $d \leq s$.

The local socle and its normal closure. By [Lemma 14.4](#), the socle S_B of the primitive part-stabilizer group is A_ℓ^e in one of [Equation \(14.3\)](#). It is the unique minimal normal subgroup, has trivial centralizer, and its quotient has order at most eight. Let K be the part kernel. Its projection to B is a nontrivial normal subgroup of the primitive group, so contains S_B . Set

$$T_0 = K \cap \prod_{B \in \mathcal{B}} S_B.$$

This intersection projects onto every S_B . To verify this without a dependence on the number of parts, take the $8!$ -th powers of all elements of K ; these lie in T_0 . In an individual simple site, every 11-cycle is an $8!$ -th power, since $11 \nmid 8!$. Such cycles generate the large alternating group. Since the projection of K contains the whole local socle, these powers project to a generating set of every individual site, hence onto S_B .

Thus T_0 is a subdirect product on all the simple sites, and is a product of full twisted diagonals. Uniqueness of the local socles makes it normal in U . The local slots are transitively permuted by a part stabilizer, as required by primitivity in the two-slot case; U is also transitive on parts. Hence U is transitive on all simple sites and on the diagonal simple factors of T_0 . The kernel axis in X found above has a nontrivial local projection, and the local centralizer of S_B is trivial. Commuting that axis with T_0 therefore gives a nonidentity element in $T_0 \cap \langle X^U \rangle$. This intersection is normal in U , so transitivity on the diagonal factors gives

$$T_0 \leq \langle X^U \rangle. \tag{14.7}$$

Encoding structures and diagonal ties. We next count the local structures and their diagonal ties without assuming that $|I'|$ is bounded. In the natural case $(1, 1)$, the site alphabet is the set of part letters. In case $(1, 2)$, choose the graph in which two distinct two-subsets are adjacent when they intersect. In case $(2, 1)$, choose adjacency when two distinct points share a coordinate. Choose this structure on one part and transport it by U to all parts; the initial part stabilizer preserves it,

so transport is well-defined. The resulting relation on I is X -invariant and hence has boundedly many possibilities.

These graphs recover their natural alphabets. In the two-subset graph, more than three pairwise-intersecting distinct pairs have a common entry; the large maximal cliques are exactly the stars, one for each alphabet entry. An ordered adjacent pair determines the star at its common entry. In the sharing graph, a clique of size at least two lies on a unique coordinate line, so an edge determines that line. Disjointness groups the lines into two directions; the lines in each direction form one recovered site alphabet. Thus the recovered alphabet points are stars in the two-subset case and coordinate lines in the product case. Their disjoint union is an equivariant image of the X -invariant subset of I^2 consisting of ordered edges within the parts. In the natural case use the diagonal pairs (p, p) instead. Relations on pairs of recovered alphabet points therefore pull back to relations on I^4 , whose orbit count is bounded by Equation (14.1).

An isomorphism between these part structures is induced by natural alphabet bijections, with a possible permutation of the two slots. Isomorphisms between high alternating simple factors are likewise induced uniquely by alphabet bijections. Thus record the twisted diagonals of T_0 by their matched-letter relation on the recovered alphabets. This relation is equivariant, because X normalizes T_0 , and it determines the full twisted diagonals. Its pullback is an invariant relation on I^4 , so Equation (14.1) bounds all choices of T_0 for the fixed block structure. This explicit four-tuple bound also covers ties between sites in different parts.

Bounded-group lifts. Let W be the stabilizer of the partition and chosen part structures. In fixed local coordinates its homomorphism

$$W \longrightarrow D^{I'} \rtimes S_{I'} \quad (14.8)$$

records the alphabet parities and slot permutations, where $|D| \leq 8$. Its kernel is $\prod_B S_B$. On $N_W(T_0)$ the kernel is exactly T_0 , because a subdirect product of nonabelian simple groups is self-normalizing inside their direct product, as proved in Proposition 14.6. Hence $N_W(T_0)/T_0$ embeds in the bounded-group wreath product in Equation (14.8).

We can make the image of X pure in this wreath product. A part stabilizer in X is a product of large alternating groups, allowing one natural point stabilizer A_{n_i-1} when the part orbit is natural. Since $n_i \geq 7$, each nontrivial factor is simple of order greater than eight. Its local homomorphism to D is therefore trivial. Fix a local quotient frame at one representative of each X -orbit of parts and transport it by X . The stabilizer property makes this unambiguous; changing frames amounts only to changing coordinates by local elements of D . All base components of X are now identity. These frames depend only on the structures and X , not on the subgroup U subsequently being counted.

Apply Proposition 14.6 to the rank-one part action X' on I' , of complexity at most s . It bounds its possible transitive overgroups and supplies the generators with intersection controls needed in Lemma 14.3. With this complexity bound and $|D| \leq 8$, that lemma bounds the possible images U/T_0 in Equation (14.8) and their indices modulo the normal closure of the pure image of X . For fixed structures and T_0 , the image determines U uniquely, because U contains the kernel T_0 . By Equation (14.7), these quotient indices are exactly $[U : \langle X^U \rangle]$. Combining the bounded choices of partitions, structures, matching relations and lifts proves both assertions. $\square$

Theorem 14.8 (Uniform computable template bound). *There is a total computable nondecreasing function $\Phi : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 1}$ such that every weighted template*

$$F = [S_Z \times Y, S_{Z \sqcup (\bigcup T)}]$$

of [Section 13](#), with at most t units, excluding the foundation Z from this count, and $|Z| > |\cup T|$, satisfies $|F| \leq \Phi(t)$. The bound is independent of all full-unit sizes, square sides, auxiliary axis lengths, offset sizes, and the size of Z . No stock-size hypothesis is required for this counting assertion.

Proof. On any nonempty union of units, the prescribed product Y contains a control with private squares whose complexity is bounded computably in the unit count. On a full unit of size at least seven, keep its natural alternating group; a smaller full unit contributes at most six fixed letters. On an auxiliary unit $(x/4) \times 2 \times 2$, if $x/4 \geq 7$, keep $A_{x/4}$ on its four repeated natural copies. Otherwise its domain has at most 24 letters and may be fixed entirely. On each primary square keep its two prescribed alternating factors; their privacy is part of the independent unit construction. Thus complexity at most $24t$ suffices.

Consider an arbitrary residual overgroup $U \geq Y_R$ on a union R of at most t units. Its orbits are unions of the transitive units, so there are at most B_t choices for this orbit partition. On every orbit O , its transitive projection U_O has at most $B(24t)$ choices, by [Proposition 14.7](#). Let N_O be the normal closure of the prescribed unit groups on O in U_O . Their contained control shows that $[U_O : N_O] \leq C(24t)$.

The subgroup N_O , supported independently on O , is actually contained in U : its initial unit groups are independently supported, and conjugating them by U realizes their normal closure in the projection U_O without adding support on another orbit. Consequently U contains $\prod_O N_O$. Once its projections are fixed, its remaining choice is a subgroup in $\prod_O (U_O/N_O)$, a group of order at most $C(24t)^t$. There are at most $2^{C(24t)^t}$ such subgroups. The empty residual set has one choice.

By the exact literal labeling of the weighted template, each lattice element consists of an absorbed subset $I \subseteq T$ and one of these residual overgroups on $T \setminus I$. For $t \geq 1$, therefore, one may take the increasing envelope of

$$2^t B_t B(24t)^t 2^{C(24t)^t}, \quad (14.9)$$

and put $\Phi(0) = 1$.

To check effectivity of the whole construction, every invariant-relation choice used above is bounded by a power of two with exponent a tuple-orbit bound from [Equation \(14.1\)](#). Every bounded quotient has at most as many subgroups as subsets. Trimming and small degree branches use computable finite ranges; equal profiles use maxima over finitely many bounded coefficient systems and the terminating Presburger procedure in [Lemma 14.5](#). The recursion of [Proposition 14.6](#) strictly decreases the equivalence chain parameter. These operations define total computable bounds and never take a maximum by a finite-group search over unbounded unknown degrees. $\square$

15 The Diophantine reduction

We combine the effective counting theorem with the size readings of [Section 13](#). The distinction between their quantifiers is essential: the counting bound is computable from the number of units, whereas the literal numerical sizes used to exhibit a template need only exist when the Diophantine instance has a solution.

Definition 15.1 (Positive circuits). A positive circuit is a finite arithmetic circuit with positive integer input variables, constant 1, and addition and multiplication gates. A *value variable* records either an input or a gate value. A circuit equation requires equality of two designated outputs.

Solvability of positive circuit equations is undecidable. Indeed, the Davis–Putnam–Robinson–Matiyasevich Theorem gives undecidability of polynomial solvability over the nonnegative integers [\[30, 31\]](#). Replace each nonnegative variable x by $y - 1$ with y positive, expand the resulting integer

polynomial, and move its positive and negative coefficient terms to opposite sides. Add 1 to both sides, so each side is positive even when one collection of terms was empty. Positive integer coefficients are made by sums of 1, powers by products, and the resulting polynomials by positive circuits. All these transformations are effective and preserve existence of a solution.

Lemma 15.2 (Encoding a circuit by weighted comparisons). *There is an absolute stock integer S with the following property. For every positive circuit equation $\mathcal{C}$, one can compute an integer $t_S(\mathcal{C})$ from its syntax such that every positive solution determines a weighted template F with at most $t_S(\mathcal{C})$ units. In every size reading of this template supplied by [Theorem 13.26](#), the actual raw sizes give a positive solution of $\mathcal{C}$ in the same cell alignment. If all cells in that reading have size at most Q , the resulting solution has all its value variables at most Q .*

Proof. For every value variable v , including all gate values, prescribe a full *raw cohort* R_v of size v per unit and a full *side cohort* E_v of size $16v$ per unit. These sizes are literal integers when a solution is given. The relation between the two cohorts is recorded by a comparison of one side unit with sixteen distinct raw units. For a constant value $v = 1$, compare one side unit with sixteen distinct singleton markers.

For each required square side, include the primary square cohort Q_v of side $16v$, its auxiliary cohort of shape $(4v) \times 2 \times 2$, the full axis-reference cohorts, and all required full offset cohorts. A unit from E_v itself may serve as the full side reference for Q_v . Every extra full reference and every raw or side cohort receives its prescribed offset where required; an offset cohort requires no further offset. The formal square sides satisfy $8 \mid 16v$ and $16v \geq 16$. The needed full references for bounded slots are included as well. Distinct roles may use disjoint lots even when their numerical sizes happen to coincide.

An addition gate $c = a + b$ is recorded by the comparison

$$E_c = E_a + E_b, \quad (15.1)$$

where an equality of cohort symbols here means an equality of the indicated totals on chosen units. For a multiplication gate $c = ab$, introduce a fresh value $d = a + b$, with its own raw and side cohorts and its side-to-sixteen-raw comparison. Require [Equation \(15.1\)](#) for $d = a + b$, and use the square comparison

$$Q_d = Q_a + Q_b + 32E_c. \quad (15.2)$$

Whenever needed, add the square cohorts for a, b, d and their associated roles. The output equality is recorded using two distinct copies of each output side cohort. Thus even an equality between two individual values is read as an equality between two-unit totals.

Each comparison is between disjoint collections of a fixed bounded number of units, and at least one collection uses more than one unit. In particular, [Equation \(15.2\)](#) uses one unit against two square units and thirty-two side units, so thirty-five bounds the total support of every circuit comparison. Repeated occurrences of the same variable within a comparison use distinct copies. The pair-total reading in [Proposition 13.24](#) equalizes all the actual sizes in a cohort, so these choices of copies do not change the values being read. The same copies may be used again in another comparison. The comparisons for constants use the common marker size one.

Choose once and for all a stock S large enough for [Theorems 13.19](#) and [13.26](#), including all bounded marker losses, all marker requests in the readings, and the bounded number of distinct copies needed in any one comparison. Enlarge it by a fixed amount if necessary for the raw/side and output comparisons. With the comparison support and spare-copy needs fixed, neither the number of comparisons or cohorts nor any numerical size enters this choice. Singleton raw units may also count as markers. Include the global marker-loss bound in every cohort's stock, so enough

copies survive even if all deletions fall in one raw cohort. The same surviving copies can be used in different comparisons.

The role list is computable from the circuit: take its value variables, the fresh sums for multiplication gates, the square and auxiliary roles they require, their full axis references, and their offsets. Put the fixed stock in every role and include the fixed marker stock. There are only finitely many prescribed roles per value or shape, and offsets introduce no further offset recursion. Counting these listed lots gives $t_S(\mathcal{C})$. This calculation uses no solution values. If sizes of different roles coincide, keeping the lots separate still gives this same bound on their total unit count.

Now start with a positive solution and construct the literal weighted template using its values. All the prescribed total equalities hold. Apply the weighted size readings in their one common cell alignment. Let r_v and e_v be the actual raw and side sizes; the cohort equalization makes these well-defined positive integers. The raw/side and constant comparisons give

$$e_v = 16r_v, \quad r_v = 1 \text{ for every unit constant.} \quad (15.3)$$

In particular each side reference is even and at least sixteen, so [Theorem 13.25](#) applies to every required primary square and reads its actual size as e_v^2 . Addition comparisons give $r_c = r_a + r_b$. At a multiplication gate, the fresh sum gives $r_d = r_a + r_b$ and [Equation \(15.2\)](#) becomes

$$(16(r_a + r_b))^2 = (16r_a)^2 + (16r_b)^2 + 32(16r_c). \quad (15.4)$$

Subtracting the first two squares gives $512r_ar_b = 512r_c$, hence $r_c = r_ar_b$. This explains the factor 32: the cross term of the squared side sizes is $2 \cdot 16^2 r_ar_b$, while one side unit for c has size $16r_c$. The doubled output comparison gives the required output equality. All equations concern the same actual raw sizes, because the size readings use one alignment simultaneously. These sizes are themselves cell sizes, so their bound by Q proves the last assertion. $\square$

Theorem 15.3 (Undecidability of finite representability). *There is no algorithm which, given the tables of a finite lattice, decides whether it is isomorphic to the congruence lattice of a finite algebra. Equivalently, there is no total computable bound, in terms of lattice cardinality, for the minimum carrier size among all representable finite lattices.*

Proof. Suppose that such a decider exists. By [Corollary 2.3](#), it gives a total computable nondecreasing function $M(N)$ that bounds a representing carrier for every representable lattice with at most N elements. Explicitly, enumerate the finitely many lattice tables on at most N points, retain the YES instances using the decider, and search the finite colored-graph witnesses for each retained table until one is found. The graph criterion supplies a finite algebra on that witness carrier. Taking the maximum over these finitely many searches produces $M(N)$.

Fix the absolute integer S from [Lemma 15.2](#). Its existence is sufficient: the single integer S may be hardwired in the algorithm below. This step does not assert an algorithm for extracting a sharp, or even a specified, sufficient stock from the structural arguments. Once fixed, the function $t_S(\mathcal{C})$ is computable from the circuit syntax. For an input circuit equation $\mathcal{C}$, compute

$$t = t_S(\mathcal{C}), \quad N = \Phi(t), \quad m = M(N), \quad Q = 60^m m!. \quad (15.5)$$

We claim that if $\mathcal{C}$ has any positive solution, then it has one whose value variables are all at most Q . Given an arbitrary positive solution, use [Lemma 15.2](#) to exhibit its weighted template F . Its literal size parameters may be much larger than the bounds just computed, but [Theorem 14.8](#) still gives $|F| \leq N$. The template is representable by the coset action of its defining ordinary subgroup interval, as in [Proposition 13.6](#). Hence it has a finite-algebra representation on at most m points.

Among the representations of all nontrivial lattice images of this F , take one with least carrier size m_0 . Since F itself is one such image, $m_0 \leq m$. The nontrivial-image reduction of [Proposition 13.6](#) gives an ordinary interval for the dual of a nontrivial image, with top group order at most

$$60^{m_0} m_0! \leq 60^m m! = Q.$$

The product-success and size-reading [Theorem 13.26](#) then provides the common actual cell alignment, after only the allowed bounded marker deletions. Every read cell size is positive and at most this top-order bound; markers have size one. [Lemma 15.2](#), with its preserved stocks, now extracts a positive solution of $\mathcal{C}$ all of whose value variables are at most Q . The side sizes are multiples of sixteen by [Equation \(15.3\)](#), so every use of the square reading in this extraction satisfies its actual numerical hypothesis.

Thus the complete quantifier implication is

$$\begin{aligned} \exists \mathbf{v} > 0 \quad \mathcal{C}(\mathbf{v}) \\ \implies \quad \exists F \quad |F| \leq N, \quad F \text{ representable} \\ \implies \quad \exists A \quad \text{Con } A \cong F, \quad |A| \leq m \\ \implies \quad \exists [H, G] \text{ as above with } |G| \leq Q \\ \implies \quad \exists \mathbf{r} \in \{1, \dots, Q\}^k \quad \mathcal{C}(\mathbf{r}). \end{aligned} \tag{15.6}$$

where the circuit determines k and all quantities in [Equation \(15.5\)](#) are computed before any solution is known. Auxiliary fresh sums can be included among the k value variables; restricting back to the original values preserves the conclusion. The reverse implication from a bounded solution to some solution is immediate.

An algorithm may therefore compute [Equation \(15.5\)](#) and search all positive assignments up to Q to the finitely many value variables, checking each gate and the output equality. It terminates, and [Equation \(15.6\)](#) makes its answer correct. It never needs to construct a weighted template with unknown sizes: that template was used only to prove that an existing solution has a bounded replacement. This would decide positive circuit solvability, contradicting the Davis–Putnam–Robinson–Matiyasevich Theorem. The equivalent carrier-bound formulation follows again from [Corollary 2.3](#). $\square$

The structural input to this reduction is precisely the weighted success theorem, whose exclusion arguments use the sufficiently-general subgroup theorem in the scope established earlier. The reduction introduces no additional uniformity in the numerical weights beyond the computable counting bound proved in [Theorem 14.8](#).

Corollary 15.4 (Undecidability of finite subgroup-interval recognition). *There is no deterministic Turing machine which, given the order table of a finite nonempty lattice L , always halts and decides whether $L \cong [H, G]$ for some finite group G and subgroup $H \leq G$, where $[H, G]$ consists of every subgroup K with $H \leq K \leq G$, ordered by inclusion. Equivalently, there is no total computable bound, in terms of lattice cardinality, on the least possible order of the top group G among lattices admitting such a realization.*

Proof. Suppose first that such a decider exists. For each $N \geq 1$, enumerate all order-relation tables on sets of at most N points, retain those which are nonempty lattice tables, and use the decider to select the affirmative ones. For each selected table, enumerate finite multiplication tables in increasing carrier size. Test the group axioms, enumerate the subgroups H , and for each H enumerate every subgroup K containing it. Testing the finitely many bijections then

decides whether the resulting full inclusion interval is isomorphic to the selected lattice. Each selected search terminates. The maximum of the witness group orders found and 1, followed by an increasing envelope if necessary, gives a total computable nondecreasing function $B(N)$ such that every affirmative lattice with at most N elements has a realization $[H, G]$ with $|G| \leq B(N)$. Conversely, any total computable bound on the least possible $|G|$ gives a decider by the same finite tests restricted to group orders up to that bound. Thus it suffices to contradict the existence of B .

Fix the absolute stock integer S of Lemma 15.2; as in the preceding proof, this one integer may be hardwired. For a positive circuit equation $\mathcal{C}$, compute from its syntax

$$t = t_S(\mathcal{C}), \quad N = \Phi(t), \quad Q = B(N).$$

These quantities are computed before any solution is known. If $\mathcal{C}$ has a positive solution, Lemma 15.2 exhibits a weighted template F with at most t units, and Theorem 14.8 gives $|F| \leq N$. This literal template is itself a full ordinary subgroup interval by Definition 13.2. It is nontrivial: the circuit construction includes side units of size $16v \geq 16$. The bound B therefore supplies an ordinary realization $F \cong [H, G]$ with $|G| \leq Q$.

Use the identity $F \rightarrow F$ as the nontrivial weighted image allowed by Convention 13.4. Apply the ordinary-input part of Theorem 13.26 to this realization with its actual top order $M = |G|$. It gives one common size reading whose positive cell sizes satisfy $u_j \leq M \leq Q$. By Lemma 15.2, that reading yields a positive solution of the same equation $\mathcal{C}$ with every value variable at most Q .

Consequently, if $\mathcal{C}$ has any positive solution, it has one in $\{1, \dots, Q\}^k$, where k is determined by the circuit and may include its finitely many auxiliary fresh sums. The reverse implication is immediate. Computing Q and testing these finitely many assignments would decide positive-circuit solvability, contradicting the undecidability established at the start of this section. The argument uses the existence of F and of its reading only to prove a bounded replacement for a solution; it does not require either to be constructed without that solution. $\square$

16 An explicit computable size ceiling

We give an ordinary computable upper bound for the cardinality of one nonrepresentable lattice in the affine-chart family. The bounded-rank exclusion in Section 8 used a sufficiently-general-subgroup theorem with an unspecified uniform threshold. To choose explicit finite parameters here, we first prove a quantitative substitute: a finite subgroup avoiding proper closed sets of a specified computable format lies between the derived group and the full fixed group of a Steinberg map. We then bound the pruning and protected-path parameters and count the resulting lattice.

These bounds concern the negative affine family, independently of the weighted-template counting bound and of any hypothetical representation decider. They provide the fixed initial segment of lattice tables used in Section 17.

16.1 A uniform expanding function

Definition 16.1 (Affine ring format). For an integer $x \geq 10^9$, an affine ring formula has *format at most x* if it has at most x occurrences in total of polynomial-equality atoms and the operations $\neg, \wedge, \vee, \exists, \forall$, at most x variables, and polynomial atoms of total degree at most x . Occurrences are counted in the formula tree, with a separate variable for each quantifier binder. The variable count includes the entire ambient affine coordinate list, even unused coordinates. Coefficients are arbitrary field elements, and a polynomial is counted through its atom and degree, rather than its expanded arithmetic expression.

Let $a(x)$ be the least positive simultaneous bound, over all algebraically closed fields and formulas of format at most x , for both the format of an equivalent quantifier-free formula in the same ambient coordinates and the number and degrees of polynomial equations defining its Zariski closure.

Lemma 16.2. *The integer $a(x)$ exists, is nondecreasing, and is total computable. This holds uniformly over the characteristic and the specialization of every coefficient.*

Proof. For fixed x , canonically rename the coordinate and bound variables. There are finitely many possible formula trees. Replace each polynomial atom by the polynomial containing every allowed monomial of degree at most x , with an independent coefficient parameter. This produces finitely many schemes. Their parameter lists are finite and computable, although the specialized coefficients need not have effective descriptions. These parameters belong to the tests of uniformity, not to the variable count of the specialized input formula.

Effective quantifier elimination in the theory ACF, without fixing the characteristic, supplies a quantifier-free formula for each scheme [6]. Write it as a finite disjunction of conjunctions. For a conjunction $f_i(z) = 0, g_j(z) \neq 0$, introduce one inverse variable u and the ideal

$$I = (f_i(z), u \prod_j g_j(z) - 1).$$

The Nullstellensatz gives the closure of that conjunction as $V(I \cap k[z])$ [48, Theorem 10.34.1]. An elimination Gröbner basis gives equations for this closure. The closure of the original finite union is the union of these closures, which again has a finite equation list, for example from the product of their defining ideals. Characteristic-independent Gröbner bounds are available [11]; we give also the uniformity argument needed to compute the *least* simultaneous bound.

For given bounds on variables, generators, and degrees, the assertion that a tuple of generators has a Gröbner basis with certificates of size at most b is first order in its coefficients. The certificates consist of bounded lists of polynomials, bounded ideal combinations in both directions, choices of leading monomials from finite lists, and Buchberger reductions or standard representations. The required monomial comparisons for a fixed elimination order form finite tests; polynomial identities become coefficient equations. We can make these assertions increasing in b . Every individual tuple in every algebraically closed field has some certificate. If no uniform b existed, the Compactness Theorem [27, p. 26], applied to ACF with constants for the coefficients and the negations of all these increasing assertions, would give a tuple with no certificate at all. Thus a uniform b exists. First-order ACF tests of successive b find one.

In such a certified elimination basis, the members involving only the retained coordinates define the projected closure. Independence of the eliminated variables is itself a finite coefficient test. Existentially quantifying a bounded certified basis therefore gives an effectively found formula $C_\phi(c, z)$ for the closure of the fiber defined by $\phi(c, z)$. The coefficient specialization precedes the certificate test. In particular, this constructs each specialized fiber's closure, rather than specializing the closure of the total parameter family. Vanishing leading coefficients, empty pieces, and empty equation lists are handled by finite branches.

For a candidate bound A , enumerate the finitely many quantifier-free templates $Q(d, z)$ of format at most A in the same coordinates. For each input scheme, the assertion

$$\forall c \bigvee_Q \exists d \forall z (\phi(c, z) \longleftrightarrow Q(d, z))$$

tests the quantifier-free part of the bound. Let $P_1(e, z), \dots, P_A(e, z)$ be dense polynomials of degree at most A with coefficient tuple e . The additional assertion

$$\forall c \exists e \forall z \left(C_\phi(c, z) \longleftrightarrow \bigwedge_{i=1}^A P_i(e, z) = 0 \right)$$

tests the closure bound; zero polynomials pad shorter lists. Conjoin these tests over the finitely many schemes and ambient dimensions. Validity of the resulting sentence in all algebraically closed fields is decidable: after effective elimination a closed sentence is a Boolean combination of integer equalities, whose behavior is checked in characteristic zero and at the finitely many prime divisors of the nonzero integers occurring. Testing $A = 1, 2, \dots$ consequently finds exactly $a(x)$. The nested input classes make a nondecreasing. Finally, Bézout bounds give elementary bounds on the number of components and their sum of degrees from these equation bounds. All these statements allow ambient affine matrix coordinates, including equations specifying the group itself. $\square$

Definition 16.3 (The function J). Let $R(c, h)$ be the least positive set size forcing a monochromatic h -element subset in every c -coloring of its unordered pairs of distinct elements. Let $\text{HJ}(a, c, s)$ be the least positive word length forcing a monochromatic s -slot word family over an a -letter alphabet in every c -coloring. Here some positions have fixed letters, possibly no positions do, and the others are partitioned into s disjoint nonempty sets; each set carries one independently arbitrary common letter. Put

$$J(x) = 1 + \max(\{2^{2^{x!}}, a(x)\} \cup \{R(c, h) : 1 \leq c, h \leq x\} \cup \{\text{HJ}(a, c, s) : 1 \leq a, c, s \leq x\}). \quad (16.1)$$

The notation J^d denotes iteration, not an ordinary power.

The finite Ramsey and multidimensional Hales–Jewett theorems [17, 43] ensure that these least bounds exist. For each parameter tuple they can be found, in principle, by finite coloring tests at successively larger sizes. Thus J is nondecreasing and total computable. Its double-exponential factorial term ensures that an additional iteration absorbs the elementary polynomial and exponential costs used below. No evaluation of these finite searches is required.

16.2 Formats for bounded root systems

We make the genericity bound quantitative in the affine coordinates of the adjoint action. The conventions for positive standard point maps, including the parameter P for the special twisted types, are those of Section 8.

Lemma 16.4 (Root words and fixed points). *Let $10^9 \leq x \leq y$ be integers, and let G be split pinned, adjoint simple of rank at most x over an algebraically closed field of positive characteristic. Its adjoint representation embeds G as a closed subgroup of GL_d , where $d = \dim G = O((x + 10)^2)$. For rational simple modules of the simply connected cover, of dimension and highest-weight coefficients at most y , graphs of the induced projective actions have format at most $J(y)$. Stabilizers of a subalgebra or of at most y subspaces of the corresponding matrix algebra, and normalizers of projective subgroups of order at most y , have format at most $J^3(y)$.*

For a positive standard map of parameter P , the fixed-point test has a Boolean description with at most $J^5(y)$ atoms and degrees at most $J^5(y)P$. A proper closed subset of G of format at most y therefore contains at most

$$J^{10}(y)P^{\dim G - 1} \quad (16.2)$$

fixed points. For $P > J(y)$ the derived simple fixed group has order at least $P^{\dim G}/(10y)^4$. The same bounds apply to the other adjoint types in the stated rank range.

Proof. The embedding. Every root is primitive in the character lattice of the adjoint torus, which is the root lattice: it is Weyl-conjugate to a simple root. Its differential is therefore nonzero in

every characteristic. An element central in the Lie algebra has no root-space components, by commuting with toral directions. Its remaining toral component vanishes because the simple-root differentials form a basis. Thus the adjoint kernel has zero Lie algebra. It is finite reduced, is central by connectedness, and is trivial because G is adjoint. The separable isomorphism theorem for algebraic groups identifies G with its closed adjoint image. The image has determinant one. We use a Frobenius-standard basis.

Word and stabilizer formats. Bruhat decomposition and the rank-one torus formulas cover both simply connected and adjoint points by root-element words of length less than $1000(y+10)^4$. This includes zero parameters and words for Weyl representatives. Root matrices on the modules in question have polynomial entries of bounded degree: a coefficient of t^h shifts weight by $h\alpha$, and the weights lie in the Weyl polytope. Coroot heights in the irreducible root lists are less than $100(y+10)$, so pairing with the coroot bounds the rootwise weight range. The number of word choices needed is at most

$$(1000(y+10)^2)^{1000(y+10)^4+1}. \quad (16.3)$$

Use entries of intermediate products as variables and the coefficients of the root maps as parameters. Describing a group element and its action by the same word gives the asserted $J(y)$ format, using invertible matrices modulo scalar matrices for a projective action. The same bound also dominates Weyl dimensions for the indicated highest-weight coefficients, by the Weyl dimension formula. Conditions that a matrix preserve a subspace, a subalgebra, a collection of at most y subspaces, or normalize a projective group with at most y listed elements add only bounded lists of variables and equations. Allowing matrices and translates as parameters gives the $J^3(y)$ bound. Images, preimages, and intersections combine these formulas; closures use [Lemma 16.2](#).

Point maps and the fixed-point estimate. Pinned diagram maps and special isogenies in their fixed characteristics have bounded graph format by the same words: their root parameters are sent to t or t^p . Split field powers, including their inverses on points, only twist the coefficients of sets. Inverses of special maps use also an inverse field power. Images of bounded sets under these point maps thus cost only a few further iterations of J .

For a positive standard map, first eliminate the bounded combination of inner, pinned diagram, or special maps, whose graph has format at most $J^2(y)$, and only then substitute the single split field power. This gives at most $J^5(y)$ atoms of degree at most $J^5(y)P$ in the fixed-point test. On a conjunction piece, the inequalities specify an open subset. Since the fixed set is finite, only isolated components after the polynomial cuts can contribute. On a variety of dimension d and sum of component degrees δ , cuts of degree at most E contribute at most δE^d such points: choose at most d generic linear combinations of the cuts retaining the finitely many points as isolated points, and apply affine Bézout for locally closed sets [[38](#), Definition 3, Proposition 2.9 and Theorem 2.14]. The generic-combinations step supplies the exponent d locally. For a proper subset of G , $d \leq \dim G - 1$. The degree bound obtained from $a(y)$, the number of Boolean pieces, and the factor $(J^5(y))^{\dim G}$ are absorbed by $J^{10}(y)$, proving [\(16.2\)](#). The ordinary order and center formulas give the stated lower bound for the derived simple when $P > J(y)$. $\square$

Proposition 16.5 (Quantitative genericity). *Let $x \geq 10^9$, and let G be adjoint simple of rank at most x over an algebraically closed field of characteristic $p > 0$. If a finite subgroup $\Gamma \leq G$ is not contained in any proper closed subset of affine ring format at most $J^{1000}(x)$, then there is a positive-height standard point map Ψ such that*

$$(G^\Psi)' \leq \Gamma \leq G^\Psi. \quad (16.4)$$

Proper closed subsets in this hypothesis need not be irreducible.

Proof. Choose a fundamental simple module of the simply connected cover of G . Its dimension n is at most $J(x)$. Its weights lie in one coset of the root lattice, so the center acts by a scalar character, giving an irreducible projectively point-faithful action of G .

Claim 1. *The matrix lifts of every nontrivial normal subgroup of Γ span the full matrix algebra, and Γ is almost simple.* This connected action preserves no nontrivial tensor splitting individually. Indeed the projective actions on the factors are algebraic, as can be seen from their conjugation actions on the factor matrix algebras. Pull back $\mathrm{SL} \rightarrow \mathrm{PGL}$ along each map from the simply connected source. The reduced identity group of the pullback still surjects onto that source: the full map is finite surjective and has only finitely many components. Its kernel is central of multiplicative type. Its connected solvable radical has trivial image and hence dimension zero, so it is semisimple. The resulting central isogeny onto the simply connected source is an isomorphism, providing a linear lift. Perfection leaves no scalar character in the tensor comparison. Irreducibility would make both tensor factors nontrivial irreducible modules, but their dominant highest weights cannot add to a single fundamental weight.

An invariant unital matrix subalgebra is accordingly scalar or full. Its radical would otherwise give a proper invariant annihilator. If it is semisimple, connectedness fixes its central idempotents, so irreducibility leaves one matrix factor. A proper nonscalar such factor would yield the excluded tensor splitting.

We may use $y = J^2(x)$ in the stabilizer tests of [Lemma 16.4](#); it covers n^2 and all the lists of matrix data needed here. Their stabilizers and the bounded unions of translates used below have format well within $J^{1000}(x)$. Avoidance implies $\Gamma \neq 1$. For every nontrivial normal subgroup of Γ , the algebra spanned by its matrix lifts has a bounded-format stabilizer containing Γ , so is G -invariant. It is nonscalar by projective faithfulness, and hence is the full matrix algebra.

If such a normal subgroup were abelian, its distinct projective elements would have lifts with distinct scalar conjugation characters. Equal characters would make the quotient of two lifts centralize the full span, forcing equal projective elements. Independence of the corresponding eigenvectors bounds the order by n^2 . Its normalizer is then a bounded proper closed subgroup: a finite nontrivial subgroup normal in connected adjoint simple G would be central, which is impossible. This contradicts avoidance.

A minimal normal subgroup is therefore a power of a nonabelian simple group. Lifts of its factors commute linearly across factors, by perfection, and their algebras jointly generate the full matrix algebra. A radical in one algebra would generate a nilpotent ideal in that full algebra, so vanishes; the center is scalar for the same commuting-generation reason. Each factor algebra is consequently a full matrix algebra, and their tensor product is simple and maps onto the full matrix span. These nontrivial tensor factors are permuted by Γ . The bounded stabilizer of their collection must contain G , whose connectedness fixes each factor. Tensor indecomposability leaves exactly one factor. Denote the resulting simple normal subgroup by S . Fullness and perfection give it trivial projective centralizer, and conjugation identifies

$$S \leq \Gamma \leq \mathrm{Aut}(S). \quad (16.5)$$

Claim 2. *The simple normal subgroup S is of Lie type in characteristic p , of rank at most $100(n+1)$ and parameter $P > J^{100}(x)$.* The classification of finite simple groups, together with the usual covering and automorphism descriptions [\[13\]](#), now reduces to a large-parameter group of Lie type. We record the numerical exclusions. A faithful projective degree- n action of A_5^u forces $n \geq 2u$. Lift to the product of the universal covers. Each perfect factor acts nontrivially on some composition factor, since a nonsoluble image cannot act trivially on all composition factors. Irreducible modules of a finite direct product over an algebraically closed field are outer tensors,

also in modular characteristic. A composition factor on which $a \geq 1$ direct factors act nontrivially has dimension at least $2^a \geq 2a$; summing proves the claim.

In alternating degree d there are $\lfloor d/5 \rfloor$ disjoint copies of A_5 . In high classical rank e there are at least $\lfloor (e-1)/5 \rfloor$ such factors on ordinary or paired hyperbolic coordinates; nonsplit forms still have enough coordinate pairs, and unitary groups can use orthonormal coordinates. These perfect permutation groups lie in the projective special group. Thus the relevant Lie rank is at most $100(n+1)$, and the alternating degree is bounded as well. For Lie parameter at most $J^{100}(x)$ in this rank range, or for these bounded alternating groups, sporadics, and small exceptions, the ordinary order and automorphism bounds give $|\text{Aut}(S)| < J^{110}(x)$. A finite set of this size itself has format below the avoidance ceiling, contradicting (16.5). Hence the parameter satisfies

$$P > J^{100}(x). \quad (16.6)$$

Suppose its defining prime p_0 differs from p . At this parameter the standard multiplier theorem identifies the simply connected fixed group $\mathbf{H}^{F_0}$ of a pinned Steinberg map with the universal central cover of S . Our projective representation lifts linearly to it and is projectively nontrivial on noncentral unipotents. In ordinary field/diagram type, put $q = P$. The highest long root θ is diagram-stable, also in the simply laced case. The fixed root elements have parameters $b = cb^q$ for a fixed $c \neq 0$, a one-dimensional additive $\mathbf{F}_q$ -space. The fixed coroot elements $\theta^\vee(a)$, $a \in \mathbf{F}_q^\times$, act by multiplication by a^2 .

For a special twisted type write

$$F_0 = \sigma \text{Fr}_{p_0}^m, \quad \sigma^2 = \text{Fr}_{p_0}, \quad q = p_0^{2m+1} = P^2, \quad p_0 \in \{2, 3\}.$$

The highest long and short roots θ, θ' are paired by the special root correspondence, which preserves the positive chamber and exchanges lengths. The pinned root-power rules and square-Frobenius identity hold in each of B_2, F_4 in characteristic two and G_2 in characteristic three [7, Sections 2–3 and Definition 5.4]. Their root groups commute because no positive combination involving both is a root. In pinned coordinates over the split prime field,

$$u(b) = x_\theta(b)F_0(x_\theta(b)), \quad b \in \mathbf{F}_q,$$

is a fixed additive copy of $\mathbf{F}_q$. On these rank-one parameters the exponent of F_0 is p_0^r , where $r = m$ or $m+1$, and the root datum also gives

$$F_0(\theta^\vee(a)) = (\theta')^\vee(a^{p_0^r}).$$

For $a \in \mathbf{F}_q^\times$, the coroot product $t(a) = \theta^\vee(a)F_0(\theta^\vee(a))$ is fixed, since the two coroot factors commute and $F_0^2 = \text{Fr}_q$. The same identity proves fixedness of $u(b)$. On its θ -coordinate, $t(a)$ acts by the two root–coroot pairings. Equivariance makes the paired coordinate the F_0 -image of the transformed θ -coordinate. Thus $t(a)$ sends $u(b)$ to $u(cb)$, where

$$c = a^{2+p_0^r \langle \theta, (\theta')^\vee \rangle}.$$

The exponent is nonzero at our large m and has absolute value at most $2 + 3p_0^{m+1}$. In ordinary type the scalar subgroup has size at least $(P-1)/2$; in special type its size is at least

$$\frac{P^2 - 1}{2 + 3\sqrt{p_0}P} > \frac{P}{20}.$$

Thus either case gives a noncentral additive p_0 -group with a scalar action subgroup of order at least $P/20$. It acts diagonalizably in the linear lift, with a nontrivial character. Finite-field

trace parametrization shows that the orbit of such a character under the scalar subgroup is free. Consequently $n \geq P/20$, contrary to $n \leq J(x)$ and (16.6). The defining characteristic is p .

Claim 3. *The outer index satisfies $[\Gamma : S] \leq J^3(x)$.* This bound will control the unions of S -cosets used in Claim 4. Lift to the simply connected universal cover of this large-parameter group, now with pinned Steinberg map F . The lift is unique by perfection and irreducible by fullness. Steinberg's Restriction Theorem, including its twisted versions, realizes it as the restriction of a rational simple $L(\lambda)$ of F -restricted highest weight [19, 45, 46]. Write its ordinary digits as $\lambda = \sum_h p^h \lambda_h$. The Tensor Product Theorem shows that at most $\log_2 n$ digits are nonzero. Each digit coefficient is less than n : on a highest simple-root face a coefficient $c < p$ gives the simple rank-one module of dimension $c + 1$.

The cyclic subgroup of $\text{Out}(S)$ induced by the pinned Fr_p has index at most $100(100(n+1)+10)^2$. Every class of this cyclic subgroup realized by Γ preserves the lifted module. Indeed remove its inner discrepancy in S ; a projective intertwining becomes a linear intertwining on the perfect cover. For $F = \text{Fr}_p^f \tau$, with τ a pinned diagram map, such a shift preserves the occupied digit positions modulo f , by uniqueness of F -restricted restriction. Wrapping a digit past f only applies a diagram to it. Fixing one occupied position bounds the preserving shifts modulo f by $\log_2 n$. At most three representatives per shift are needed, since the actual order divides $f \text{ord}(\tau)$.

For special types the unequal root lengths require a different slot count. For $F = \text{Fr}_p^m \sigma$ with $\sigma^2 = \text{Fr}_p$ and $f = 2m + 1$, write

$$\sigma^* \omega_i = p^{c_i} \omega_{i'}, \quad c_i, c_{i'} \in \{0, 1\}, \quad c_i + c_{i'} = 1. \quad (16.7)$$

Here the restricted parametrization is Steinberg's special-factor version [45, Sections 11–12, especially Theorem 12.2]; the conversion to the following unequal coefficient boxes and the slot rotation are deductions from that parametrization and the displayed root-datum pullback. The independent universal-cover input used to lift the projective representation has already been imposed above. For a short node a paired with a long node a' , the pullbacks are $\sigma^* \omega_a = \omega_{a'}$ and $\sigma^* \omega_{a'} = p \omega_a$. On the finite fixed group, σ agrees with Fr_p^{m+1} , by applying σ to $F(g) = g$. The special tensor factorization therefore converts paired coefficients

$$0 \leq A_a < p^{m+1}, \quad 0 \leq B_{a'} < p^m \quad \mapsto \quad C_a = A_a + p^{m+1} B_{a'} < p^f$$

in Steinberg's short-supported parametrization. The ordinary digit factors occupy the disjoint ranges $0, \dots, m$ and $m + 1, \dots, 2m$. This is a bijective change of parameters, simultaneously over all paired nodes; it gives both existence and uniqueness for the unequal boxes. Put

$$b_i = m + c_{i'}, \quad b_{i'} = m + c_i, \quad b_i + b_{i'} = f.$$

The coefficient at i is less than p^{b_i} , so the two node boxes have b_i and $b_{i'}$ slots. Their concatenation is a circle of length f . The first exponent beyond the slots at i is the F^* -image of slot zero at i' .

Take shifts s modulo f , discarding those for which any occupied ordinary digit position h satisfies $h + s = m \pmod{f}$. At most $\log_2 n$ shifts are discarded. A nonzero digit newly at $e = h + s \pmod{f}$ is kept when $e < m$. When $e > m$, replace it by the module with highest weight

$$p^{e-m-1} \sigma^* \lambda_h. \quad (16.8)$$

The pinned maps commute, and

$$\text{Fr}_p^e = \text{Fr}_p^{e-m-1} \sigma F.$$

Since F fixes the finite group pointwise, this replacement preserves the finite restriction. Isogeny pullback preserves irreducibility. On each node pair the crossing exponent becomes

$$e - m - 1 + c_i = e - b_i$$

at the other node, exactly rotating the concatenated slots by s . Excluding $e = m$ lets the whole digit be kept or crossed simultaneously on all its nodes. There are no overlapping slots. The resulting rational tensor has irreducible finite restriction, hence is itself irreducible, and its sum highest weight is F -restricted. Uniqueness of restricted simples on the simply connected fixed group forces the rotation to preserve support whenever the shift is realized by Γ . On a nonempty paired-node cycle at most $2 \log_2 n$ slots are occupied, so at most this many rotations preserve its support. Including the discarded shifts and the bounded diagonal/graph factor gives, in all types, the ample bound

$$[\Gamma : S] \leq J^3(x). \quad (16.9)$$

Claim 4. *There is a positive-height standard point map Ψ of G such that $S = (G^\Psi)'$ and $N_G(S) \leq G^\Psi$. If several ordinary digits occur, S preserves a nontrivial tensor splitting. Its stabilizer in G is proper by the fundamental-module argument. At most $[\Gamma : S]$ translates cover Γ , giving a proper bounded-format closed set, a contradiction. The union remains proper because the ambient connected group variety is irreducible.*

For a single digit, untwist its field power, which permutes the pinned finite points. The same projective S is then the finite-point image of a rational module with highest-weight coefficients below n . Let C be its adjoint algebraic domain. The preimage in C of the projective image of G is closed, since algebraic-group images are closed, has format at most $J^{10}(x)$, and contains $(C^F)'$. To obtain the format bound use the two common-word action graphs with $y = J^2(x)$, which covers both ranks, module dimensions, and highest-weight coefficients. Apply Lemma 16.4 with $y = J^{11}(x)$. The inequality $P > J^{100}(x)$ makes the derived-fixed-group order exceed the number of fixed points allowed in a proper subset, so this preimage is all of C . The projective image of C cannot be strictly smaller than that of G : its pullback in G would be a proper bounded-format closed subgroup containing S , whose at most $[\Gamma : S]$ translates again cover Γ . The word-graph estimates and (16.9) keep both this test and the tensor test below $J^{1000}(x)$.

The adjoint domains are therefore identified on points through these projective isogenies. By the isogeny classification, transporting F gives a positive-height standard point-map element Ψ . Up to inner maps, self-identifications use diagrams and positive or negative split-field or special powers, and transport conjugates within that point-map group. For the dual pair B_e, C_e in characteristic two, $e \geq 3$, the pinned cross isogenies commute with split powers and need no diagram twist. We may compose the two projective isogenies with the adjoint quotient of their common image, which changes nothing on points, as the maps from the adjoint domains are already point-injective. Even if both maps pass through a cross-type target, the cross maps and their pointwise inverses commute with split powers. Hence $S = (G^\Psi)'$. If $g \in G$ normalizes S , then $g^{-1}\Psi(g)$ centralizes S , whose inner centralizer is trivial; thus $\Psi(g) = g$. Since Γ normalizes S , this proves (16.4). $\square$

16.3 A simultaneous parameter choice

Definition 16.6. Set

$$x_0 = 10^9, \quad x_{j+1} = J^{x_j}(x_j) \quad (0 \leq j < 5), \quad M_* = x_5. \quad (16.10)$$

These are ordinary computable integers.

Theorem 16.7 (Size ceiling). *The negative affine-chart construction has a choice of finite parameters for which its lattice L is nonrepresentable and*

$$|L| < M_*.$$

In particular a nonrepresentable lattice exists among the tables of cardinality at most the ordinary computable integer M_ .*

Proof. We verify the quantitative requirements of the structural pruning, protected-path, and bounded-root arguments. The parameters are selected in the order

$$x_0 ; \quad D, q ; \quad \ell, r ; \quad \text{backward trial increments} ; \quad m, n ; \quad N, g.$$

This order matters because only the initial graph-pattern round uses a slot count depending on n . The later-round slot counts and the chart losses are fixed before the supply of charts is chosen, as required by [Proposition 5.5](#).

High-type thresholds and structure counts. Use 10^8 as the high cutoff for alternating degree or classical rank; the remaining ranks are at most x_0 . The high automorphism and covering assertions have only the usual low exceptions. In the notation of the direct-system pruning in [Lemmas 6.9](#) and [6.11](#), supported classical groups of natural dimension h have full special order, also after projectivizing, at least

$$s_0^{ch^2-20h},$$

whereas the full isometry or linear order is at most $s_0^{ch^2+20h}$, including in small dimensions. The direct-split upper bound

$$h!s_0^{ch^2/2+20h}$$

is strictly smaller than the supported full special order for $h > 10^6$. The linear comparison in [Lemma 6.14](#) likewise follows from

$$h!|F|^{h^2/4+4} < |F|^{h^2/2-20h} \quad (h > 10^6).$$

Perfection, natural irreducibility, and existence of the supported classical groups already hold there. In the initial subspace pruning of [Propositions 6.4](#) and [6.5](#), the retained small-space irreducibility cutoff may be 100. Completing two such spaces costs dimension at most 400. Summing the endomorphism dimensions at the successive span-growth steps, including the two spans in linear type, bounds the independent-list requirement by fewer than 1000^5 strict steps.

In the direct-block, field, and tensor rounds, the small-special-group, rotation, and bounded-field cases can all use 10^6 as the large cutoff for the indicated dimension or field size. In particular plane rotations then have projective order greater than four when required. Transfer avoids the ordinary small-special exceptions once total ground dimension exceeds 10^6 . This remains so on the complement of a nuisance-plane tensor slot, by either a high transferred dimension or a large extension degree. The compositum-torus test in [Proposition 6.22](#), in total dimension $4s > 10^6$, uses only

$$\frac{|F|^s + 1}{2s} > 2 + |F| + 1.$$

Here $s > 250000$, so the inequality holds already at $|F| = 2$ and continues to hold as $|F|$ increases. The counts here never bound arbitrary matrices over an unbounded field merely from bounded dimension. Instead, the order comparisons bound refinement fibers, comparable split lengths, or the number of distinct elementwise-commuting simple tensor slots, of which at most two are then designated. When matrices inside a bounded plane algebra are counted, the field is bounded too. Exceptional matching counts use the bounded aligned types of a matched pair; only the plane case needs isometry choices per edge orbit, and that case has bounded field.

For the top-action orbit count in [Lemma 6.7](#), take the six successive cutoffs $x_0, J(x_0), \dots, J^5(x_0)$. There are at most four class sizes, so one of the five gaps has no class size strictly between its endpoints. Call sizes at most its lower endpoint small. Small sizes are then at most $J^4(x_0)$, while each large alternating order exceeds 1000 times the product of the small factorials, using J at the actual lower endpoint. The joint action intersected with the product of the large alternatings has

index at most $2^4 \cdot 2^2$ times those factorials: the factors account for parities and the two possible class swaps. This is enough for the subdirect projection test. Equality patterns on at most four large-index occurrences, up to the diagonal identifications, together with class choices and the individual small indices, bound the orbit count by, for example,

$$2^4 \cdot 15^4 (J^4(x_0))^4. \quad (16.11)$$

For fine-cell pairs multiply by at most $(10^6)^2$, from the aggregate split bounds. For the tie-chain test on simple-site pairs, fibers of size at most 2^2 suffice. Point cells in the alternating partition argument have independent supported alternating groups, giving bounded orbits in every ordered-cell-pair fiber as well.

Invariant partition and subset counts exponentiate these orbit bounds once. A further isometry choice on each bounded-field plane edge orbit costs base at most $(10^6)^4$ per orbit. A count inside a single bounded-field plane matrix algebra may even use the entire power-set size. In the shared-component ordered-factor arguments, a matrix centralizer factor of natural degree at most 10^6 permits at most $\log_2(10^6)$ nontrivial elementwise-commuting new tensor slots, over any field. A large-space factor permits only possibly the whole factor as a new one there. Designating one or two components among the original at most four and these additions has bounded cost. Therefore $J^{50}(x_0)$ exceeds all line-count and rich-top tie-chain bounds, including the small-field and matching choices. It also bounds the retained/missing, support, group-type, large/small, and exceptional-case subdivisions, and the categories and rounds per ancestor. Recording even a thousand such tests or small values up to x_0 is within this bound. No color records an arbitrary degree or an arbitrary matrix.

The protected path. For a nontrivial projective representation in defining characteristic and high classical natural dimension v , the Weyl-orbit lower bound $v/4$ from [Lemma 9.3](#) defeats both the smallest tensor slots and the powers $v^{1/s}$, $s \geq 2$. These comparisons already hold at dimension 10^6 in [Proposition 9.8](#), leaving fewer than 10^6 primes when that argument stops. The positive-height fixed-order comparison used in [Proposition 9.9](#) follows from the bounded product factors in high rank: a height step contributes at least $2^{\dim G}$.

For a change of defining characteristic in [Lemma 9.5](#), the paired linear subgroup has dimension $h \geq v_j/3$. Its column-shear characters give $v_i \geq P_j^{h-1} - 1$, and hence, for $v_j \geq 10^6$,

$$\log_2 v_i \geq \frac{v_j}{5} \log_2 P_j. \quad (16.12)$$

A second change would give $v_l \log_2 P_l \leq 5 \log_2 v_j$. Using the uniform order bound $|S_l| \leq P_l^{3v_l^2}$, we obtain

$$\log_2 |S_l| \leq \frac{75(\log_2 v_j)^2}{\log_2 P_l} \leq 75(\log_2 v_j)^2 < \frac{2v_j}{5} \leq 2 \log_2 v_i,$$

contradicting the required representation-degree comparison. The strict inequality holds at $v_j = 10^6$ and thereafter because $v_j/(\log_2 v_j)^2$ increases in this range. For [Lemma 9.4](#), the block count and Bochert bound give index at least $2^{e/7}$ for a proper transitive subgroup in large degree e . The numerical comparison is

$$e! < 2^{(\binom{e}{2}-1)/7},$$

since $e! \leq e^e$ and $14 \log_2 e < e - 1 - 2/e$ for $e \geq 10^6$. The classification of 2-transitive groups has no nonnatural alternating exception at these degrees.

The word tests in [Propositions 9.18](#) and [9.20](#) require only a high constant rank: there must be enough prefix positions and neighbors, and in the spin case the exponentially many sign choices

must exceed the rank. Fewer than 100 pair colors at a fixed series and a homogeneous set of size ten suffice. Thus a high protected path has at most two high alternating nodes and at most one characteristic switch on its classical subsequence, giving at most four consecutive same-characteristic classical runs. Within each run pigeonhole among the four series, and then apply pair Ramsey to the shapes of the weight digits in that series. The total bounds are far below $J^2(x_0)$. We may therefore take

$$D = J^{1000}(x_0) < x_1, \quad q = 7^D < x_1. \quad (16.13)$$

The extra stages allow a displaced point start and a stage with a bounded-type parent. The field order q exceeds every absolute line-count requirement, and $q - 1$ has the distinct prime divisors 2 and 3, as required by the affine-chart construction.

Bounded-type descents and residual length. Use x_1 as the rank bound in [Proposition 16.5](#). If its avoidance hypothesis fails, there is a proper closed X of format at most $J^{1000}(x_1)$. By [Convention 8.3](#), every point map used in [Lemma 8.5](#) has one of the forms

$$\text{Int}(g)\tau \text{Fr}_p^h, \quad \text{Int}(g)\sigma^\epsilon \text{Fr}_p^h, \quad h \in \mathbf{Z}, \quad \epsilon \in \{0, 1\},$$

where τ is a pinned diagram map. Indeed $\sigma^2 = \text{Fr}_p$ and $\sigma^{-1} = \sigma \text{Fr}_p^{-1}$. In the fixed split models, positive and negative field powers change only equation coefficients. The remaining inner, diagram, and special maps have bounded-format graphs, independently of h ; special maps occur only in characteristics two and three. Thus images and inverse images are given by bounded formulas after the coefficient twist. These point maps are Zariski homeomorphisms, so their images of X are closed. Applying the fiberwise equation bound of [Lemma 16.2](#) gives equations of degree less than $J^{1100}(x_1)$ for all the translates and transforms defining Z .

For u ambient coordinates and equation degree at most d_0 , their entire equation span has dimension at most $\binom{u+d_0}{u}$. Selecting a basis therefore replaces the possibly infinite intersection defining Z by that many equations. This elementary cost keeps its format below $J^{1200}(x_1)$. Its conjugation normalizer is defined by the bounded-format condition

$$\forall z \in G \quad (z \in Z \iff gzg^{-1} \in Z).$$

Taking equations for this closed subgroup and its monomial-space dimension keeps the latter below $J^{1400}(x_1)$. The independent-list strict descents securing the sandwich are bounded by this dimension; one or two more list members suffice.

All large-parameter requirements may be imposed by

$$P > J^{2000}(x_1). \quad (16.14)$$

This follows from the fixed-point versus degree comparison, including the zero-height fixed loci. To compare a smaller positive height whose parameter is at least $\sqrt{P}$, the usual product formulas, including the actual central index, give relative errors at most

$$\frac{100(x_1 + 10)^4}{\sqrt{P}}.$$

The minimal leading full-order growth factor per height has integer square and exceeds the relevant center bound $z \leq 10(x_1 + 10)^2$. The displayed error is therefore small enough for the strict order ratio. For parameters below $\sqrt{P}$ the gross order comparison suffices.

All fixed-group orders below the cutoff in [\(16.14\)](#), as well as bounded alternating, sporadic, and exceptional-small socle orders, are below $J^{2010}(x_1)$. Reserving residual length $J^{2020}(x_1)$ enforces

the cutoff by the available index over the base, both below current path nodes and below tested bounded-type flats. In the final bounded-type label estimate $C2^t$, the sandwich quotient permits $C = 2^z$. Hence the increment

$$t = J(x_1) \quad (16.15)$$

suffices.

Flag selection and backward pattern increments. Only the two continuation colors in [Proposition 9.12](#) are needed. Set

$$\ell = J^{2030}(x_1), \quad r = 3\ell. \quad (16.16)$$

Then $Dr < x_2$ and $t < r$. The tensor-prime and root counts give the failure union bound

$$2q^{-\ell}D(3Dr + 10^6), \quad (16.17)$$

which is smaller than the two-color matching fraction $1/3$. Indeed $q^\ell > 6D(9D\ell + 10^6)$ at the displayed choices.

The total number of successive high-pruning rounds in [Proposition 5.4](#) at one stage is less than $x_2/100$: these comprise the simultaneous initial subspace round and the ordered categories and cases relative to at most D ancestors. Every subsequent round uses at most x_2 slots. Compute the required increments backwards, beginning with $s = r$. Suppose the desired next increment satisfies $s \leq y$, where $y \geq x_2$. The current path rank is at most Dr , and the alphabet of coordinate functionals has size at most q^s . A failure color records an ancestor, a bounded category and case, and a pullback subset of at most q^{Dr+s} points. These use fewer than $J(y)$ colors. For example, the subset contribution alone is bounded by

$$2^{q^{Dr+s}} \leq 2^{y^{2y}} < 2^{2^{y!}} < J(y),$$

with ample room for the bounded ancestor and case factors. The alphabet and number of slots also fit below $J(y)$. The definition of J thus bounds the Hales–Jewett vertical length by $J^2(y)$, and $J^3(y)$ accommodates the preceding increment including the horizontal directions. Fewer than $x_2/100$ backward rounds require fewer than $3x_2/100$ iterations, strictly below the x_2 iterations defining x_3 . In particular the increment sought in the initial round is below x_3 .

Truncation, initial slots, and chart supply. Choose

$$\begin{aligned} m &= 2x_3, & n &= 4x_3, & y &= J(x_3), \\ N &= J^4(y), & g &= J^4(y). \end{aligned} \quad (16.18)$$

Path ranks are below x_2 and trial increments below x_3 , so all path and trial ranks lie strictly below m . The additional at most x_2 independent-list and pair-span allowance still lies below m ; residual heights are greater than $2x_3$. These exceed the reserved bounded-type residual length. Initial subspace pruning may ask for more labels than the interval height, but its argument never requires the whole list to have a proper span: only pair spans and the bounded independent lists must fit the truncation.

After m, n are fixed, the initial graph patterns may use $n + x_2$ slots. This fits below $y = J(x_3)$; the color and alphabet bounds fit below $J(y)$ as before. These bounds also cover the direct trial at a bounded-type parent, without any earlier anchor. Thus the value of N in (16.18) suffices. The chart loss in any round is less than $J(y)$, since it is bounded by the ancestor/category/case colors and two matching failed charts suffice for contradiction. Keep one successful flat in each surviving chart for the next round. There are fewer than $x_2/100$ rounds per node, so the choice of g leaves successful charts after all of them. The supply is private to each path node, rather than a common pool for the whole tree. Both N and g are less than x_4 .

These choices implement the pattern procedure: in a failed chart use graphs of horizontal-to-vertical linear maps, or affine maps at the empty start, and color them by the failure data and pullback flat. The Hales–Jewett family fixes the pullback. Varying its slots by scalar multiples of a horizontal functional nonzero at a point of that flat gives distinct witness flats indexed by an affine parameter space. Pair spans add one dimension and contain the whole parameter line; bounded independent lists add one dimension per new point. The chart intersection rules therefore turn these patterns into the line and descent configurations used in the exclusions. Witnesses from two distinct failed charts have label intersection exactly the base group. Successive rounds retain earlier exclusions because the later chosen labels contain the earlier successful anchor. Consequently at each high node a pruned increment- r flat gives the next almost simple socle with its ancestor protections, and the flag selection preserves the required root and tensor data. A bounded-type parent instead gives the direct bounded-complexity contradiction. Thus a putative representation either has that contradiction or forces a protected high path beyond the chosen D . The resulting finite lattice is nonrepresentable.

Counting every lattice element. Take one leaf for each required leaf attachment. A chart has at most

$$F_0 = 2^{q^N} \quad (16.19)$$

flat subsets. There are g initial charts, branching at most gF_0 at each of D generations, and at most F_0 leaf attachments per trunk chart. Thus the number of all charts is at most

$$C = (D + 1)g(1 + gF_0)^D(1 + F_0). \quad (16.20)$$

Both orientations together have at most $2CF_0$ interior labels, and there are two common endpoints. Put $X = x_4$ and $A = J(X)$. Since $q, N < X$, we have

$$q^N \leq X^X < 2^{X^1}, \quad F_0 < 2^{2^{X^1}} < A.$$

Also $D, g < X < A$, so $1 + gF_0 < A^2$ and $(D + 1)g(1 + F_0) < A^4$. Hence $C < A^{2X+4}$ and

$$|L| \leq 2CF_0 + 2 < A^{2X+7} < J(A) = J^2(X) < x_5. \quad (16.21)$$

The middle inequality follows from $J(A) > 2^{2^{A^1}}$ and $A > X \geq 10^9$; the last follows from the x_4 iterations defining x_5 . This counts the endpoints as well as every interior label and proves the claimed ceiling. $\square$

17 A finite description of a minimum lattice

The computable ceiling M_* from [Theorem 16.7](#) can be combined with one ordinary Busy Beaver constant to specify a nonrepresentable lattice of minimum cardinality. The constant is defined independently of lattices and representations. Once it is given, the carrier and every order-table bit are specified by bounded finite arithmetic. All arithmetic and all assertions of termination in this section refer to the ordinary standard natural numbers.

17.1 The register language and its constant

Definition 17.1 (Register programs and size). Registers v_i , $i \geq 0$, hold natural numbers and initially all contain zero. Terms use these registers, the constants 0, 1, addition, truncated subtraction $u \dot{-} v$, multiplication, exponentiation u^v , and integer floor quotient. We set $0^0 = 1$ and give quotient

by zero the value zero. Bounded formulas use the binary predicates $=, <$, the connectives $\neg, \wedge, \vee$, and quantifiers $\exists v_i < t, \forall v_i < t$, with v_i absent from the bound term t . A variable bound within a formula is local to that formula and does not write to a register.

Programs are formed by assignments of terms to registers, binary sequencing, if-then-else with a bounded condition and two program branches, and while with a bounded condition and a program body. They have the usual deterministic semantics. A terminating program outputs its final value of v_0 .

The size of a program is its syntax-tree cost: each operation, constant, predicate, connective, quantifier, assignment, sequencing, if-then-else, or while node costs one. Each occurrence of v_i costs $1 + i$, including assignment targets and quantifier binders. Parentheses cost nothing, and no extra cost is charged for the use of a quantifier's term as its bound.

Definition 17.2 (The Busy Beaver cutoff). With $M_* = x_5$ as in (16.10), set

$$\begin{aligned} k &= 2^{(M_*+10)^4}, \\ B &= 1 + \max\left(\{0\} \cup \{\text{output}(p) : p \text{ terminates and has size} \leq k\}\right). \end{aligned} \quad (17.1)$$

The programs in this definition use exactly Definition 17.1; in particular B is not a symbol in their language.

This is a maximum-output variant of Radó's finite-program extremal construction [42], with the register syntax and cost fixed here. It is not the classical two-symbol state-count function. There are finitely many programs of bounded size, since the cost of a register occurrence bounds its index. Thus B is a definite finite integer. It is an ordinary, unevaluated Busy Beaver constant, defined by termination and output of programs without an oracle.

17.2 Bounded arithmetic for lattice tables and witnesses

Definition 17.3 (Digits and order tables). For $b \geq 1$, put

$$\text{dig}_b(z, i) = \left\lfloor \frac{z}{b^i} \right\rfloor \dot{-} b \left\lfloor \frac{z}{b^{i+1}} \right\rfloor. \quad (17.2)$$

Base 1 is used only where the intended digit is zero, in which case this formula indeed gives zero. For $n \geq 1$ and $0 \leq t < 2^{n^2}$, let $T_{n,t}$ be the relation on $\{0, \dots, n-1\}$ defined by

$$a \preceq b \iff \text{dig}_2(t, an + b) = 1. \quad (17.3)$$

Write $E(a, b)$ for the right side, with n, t understood. The bounded predicate $P(n, t)$ says that $T_{n,t}$ is a lattice: it is the conjunction of

$$\begin{aligned} \forall a < n \ E(a, a), \\ \forall a, b < n \ ((E(a, b) \wedge E(b, a)) \implies a = b), \\ \forall a, b, c < n \ ((E(a, b) \wedge E(b, c)) \implies E(a, c)), \end{aligned}$$

and the following two assertions for every $a, b < n$:

$$\begin{aligned} \exists u < n \ (E(a, u) \wedge E(b, u) \wedge \forall v < n \ ((E(a, v) \wedge E(b, v)) \implies E(u, v))), \\ \exists l < n \ (E(l, a) \wedge E(l, b) \wedge \forall v < n \ ((E(v, a) \wedge E(v, b)) \implies E(v, l))). \end{aligned}$$

Repeated bounded variables denote successive quantifiers with the displayed bound. Implication, equivalence, and non-strict comparison throughout this section abbreviate their expressions using the primitive connectives and $=, <$.

The next definition gives a bounded formula for the graph criterion, including the admissible placements and the paths. All bound variables are renamed when abbreviations are expanded, to avoid capture.

Definition 17.4 (The witness predicate). For $j \geq 1$, the formula $Q(n, t, j)$ asserts that there is

$$z < n^{j^2}, \quad d(p, q) = \text{dig}_n(z, pj + q) \quad (p, q < j), \quad (17.4)$$

with the following properties.

First, the colors are symmetric, bottom occurs exactly on loops, and the triangle inequality holds. Explicitly, set $\text{Bot}(a) \equiv \forall b < n \ E(a, b)$ and require

$$\begin{aligned} \forall p, q < j \quad d(p, q) &= d(q, p), \\ \forall p, q < j \quad (\text{Bot}(d(p, q)) &\longleftrightarrow p = q), \\ \forall p, q, u < j \quad \forall b < n \quad (E(d(p, u), b) \wedge E(d(u, q), b)) &\implies E(d(p, q), b). \end{aligned} \quad (17.5)$$

Second, require order reflection:

$$\forall a, b < n \quad \neg E(a, b) \implies \exists p, q < j \ (E(d(p, q), a) \wedge \neg E(d(p, q), b)). \quad (17.6)$$

For the third requirement, let $0 < s < 2^{j^2}$ be a nonempty ordered-pair mask and write $\text{Sel}_s(p, q) \equiv \text{dig}_2(s, pj + q) = 1$. A number $f < j^j$ specifies a vertex map by $f[p] = \text{dig}_j(f, p)$. Set

$$\text{Adm}(f) \equiv \forall p, q < j \ E(d(f[p], f[q]), d(p, q)). \quad (17.7)$$

The derived undirected adjacency is the bounded predicate

$$\begin{aligned} \text{Adj}_s(r, r') \equiv \exists f < j^j \quad (&\text{Adm}(f) \wedge \exists a, b < j \ (\text{Sel}_s(a, b) \wedge \\ &((f[a] = r \wedge f[b] = r') \vee (f[a] = r' \wedge f[b] = r))))). \end{aligned} \quad (17.8)$$

For $h < j^{j+1}$, put $h[i] = \text{dig}_j(h, i)$, and define

$$\begin{aligned} \text{Path}_s(p, q) \equiv \exists h < j^{j+1} \quad (&h[0] = p \wedge h[j] = q \wedge \\ &\forall i < j \ (h[i] = h[i+1] \vee \text{Adj}_s(h[i], h[i+1]))). \end{aligned} \quad (17.9)$$

Write

$$\begin{aligned} \text{Up}_s(b) &\equiv \forall a, c < j \ (\text{Sel}_s(a, c) \implies E(d(a, c), b)), \\ \text{BelowJoin}_s(p, q) &\equiv \forall b < n \ (\text{Up}_s(b) \implies E(d(p, q), b)). \end{aligned} \quad (17.10)$$

The third requirement is

$$\forall s < 2^{j^2} \quad s \neq 0 \implies \forall p, q < j \ (\text{BelowJoin}_s(p, q) \implies \text{Path}_s(p, q)). \quad (17.11)$$

The conjunction of all three initial color requirements—symmetry, the bottom condition (17.5), and the triangle condition— with (17.6) and (17.11), under the existential quantifier in (17.4), is $Q(n, t, j)$.

Every expression just given expands to the register language's bounded arithmetic. All element and vertex variables have bounds n and j ; the remaining bounds n^{j^2} , 2^{j^2} , j^j , and j^{j+1} are terms. In particular the definition uses neither an unbounded path quantifier nor a quantifier over a function as a separate sort.

Lemma 17.5 (Meaning of the arithmetic witness). *For every table satisfying $P(n, t)$,*

$$\mathcal{R}(\{0, \dots, n-1\}; \preceq) \iff (\exists j \geq 1) Q(n, t, j). \quad (17.12)$$

Moreover $Q(n, t, j)$ holds exactly when the colored-graph criterion has a witness with j vertices.

Proof. For a table satisfying $P(n, t)$, the digit encoding lists all color tables on j vertices. The predicate Bot identifies the least lattice element regardless of its numerical label. Being below every common upper bound is equivalent to being below the join, so the triangle clause and BelowJoin_s express the corresponding lattice inequalities in [Theorem 2.2](#).

The placement codes list all selfmaps of the vertex set. The nonzero ordered-pair masks select exactly the same edge sets as undirected masks: colors are symmetric, and Adj_s permits either ordering of the endpoints. Every connected pair has a path of at most $j - 1$ edges; equality steps pad it to the j steps encoded by h . Conversely, every encoded path certifies connectivity. The base-one digit convention covers $n = 1$ and $j = 1$, and the masks include loops. Thus $Q(n, t, j)$ holds exactly when there is a graph witness with j vertices. The representability equivalence now follows from [Theorem 2.2](#). $\square$

17.3 The finite table selection

Definition 17.6 (Carrier and order). Let

$$\mathcal{I} = \{(n, t) : 1 \leq n \leq M_*, 0 \leq t < 2^{n^2}\}$$

have increasing pair-lexicographic order, with n compared first. Define

$$z_{n,t} = \begin{cases} 1, & P(n, t) \text{ and } (\forall j = 1, \dots, B) \neg Q(n, t, j), \\ 0, & \text{otherwise,} \end{cases} \quad (17.13)$$

and, with the empty product equal to one,

$$w_{n,t} = z_{n,t} \prod_{\substack{(m,u) \in \mathcal{I} \\ (m,u) < (n,t)}} (1 - z_{m,u}). \quad (17.14)$$

Set

$$a_0 = \sum_{(n,t) \in \mathcal{I}} n w_{n,t}, \quad t_0 = \sum_{(n,t) \in \mathcal{I}} t w_{n,t}. \quad (17.15)$$

The structure L_0 has carrier $\{i : 0 \leq i < a_0\}$ and order

$$i \leq_{L_0} l \iff \text{dig}_2(t_0, i a_0 + l) = 1. \quad (17.16)$$

All products, sums, and tests in this definition are bounded finite constructions once B is fixed. We next prove that the cutoff is exhaustive on this entire initial segment, so that a failed bounded test selects a negative instance.

Lemma 17.7 (Size of a graph-witness search). *For $1 \leq n \leq M_*$ and $0 \leq t < 2^{n^2}$, there is a program of [Definition 17.1](#), of size less than k , which terminates exactly when $Q(n, t, j)$ holds for some positive j . On termination it outputs the least such j . The program contains neither B nor any oracle instruction.*

Proof. Fix this particular pair n, t . Consider the following program schema, where the first two right-hand sides are arithmetic terms for the specified binary numerals and Q is expanded as in Definition 17.4:

$v_1 := \text{the numeral for } n;$
 $v_2 := \text{the numeral for } t;$
 $v_3 := 1;$
while $\neg Q(v_1, v_2, v_3)$ **do** $v_3 := v_3 + 1;$
 $v_0 := v_3.$

Binary sequencing makes this a program of Definition 17.1. Each test of Q is bounded and therefore terminates. The loop tests the positive integers in order, proving the claimed termination and output properties without assuming $P(n, t)$. Writing the numerals by binary Horner expressions uses only 0, 1, addition, and multiplication. Since $t < 2^{n^2}$, the two numeral terms cost at most $100(n+1)^3$ nodes, with ample allowance for their fixed register targets.

The bounded formula templates and the rest of the program schema are fixed independently of n, t . Their expansion can be counted directly. Use $2 = (1 + 1)$, replace $A \Rightarrow B$ by $\neg A \vee B$, replace $A \Leftrightarrow B$ by $(\neg A \vee B) \wedge (\neg B \vee A)$, and use binary conjunctions and one binder for each successive quantifier. Substitute every abbreviation separately, without shared subexpressions or loop unrolling. Counting a variable occurrence initially as one gives:

Clause	Nodes	Variable occurrences	Binders
Symmetry	53	26	2
Bottom condition	173	68	4
Triangle condition	240	92	4
Order reflection	198	72	4
Mask and connectivity	717	281	13
Full Q	1394	542	28

The last line adds four conjunction nodes and the outer existential: its binder and the seven-node bound $n^{j^{1+1}}$ contribute nine nodes with the quantifier itself. Renaming the expanded binders consecutively as $v_{100}, \dots, v_{127}$ avoids all capture; the bound of each quantifier is renamed in the preceding environment. Only v_1, v_2, v_3 remain free. Every node then costs at most 128, so Q costs at most $128 \cdot 1394 = 178432$. Outside the two numeral assignments, the rest of the program costs 29: initialization costs 6, the while and negation nodes cost 2, increment costs 11, final output costs 6, and the four sequencing nodes cost 4. In particular the following much larger allowance bounds the entire program:

$$100(n+1)^3 + (2 \cdot 10^6)^{204} < 2^{(M_*+10)^4} = k. \quad (17.17)$$

The second summand exceeds 178461. For the strict inequality, $M_* + 1 \leq 2^{M_*}$ bounds the first summand by 2^{3M_*+7} , and $2 \cdot 10^6 < 2^{21}$ bounds the second by 2^{4284} . Since $M_* > 10^9$, their sum is less than 2^{3M_*+8} , which is less than $2^{(M_*+10)^4}$. □

Lemma 17.8 (Uniform Busy Beaver witness cutoff). *If $1 \leq n \leq M_*$, $0 \leq t < 2^{n^2}$, and $P(n, t)$ holds, then*

$$\mathcal{R}(T_{n,t}) \iff (\exists j \in \{1, \dots, B\}) Q(n, t, j). \quad (17.18)$$

In fact every representable such table has a witness of size strictly less than B .

Proof. For a representable table, Lemma 17.5 makes the program in Lemma 17.7 terminate with a witness size j . Its size is less than k , so the definition of B gives $j < B$. Conversely, any $j \leq B$ satisfying $Q(n, t, j)$ supplies a representation by Lemma 17.5. $\square$

Theorem 17.9 (Finite description of a minimum example). *The bounded formulas (17.13)–(17.16), with the ordinary unevaluated Busy Beaver constant (17.1), specify the full finite order table of a nonrepresentable lattice L_0 of minimum cardinality. Its cardinality is the integer a_0 in (17.15), and*

$$7 < a_0 \leq M_*.$$

This is a finite order description and a proof of minimum cardinality; it does not supply a numerically evaluated value of that cardinality.

Proof. By Lemma 17.8, for every index in $\mathcal{I}$ the bit $z_{n,t}$ is one exactly when the table is a nonrepresentable lattice. By Theorem 16.7, some such bit is one. Let (n_*, t_*) be the first index for which it is one. The product in (17.14) is one at that index; at each later index it contains the zero factor $1 - z_{n_*, t_*}$, and at each earlier index the factor $z_{n,t}$ is zero. Hence exactly one $w_{n,t}$ equals one. Formula (17.15) gives $a_0 = n_*$ and $t_0 = t_*$, so (17.16) is precisely this lattice’s full order table. It is nonrepresentable.

Every nonempty lattice of cardinality smaller than a_0 has an order table among earlier indices in $\mathcal{I}$. Its selection bit is zero and its lattice predicate is true, so the cutoff lemma makes it representable. Thus L_0 has minimum cardinality among all finite nonrepresentable lattices. The represented range through seven elements in Theorem 3.4 gives $a_0 > 7$, and its index gives $a_0 \leq M_*$. Nothing in this argument evaluates B : all its uses concern the outputs of the terminating programs in its definition. The result is therefore the claimed finite description with an exhaustive bound, rather than a numerical enumeration. $\square$

The constants and entries above are unambiguous in the standard natural numbers. Their definition supplies no ordinary computable procedure for evaluating the Busy Beaver constant, no evaluated decimal value of a_0 , and no conventional small diagram of the minimum lattice. Those would be stronger forms of identification than the finite description proved here.

Proof of Theorem 1.1. Part (i) is Theorem 2.2; Part (ii) is Theorem 15.3. The finite description and its minimum-cardinality property are proved in Theorem 17.9. The strict lower bound of seven follows from Theorem 3.4. Part (iv) is Corollary 15.4. $\square$

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